

As in the first edition, this book has been carefully edited to make it accessible to the English-speaking student. I have translated the new material myself, taking pains to ensure that the vocabulary is appropriate for an undergraduate student. In order to bring out this new edition as rapidly as possible, the translation was made directly from the German manuscript.

Students will certainly find this text useful. Research scientists and technicians already employed in industrial microbiology will also benefit from this practical overview of the field. In addition to industrial microbiologists, others who may find this book valuable include food scientists, environmental scientists, chemical engineers, and organic chemists.

The authors and I are pleased that we are able to offer this modern, attractive, accessible book, which should be of great value to both students and practicing researchers.

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Biotecndology. Text book of Contents Industrial Microbiology

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Biotechnology is the use of microbiology, biochemistry, and engineering in an integrated fashion with the goal of using microorganisms and cell and tissue cultures (or their parts) to manufacture useful products. Biotechnology can be divided into two categories which are sometimes called "traditional biotechnology" and "new biotechnology". The major products of the traditional biotechnology industry are food and flavor ingredients, industrial alcohol, antibiotics, and citric acid. These products amount, on a worldwide basis, to about 300 billion dollars annually. The new biotechnology, which involves the use of the newer techniques of genetic engineering and cell fusion to produce organisms capable of making useful products, provides at present products with a total value of less than a billion dollars. In the future, however, it is predicted that the new biotechnology will account for a much larger fraction of the total biotechnology industry.

Industrial microbiology, the major foundation of biotechnology, arose out of empirical de-

velopments in the production of wine, vinegar, beer, and sake, and with the traditional fungal fermentations used in Asia and Africa for the production of food. An experimental approach to the production of microbial metabolites only began at the beginning of the 20th century. Up until the time of World War II, the main microbial products that had developed from this experimental approach were enzymes such as proteases, amylases, and invertase.

A major breakthrough in biochemical and microbial engineering occurred after World War II as a result of the large-scale production of the first antibiotic, *penicillin*. In order to produce this antibiotic economically, important engineering developments had to be made, including the development of techniques for large-scale sterilization, aeration, and growth of microorganisms. In addition, genetic methods for microbial strain improvement were perfected.

From World War II up until about 1960, the major new biotechnology products were antibiotics. Through intense efforts of the pharma-

1 Introduction

1

ty, numerous new antibiotics were developed of these around 1970 were put into production. In addition, in this early period, processes were developed for the chemical transformation of the culture of animal cells for the production of vaccines was perfected.

From 1960 through 1975, new processes for the production of amino nucleosides as flavor enhancers developed, primarily in Japan. In addition, processes for enzyme production for lytic, and medical purposes were developed in this same period, successful in the immobilization of enzymes. During this time a further development was the use of continuous fermentation for the production of single-cell protein and bacteria for use as human food. Single-cell protein processes using microorganisms capable of producing starting materials such as polyols and methanol. In this same period, polymers such as xanthan and food additives, were also developed. Commercial processes, somewhat advanced during this period, use of microorganisms for the production of antibiotics (an aspect of geomicrobiology) and techniques for anaerobic microorganisms, derived out of sewage treatment process.

Biotechnology has entered some phases. First was the development of the recombinant DNA technique for the production of antibodies, of interest in the diagnosis field. Soon after the production of human proteins using *Escherichia coli*. The first insulin was introduced in 1982, Factor VIII, human growth hormone, and urokinase. At present, a number of human proteins are in the development stage.

production of human proteins and bacteria is generally recognized as

Table 1.1 Patent applications in 1984 for three countries with major biotechnology industries.

Country	Molecular biology patents	Fermentation patents (including enzymes)
USA	100	160
Japan	90	700
Federal Republic of Germany	12	55

the major "highlight" of the period since 1975, in actuality other products are economically more important. For instance, the production of ethanol by immobilized cells has become a major process. The enzyme *glucose isomerase* has become a 27 million dollar industry and is used to produce high-fructose syrup which itself has a value of 2.5 billion dollars. *Aspartame*, a major artificial sweetener, is produced microbially. Many new antibiotics have been introduced. Cheap fats are being increased in value by enzymatic esterification, the enzymes being microbial products. The biodegradation of persistent chemicals using specially developed microbial strains as starter cultures is being field-tested.

To place in perspective research activities in traditional biotechnology and the "new" biotechnologies (using genetic engineering, etc.), Table 1.1 provides a comparison of the number of patent applications in the whole field of biotechnology for three major industrial countries, U.S.A., Japan, and the Federal Republic of Germany. As can be seen, traditional biotechnologies still dominate, especially in Japan.

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ceutical industry; numerous new antibiotics were discovered and of these around 20 were put into commercial production. In addition, in this early post-World-War-II period, processes were developed for the chemical transformation of steroids, and the culture of animal cells for the production of virus vaccines was perfected.

In the period from 1960 through 1975, new microbial processes for the production of amino acids and 5'-nucleosides as flavor enhancers were developed, primarily in Japan. In addition, numerous processes for enzyme production for industrial, analytic, and medical purposes were perfected. During this same period, successful techniques for the immobilization of enzymes and cells were developed. During this time a further development was the use of continuous fermentation for the production of single-cell protein from yeast and bacteria for use as human and animal food. Single-cell protein processes were developed using microorganisms capable of using petroleum-based starting materials such as gas oil, alkanes, and methanol. In this same period, microbial biopolymers such as xanthan and dextran, used as food additives, were also developed into commercial processes. Somewhat distinct processes that were advanced during this period were the use of microorganisms for tertiary oil recovery (an aspect of geomicrobiology) and the perfection of techniques for anaerobic cultivation of microorganisms, derived out of studies on the sewage treatment process.

Since 1975 biotechnology has entered some important new phases. First was the development of the hybridoma technique for the production of monoclonal antibodies, of interest primarily in the medical diagnosis field. Soon after was the production of human proteins using genetically engineered *Escherichia coli*. The first product, human insulin was introduced in 1982, followed soon by Factor VIII, human growth hormone, interferons, and urokinase. At present, a vast array of human proteins are in the development stage.

Although the production of human proteins by engineered bacteria is generally recognized as

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2

Screening for new metabolites

2.1 GENERAL

The biochemical capabilities of microorganisms are vast, and a wide variety of new or unusual compounds may be produced by various microbial isolates. One of the main tasks of the industrial microbiologist is to develop procedures for obtaining new microbial metabolites. There are five distinct approaches:

1. Screening for the production of new metabolites with new isolates and/or new test methods. This is the only way to obtain completely new classes of substances.
2. Chemical modification of known microbial substances.
3. Biotransformation (Chapter 15), which results in change in a chemical molecule by means of a microbial or enzymatic reaction.
4. Interspecific protoplast fusion (Section 3.4), which is a means of recombining genetic information from rather closely related producer strains. New or hybrid substances are ex-

pected and the method is widely used in the antibiotic industry.

5. Gene cloning (Section 3.6), in which genes may be transferred between unrelated strains which are producers of known substances. Alternatively, transfer may be to nonproducers which contain "silent" genes, leading to the generation of modified or even new substances. This will undoubtedly be the method of choice in the future.

In order to be successful, screening must be an interdisciplinary activity, combining the activities of microbiology, chemistry, biochemistry, engineering, and bibliographics. Microbiology is involved in the isolation and identification of microorganisms, strain preservation, testing for biological activity, and fermentation practice. Biochemistry provides the analytical procedures needed as well as the approaches to purification of the biologically interesting molecules whereas synthesis of substrates and inhibitors falls under the purview of the chemist. The bibliographic

specialist searches the literature as well as the various computerized data bases. The engineer's activity focuses on the development of technical equipment needed for the successful process.

Screening can never be considered a routine activity, since the methods must always be adapted to the newest techniques and knowledge. The goal is always to detect and identify new substances of commercial interest and to separate them in the quickest possible way from the numerous easily detected substances that are of no commercial interest. Some of the most intelligent screening methods that are currently used are the products of Japanese scientific activity, particularly the group of Omura. For instance, 42 completely new compounds were found by Omura (1986) using systems that detected microbially produced substances with antibacterial, antimycoplasmal, antianaerobe, antifungal, antiparasite, and antitumor activity. Substances were also found that acted as herbicides and as inhibitors of penicillin, elastase, and adenosine deaminase.

Further overviews of the newer screening methods are given by Vandamme (1984), Box (1985), Verral (1985), Cheetham (1987), and Elander (1987).

2.2 PRIMARY AND SECONDARY METABOLITES

BuLock borrowed the term "secondary metabolite" from plant physiology in 1961 and applied it to microbiology. While primary metabolites are essential for life and reproduction of cells, and primary metabolism functions similarly in all microorganisms, the following is true for secondary metabolites:

- Every secondary metabolite is formed by only a few organisms.
- Secondary metabolites are seemingly not essential for growth and reproduction.
- Their formation is extremely dependent on environmental conditions.
- Some secondary metabolites are produced as

2.2 PRIMARY AND SECONDARY METABOLITES / 5

a group of closely related structures; one strain of *Streptomyces*, for example, produces 32 different anthracyclines.

- Some organisms form a variety of different classes of substances as secondary metabolites.
- The regulation of the biosynthesis of secondary metabolites differs significantly from that of the primary metabolites.

There are several hypotheses about the role of secondary metabolites, of which Hans Zahner's, illustrated in Figure 2.1, is the most elegant. Besides the five phases of the cell's own metabolism (intermediary metabolism, regulation, transport, differentiation and morphogenesis), secondary metabolism is considered a "playing field" for the evolution of further biochemical development, which can proceed without damaging primary metabolism. Genetic changes leading to the modification of secondary metabolites would be expected not to have any major effect on normal cell function. If a genetic change leads to the formation of a compound that in some way is beneficial, then this genetic change would be fixed in the cell's genome, perhaps becoming essential. In this way the former secondary metabolite would be converted into a primary metabolite.

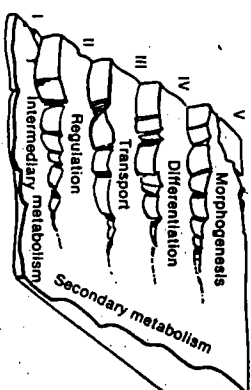


Figure 2.1 The five levels of primary metabolism with the "playing field" of secondary metabolism (Zähner, 1979).

Screening for new metabolites

There are no universal screening methods. The success of a screening program depends upon the selection of appropriate tests as well as appropriate microorganisms to be tested. The capacity of an industrial screening group for isolation of microorganisms and thorough testing is around 100-2000 strains per year.

Today, most screening programs focus on hemotherapeutically useful products for the following areas: activity against antibiotic-resistant strains, tumors, and viruses, as well as a search for enzyme inhibitors and pharmacologically active substances (hormones, etc.). Better starter cultures for the food industry as well as microorganisms that are capable of degrading hazardous and persistent chemicals are also sought.

Of the 10,000 antibiotically active compounds known in the late 1980's, 67% are produced by microorganisms (67% by actinomycetes, 9% by other bacteria, and 15% by fungi). Additionally, about 2000 other biologically active secondary metabolites are known, as well as a large number of enzymes.

3 STRAINS USED IN SCREENING

The success of a screening program depends on both the kinds of organisms used and the methods for detection of activity. Currently, the choice of strain has a 30-40% influence on the outcome, the test procedure a 60-70% influence.

A gram of soil contains between 10^4 - 10^6 bacteria, 10^4 - 10^5 actinomycete spores, and 10^3 - 10^4 fungal spores. Less than 1% of the world's microorganisms have been intensively studied. Above all, the approximately 100,000 known fungi have been poorly studied, so that a vast number of new natural products can be expected from this group in the future.

In the isolation of new metabolic products, researchers try to isolate strains from extreme or unusual environments in the hope that such strains may be capable of producing new metabolites. For instance, microorganisms from high

altitudes, cold habitats, sea water, deep sea, deserts, geysers, and petroleum fields are being examined. Depending on the inoculum source and enrichment procedure, specific groups of organisms may be isolated. Table 2.1 presents some examples of the kinds of organisms that can be isolated with various enrichment methods. The isolation of strains can be carried out with the following scheme: 1) The soil or water sample is suspended in a definite amount of sterile water to which Tween has been added as an emulsifying agent. The sample is vigorously agitated. 2) The supernatant is diluted 10^{-1} - 10^{-10} . 3) Samples from this dilution series are plated on various culture media and then incubated. 4) Single colonies from the plates are picked and purified by restreaking. 5) The pure strains are maintained as agar cultures in test tubes.

The screening procedure can often be speeded up by testing the initial isolates directly for biological activity. Some examples of procedures that can be carried out directly on agar plates are given in Table 2.2. The soil or water samples are diluted directly onto the test plates and only those colonies showing activity are isolated.

The variability of metabolites produced by individual genera is somewhat limited except in the streptomycetes. For instance, except for industrial enzymes, *Bacillus* strains almost exclusively produce peptide antibiotics.

In one extensive study, 20,000 *Actinoplanes* strains were isolated, of which 13,000 were screened for the formation of antibiotics. Within

Table 2.1 Enrichment of microorganisms by selection of appropriate culture conditions

Enrichment methods	Type of isolate
Extreme pH values (pH 2-4)	Acidophiles
Low temperatures (4-15°C)	Psychrotrophs
High temperatures (42-100°C)	Thermotrophs
High NaCl concentrations	Nocardia, halophiles
N ₂ atmosphere	Anaerobes
Chitin as growth substrate	Lysobacter
Wood bark, roots	Mycobacteria
Pollen grains	Actinoplanes

Table 2.2 Test systems for screening of metabolites

Product sought	Test system
Antibiotics	Agar plates with strains of test microorganisms*. Inhibition zones as indicator of activity
β -Lactamase-resistant antibiotics	Agar plates with test microorganisms to which β -lactamase has been added
Proteases	Agar plates with casein, selection of colonies which produce clear zones on the turbid plates
Amylases	Agar plates with starch, selection of colonies after staining with iodine
Lipases	Agar plates with oil emulsion, selection of colonies after precipitation of free fatty acids with Ca ²⁺
Phosphatases	Agar plates with phenolphthalein-diphosphate and pH indicator, selection based on color change
NAD	Agar with microorganism auxotrophic for NAD

*For example, *Staphylococcus aureus*, *Proteus vulgaris*, *Candida albicans*, *Penicillium notatum*, bacteria resistant to or hypersensitive to aminoglycoside, macrolides, or β -lactam antibiotics.

a ten year span, 41 new antibiotics were isolated and characterized from these strains. These antibiotics turned out to be almost all either acetylmalonyl or amino acid derivatives.

2.4 TEST SYSTEMS

The success of a screening procedure is quite dependent on the development of "intelligent" tests with which known or undesirable antibiotics can be eliminated and compounds with the required properties can be recognized. For instance, in one screening program (Omura et al., 1979), a procedure was set up to discover inhibitors of cell wall biosynthesis, because all known antibiotics with this mode of action have low toxicity. In the first phase, screening was done for metabolites which inhibited *Bacillus subtilis*, but did not inhibit *Acholeplasma laidlawii* (which lacks a cell wall). In the second phase, substances were sought which inhibited the synthesis of

meso-diaminopimelic acid, a component of the cell wall of bacteria, but did not inhibit the incorporation of leucine, an indicator of protein synthesis. In the third phase, substances whose molecular weights were greater than 1000 were eliminated by use of membranes, because larger molecules often elicit undesirable side effects when used therapeutically. With these three procedures, culture filtrates of 10,000 strains (fungi, bacteria and actinomycetes) were screened. A new antibiotic, azureomycin, was discovered, and six known antibiotics were also reisolated.

The antibiotic penicillin is a β -lactam, and one mode of resistance is through the production of β -lactamase, an enzyme which splits the β -lactam ring. Inhibitors of β -lactamase might thus prove useful in permitting penicillin therapy against resistant organisms. For the screening of microbial β -lactamase inhibitors, supernatants of the cultures were placed on agar plates containing penicillin or cephalosporin and one of the β -lactamase-producing microorganisms. Thus it could be determined during the first screening whether different β -lactamases could be inhibited.

Nozaki et al. (1987) used a screening procedure with β -lactam-hypersensitive mutants to isolate the antibiotic lactvicin, an antibiotic which combines with the penicillin-binding site of gram-negative and gram-positive bacteria even though it lacks a β -lactam ring.

Fleck and Strauss (1975) used molecular biology tests to discover an antitumor metabolite. Continuous fermentation can provide a method to isolate from mixed cultures strains with improved properties. By raising the temperature or the alcohol concentration, microorganisms can be selected that are either thermophilic or alcohol tolerant. Similar tests can be used to isolate strains that produce temperature-stable extracellular enzymes.

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3.1 GENERAL

With the exception of the food industry, only a few commercial fermentation processes use wild strains isolated directly from nature. Mutants or recombinants which are specifically adapted to the fermentation process are used in the production of antibiotics, enzymes, amino acids, and other substances. The objective of a genetic strain improvement program depends on the process.

In general, the major motivation for industrial strain development is economic, since the metabolite concentrations produced by wild strains are usually too low for economical processes. Through an extensive strain development program (which may require several years), yield increases up to 100 times or more can usually be attained. The success of these programs depends greatly on the substance to be examined. For instance, the yield from products involving the activity of one or a few genes, such as enzymes, can be increased simply by raising the gene dose. However, with secondary metabolites, which are

frequently the end result of complex, highly regulated biosynthetic processes, a variety of changes in the genome may be necessary to permit the selection of high-yielding strains.

For a cost-effective process, strains with improved fermentation properties may also be needed. Depending on the system, it may be desirable to isolate strains which require shorter fermentation times, which do not produce undesirable pigments, which have reduced oxygen needs, with lower viscosity of the culture so that oxygenation is less of a problem, which exhibit decreased foaming during fermentation, which are able to metabolize inexpensive substrates with tolerance to high concentrations of carbon or nitrogen sources, or with resistance to bacteriophage.

Wild strains frequently produce a mixture of chemically closely related substances. Mutants which synthesize one component as the main product are preferable, since they make possible a simplified process for product recovery. Changes in the genotype of microorganisms

Strain development

3

ad to the biosynthesis of new metabolites, mutants which synthesize modified antibiotics may be selected.

One of the most significant approaches to strain improvement can be anticipated from the use of recombinant DNA techniques. Bringing together in one organism genes from several organisms has the potential for not only increasing yields but also for producing entirely new substances. Perhaps of even greater significance is the use of recombinant DNA techniques for the production by microbes of nonmicrobial products, such as insulin, somatostatin, human growth hormone, virus vaccines, and interferon. However, one difficulty in applying the newer genetic techniques to improvement of existing processes is that the organisms of widest use in industry are unfortunately not the organisms for which the greatest amount of basic genetic information is available. There is an apparent gap between basic research and industrial application. Biosynthesis and regulation, along with the genetic fundamentals of industrially important microorganisms, must be understood before the combination of empirical procedures currently used can be replaced by appropriate new approaches. In the last ten years in several industrial firms, steps to bridge the gap between knowledge and industrial application have been made, and we can anticipate a marked increase in the effectiveness of strain improvement programs. Protoplast fusion, site-directed mutagenesis, or recombinant DNA methods are examples of the use of newer technologies which have been especially useful in the production of primary metabolites such as amino acids, but are finding increasing use in strain development programs for antibiotics.

In the present chapter we present the fundamental genetic approaches to strain improvement and show how these approaches are used in practice.

MUTATION

Introduction

Changes in the genotype are caused by **mutagenetic recombination**. In a balanced strain

development program each method should complement the other. In the past, procedures for achieving genetic recombination in industrially important production strains hardly existed, so that the spectacular successes of strain development in industry are basically due to the extensive application of mutation and selection. Table 3.1 shows that the success of this method has been impressive. However, current data on the production levels of industrial high-performance mutants are rarely published, so that the data in Table 3.1 may not show the full extent of development.

Although yields of penicillin in the early days of industrial production were less than 100 units/ml, today the penicillin yield is around 85,000 units/ml (approx. 50 g/l). Because of the low yield per weight of substrate used (weight of penicillin produced per weight of glucose used is around 0.12), continued increases can be expected in the future. The success of these purely empirical strain development programs depends on an optimal use of mutagenesis procedures in combination with an effective system for selecting high-yielding strains.

Spontaneous and induced mutations

Mutations occur *in vivo* spontaneously or after induction with mutagenic agents. Mutations can also be induced *in vitro* by the use of genetic engineering techniques. The rate of spontaneous mutation depends on the growth conditions of the organism and is between 10^{-10} and 10^{-7} per generation and per gene; usually the mutation

Table 3.1 Increase in antibiotic production through mutation and selection between 1943-1961

Antibiotic	Productivity at time of discovery units/ml	Productivity of high yield mutants units/ml
Penicillin	20 (1943)	8000 (1955)
Streptomycin	50 (1945)	5000 (1955)
Erythromycin	100 (1955)	2000 (1961)
Chlortetracycline	200 (1948)	4000 (1959)
Oxytetracycline	400 (1950)	6000 (1959)

(Alikhanian, 1962)

rate is between 10^{-7} and 10^{-8} . All mutant types are found among spontaneous mutations, although deletions are relatively frequent. The causes of spontaneous mutations which are thus far understood include integration and excision of transposons, along with errors in the functioning of enzymes such as DNA polymerases, recombination enzymes, and DNA repair enzymes. Because of the low frequency of spontaneous mutations, it is not cost-effective to isolate such mutants for industrial strain development. The mutation frequency (proportion of mutants in the population) can be significantly increased by using mutagenic agents (mutagens); it may increase to 10^{-5} - 10^{-3} for the isolation of improved secondary metabolite producers or even up to 10^{-2} - 10^{-1} for the isolation of auxotrophic mutants.

Spontaneous and induced mutants arise as a result of structural changes in the genome:

- **Genome mutation** may cause changes in the number of chromosomes.
- **Chromosome mutation** may change the order of the genes within the chromosome, e.g., by deficiency, deletion, inversion, duplication, or translocation.
- **Gene or point mutations** may result from changes in the base sequence in a gene.

Most information is available about point mutations. Through base substitution, a base pair in the wild type allele may be replaced by another base in the mutant allele. Several kinds of changes are recognized. A **transition** is an exchange of a purine with another purine or a pyrimidine with another pyrimidine. A **transversion** refers to the substitution of a pyrimidine with a purine or vice versa. One characteristic of point mutants is that they can revert. Another category includes those called **frameshift mutations**, which result when one nucleotide or more is inserted or deleted, thus altering the reading frame in the following transcription and translation processes, and leading to a changed amino acid sequence in the resulting protein.

Although genome mutations are important in plant genetics, mutations used in microbial strain improvement usually are point mutations; chromosome mutations also occur (e.g., deletions, duplications) but are of minor significance.

Repair mechanisms

Mutant formation is a complex process. Premutational lesions of DNA occur spontaneously or through the action of mutagenic agents; only part of these result in stable mutants during subsequent replication. The following structural changes occur in DNA:

- Pyrimidine dimers, in which two adjacent pyrimidines on a DNA strand are coupled by additional covalent bonds and thus lose their ability to pair.
- Chemical changes of single bases, such as alkylation or deamination, thus causing changes in the pairing properties of the DNA.
- Crosslinks between the complementary DNA strands, which prevent their separation in replication.
- Intercalation of mutagenic agents into the DNA, causing frameshift mutations.
- Single-strand breaks.
- Double-strand breaks.

These premutational structural changes can lead to mutations directly by causing pairing errors in replication or indirectly by error-prone repair in the next following round of DNA replication.

Repair systems play a significant role in the mutation process. As a result of repair, potentially lethal changes in the DNA may be eliminated. If the repair systems function in an error-free manner, potentially mutagenic lesions are eliminated before they can be converted into final mutations. A number of repair systems have thus far been discerned in microorganisms, particularly bacteria. In the following, these repair mechanisms are discussed briefly.

Photoreactivation Short-wavelength ultraviolet irradiation (254 nm) affects DNA in a number of

ways, but a well-established action is the formation of thymine dimers, a state in which two adjacent thymine molecules are chemically joined, so that replication of the DNA cannot occur. When such an ultraviolet-irradiated population is subsequently exposed to visible light of a wavelength of 300 to 450 nm, the survival rate increases and the frequency of mutation decreases. This is due to the activation of a photoreactivating enzyme (photolyase) which splits thymine dimers. In the dark, this enzyme bonds with thymine dimers; in the presence of light the enzyme splits the dimers into monomer pyrimidines. Up to 80% of the thymine dimers existing in the genome can be photoreactivated. UV-induced DNA crosslinks can also be photoreactivated. This repair system functions in an error-free manner and thus does not allow mutations to occur.

Excision repair In contrast to photoreactivation, which is possible with single-stranded DNA, the complementary strand is required for excision repair. In a dark reaction, damages to the DNA, such as ultraviolet-induced pyrimidine dimers, alkylation, or deamination, are recognized by specific DNA endonucleases. Different repair paths are involved according to each DNA lesion. Excision repair can be partially prevented by inhibitors such as caffeine, acriflavin and 8-methoxypsoralen.

The mechanism of excision repair for ultraviolet induced lesions is different than that for lesions induced by alkylation or deamination. In one mechanism of repair of ultraviolet lesions, a nucleotide-excision repair mechanism is operable. In effect, defective nucleotides are cut out and replaced, according to the mechanism illustrated in Figure 3.1. For this mechanism to operate, the normal DNA replication process is not required.

In some bacteria, an ATP-dependent endonuclease has been demonstrated in ultraviolet-exposed cells. This endonuclease is controlled by three genes, *uvrA*, *uvrB* and *uvrC*. The endonuclease splits a phosphodiester bond on the 5' side

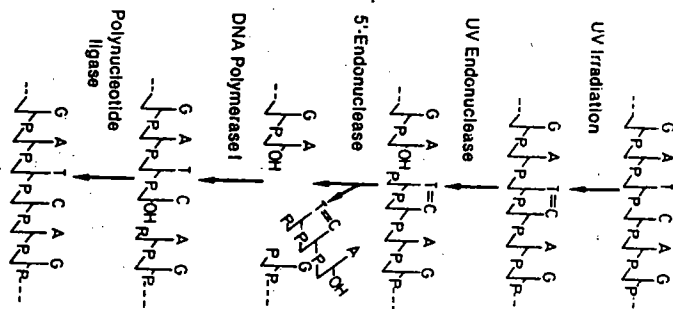


Figure 3.1 Repair of DNA containing pyrimidine dimers by nucleotide excision

of the pyrimidine dimer and 3' hydroxyl and 5' phosphoryl termini are produced. With the help of a 5' exonuclease and DNA Polymerase I, a 6- to 7-nucleotide-long oligonucleotide is cut out along with the dimer and the resulting gap is expanded to approximately 30 nucleotides. The missing nucleotides are filled by DNA polymerase I (*polA*) starting from the 3' end and are connected by a polynucleotide ligase (*lig*). This repair mechanism is almost error-free and mutation is thus avoided.

Another mechanism for repair of ultraviolet damage is termed *recA*-dependent repair. This second repair path requires DNA replication and seems to be mutagenic since large gaps of up to 1500 nucleotides are cut out. For this repair

mechanism to function *recA*, *recB*, *recC*, *lexA*, *uvrD* and *polC* genes are needed. This type of repair is a form of the so-called SOS response (see later).

Excision repair of alkylated and deaminated DNA operates in a different manner. In this repair path, modified bases are recognized and cut out. It is therefore also known as base excision repair (Figure 3.2). In alkylated DNA of *Micrococcus luteus* it has been shown that the 3-methyladenine formed is recognized by an N-glycosylase, which splits the N₂C₁ bond between the base and deoxyribose, so that a purine-free site is formed in the DNA. 7-Methylguanine is tolerated by the cells so that formation of this alkylated base is not mutagenic. The lesion is recognized by a special AP endonuclease (AP = apurinic or apyrimidinic site), which splits the

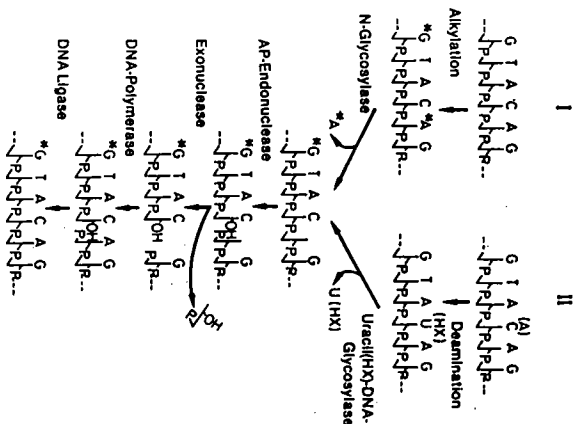


Figure 3.2 Base excision as a mechanism for the repair of alkylated DNA (I) or deaminated DNA (II) with uracil (U) or hypoxanthine (HX). *G = 7-methylguanine; *A = 3-methyladenine

DNA strand. The sequence following is similar to nucleotide excision repair, but it has not yet been determined whether identical enzymes are involved. Uracil DNA glycosylase and hypoxanthine DNA glycosylase, both involved in the repair of deaminated DNA, have been found in various bacteria.

Postreplicative recombination repair If a DNA strand contains a lesion which hinders base pairing, a gap in the daughter strand of approximately 1000 nucleotides is formed during replication. According to Figure 3.3, which is partially hypothetical, these gaps are filled with material from the parent strands through recombination processes (by the action of the *recA* gene product). The repair of the parent strands occurs through repair replication with the daughter strands as a matrix by means of DNA Polymerase I. Several replication and recombination stages are involved in this repair process.

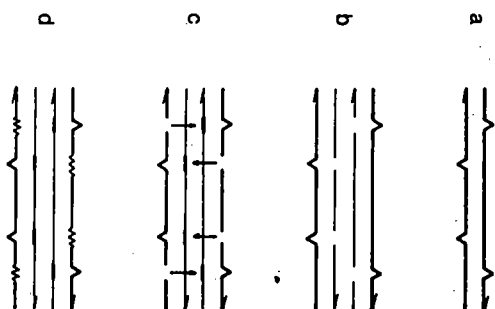


Figure 3.3 Model of post-replicative repair. a. DNA double strand with premature lesions (---). b. Gaps in daughter strands which have resulted during replication. c. and d. Hypothetical exchange of parent strand DNA and replication repair (Smith, 1976)

induced condition, the product of the *lexA* gene binds to the identical operator sequence of several unlinked genes: *recA*, *umuDC*, and *uvrA*. After initiation by a regulatory signal, several SOS functions are derepressed (for example, DNA exonuclease V), and DNA replication can restart at the chromosomal origin. The repair is performed by a DNA polymerase which differs from the constitutive DNA polymerases I, II and III in that it continues to carry out DNA synthesis in spite of the lesion in the template, which takes place mainly when "false" bases are incorporated. It is assumed that a new polymerase is not induced but rather a factor is formed which lowers the proof-reading of one or more DNA polymerases.

Adaptive repair In *Escherichia coli*, increased resistance to the mutagenic and lethal effects of high doses of alkylating agents has been found to occur after lengthy treatment with sublethal concentrations of such agents. This effect, which

cannot be elicited by ultraviolet radiation, can be traced to a further **inducible repair system**. The number of alkylated bases, particularly O⁶-methylguanine, in the genome is reduced by this adaptive repair mechanism which works almost without error. In this way, the synthesis of an O⁶-methylguanine DNA methyl transferase is induced. In uninduced *E. coli* between 13–60 molecules of this methyl transferase are present whereas induced cells contain more than 3000 molecules. Another enzyme, 3-methyladenine DNA glycosylase II, is involved in the breakdown of 3-methyl adenine, 3-methyl guanine, 7-methyl adenine, and 7-methyl guanine. In the induced cell the glycosylase content is increased by a factor of 20. The mechanism of the induction of these enzymes is not understood. The capacity of this repair system is limited however. O⁶-methylguanine accumulates when larger mutagen doses are used, and the mutation frequency is directly proportional to the O⁶-methylguanine concentration. The enzymatic mechanism involved in the elimination of the compound is not yet understood.

Figure 3.4 SOS repair in *Escherichia coli* (based on Witkin, 1976)
A. Excision repair: a. DNA with pyrimidine dimers (—/—). b. Excision. c. Repair replication stops at the second dimer. Exonuclease activity continues. d. SOS repair is induced. It polymerizes the DNA above and beyond the dimer and causes a mutation through erroneous incorporation (X). e. The second dimer is repaired through nucleotide excision so that both DNA strands carry the error.
B. Post-replication repair: a. DNA with pyrimidine dimers; b. During DNA replication, gaps result in the daughter strands, which overlap and cannot be closed by recombination repair. c. SOS repair is induced, as in Ad; d. The second gap is closed by recombination. In the following replication, a daughter strand produces overlapping gaps again. See Bb; e. The second gap is also closed through SOS polymerase

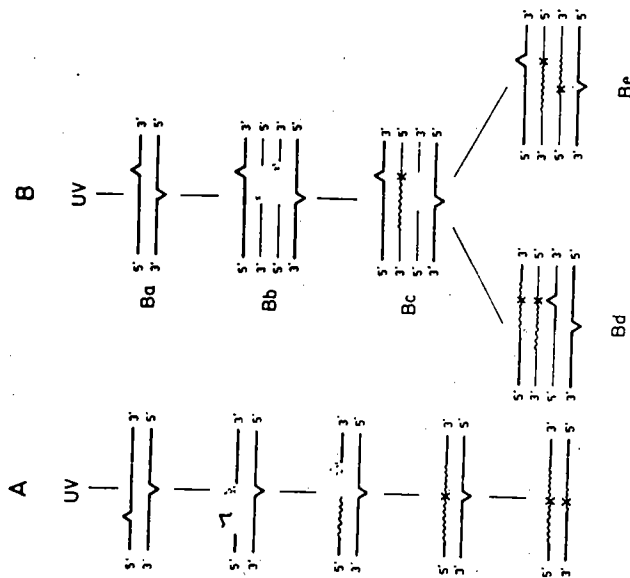
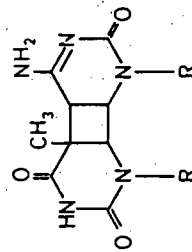


Figure 3.5 Thymine-cytosine-cyclobutane dimer, the photoproduct formed as a result of ultraviolet radiation



Long-wavelength ultraviolet radiation Radiation at wavelengths of 300–400 nm has less lethal and mutagenic effects than short-wavelength UV. However, if the exposure of cells or bacteriophages to long-wavelength UV is carried out in the presence of various dyes which interact with DNA, greater death rates and increased mutation frequency result. Especially effective activators of long-wavelength UV are the psoralen derivatives, 8-Methoxypsoralen (Figure 3.6, Structure

Mutagenesis through radiation

Both ultraviolet radiation and ionizing radiation are used in mutagenesis studies. The mechanisms of mutagenesis are quite different for each type of radiation, however.

Short-wavelength ultraviolet One of the more effective mutagenic agents is short-wavelength

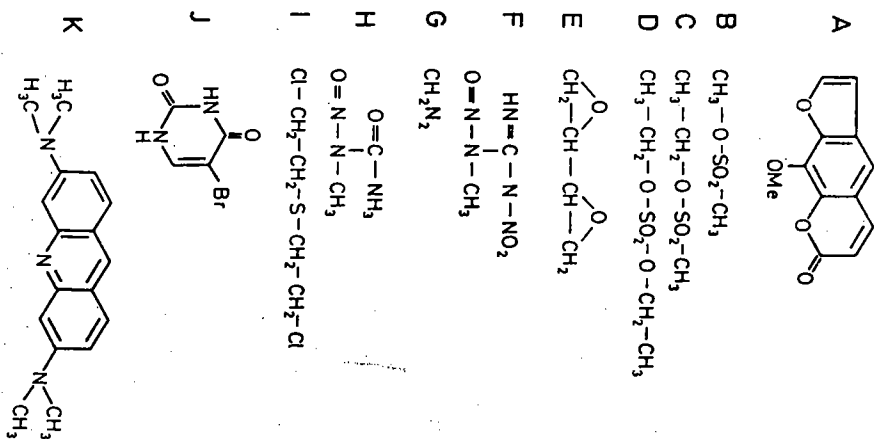


Figure 3.6 Structure of different mutagens
 A. 8-Methoxypsoralen; B. Methylmethanesulfonate (MMS); C. Ethylmethanesulfonate (EMS); D. Diethylsulfate (DES); E. Dipropylsulfate (DEP); F. N-Methyl-N'-nitro-N-nitrosoguanidine (NTG); G. Diazomethane; H. N-Methyl-N-nitrosourea (NMU); I. Di-(2-chloroethyl)-sulfide (mustard gas); J. 5-Bromouracil (BU); K. Acridine orange (AO)

A) intercalates between the base pairs of double-stranded DNA and after the absorption of long-wavelength UV, an adduct is formed between the 8-methoxypsoralen and a pyrimidine base. Ab-

sorption of a second photon causes the coupling of the pyrimidine-psoralen monoadduct with an additional pyrimidine. Biadduct formation between complementary strands of nucleic acid results in crosslinks. These lesions cannot be photoreactivated, although they are eliminated through nucleotide excision repair in conjunction with the mutation-causing SOS repair system.

Ionizing radiation Ionizing radiation includes X-rays, γ -rays, and β -rays, which act by causing ionization of the medium through which they pass. These rays are usually used for mutagenesis only if other mutagens cannot be used (e.g. for cell material impenetrable to ultraviolet rays). Single- and double-strand breaks occur with a significantly higher probability than with all other mutagens. Ninety percent of the single-strand breaks are repaired by nucleotide excision; the *recA*-dependent repair mechanism is also involved. Double-strand breaks result in major structural changes, such as translocation, inversion or similar chromosome mutations. Therefore ultraviolet radiation or chemical agents are normally preferable for mutagenesis in industrial strain development.

Mutagenesis with chemical agents

A variety of chemicals are known which are mutagenic, and these may be classified into three groups according to their modes of action:

- Mutagens which affect nonreplicating DNA
- Base analogs, which are incorporated into replicating DNA due to their structural similarity with one of the naturally occurring bases.
- Frameshift mutagens, which enter into DNA during replication or repair and through this intercalation cause insertion or deletion of one or a few nucleotide pairs.

Chemicals which affect nonreplicating DNA A number of chemicals are known which cause direct damage to nonreplicating DNA. Nitrous

acid (HNO_2) deaminates adenine to hypoxanthine and cytosine to uracil. Through the changed pairing properties of the deamination products (thymine pairs with cytosine, uracil with adenine) $\text{AT} \rightarrow \text{GC}$ and/or $\text{GC} \rightarrow \text{AT}$ transitions occur. In addition, nitrite induces crosslinks between the complementary strands. Figure 3.7 shows the establishment of the mutation after two generations. Excision and recombination repair are involved in the elimination of deamination products. Besides point mutations, deletions occur relatively frequently with nitrous acid treatment.

Hydroxylamine (NH_2OH) reacts with pyrimidines, but only the reaction with cytosine is mutagenic, whereby the amino group is replaced with a hydroxylamino group. The hydroxylamine derivative from cytosine shows tautomer-

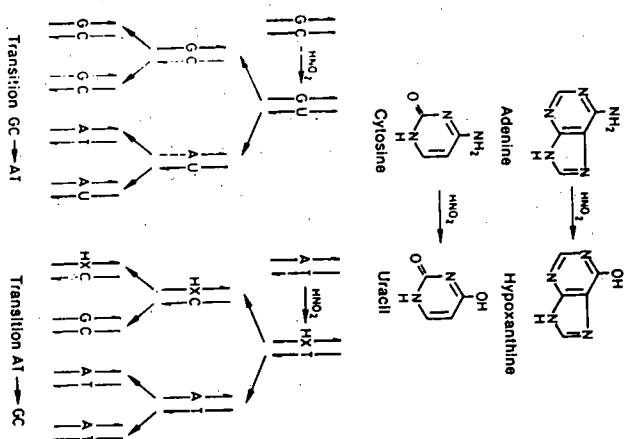


Figure 3.7 Mutation caused by nitrous acid

ization and pairs then with adenine, so that through hydroxylamine action $\text{GC} \rightarrow \text{AT}$ transitions are caused.

Another group of chemicals affecting nonreplicating DNA are the **alkylating agents**. Except for ultraviolet radiation, alkylating agents are the most potent mutagenic system for practical application. Compounds frequently used include ethyl methanesulfonate (EMS), methyl methanesulfonate (MMS), diethylsulfate (DES), dipropylsulfate (DEP), N-methyl-N'-nitro-N-nitrosoguanidine (NTG), N-methyl-N-nitrosourea and mustard gas (see structures B-I, Figure 3.6). Transitions, transversions, deletions and frameshift mutations occur as a result of the action of alkylating agents.

Mutagenesis with alkylating agents occurs via various pathways. These compounds cause the formation of a whole spectrum of alkylated bases in DNA, along with phosphotriester, purine-free sites and single-strand breaks. Although 7-alkylguanine is in all cases the most common alkylation product, it does not result in mutations. O-alkylguanine and O-alkylthymine are the most important premutational lesions, and as a result of pairing errors, mainly $\text{AT} \rightarrow \text{GC}$ transitions are elicited (direct mutagenesis). A second process which also results in mutation is the induction of error-prone SOS repair (see above when relatively high doses of mutagen are used).

It has been suggested that the occurrence of mutations in *Escherichia coli* is dosage-dependent in relation to its repair system: At a low level of alkylation of DNA the constitutive error-free systems perform the repair and mutations seldom occur. At higher mutagenic doses, on the other hand, the performance of the constitutive repair systems is not sufficient and the adaptive repair enzymes are induced. At even higher doses, the enzymes involved in SOS repair are also induced. Between the error-free adaptive repair and the error-prone SOS system there is competition for the repair of DNA lesions. The frequency of mutation is critically dependent upon which of these repair systems is working.

The use of N-methyl-N'-nitro-N-nitrosoguanidine (NTG) in a mutation program is difficult because of its carcinogenic effects, but it is one of the most effective chemical mutagens. A large proportion of mutants is found under optimal conditions with a low killing rate. In *Streptomyces coelicolor*, for instance, 8–10% of the survivors are found to be auxotrophs, and in *Escherichia coli* up to 50% of the surviving population consists of mutants. Ninety percent of the mutations induced by NTG are GC → AT transitions; to a small extent deletions and frameshift mutations are also found as a result of the deletion of GC pairs.

The exact molecular reaction mechanism of NTG is not yet understood. The compound is easily decomposed *in vivo*, and in acidic solutions nitrous acid is formed. Although nitrous acid is a mutagen, it is not effective in the pH range where NTG is active (pH 6–9); diazomethane (Figure 3.6, Structure G), a strongly methylating agent, is formed under alkaline conditions.

Besides the alkylation of nonreplicating DNA, the main point of action of NTG is at the replication point of DNA, through a change in DNA polymerase III during DNA replication. In this process, there is incorrect duplication in a short segment of the DNA until the defective polymerase is replaced by an intact molecule. This explains the observation that NTG mutations frequently occur in gene clusters.

Base analogs Because of their structural similarity, base analogs such as 5-bromouracil (BU, Figure 3.6 J) or 2-aminopurine (AP) are incorporated into replicating DNA instead of the corresponding bases thymine and adenine. The analogs tautomerize more frequently than the natural bases: BU in keto form pairs with adenine, whereas BU in enol form pairs with guanine. If the keto form of BU is incorporated, there is a AT → GC transition caused by tautomerization; if the incorporation takes place in the enol form a GC → AT transition is caused (Figure 3.8).

Base analogs are of minor importance for practical application, because for industrially important strains which are poorly understood genetically it can be rather costly to set up the optimal conditions for mutagenesis. As an example of the difficulties, the incorporation of BU in DNA takes place only if the organism is growing under thymine deficiency. Thus, thymine auxotrophs are frequently used; however, use of such auxotrophs can lead to complications in strain development, since in many cases auxotrophic mutation results in reduced production of the desired substance.

Frameshift mutagens Frameshift mutagens intercalate into the DNA molecule and cause errors which result in an alteration of the reading frame, resulting in the formation of faulty protein or no protein at all. The most commonly used frameshift mutagens are the acridine dyes, such as acridine orange (structure in Figure 3.6 K), proflavine and acriflavine. The induction of insertions or deletions is dependent on the ability of an acridine dye to become inserted between two neighboring bases of a DNA strand. The size of the acridine molecule, 3.4 Å, is exactly equivalent to that of a mononucleotide. Intercalation occurs most likely in DNA segments with identical base pairs in areas of strand irregularities, such as at the end of a chromosome, near the replication fork, or at a site undergoing recombination. A proposed model for the mode of action is illustrated in Figure 3.9.

Although acridines are useful in research, they are not very suitable for a routine isolation of mutants in strain development: although they are strong mutagens for bacteriophages T2 and T4, they have little or no mutagenic effect in bacteria.

Further methods of mutagenesis

Strains used industrially are usually less well characterized genetically than the organisms commonly used in basic research and there are practically no applicable processes other than

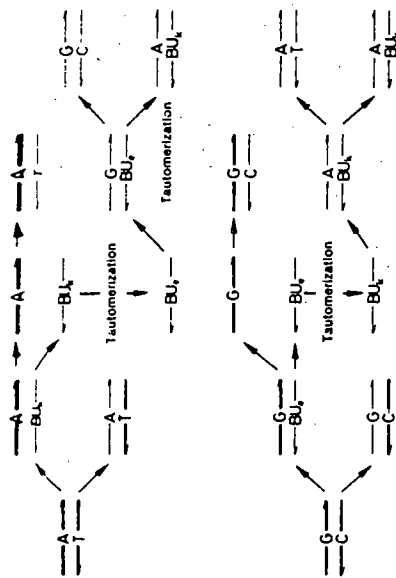


Figure 3.8 Mutation via action of 5-bromouracil (BU). A, BU is incorporated in keto form (BU_k). Tautomerization during replication causes an AT → GC-transition. B, BU is incorporated in enol form (BU_e). Tautomerization during replication causes a GC → AT-transition

mutagenesis with radiation or chemical agents. For the sake of completeness, several interesting possibilities are mentioned here which have thus far found only limited application.

Mutator genes In *Escherichia coli* the frequency of mutation can be increased by a factor of 100 through the introduction of a mutator gene. The cause of this effect is an error-prone DNA polymerase which frequently makes mistakes copying a template, resulting in either transitions or

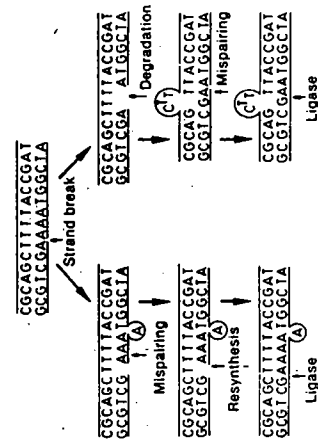


Figure 3.9 Possible mechanism for the production of frameshift mutations

transversions, depending on the mutator gene. Three such mutator genes have been demonstrated in *Escherichia coli*. Because of the high rate of mutation, the handling of such mutator strains in production may present difficulties.

IS-Elements, transposons, and bacteriophage Mu. The occurrence of mutations through integration into DNA of prophages or IS-elements and transposons is well known in *Escherichia coli* and *Salmonella typhimurium*. IS-elements are DNA sequences of variable length (800–1400 base pairs) which can be incorporated in different sites of the genome and released again. Integration and excision take place in *recA*-independent recombination. This applies also to transposons (genetic elements containing flanking IS-elements in inverse orientation, often with antibiotic-resistance genes) and the temperate bacteriophage Mu.

These elements destroy the function of the gene at the site of their integration. The incorporation of IS1 also has a polar effect on the genes distal with respect to the promoter. The genes are barely or not at all transcribed, probably due to the blocking of mRNA synthesis. IS2 bears a promoter, which, when incorporated in the appropriate orientation, results in the constitutive expression of genes located downstream. More-

over, IS-elements cause chromosome aberrations. In particular IS1 causes deletions, whereas IS2 causes duplications. Transposon mutagenesis is discussed in a later section.

Comutation and sequential mutagenesis

Nitroguanidine (NTG), which causes multiple mutations at the replication point through its effect on DNA polymerase III, can be used to induce mutations in certain sites of the genome by means of a process called comutation or sequential mutagenesis.

Comutation When a mutation is induced in a specific locus, a large number of further mutations, so-called comutations, may be found in closely linked genes. In *E. coli* 40% of comutations are concentrated in a region of about 50,000 base pairs (about 1/60 of the genome). In *Streptomyces coelicolor* the comutation region is about twice as large. By using a selective marker, such as the reversion of an auxotrophic mutant, clones can be isolated after NTG mutagenesis which carry mutations in neighboring genes of the selective marker. In some cases this mutagenesis in a specific operon can be used in strain development, provided the genes controlling production have been mapped.

Sequential mutagenesis The replication of circular chromosomes is a sequential process. In *Escherichia coli*, the replication point moves from a fixed starting point O (origin) to an end point T (terminus) in a bidirectional fashion. In an exponentially growing, nonsynchronized bacterial culture, the replication of individual chromosomes is quite varied. In synchronized DNA replication, a specific genome segment is replicated in the vast majority of the chromosomes at a specific time. If mutation is induced by use of NTG-pulses and the mutation frequency of certain markers is subsequently plotted against time, maxima are found at certain times during a replication cycle (Figure 3.10). The time between start of replication and the appearance of such a

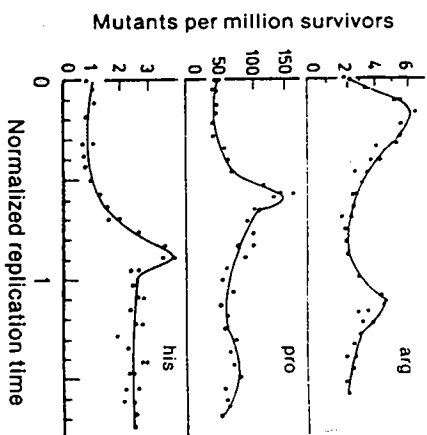


Figure 3.10 Sequential mutagenesis of *Escherichia coli* TAU-bar (Cerdeña-Olmedo et al., 1968). Samples of a synchronized culture (25°C) were mutagenized at 5 min. intervals with nitroguanidine (0.1 mg/ml; pH 5.5) and assayed for revertants to *arg*, *pro* and *his*.

maximum is characteristic of the position of a specific marker on the genetic map. Sequential mutagenesis has already been used in several organisms. By this means, genetic maps can be drawn up; moreover, specific genes can be mutated as desired, provided the time of replication is known.

Directed mutagenesis

The methods of mutagenesis which have been discussed up to this point are completely undirected. The development of gene technology has led to revolutionary new methods which make it possible to isolate mutants of specific genes of interest. Although the details of gene technology are discussed in Section 3.6, some of the approaches to use for mutagenesis purposes are discussed here.

Deletions A circular DNA molecule that has only a single recognition site for a particular restriction endonuclease is linearized when it is

treated with this restriction enzyme. The single-strand regions at the site of cutting are then digested with the specific nuclease S1, producing blunt-end fragments. When the linear molecule is introduced into a cell, recircularization can occur, although the shortened ends are generally removed by a polynucleotide ligase (Figure 3.11), thus leading to the formation of a small deletion. Larger deletions are obtained if the DNA contains two recognition sites for the restriction enzyme. For example, after separation of the fragments the larger fragment can be closed into a ring and cloned.

Insertions By use of a linker-adaptor approach, an adaptor DNA or a linker DNA can be inserted at the recognition site where the restriction enzyme has acted. An adaptor molecule is a chemically synthesized double-stranded DNA that can be used to connect together the ends of two DNA molecules. A linker molecule is similar but possesses a recognition site for a restriction enzyme.

Transposon mutagenesis is another method for inducing mutation via insertion. Transposons are known in both prokaryotes and eukaryotes and can insert at arbitrary sites in the genome.

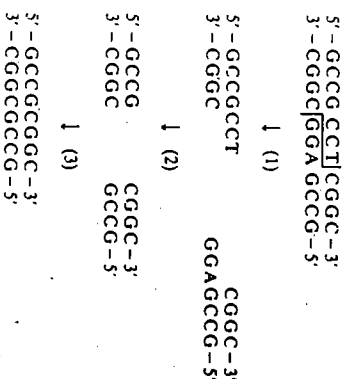


Figure 3.11 Directed mutagenesis by site-specific deletion in the region of the recognition site for the restriction enzyme BglI in the genome of virus SV40. 1. Cut with enzyme BglI. 2. Removal of the single-stranded region through the action of the single-strand specific nuclease S1. 3. Joining of the DNA ends by a ligase.

Transposon Tn5 contains the gene for resistance to an aminoglycoside antibiotic which can be expressed in a wide variety of both prokaryotes and eukaryotes. Several transposons have been integrated into plasmids (for instance, Tn1 and Tn3), others in either plasmids or chromosomes (for example, Tn5). Thus, transposons are available for a wide variety of purposes in gene technology.

Transposon mutagenesis offers a wide variety of advantages. A mutant phenotype with a very low reversion rate can be obtained. In addition insertion mutations are relatively easy to isolate since the transposons contain antibiotic resistance markers. All one needs to do is plate on a medium containing the antibiotic; only cells containing the transposon will be able to grow and form colonies. The integration of a transposon causes an interruption in transcription, so that transposon mutagenesis exhibits a polar effect. Because of this, the site at which the transposon has been integrated into the operon can be readily determined by assaying for enzyme activity or measuring the accumulation of an intermediate product.

Point mutants at specific locations in the DNA Bisulfite mutagenesis can be used to convert cytosine residues in the single-stranded DNA into uracil residues, since treatment of the DNA with sodium bisulfite causes deamination of cytosine. After synthesis of the complementary strand, a transition from GC → AT results. The production of single-strand regions can be brought about using certain restriction enzymes which, in the presence of ethidium bromide, split only a single strand of the DNA. The single-stranded site can be extended by treatment of the DNA with an exonuclease enzyme.

Nucleotide analogs can be incorporated in vitro into either RNA or DNA, using an enzyme which will replicate the nucleic acid in a synchronous fashion. Suitable nucleotide analogs include N⁶-hydroxyadenine triphosphate (N⁶-hydroxy-CTP) or N⁶-hydroxydeoxycytidine triphosphate (N⁶-hydroxy-dCTP). An example of

use of this procedure is shown in Figure 3.12. 1⁴-hydroxycytosine is incorporated by means of an *in vitro* RNA synthesis using the RNA replicase from bacteriophage Q β . The incorporated 1⁴-hydroxycytosine pairs in the next round of synthesis with both G and A, leading to the formation of GC $\rightarrow$ AT transitions.

Oligonucleotide mutagenesis The availability of synthetic oligonucleotides makes possible a very specific method for mutagenesis, since a nucleotide in any desired position in the DNA sequence can be substituted with any of the other three nucleotides. First the source gene is cloned into a single-stranded vector such as bacteriophage λ . To the system is added a synthetic oligonucleotide of 15 to 100 bases which contains a sequence complementary to the region of interest, but with one base that is mismatched. The mismatched oligonucleotide then serves as a primer for the action of DNA polymerase and a complete

second strand is formed. The completed double-stranded heteroduplex DNA, containing a single incorrect nucleotide, is then incorporated into a cloning vector such as *E. coli* by transformation. After replication of the cloning vector, either a mutant or wild type is obtained. If the result is a mutant, the mutation is at the desired location in the gene.

Phenotypic expression of mutations

Many mutations which result in increased formation of metabolites are recessive. When a recessive mutation takes place in a uninuclear, haploid cell (e.g. bacteria and actinomycete spores, asexual conidia of fungi), a heteroduplex results from it (Figure 3.13a); the mutant phenotype can only be expressed after a further growth step. This also applies to exponentially growing bacterial cells, which can contain 2-8 chromosomes (Figure 3.13b); not until several steps in repro-

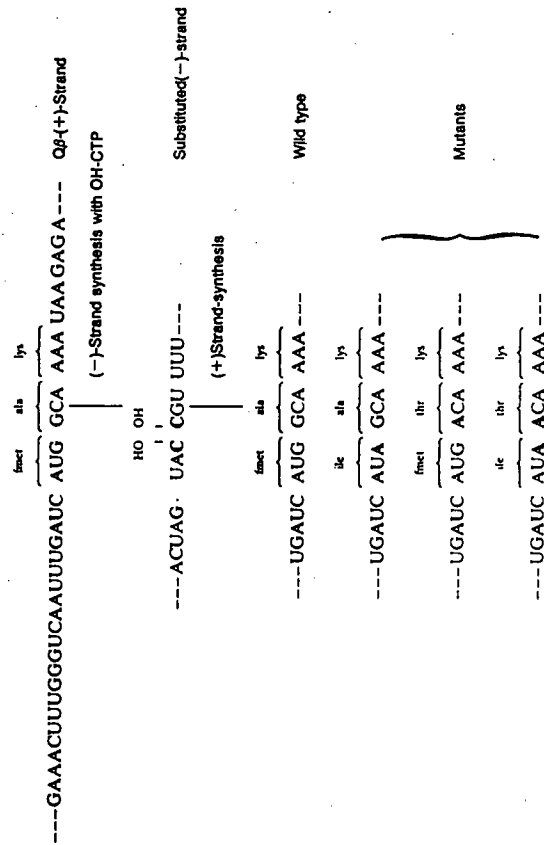


Figure 3.12 Mutagenesis through the incorporation of hydroxy-CTP during the synchronized synthesis of RNA by the Q β replicase of bacteriophage Q β . Based on Tamiguchi and Weissman (1978).

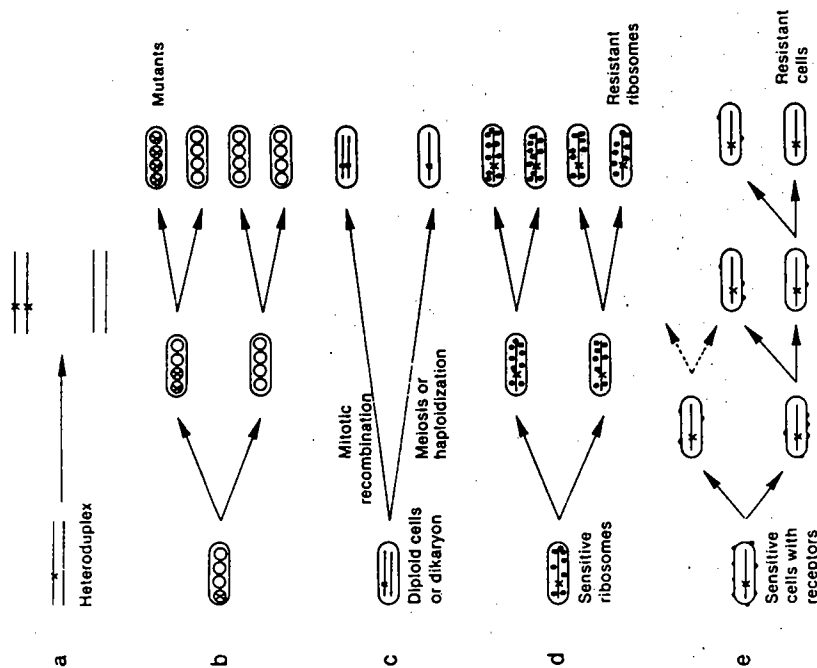


Figure 3.13 Phenotypic expression of mutants (Clarke, 1975). For explanation see text

duction have taken place do pure mutant clones appear.

With the filamentous actinomycetes, special procedures for mutant expression must be used. In the course of strain development, actinomycetes can lose their sporulation ability. To obtain cells for plating, the heterokaryotic mycelium which results from mutagenesis is grown and then fragmented by ultrasonic treatment or shaking with glass beads. After filtration through paper, cotton, or an 8 μ m membrane filter, mycelium fragments containing only one or a few nuclear bodies are used for plating. Homokaryotic material can be ultimately selected by re-

peating this segregation process. Another way for attaining segregation is the preparation of protoplasts containing one or few nuclei.

In diploid or heterokaryotic eucaryotes, recessive mutations are allowed to undergo phenotypic expression after meiosis, haploidization, or mitotic recombination (Figure 3.13c).

Delays in expression which are not directly the result of genetic effects are observed, such as mutations which cause changed ribosomes (Figure 3.13d) or mutations resulting in the loss of surface receptors (as in the development of bacteriophage resistance, Figure 3.13e). In both cases, the wild type structures must be diluted

out during growth before the mutation is recognizable phenotypically.

Optimizing mutagenesis

Although the molecular mode of action of some mutagens is quite well known, what can never be predicted is the effect of a mutagen on a specific gene or the effect of a mutation on a complex process, such as the biosynthesis of a secondary metabolite. The appearance of mutants, that is, strains in which a mutation has phenotypically resulted in a change, depends on several factors.

- The appearance of mutations is dependent on the base sequence of the gene to be mutated. Mutations are not distributed evenly around the genome; there are areas with high mutation frequency, the so-called **hot spots**. Different mutagens cause hot spots at different sites in the genome.
- The repair systems of the cell also play a role. In strains with partially defective repair mechanisms, organisms may be killed without having induced mutations, so that specific mutagens can be ineffective.
- A gene activity which has become lost through mutation can be restored at least partially through a second mutation, a **suppressor mutation**.

Suppressor mutations act in several different ways. Suppressors can occur in the same gene that already carries the primary mutation (intragenic suppressors). The primary missense mutation is compensated through the exchange of an amino acid or an additional deletion or insertion which corrects a primary frameshift mutation so that the reading frame remains intact. Suppressor mutations which occur in another gene (extragenic suppressors) compensate the primary mutation particularly at the level of translation, by the formation of mutant transfer RNAs or ribosomes. In strains with suppressor mutations, the function of the enzyme in question is restored up to 10% of the wild type activity. This is enough actively so that the existence

of specific mutant types (such as auxotrophy) might be overlooked.

In addition to these strain-specific factors, the **treatment conditions** have a critical effect on mutagenesis. Such factors as the pH, buffer composition, mutagen concentration, exposure time, temperature, and growth phase of the organism may greatly affect the efficiency of the process. By plotting **dose-response curves** (Figure 3.14), all of these factors may be optimized. Action on DNA causes not only mutation, but also killing, due either to irreparable damage to the DNA or to formation of lethal mutations. Therefore mutants are sought from among the few surviving members of the population which had been exposed to a strong mutagenic treatment (death rate > 99%), because there is thus a certain probability that each of the surviving cells carries one or several mutations. A high death rate alone,

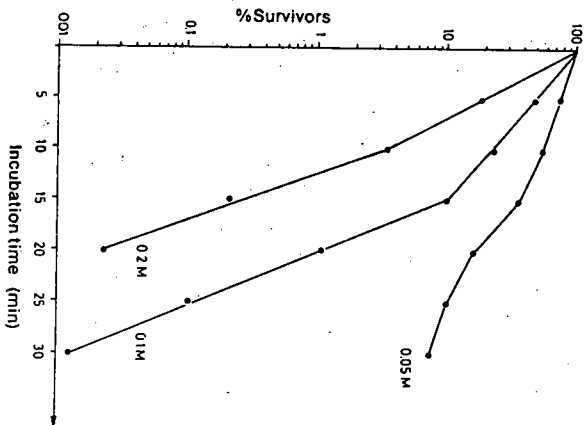


Figure 3.14 Killing of *Micromonospora injerensis* by nitrous acid

however, is no guarantee of the occurrence of mutations in specific genes. These mutations can only be reliably determined by assessing qualitatively or quantitatively changes in the product of this gene. In the case of an antibiotic, **production** should be considered as a criterion for mutagen influence. To assess this, a random selection of survivors in a population treated with mutagens is assayed for antibiotic formation in laboratory fermentations. The antibiotic titers of mutagen-treated isolates are plotted in a histogram and compared to a control group. Figure 3.15 clearly shows the differences in the variability of the population according to mutagen concentration. Since studies of this type are quite

costly, easily detectable changes such as mutations for resistance or reversion to auxotrophy are frequently used to optimize conditions for mutagenesis. However, the results of these latter experiments need not have any bearing on optimal conditions for increased formation of a desired product.

3.3 SELECTION OF MUTANTS

Besides an optimal mutagenesis, the **method of selection** is crucial for the effective screening of mutants. Looking for a desired mutant is analogous to the famous "search for a needle in a haystack". There are basically two ways of

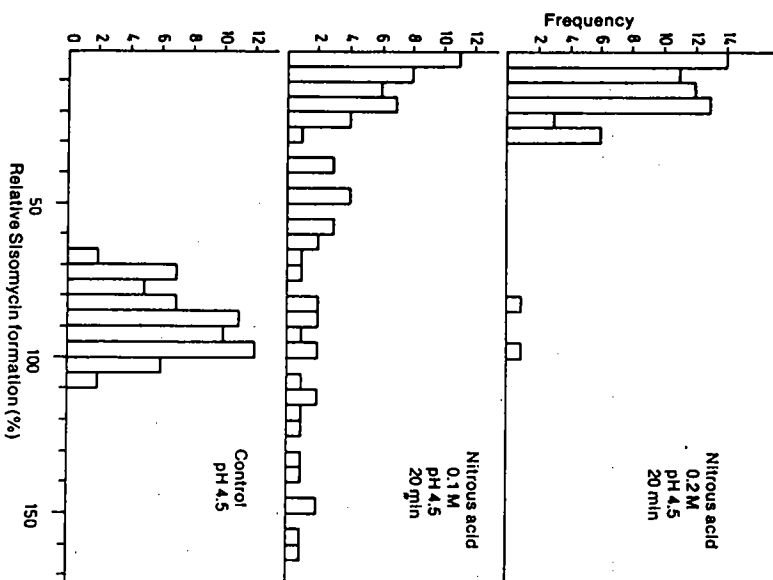


Figure 3.15 Histogram of sisomicin production by *Micromonospora injerensis* (control strain = 100%) in relation to mutagen treatment with nitrous acid

screening. A random selection of survivors from a mutagenized population can be examined for antibiotic production or other properties in a fermentation process that closely mimics the large-scale process. This procedure is very costly, but is often the only way to find mutants with increased productivity in industrial strains. Whenever possible, a screening method is used in which selective conditions are chosen which promote the growth or early detection of the mutants.

Random screening

The high yields obtained with industrial microorganisms have been possible largely through the process of empirical selection after mutagenesis. After mutagen treatment and expression of induced mutations, a random selection of surviving clones is inspected for ability to produce the product of interest. This is done in model fermentations which are carefully adapted to the medium and fermentation parameters of the large-scale procedure, in order to maximize the likelihood that the strains will be suitable for industrial production. The best strains from such a mutation cycle are repeatedly mutated and selected. A gradual increase in the yield is attained by continuing with these steps. In this mutation and selection program, study does not center only on the strain exhibiting the best yield. This is because multiple mutations usually occur due to the high mutagen doses and in the course of strain development these unrecognized mutations can cause certain strains to show no rise in productivity. Therefore, depending on the capacity of the screening program, the 5-10 best strains of a mutation-selection cycle should be used as parent strains for future mutagenesis. These strains are normally treated with mutagens different from those used in the initial isolation. Many factors determine how many isolates must be screened to obtain strains with increased productivity. Factors which influence the size of the screening program are: frequency of mutation, extent of yield increases, the amount of time re-

quired for a mutation-selection cycle, the available test capacity of the screening program, and the accuracy of the screening test (e.g. antibiotic assay). As a rule, mutants with high yields are much rarer than those with only slight improvements. Moreover, the variability of mutagen treated populations is quite high even when mutagenesis is performed under identical conditions. Thus it is usually more economical to screen a small number of survivors (about 20-50) after many different mutagen treatments and to continue mutating strains having small yield increases as quickly as possible, than it is to test a large number of isolations after a few mutagen treatments and to hope for a one-step large yield increase.

The number of strains which must be screened to obtain mutants with a yield increase depends on the strain, the conditions of mutagenesis, the biosynthesis pathway, and the regulation of the product which is being optimized. Normally, several hundred to several thousand isolates per mutation cycle must be tested. In practice, with nonautomatic methods the number of isolates that can be tested per unit time is usually limited to 1000-2000 per week. Thus the screening capacity determines the speed of the progress to be expected. In the first stage of mutant screening, only one fermentation sample per isolation is usually assayed, provided that the test error is smaller than the yield increase expected. The best isolates of the first series (usually 10-30%) are then tested in a second fermentation. Since the best strains of this second screening are then used in a still further mutation cycle, the yield increase must be statistically significant when compared to the parent strain. The number of replicates from the reference strain and the mutants should be chosen statistically. An optimal increase in yield per test period can then be calculated, if the number of isolates required to attain a specific yield increase can be tested within the time period needed for mutagenesis, colony selection, and assay of the isolates.

Several industrial companies are developing ways to automate mutant screening procedures to increase the screening capacity.

Selective isolation of mutants

Several examples of the many selective methods used in strain development are mentioned here. *Isolation of resistant mutants* A high cell density of a mutagenized population can be plated on a selective medium containing a concentration of a toxic substance that prevents the wild type from growing. Only the resistant clones can develop. In this way, mutants may be isolated which are resistant to antibiotics or antimetabolites. The antibiotic resistance character can not only be used as a genetic marker, but mutants isolated may also have an increased cell permeability or a protein synthesis with a higher turnover, making them useful for industrial purposes.

Antimetabolite resistance can be used to select mutants which exhibit defective regulation. Altered regulation may occur in such mutants. Antimetabolites, because of their structural similarity to metabolites, may cause feedback inhibition, but are unable to substitute for normal metabolites. Antimetabolites cause death of normal cells, but analog-resistant mutants can form

an excess of metabolites, in some cases through changed regulatory mechanisms (elimination of allosteric inhibition; constitutive product formation). Table 3.2 shows some antimetabolites frequently used in screening programs.

Isolation of auxotrophs By using certain blocked mutants, desired products such as amino acids and nucleotides may be formed via branching biosynthetic pathways (see Chapters 9 and 10). Auxotrophic mutations in antibiotic-producing organisms frequently result in reduced product formation. Improved strains can be obtained in some cases (e.g. in tetracycline) by isolation of prototrophic revertants (suppressor mutants) from auxotrophs. In addition, auxotrophic mutations can frequently be used as genetic markers.

The isolation of auxotrophs is done by plating of the mutagenized population on a complete agar medium, on which the biochemically deficient mutants can also grow. By means of Lederberg's well-known replica plating technique, the clones are transferred to minimal medium where the auxotrophic colonies cannot grow. These mutants are picked up from the master plates and their defect is characterized. Since in this method a large number of plates must be observed, various procedures have been devel-

Table 3.2 Frequently used antimetabolites

Natural metabolite	Antimetabolite	Natural metabolite	Antimetabolite
Adenine	Psicouanine 2,6-Diaminopurine	Leucine	5,5,5-Trifluoroleucine 4-Azaleucine
Guanine	8-Azaxanthine	Methionine	α -Methylmethionine Norleucine, ethionine
Uracil	5-Fluorouracil	Phenylalanine	p-Fluorophenylalanine Thienylalanine
p-Aminobenzoic acid	Sulfonamide	Proline	3,4-Dihydroproline
Nicotinic acid	3-Acetylpyridine	Tryptophan	5-Methyltryptophan 6-Methyltryptophan
Pyridoxine	Isoniazid	Tyrosine	p-Fluorophenylalanine
Thiamine	Pyrimethamine	Threonine	β -Hydroxythreonine
Arginine	Canavanine	Valine	α -Aminobutyric acid Isoleucine
Histidine	2-Thiazolalanine 1,2,4-Triazol-3-alanine		

oped to enrich for auxotrophic mutants by removing or killing prototrophic organisms. In a process known as **filtration enrichment**, after mutagenesis the spores of filamentous organisms (actinomycetes, fungi) are allowed to develop in a liquid minimal medium. The developing microcolonies of prototrophs are then separated by filtration, leaving behind in the filtrate spores of auxotrophs which have been unable to grow. The filtrate is then plated and the resulting colonies are checked for auxotrophic characteristics.

Another procedure for selection of auxotrophs, which can be used also for unicellular organisms, makes use of the fact that penicillin kills growing cells but not nongrowing cells. In this **penicillin-selection procedure**, growing cells are selectively killed by antibiotic treatment, thus enriching for auxotrophs which cannot grow on minimal medium. Depending on their mode of action several inhibitors other than penicillin can also be used in this procedure: dihydrostreptomycin for *Pseudomonas aeruginosa*, naldixic acid for *Salmonella typhimurium*, colistin for the penicillin-resistant *Hydrogenomonas* strain H16, and nystatin for *Hansenula polymorpha*, *Penicillium chrysogenum*, *Aspergillus nidulans*, and *Saccharomyces cerevisiae*.

An enrichment procedure with sodium penicillorphenolate makes use of the greater toxicity of this compound against germinating spores than against vegetative cells. The method has been successfully applied with *Penicillium chrysogenum*, *Streptomyces aureofaciens*, *S. olivaceus*, and *Bacillus subtilis*.

By these methods, enrichments for auxotrophs of 10- to 100-fold can be attained, thus increasing the probability of obtaining mutants. However, it should be remembered that the types of mutants present in the original population may be shifted; for instance, an increased proportion of proline auxotrophs has been found in *E. coli* after auxotroph enrichment.

Other procedures The presence or absence of specific enzyme activities can be observed di-

rectly in colonies growing on plates by spraying with suitable reagents or by incorporating indicator dyes into culture medium. Antibiotically active substances may be detected by measuring the inhibition of sensitive assay organisms. By using this method, the antibiotic content of a solution can also be determined. A frequently used variant of this method is the "agar plug method," in which agar cylinders with single colonies are transferred to test plates after incubation in a moist chamber (Figure 3.16). The diameter of the resulting inhibition zones serves as a measure of the antibiotic production of each strain. However, one problem with this agar plug method is that frequently there is only a slight correlation between antibiotic formation in plate culture and the antibiotic production in submerged fermenta-

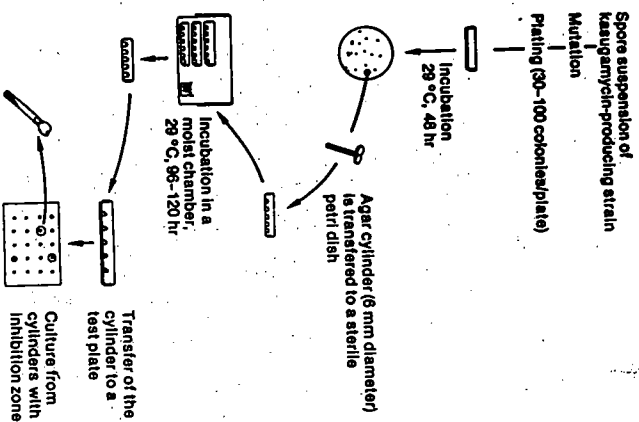


Figure 3.16 Use of the "Agar plug" method in kasugamycin strain development (Ichikawa et al., 1971)

tion. Strains which produce at high yields when grown on plates may produce at only low yields or not at all in liquid culture. Therefore the procedures mentioned are suitable for processes where simply a differentiation between productivity and nonproductivity is sufficient, such as for detecting the formation of constitutive enzymes. If screening is initiated using high-yielding strains, further increases in yield often cannot be detected by this method.

3.4 RECOMBINATION

The genetic information from two genotypes can be brought together into a new genotype through genetic recombination, which is thus another effective means of increasing the genetic variability of a population. As an example of how genetic variability can be increased, consider the following: In mutant screening, each high-yielding mutant is derived ultimately from the wild strain after a series of mutation and selection steps. A cell line with 10 mutations, for example, contains 10 new genotypes. By crossing the last high-yielding mutant with the wild type strain, 2¹⁰ (or 1024) different genotypes can be elicited. This same reasoning also applies to the crossing of high yielding strains from two different lines.

The advantages of genetic recombination are:

- Different alleles of the parent strains with increased metabolite production can be brought together in one strain, so that the cumulative effect of these mutations can be greater than the effect of the single mutation. However, the original hope of attaining a significant yield increase by merely recombining two high-yielding mutants has only been fulfilled in a few cases. In most cases, the productivity of the recombinants usually is intermediate between the values of the parent strains (see Chapter 13, for examples).

In the course of strain development, there is frequently a decline in the increase in yield after each stage of mutation. Besides mutants which are selectively enriched because of

their increased productivity, there is the development of inapparent mutations which prevent a further increase in the metabolite production through pleiotropic influences. With genetic crosses, these unfavorable mutant alleles may be replaced with alleles of one of the parents in the cross.

- High-yielding strains can actually increase the cost of the fermentation because of changed physiological properties (greater foaming, changed requirements for culture medium, etc.). By crossing back to wild-type strains, high-yielding strains with improved fermentation properties may be formed.

Hence an effective strain development approach should involve the use of sister-strain, divergent strain, and ancestral crosses at specific intervals, besides use of careful mutagenesis to ensure the maintenance of genetic variability.

Sexual and parasexual cycles in fungi

The fungi have two distinct types of genetic recombination processes that can be used in a strain improvement program. These are known as the sexual and parasexual cycles. We consider each of these processes in turn.

Sexual recombination Some fungi used industrially (e.g., strains of the genera *Aspergillus*, *Claviceps*, *Emmenthospora*, and *Saccharomyces*) have complete sexual cycle. In these organisms, nuclear fusion (karyogamy) results after fusion of hyphae has led to a mingling of nuclei in the heterokaryotic mycelium. After diploid formation, recombination takes place during the subsequent meiosis process. A new genotype results either from the combination of parent chromosomes or through crossing over as a result of segment exchange of paired homologous chromosomes.

Parasexual recombination Some of the most economically useful fungi, such as *Penicillium chrysogenum* (producer of penicillin) and *Cephalosporium* (producer of cephalosporins), lack a

Escherichia coli and *Pseudomonas* are Gram-negative bacteria of industrial interest in which conjugation is well developed and in which transduction systems are present. Pseudomonads have sex plasmids similar to F, the factor of *Escherichia coli*. Transformation systems exist for *Bacillus* strains (*B. subtilis*, *B. licheniformis*, *B. pumilus*); in addition, transduction systems have been established for *B. subtilis* with phages PBS1 and SP10.

Recombination in actinomycetes

Among industrial microorganisms, actinomycetes (filamentous Gram-positive bacteria) are economically significant as antibiotic producers. In streptomycetes, transduction has been sought unsuccessfully in *S. olivaceus* and *S. griseus* and in *S. aureofaciens* and *S. griseus*. In the thermophile *Thermactinomyces vulgaris*, however, transformation has been definitely shown.

Conjugation is the most common form of genetic exchange in actinomycetes in vivo (Table 3.4). On the average, one-fifth of the genome is transferred to the recipient cell; the frequency of recombination is between 10^{-3} and 10^{-4} . It has been determined that fertility factors are involved in conjugation in three streptomycetes (*S. reticuli*, *S. rimosus* and *S. coelicolor*). In *S. coelicolor* A3(2), the best understood actinomycete, two plasmids have been intensively studied: SCP1 with a molecular weight of approximately 100×10^4 , which contains the genes for methylenomycin synthesis; and SCP2 (molecular weight, $18-20 \times 10^4$), of which the mutant form SCP2 causes greater recombination frequency.

Recombination has not been of great importance in industrial strain development of actinomycetes. This is because parent strains with selective markers must be used to identify low frequency recombinants and the construction of doubly marked parent strains (such as auxotrophy and antibiotic resistance) is time-consuming. Singly marked strains can not be used in a genetic recombination study because the spontaneous

Table 3.3 Antibiotic-producing fungi in which recombination has been discovered

Species	Antibiotic	Recombination type
<i>Aspergillus nidulans</i>	Penicillin G	sexual, parasexual
<i>Cephalosporium acremonium</i>	Cephalosporin C	sexual, parasexual
<i>Emicelopsis salmosynemala</i>	Penicillin N	sexual, parasexual
<i>Emicelopsis terricola</i>	Cephalosporin C	sexual, parasexual
<i>Penicillium chrysogenum</i>	Penicillin G, O, V	parasexual
<i>Penicillium patulum</i>	Griseofulvin	parasexual
	Patulin	

(Hopwood and Merrick, 1977)

which brings about generalized transduction in *Escherichia coli*. In specialized transduction, only the loci which are adjacent to the attachment site of the phage in the bacterial chromosome are transferred. The insertion of the phage into the chromosome results in the further incorporation of the attached piece of DNA into the genome of the host cell. Phage λ is an example of a phage which brings about specialized transduction. Conjugation generally involves the participation of plasmids. In this process, single-stranded DNA is transferred from the donor cell to the recipient cell after the two cells have come into contact. In the *Escherichia coli* system, F^+ cells are donor cells in which plasmid exists in free form. After contact with F^- cells which do not contain the fertility factor, a copy of the F factor is transferred. Thus an F^- population is almost completely transformed into an F^+ type. After the F factor is integrated into the chromosome, the resulting Hfr strains show a significantly higher frequency of recombination (up to 10^{-1} compared to 10^{-3} previously). In conjugation, a fragment of the F factor is first transferred, then the bacterial genome, and finally the remaining fragment of the F plasmid. Since the conjugation process is almost always interrupted before completion, the recipient cell usually does not become an Hfr cell.

produce strains with increased penicillin titers. The amount of penicillin formed by the diploid strains and their segregants was, however, no greater than that of the parent strains. Only when haploid recombinants were used could stable strains with improved antibiotic production be isolated.

Recombination in bacteria

The parasexual mechanisms established in vivo in bacteria include: transformation, transduction and conjugation. In each case only a fragment of the genome of the donor cell is transferred into a recipient cell which thus becomes a partial diploid (merozygote). After homologous pairing, recombination occurs, but not every DNA transfer results automatically in recombination. In transformation, short pieces of DNA are taken up by competent recipient cells. In generalized transduction, temperate phage particles which have lost a piece of their own genomes transfer a DNA fragment of the host bacteria at the optimal rate of 10^3 per phage and per characteristic. Phage P1 is an example of a phage

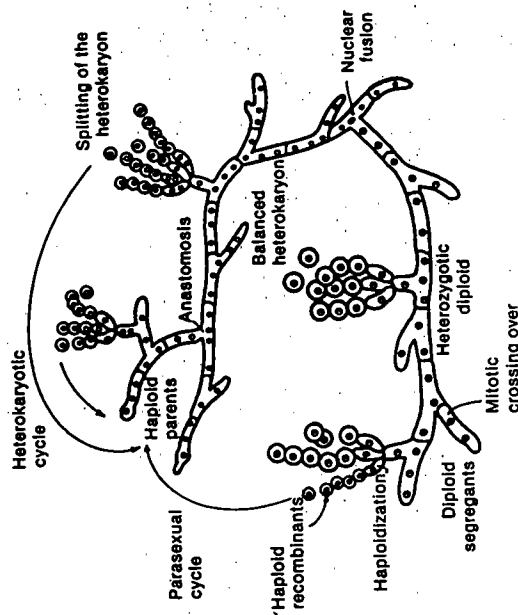


Figure 3.17 Parasexual cycle of *Penicillium chrysogenum* (Sermoniti, 1959)

rium acremonium (producer of cephalosporin), do not have a sexual cycle. Fortunately, the discovery of parasexual processes in imperfect fungi has led to the development of suitable breeding techniques. In parasexuality, the fusion of two hyphae of equal- or different polarity results in a mycelium with nuclei of both parent strains. This heterokaryon is normally stable with the nuclei mingling but not interacting. However, in rare cases (10^{-2} – 10^{-4} in *P. chrysogenum*), nuclear fusion occurs and a diploid nucleus is formed. In such diploid nuclei, mitotic crossing over between chromatids of homologous chromosomes may occur, resulting in genetic recombination (Figure 3.17). To obtain a recombinant, the formation of haploid cells or spores must occur. Spontaneous haploidization is relatively rare (10^{-3}), but can be induced with p-fluorophenylalanine. Haploidy occurs not through meiosis but through random distribution of the chromosomes to the progeny nuclei.

Table 3.3 shows the industrial fungi for which genetic recombination has been established. Heterozygotic diploids from parent strains with different cell lines have been used in attempts to

Table 3.5 Frequency of recombination after conjugation and protoplast fusion in streptomycetes

Strain	Mechanism of recombination	Recombination frequency
<i>S. coelicolor</i> A3(2) (SCP1-SCP2) X SCP1-SCP2 ⁺	Conjugation	1.7×10^{-4}
<i>S. coelicolor</i> A3(2) (SCP1-SCP2 ⁺) X SCP1-SCP2 ⁻	Protoplast fusion	4.7×10^{-3}
<i>S. parvulus</i> ATCC 12434	Conjugation	1.7×10^{-7}
<i>S. lividans</i> 66	Protoplast fusion	6.2×10^{-4}
<i>S. griseus</i> CUB 94	Conjugation	6.3×10^{-4}
	Protoplast fusion	0.6×10^{-3}
	Conjugation	1×10^{-3}
	Protoplast fusion	6.0×10^{-3}
	Conjugation	2.3×10^{-4}
	Protoplast fusion	1.0×10^{-3}

(Hopwood et al., 1977)

niques, not only mutation and selection but also recombination and selection may be considered equally useful methods of strain development.

Interspecific hybridization This approach allows genetic information from different species to be combined *in vivo* in order to create new or modified products. The use of protoplast fusion for this objective is being examined by different antibiotic manufacturers.

Among the fungi, interspecific crosses have been attempted between *Aspergillus nidulans* X *A. rugulosus*; *A. nidulans* X *A. fumigatus*; *Penicillium chrysogenum* X *P. cyanofulvum*; *P. cycloclum* X *P. citrinum*; *Saccharomyces cerevisiae* X *S. diastaticus*; *Kluyveromyces fragilis* X *K. fragilis*. Correct heterokaryons and recombinants have only been obtained in the cross between *P. cyanofulvum* X *P. citrinum*, in which a fusion frequency of less than 10^{-5} was obtained as compared to an intraspecific frequency of 40–60%. Interspecific crosses among streptomycetes have given similar results. The low frequency of heterospecific fusion may be because of genome incompatibilities which prevent recombination from taking place or because of the presence of restriction/modification systems that lead to physiological incompatibility.

The fusion frequency of intergeneric crosses between *Candida tropicalis* X *Saccharomyces fi-*

buligera is also quite low, around 10^{-5} . Crosses between *Saccharomyces cerevisiae* X *Lipomyces kononenkoae* showed several fusion products which exhibited metabolic properties of both parental strains, as well as the formation of nonparental segregants after haploidization.

Transformation and transfection of protoplasts Recombinant DNA technology made possible for the first time the development of methods for transformation of protoplasts with plasmid, chromosomal, or viral DNA. In the basic method, which is simple and generally applicable, protoplasts are treated with DNA in the presence of PEG and Ca^{2+} . In this way, all of the numerous techniques of genetic engineering become available for use with microorganisms of industrial interest (fungi, actinomycetes, other bacteria): amplification of genes, restoration of metabolism in deficient mutants, *in vitro* mutagenesis with protein engineering. Thus, even microorganisms for which conventional host-vector systems are not available can be handled. For example, excellent transformation systems have been developed for microorganisms of interest to the dairy industry such as *Streptococcus lactis* and various lactobacilli, for *Staphylococcus carnosus* and *Bacillus subtilis*, and vitamin B_{12} -producing *Pseudomonas putida* and *Streptomyces clavuligerus* systems have also been developed for β -lactam-producing strains of *Streptomyces clavuligerus* and *S. wachayensis*, vancomycin-producing *S. erythraeus*, and gentamicin-producing *Micromonospora purpurea* subspecies *lurida*. Using virus vectors, transformation (transfection) systems have been described for *Streptomyces*, *Thermomonospora*, *Mycobacterium smegmatis*, and *Brevibacterium lactofermentum*.

3.5 REGULATION

Both catabolic and anabolic processes are regulated and metabolism is generally so efficient that excess products are not formed. Strains with less efficient regulation can be selected in a screening

process. It is well established that strain development and the optimization of fermentation conditions lead to a relaxation of regulation in the producing strains. A broad understanding of biosynthesis, the enzymes involved in these processes, and their regulation is necessary for developing a rational approach to the alteration of the regulation of a fermentation process.

Microbial metabolism is controlled by the regulation of both enzyme activity and enzyme synthesis. We discuss both types of regulatory phenomena here.

Regulation of enzyme activity

Extensive research over the past several decades has shown that the activity of enzymes can be controlled by different mechanisms. We discuss here those mechanisms which are thought to be of significance for industrial process development.

Feedback inhibition In an unbranched biosynthetic pathway, the end product inhibits the activity of the first enzyme of the pathway, a process called *feedback inhibition*. A conformation change and hence inactivation (allosteric effect) occurs when an effector (end product) is attached to a specific site of the enzyme (allosteric site). The end product thus inhibits the activity of the enzyme noncompetitively.

In a branched biosynthetic pathway, feedback inhibition of the first common enzyme by means of one of the end products would cause more than one end product to be affected. In branched biosynthetic pathways, different kinds of feedback inhibition are found:

- The end product inhibits the first enzyme in each case *after* the branch point.
- The first step in the common synthesis path is catalyzed by several isoenzymes, each of which can be regulated independently.
- The first common enzyme in a branched biosynthetic pathway is influenced by each end product only slightly or not at all; there must

be an excess of all end products for inhibition to occur (a phenomenon called multivalent inhibition).

- Each end product of a branched pathway acts as an inhibitor; cumulative inhibition is the effect of all the inhibitors.

Examples of various feedback-inhibition reaction types are given in Chapters 9 and 10.

Energy charge Feedback inhibition also regulates catabolic pathways in which ATP is the main product. The relative concentrations of AMP, ADP and ATP in the cell are used in the following formula to calculate the energy charge (EC):

$$EC = \frac{(ATP) + 0.5 (ADP)}{(AMP) + (ADP) + (ATP)}$$

The values of the EC calculated in this way lie between 0 and 1.0.

If the EC is high, the activities of enzymes involved in ATP synthesis are inhibited (for example, isocitrate dehydrogenase is inhibited when the energy charge is ≥ 0.8). Conversely, the activities of anabolic enzymes (e.g., aspartokinases), which consume ATP, are stimulated by a high energy charge. Thus, an alteration in the rate of catabolism, since it affects ATP level and hence energy charge, may cause an increase or decrease in the activity of a variety of enzymes.

Breakdown of enzymes Enzymes which are no longer needed in metabolism may be broken down through the action of highly specific proteases. One of the best-known examples is the enzyme tryptophan synthetase in *Saccharomyces cerevisiae*, which is broken down specifically when the cells go into the stationary phase.

Modification of enzymes The activity of some enzymes (such as glutamine synthetase in *Escherichia coli*) is controlled by conformational changes, such as phosphorylation or adenylylation.

biosynthesis and regulation of secondary metabolite production, on the supply of intermediates of primary metabolism for secondary metabolite synthesis, and on the energy links between catabolic, anabolic and amphibolic pathways.

Regulatory mechanisms that affect the products of secondary metabolism are outlined below.

Induction In batch fermentations with readily metabolizable carbon and nitrogen sources, secondary metabolites are formed primarily after growth has ceased. The logarithmic growth phase is called the trophophase, and the subsequent phase, in which the secondary metabolite may be produced, is called the *idiophase*. Secondary metabolites are therefore also sometimes referred to as *idiolites*. In all cases studied thus far, the synthesis of enzymes involved in secondary metabolism is repressed during the trophophase (Table 3.7). The composition of the culture medium can however also be so arranged so that a significant fraction of a slowly metabolizable substrate is used, the organism thus growing under suboptimal conditions, leading to a situation where growth and secondary metabolite formation occur in parallel.

There is only scanty information on the nature of the induction of enzymes of secondary metabolism: Methionine induces cephalosporin and fosfomycin synthesis, factor A induces streptomycin production (see Section 13.4) and tryptophan is a precursor and regulator of ergot alkaloid biosynthesis. (see Section 14.6).

Table 3.7 Key enzymes of secondary metabolism which are induced at the end of the trophophase

Enzyme	Secondary metabolite
Amidinotransferase	Streptomycin
Acyltransferase	Penicillin
Phenylacetate-activating enzyme	Tylosin
Oxidoreductase	Gramicidin S
Transmethylase	Actinomycin
Synthetase I and II	Ergot alkaloids
Phenoxazinone synthetase	
Dimethylallyl tryptophan synthetase	
Chanoavrin-lyase	

Endproduct regulation It is known that antibiotics inhibit their own biosynthesis (e.g. penicillin, chloramphenicol, virginiamycin, ristomycin, cycloheximide, puromycin, fungicide, candicidin, streptomycin). The mechanism of feedback regulation has only been explained in a few cases: chloramphenicol represses arylamine synthetase, which is the first enzyme in the biosynthetic pathway which branches off from chloramphenicol and penicillin, it has been shown that the concentration of the end product which inhibits corresponds to the production level. Thus, if strains could be isolated which were less sensitive to endproduct inhibition by these antibiotics, they might produce higher yields.

Catabolite regulation Catabolite regulation is a general regulatory mechanism in which a key enzyme involved in a catabolic pathway is repressed, inhibited, or inactivated when a commonly used substrate is added. Substrates which have been found to bring about catabolite repression include both carbon and nitrogen sources.

Carbon sources: Biosynthesis of different secondary metabolites (antibiotics, gibberellins, ergot alkaloids) is inhibited by rapidly fermentable carbon sources, particularly glucose (Table 3.8). Depending on the organism and metabolite, the basic mechanism of this carbon catabolite regulation is different. One is the well-known **carbon catabolite repression** found in many bacteria, yeasts and molds which involves a catabolite activator protein (CAP) that must combine at the promoter site before RNA polymerase can attach. The CAP will only bind if it is first complexed with cyclic adenosine monophosphate, *cyclic AMP*. Readily utilizable carbon sources such as glucose stimulate an enzyme which causes the breakdown of cyclic AMP, thus rendering CAP inactive. Thus, glucose inhibits the synthesis of the mRNA for any enzyme requiring CAP for its biosynthesis.

If readily utilizable carbon sources such as glucose elicit catabolite regulation, how is it pos-

Table 3.8 Catabolite regulation in antibiotic biosynthesis

Antibiotic	Carbon Source	
	Inhibitory	Not inhibitory
Penicillin	Glucose	Lactose
Cephalosporin C	Glycerol	Asparagine
Actinomycin	Glucose	Sucrose
Streptomycin	Glucose	Galactose
		Mannan and L-Rhamnose
Silomycin	Glucose	Maltose
Indomycin	Glucose	Fructose
Bactracin	Glucose	Citrate
Chlaramphenicol	Glucose	Glycerol
Mitomycin	Glucose	Glucose-feeding
Neomycin	Glucose	Maltose
Kanamycin	Glucose	Galactose
Butirosin	Glucose	Glycerol
Puromycin	Glucose	Glycerol
Novobiocin	Citrate	Glucose
Candidin	Glucose	Glucose-feeding
Candicidin	Glucose	Glucose-feeding

Listed are carbon sources which cause catabolite repression and carbon sources which do not inhibit product formation.

sible to use these substrates in an industrial fermentation? In the manufacture of antibiotics, carbon sources other than glucose may be used or glucose can be fed at low rates, to minimize catabolite repression.

Nitrogen sources: In several antibiotic fermentations it has been observed that ammonia or other rapidly utilizable nitrogen sources act as inhibitors. The fundamentals of this regulation have not yet been completely understood, although glutamine synthetase and glutamic dehydrogenase are considered key enzymes. In enteric bacteria it has been established that glutamine synthetase has a regulatory function in the synthesis of additional enzymes which are involved in nitrogen assimilation.

Phosphate regulation In a culture medium inorganic phosphate (P_i) is required within a range of 0.3–300 mM for the growth of prokaryotes and eucaryotes. However, a much lower phosphate concentration inhibits the production of many

secondary metabolites. In a number of systems studied, the highest P_i concentration which allows unimpeded production of secondary metabolites is about 1 mM; complete inhibition of production occurs at about 10 mM P_i . Phosphate regulation has been observed in the production of alkaloids (Section 14.6), gibberellins and particularly in several antibiotics (Table 3.9). The phosphate regulation mechanism is not yet fully understood. P_i controls the metabolic pathways which precede the first stage of secondary metabolite formation, but also affects the biosynthesis of secondary metabolites themselves.

Several varying phosphate effects have been described. In one type of mechanism, phosphate stimulates primary metabolism. The consumption of P_i in the culture medium seems to be one of the main causes for the metabolic change from the trophophase to the idiophase. By adding P_i at the end of the trophophase, the idiophase can be delayed; after addition of P_i in the idiophase, growth resumes and antibiotic synthesis stops. In a second mechanism, phosphate inhibits or represses phosphatases, enzymes which are involved in the biosynthesis of secondary metabolites. This mechanism has been demonstrated for the biosynthesis of the aminoglycoside antibiotics. A third mechanism leads to a shift in the

Table 3.9 Some antibiotics whose formation is inhibited by organic phosphate

Antibiotic	Producer	P_i range (mM) permitting antibiotic formation
Streptomycin	<i>Streptomyces griseus</i>	1.5 – 15
Kanamycin	<i>S. kanamyceticus</i>	2.2 – 5.7
Bactracin	<i>Bacillus licheniformis</i>	0.1 – 1.0
Gramicidin S	<i>B. brevis</i>	10 – 60
Amphotericin B	<i>S. nodosus</i>	1.5 – 2.2
Candicidin	<i>S. griseus</i>	0.5 – 5.0
Chlortetracycline	<i>S. aureofaciens</i>	1 – 5
Vancomycin	<i>S. orientalis</i>	1 – 7
Actinomycin	<i>S. antibioticus</i>	1.4 – 17
Tetracycline	<i>S. aureofaciens</i>	0.14 – 0.2
Cycloheximide	<i>S. griseus</i>	0.05 – 0.5

(from Martin, 1977)

carbohydrate metabolism from the pentose phosphate cycle or hexose monophosphate pathway to glycolysis when excess phosphate is present. NADPH, thus becomes a limiting factor in the synthesis of antibiotics.

Finally, it has been shown that phosphate restricts the induction of secondary metabolite production. For instance, dimethyl allyltryptophan synthetase, the first specific enzyme of ergot alkaloid biosynthesis, is not produced in the presence of high P_i concentrations (see Section 14.6).

Some data indicate that phosphate regulation occurs via a control system at the transcription level. A model of negative control shown in Figure 3.21 has been proposed for the induction of candidin synthesis. The repressor concentration decreases at the end of the trophophase, so that the transcription of genes that code for antibiotic synthesis is possible. It has not yet been determined whether phosphate itself functions as a corepressor or whether it regulates the content of intracellular effectors (cAMP; ATP; phosphorylated nucleotides such as ppGpp or pppGpp). In the case of candidin synthesis, an effector may be ATP. The intracellular ATP content decreases sharply before the production of the antibiotic, but increases rapidly after adding P_i .

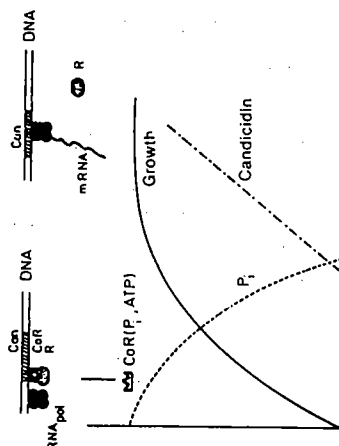


Figure 3.21 Model for control of candidin synthesis through P_i or ATP (Martin et al., 1979). RNA_{pol}, RNA polymerase; CoR, corepressor; R, repressor.

The industrial production of P_i -regulated secondary metabolites is carried out, as far as possible, under conditions of P_i limitation. The standard culture media used for production often contain relatively high proportions of P_i (around 2.5 mM) and at these levels, phosphate can decrease yields. The isolation of phosphate-deregulated (PD) mutants (as has been done for candidin producing strains) seems to be a worthwhile possibility for strain development. PD mutants, all of which are less sensitive to phosphate regulation, frequently produce candidin up to 70% better than the wild type in a normal medium.

Autoregulation In some actinomycetes it has been possible to show that differentiation and secondary metabolism are subjected to a type of "self-regulation" from low-molecular weight substances. For instance, in *Streptomyces griseus* and *S. bikiniensis* the formation of streptomycin, the development of streptomycin resistance, and the spore-formation are all affected by factor A, a substance produced by the streptomycetes themselves (see Figure 13.24 for the structure). It has been shown that the streptomycin resistance property is due to the increased transcription of the gene for the enzyme, streptomycin phosphotransferase, induced by the factor A. The effect on streptomycin formation is thought to be due to a shift in the metabolism of the carbohydrate source: although the activity of the enzyme glucose-6-phosphate dehydrogenase is high in factor A-deficient mutants, this enzyme cannot be demonstrated in high-yielding strains. Addition of factor A to mutants leads to a strong decrease in enzyme activity. It is assumed that when the pentose phosphate cycle is blocked through the absence of glucose-6-phosphate dehydrogenase, glucose is channeled into pathways involved in the formation of streptomycin units.

In a sense, factor A can be considered analogous to a hormone. Autoregulatory mechanisms similar to that of factor A have been found in other actinomycetes. For instance, a factor is hypothesized in *S. virginiae* which stimulates the

formation of the antibiotic virginiamycin. In rifamycin-producing *Nocardia mediterranei* butyryl phosphoadenosine has been characterized as a regulatory factor. Two γ -lactones (L factors) have been shown to be autoregulatory agents in leu-kaemomycin-producing *S. griseus*.

3.6 GENE TECHNOLOGY

By gene technology is meant those techniques that permit the manipulation of genes as biochemical entities. Gene technology includes in vitro recombination, gene cloning, gene manipulation, and genetic engineering. These techniques permit the introduction of specific DNA sequences into prokaryotic or eucaryotic organisms and the replication of these sequences; that is, to clone them. To carry out these procedures, the following steps are necessary:

- The DNA sequence to be cloned must be available.
- The sequence must be incorporated into a vector.
- The vector with the DNA insert must be introduced by transformation into a host cell, where the vector must replicate the insert in a stable manner.
- The clone which contains the foreign DNA must be selectable in some manner.

Methods of gene technology

The basic procedures for the first steps in gene cloning are outlined in Figure 3.22. Details can be found in specialized books on gene cloning (see References).

Isolation of DNA sequences for cloning

Genome fragments Restriction endonucleases are used to cut DNA. These enzymes belong to specific restriction and modification systems and are used by the cell to protect itself from foreign DNA. Restriction enzymes split double-stranded DNA at specific sites, usually at palindromic

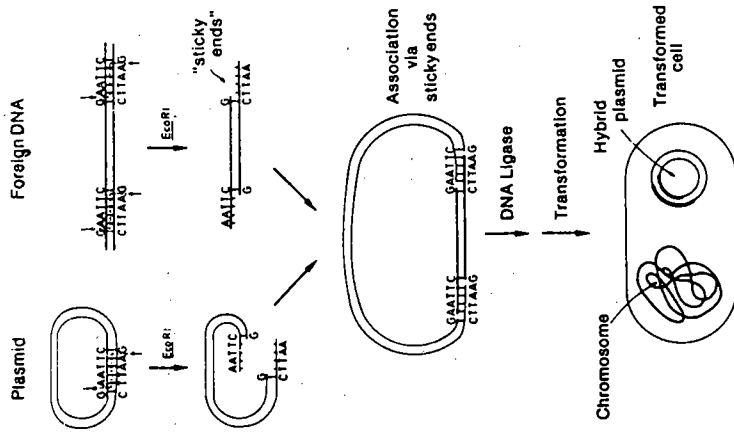
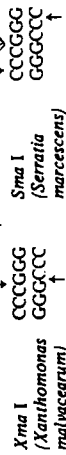


Figure 3.22 Method for the production of recombinant DNA

structures 4–11 nucleotides in length. Either blunt double-stranded ends or short, cohesive, single-stranded "sticky ends" develop, depending on the enzyme involved:



More than 600 restriction endonucleases are known in bacteria.

If the sequence of the DNA to be cloned is unknown, it is possible to use a so-called "shotgun" approach. With this procedure, a *gene bank* is produced by using suitable restriction enzymes

to fragment the total genome of the organism into pieces of about 20 kilobases in length. Each of these fragments is linked to a vector (generally a phage or cosmid) and cloned into a suitable host. With an appropriate screening method the cultures containing the clones are selected, among which the target gene should be present. There will be a 99% probability, assuming 20 kb fragments, that in *E. coli* (total genome length of about 4000 kb) all genes will be cloned if 900 separate clones are selected (in yeast, which has a total genome size of 13,500 kb, 3100 clones would be required).

It is preferable if the initial cloning is carried out with enriched fragments. Enrichment can be done either by use of sucrose-gradient centrifugation, agarose-gel electrophoresis, column chromatography, or by use of specific gene probes. Gene probes are short DNA or RNA fragments that have been "tagged" in some way which are able to hybridize with the specific genome sequence. Gene probes can be tagged with ^{32}P or ^{35}S labeled nucleoside triphosphates, permitting the probes to be detected by autoradiography. An alternative procedure is the use of biotin-labeled DNA. Because of the strong affinity of biotin for avidin (a glycoprotein obtained from egg white) or streptavidin (a protein obtained from cultures of *Streptomyces avidinii*) the presence of the biotin-labeled DNA can be detected using avidin or streptavidin which has been marked with a fluorescence dye, an enzyme, or an antibody.

To obtain expression of the cloned gene, it is necessary to understand the genetic organization of prokaryotes and eukaryotes. In eukaryotes, the coding regions of genes (exons) are generally broken with noncoding regions (introns). From the primary transcript from the nuclear gene, the introns must be removed (RNA processing) before the mature translatable mRNA is present. The RNA processing steps cannot be carried out by prokaryotes so that a correct translation of cloned genes of eukaryotes is not possible. Therefore, when bacteria are used as cloning hosts it is preferable to use synthetic DNA or complementary

DNA (a DNA copy of the mature mRNA) (see below).

Synthetic DNA In order to produce a specific DNA fragment containing the coding region of a protein, the DNA sequence is deduced by "reverse translation" from the amino acid sequence of this protein. Using an automated DNA synthesis machine, it is possible to produce DNA fragments of 20 to 100 bases, which can be connected together to make longer sequences. Examples of the use of this technique are the artificial synthesis of the gene for somatostatin, a peptide hormone with 14 amino acid residues, and the synthesis of the A and B chains of insulin, which were cloned and expressed in *E. coli*. Since the DNA synthesis procedure is completely under the control of the investigator, it is also possible to produce sequences in which one or more bases have been changed, making possible the production of highly specific mutations.

Production of complementary DNA (cDNA) Specific mRNA molecules, isolated from mRNA-rich cells or tissues, are used as templates in vitro with the enzyme *reverse transcriptase*, to produce complementary DNA. An example of the use of this method for cloning the insulin gene from rats in *E. coli* is shown in Figure 3.23.

Vectors As vectors for the transfer of DNA, a variety of genetic elements are available. Most commonly used are plasmids, temperate phages, and cosmids.

Plasmids are defined as circular, extrachromosomal, self-replicating DNA molecules. They have molecular weights between 1.5×10^6 and 200×10^6 . Large plasmids and small plasmids have different properties. Large plasmids have an average molecular weight of 65×10^6 , there are usually 1-2 copies per bacterial chromosome in the cell and they can be transferred during conjugation. Small plasmids (about 5 million molecular weight), of which there are more than 10 copies per bacterial chromosome, are not con-

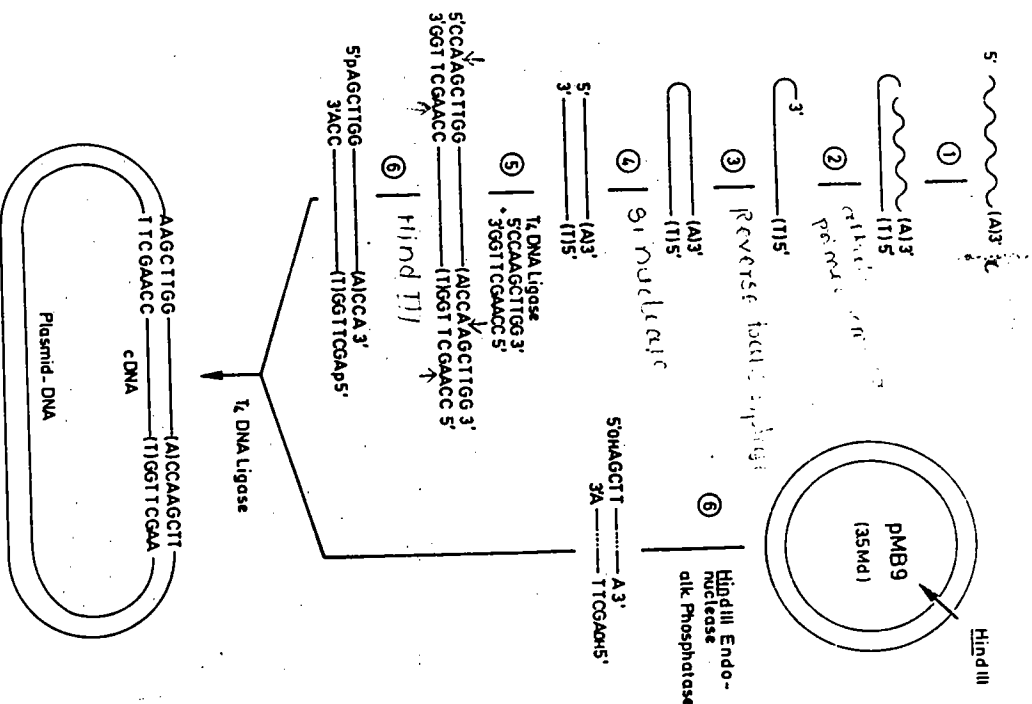


Figure 3.23 Cloning rat insulin genes (Ullrich et al., 1977). 1. mRNA is isolated from an insulin-producing tumor of the rat and transcribed with reverse transcriptase into cDNA. Simultaneously, the enzyme reverse transcriptase produces a hairpin structure (fold-back region) which serves as a primer for the completion of the double strand. 2. RNA is removed from the mRNA-cDNA hybrid molecule by treatment with alkali. 3. The single-stranded cDNA becomes double-stranded through reverse transcriptase. 4. The fold-back region of the hairpin structure is digested by S1-nuclease. 5. Through T4 ligase, chemically synthesized nucleotide sequences containing the recognition sequence for the restriction endonuclease *Hind*III are attached at both ends of the cDNA. 6. Through the action of *Hind*III, sticky ends are created which permit the attachment to a similarly-treated plasmid DNA.

jugative. Plasmids code for numerous and varied cell functions:

- Fertility: ability to transfer genetic material through conjugation.
- Antibiotic resistance: resistance to one or more antibiotics; R-plasmids are known in more than 50 species of bacteria.
- Resistance to heavy metals: Cd²⁺, Hg²⁺.
- Resistance to ultraviolet radiation.
- Production of bacteriocins, substances which inhibit or kill cells of the same species.
- Production of antibiotics: a plasmid has been shown to code for the antibiotic methylenomycin.
- Utilization of unusual carbon sources: breakdown of camphor, octane, and octanol by *Pseudomonas*, for example (see Table 3.10).
- Formation of toxins and surface antigens, such as enterotoxin and hemolysin.
- Tumor induction in plants: formation of crown gall tumors by the Ti plasmid of *Agrobacterium*.
- Involvement in sporulation in streptomycetes.

Plasmids are capable of adding foreign genes to the genetic material of the cell. If there is no homologous area in the bacterial chromosome for these foreign genes, they cannot be exchanged during crossing over, but may be maintained through plasmid replication. Some plasmids exhibit the phenomenon of *incompatibility*, in which two related plasmids are unable to be maintained together in the same cell.

Table 3.10 Catabolic plasmids from *Pseudomonas*

Plasmid	Substrate broken down	Transmissibility
NAH	Naphthalene	Conjugative
SAL	Salicylic acid	Conjugative
CAM	Camphor	Conjugative
OCT	Octane, hexane, decane	Nonconjugative
TOL	p- or m-xylene, toluene	Conjugative
PIP1	2,4-dichlorophenoxyacetic acid	Conjugative
PAC25	3-chlorobenzoate	Conjugative
PAC27	3- and 4-chlorobenzoate	Conjugative

Cosmids are plasmids which contain the *cos* region of phage λ , which makes it possible for the plasmid to be packaged within bacteriophage particles and hence transferred by infection to *E. coli*. Cosmids are capable of incorporating extremely long sequences of DNA, 32–47 kb, and hence are especially useful for the production of gene banks.

The foreign DNA is either incorporated into a specific restriction site of the cosmid (insertion vector) or replaces a fragment of equivalent length of the vector (substitution vector).

Depending on the purpose, the following kinds of vectors can be constructed:

Cloning vectors are used primarily for the amplification of foreign DNA in the host cell. With certain plasmids, the copy number per cell can be increased by growth in the presence of the antibiotic chloramphenicol, which inhibits the replication of the host genome without affecting the replication of the plasmid. In this way, recombinant plasmids can often be produced to as many as 1000 copies per cell.

Sequence vectors are those which contain an increased number of recognition sites for restriction enzymes (so-called multi-purpose cloning sites). When the foreign DNA is integrated into such a plasmid, DNA fragments of various sizes can be produced and sequenced.

Expression vectors contain not only the foreign gene but also promoter, operator, and terminator sequences so that the gene can be efficiently transcribed into mRNA. In addition, expression vectors contain the ribosome-binding site so that efficient translation can take place.

The expression of foreign DNA in the host organism is largely dependent on the system used. In *Escherichia coli*, procaryote DNA is transcribed and expressed into protein. Under favorable conditions as much as 30% of the dry weight of the cell can consist of the heterologous protein.

The expression of genes of lower eucaryotes, such as *Saccharomyces cerevisiae* or *Neurospora crassa*, has been successfully achieved in *Escherichia coli*, but in more complex eucaryotes, DNA

transcription and translation have not occurred correctly. These difficulties may be overcome by incorporating eucaryote DNA into a procaryote gene containing appropriate regulatory signals, thus permitting expression of the eucaryote DNA. This was first accomplished with somatostatin, which was incorporated into the lactose operon located on plasmid pBR322 (Figure 3.24).

The product synthesized by the *Escherichia coli* cell (a fusion protein) was β -galactosidase-somatostatin. Active somatostatin was cleaved from the chimeric protein *in vitro* by treatment with cyanogen bromide, which splits peptides at methionine residues. The yield of somatostatin obtained was one-tenth of the value calculated from the plasmid copy number but was still relatively favorable (10 mg somatostatin protein per 100 g *Escherichia coli* wet weight) considering the simplicity of the *Escherichia coli* fermentation.

In the same way, synthetic genes of the A and B chain of human insulin were incorporated into the β -galactosidase region of the pBR322 plasmid and cloned separately in *Escherichia coli*. The transformed bacteria synthesized the A and B chains separately; by mixing both compounds, an active insulin preparation was developed and has now been marketed by the Eli Lilly Company.

Another insulin production method involved use of *Escherichia coli* x1776. Complementary DNA copies of preproinsulin-mRNA of the rat were cloned into the penicillinase gene of plasmid pBR322. A penicillinase-proinsulin complex, penicillinase(24-182)-(Gly)-proinsulin(4-86), was formed by the bacteria. The synthesis rate was about 100 molecules per cell. Since *Escherichia coli* penicillinase is an extracellular protein, the complex was excreted from the cell and could be transformed into active insulin by splitting the penicillinase residue from the insulin C chain.

In order for a vector to replicate in the host cell, it must contain the appropriate origin-of-replication site (*ori*). Vectors containing *ori* of two separate host systems are able to replicate in both. Such vectors, called shuttle vectors, are

known for *E. coli* and *Bacillus subtilis*, and for *E. coli* and yeast.

Incorporation of foreign DNA This is done differently with blunt end or cohesive end fragments. With blunt-end fragments, the 5' ends are digested by λ -exonucleases and the 3' ends are extended through action of a terminal transferase enzyme with ATP or TTP. In this process, a terminal poly-A or poly-T sequence results. One strand with a poly-A sequence can combine with another strand containing a poly-T sequence to form a circle as a hybrid plasmid. When there are sticky ends, DNA fragments of any origin combine as long as the same restriction endonuclease has been used in the creation of both fragments.

Insertion of foreign DNA into the host The efficient incorporation of foreign DNA into the target host cell is a critical step in gene manipulation. It is essential that the host is able to take up the engineered DNA. Host organisms include not only microorganisms but also plant, animal, and human cells. In *E. coli* transformation of naked plasmid or phage DNA can be brought about in the presence of CaCl_2 , which renders the cell wall and cell membrane permeable to free DNA. With plasmid DNA, 10^7 – 10^8 transformants can be obtained per μg DNA. An alternative procedure is to use a transduction system consisting of phage or cosmid DNA packaged into empty phage heads. For gram-positive bacteria, actinomycetes, yeasts, and filamentous fungi, effective transformation of plasmid or phage DNA can be accomplished using the PEG system in protoplasts (see Section 3.4). H_2O CaCl_2 PEG

Analysis of recombinant clones There are several possible procedures that can be used to screen transformed cells and determine that the cloning and expression processes have been effective:

- Identification of host cells containing the foreign-DNA-containing vector;

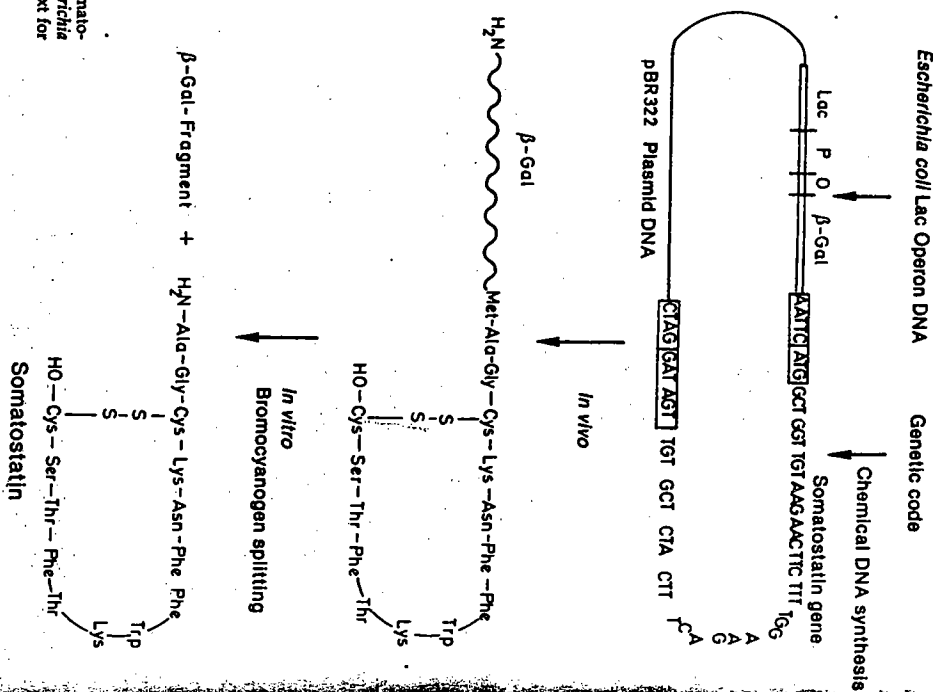


Figure 3.24 Expression of somatostatin gene sequence in *Escherichia coli* (Takura et al., 1977). See text for explanation

- Detection of the foreign DNA in the host cells;
- Detection of the foreign DNA indirectly by assaying for the expression product (foreign protein).

To select transformed cells, the **marker in-activation technique** can be used. Vectors are used containing two selectable markers (for in-

stance, antibiotic resistance), one of which contains the recognition site for the restriction enzyme used in the cloning process. If the foreign DNA becomes integrated into this antibiotic resistance gene, the activity of that gene is lost (insertional inactivation). Host cells that lack the vector are sensitive to both antibiotics, host cells containing a vector lacking the foreign DNA are resistant to both antibiotics, whereas vectors with

inserted foreign DNA are sensitive to the one antibiotic into whose resistance gene the foreign DNA has been inserted.

To assay a complete gene bank for the presence of the gene, it is anticipated that around 1000 recombinant clones must be tested. Further tests must then be used to determine that the appropriate clone has been obtained. To demonstrate the presence of cloned DNA in the cell, two types of procedures are used, colony hybridization and Southern blotting.

Colony hybridization is done by using the replica-plating technique to transfer around 200 colonies from agar plates to a nitrocellulose filter. The colonies on the filter are then lysed and the released DNA denatured with 0.5 N NaOH. After removing excess protein with the enzyme proteinase K, the single-stranded DNA is fixed to the filter by heating to 80°C. After hybridization with a ³²P-labeled probe (either DNA or RNA), the filter is subjected to autoradiography. The colonies corresponding to the active spots can then be isolated from the original plate.

After isolation of hybrid DNA molecules, the specific DNA sequence can be identified by means of the Southern blotting technique. After treatment of the DNA with a restriction enzyme, the resulting DNA fragments are separated by gel electrophoresis. The DNA bands are then transferred to paper or nylon filter by blotting. The filter provides a stable carrier for the fragments, and the fragment containing the target sequence is identified using a gene probe. (The corresponding technique for RNA is called Northern blotting, that for proteins is called Western blotting.)

A different kind of procedure for detecting the desired clones involves seeking clones in which the gene product (protein) has been expressed. Since the expression efficiency is often quite low, a very sensitive method for detecting the gene product is necessary. One of the most widely used methods is immunological, in which an antibody (marked by radioactivity or enzyme) is used as a probe.

Cloning and expression systems for various microorganisms

The selection of a suitable cloning system depends to a great extent on the protein to be synthesized. Low-molecular weight proteins with few disulfide bridges and which do not require glycosylation for activity can be cloned and expressed in bacteria. Bacteria are unable, however, to glycosylate proteins, whereas yeasts preferentially glycosylate with mannose rather than glucose (the preferred sugar in human proteins). To produce complex proteins, cell cultures are most suitable. In such cultures, not only is the protein properly glycosylated, but it is also secreted in its native configuration.

As shown in Table 3.11, when foreign proteins are expressed, much of the synthesized protein may be retained in the cells rather than being excreted into the medium. It may therefore be preferable to begin with a system in which total activity is lower in order to save costs of isolation and purification.

Gram-negative bacteria Because of its well known biology and genetics, *E. coli* is the preferred organism for experiments in genetic engineering. A large number of heterologous genes have been cloned and expressed in this organism. However, wild type plasmids such as ColEI, pSC101, and RSF2124 presented a number of disadvantages, so that suitable vectors have been constructed *in vitro*. A widely used vector is pBR322, a ColEI-like plasmid, which contains

Table 3.11 Final yield of heterologous proteins in relation to protein secretion in various cloning systems

Cloning system	Amount of het. protein excreted (mg/l)	Final yield (mg/l)
<i>E. coli</i>	5000	No
<i>B. subtilis</i>	100	Yes
<i>Saccharomyces</i>	10	Yes
<i>Cerevisiae</i>	2000	No
Animal cell culture	100	Yes

Final yield after isolation and purification. Data of Davie (1986).

the ampicillin-resistance gene (Amp^r) of RSF2124, the tetracycline-resistance gene (Tc^r) of pSC101, and the origin of replication of pMP1. In addition, pBR322 contains recognition sites for 20 restriction enzymes. Other suitable vectors for *E. coli* include phage λ and single-stranded DNA plasmids M13, f1 and fd, which are especially suitable for DNA sequencing. Details of cloning systems for *E. coli* are widely available in manuals and reference books (see References).

Large-scale production of foreign proteins in *E. coli* presents some difficulties, especially because foreign proteins are often subjected to extensive proteolysis or are produced in the cell in the form of highly insoluble protein bodies (inclusion bodies). Purification of these atypically folded and frequently denatured proteins leads in many cases to only a very tiny amount of the correct product.

To clone in other gram-negative bacteria, for instance *Pseudomonas*, derivatives have been developed of the wild type conjugative plasmids IncP and IncW as well as the nonconjugative plasmid IncQ. These plasmids possess an extremely wide host range and can be used in almost all species of gram-negative bacteria.

Gram-positive bacteria Species of the genus *Bacillus* have been used for many years in industry for the production of peptide antibiotics and exoenzymes. Their use in large-scale cultivation is therefore well understood. An additional advantage of the bacilli for gene technology is that excellent excretion of foreign proteins takes place. However, the native plasmids and phages of *Bacillus subtilis* are not suitable as cloning vectors. In vitro construction of suitable plasmids was based on the utilization of antibiotic resistance plasmids of *Staphylococcus aureus*. An artificial plasmid pHV11, containing tetracycline-resistance and chloramphenicol-resistance genes (Tc^rCm^r) was developed for *B. subtilis* which was analogous to pBR322 of *E. coli*. Addition of the origin of replication of *E. coli* led to the development of a shuttle vector for *E. coli* and *B. subtilis*. Vectors suitable for various experimental

purposes could also be derived from temperate phages p11, ϕ 105, and NP02. For the transformation of bacilli, it is necessary to begin with protoplasts and to use single-stranded DNA. One problem with *B. subtilis* is that plasmids are often unstable, leading to rearrangements of the genome or deletions.

Other gram-positive microorganisms of industrial interest for which cloning systems have been developed include streptococci (for the dairy industry), clostridia (for acetone, butanol, isopropanol, butyric acid, and acetic acid fermentations), *Corynebacterium* and *Brevibacterium*, for amino acid and nucleotide production.

A number of plasmid and phage vectors have been developed for the industrially important group of streptomycetes. Among others, these are pIJ61, a derivative of plasmid SLP1.2 of *Streptomyces lividans*, SCP2, a low-copy-number plasmid of *S. coelicolor* A3(2), bacteriophage ϕ C31 of *S. coelicolor*, and plasmid pIJ101, a conjugative multicopy plasmid of *S. violaceoruber*. Detailed procedures for cloning in *Streptomyces* have been developed (see References). With these methods, recombinant DNA techniques can be used in the study of biosynthesis and regulation of secondary metabolites in streptomycetes. With such basic information available, intraspecific or interspecific cloning can be used for strain improvement. Some successes in this area have already been obtained. The O-methyltransferase gene of *S. coelicolor* A3(2), involved in the biosynthesis of undecylprodigiosin, has been cloned. Studies on glucose repression of glycerol catabolism in *S. coelicolor* have led to clarification of the mechanism of glucose catabolite regulation.

Genes controlling the production of three antibiotically active substances of *S. coelicolor* have been cloned and expressed: actinorhodin (7 chromosomal genes), undecylprodigiosin (5–6 chromosomal genes), and methylenomycin (5–6 plasmid-controlled genes). In addition, genes involved in antibiotic synthesis and antibiotic resistance have been cloned for a variety of other streptomycetes (see Section 3.7). Because of the

extreme complexity in the biosynthetic pathways of antibiotics, much further basic research will be needed before these newer methods will lead readily to strain improvement.

Because streptomycetes readily excrete proteins, *S. lividans* has been used as a model system for the expression of bovine somatotropin, human serum albumin, and human interleukin-2, although the use of streptomycetes as expression systems for heterologous proteins is still in its infancy.

Eucaryotes The yeast *Saccharomyces cerevisiae* has replaced *E. coli* in a number of projects for the expression of eucaryotic proteins. Large-scale production of yeast is well-established. Genes for foreign proteins have been expressed and the proteins secreted into the medium. A further advantage for the production of genes from mammals is that yeast is able to glycosylate foreign proteins. A whole series of mammalian proteins have been successfully expressed and correct splicing of mRNA and correct glycosylation do not always occur.

A range of vectors have been developed for *Saccharomyces cerevisiae*, mostly hybrids between yeast DNA and *E. coli* plasmids which can serve as shuttle vectors. The principle vectors are summarized here:

- **YEp vectors** (yeast integrating plasmids) contain the ColE1 origin of replication of *E. coli* and chromosomal sequences from yeast. They replicate as plasmids in *E. coli*. Although they do not replicate in yeast they are able to integrate into the yeast genome, although the integrated sequences are relatively unstable.
- **YEp vectors** (yeast episomal plasmids) are circular plasmids (2 μ m in length, about 6 kb) that are present in yeast cells in around 50–100 copies. When connected with selectable yeast sequences and corresponding sequences from *E. coli* plasmids, they can be used as vectors. Vectors based on the 2 μ m DNA are, however, relatively unstable.

- **YRp vectors** (yeast replicating plasmids) are yeast sequences which have a chromosomal origin and contain an origin of replication. After incorporation into bacterial plasmids, they can be used as vectors. They exhibit a relatively high frequency of transformation (10^3 – 10^4 transformants per μ g DNA).

- **YCp vectors** (yeast centromere plasmids) are YRp vectors into which the centromere of chromosome III has been incorporated. These plasmids behave as circular minichromosomes, are stably inherited, and can be used as vectors.

Filamentous fungi Although a variety of antibiotics and enzymes are produced commercially with filamentous fungi, cloning systems for these organisms are not nearly so well developed as for yeast. Plasmids resembling the Yip type have been constructed by combining genes from *Neurospora* with the origin of replication of pBR322 of *E. coli*. Hybrids between pBR322 and mitochondrial DNA (mtDNA) of the ascomycete *Podospora anserina* replicate in both *E. coli* and *P. dospora*. Plasmids from *N. crassa* and *P. anserina* can perhaps be produced by insertion of the origin of replication from mtDNA sequences.

Various commercially grown aspergilli, such as *Aspergillus niger*, *A. awamori*, and *A. oryzae*, excrete large amounts of exoenzymes. However, the species *A. nidulans* is the best known genetically and has served as a model for the development of effective transformation and expression systems. Biologically active bovin chymosin, human interferon $\alpha 2$, and human tissue plasmin activator (tPA) have been expressed and secreted in *A. nidulans*. However, it is not certain whether the correct glycosylation has been obtained.

The procedures that have been developed in yeast and aspergilli have also been adapted to the important β -lactam-producing fungi *Penicillium chrysogenum* and *Cephalosporium acremonium*. In *C. acremonium* a cosmid gene bank has been constructed which has permitted the isolation of the *leuB* gene which codes for β -isopropylmalate dehydrogenase. A vector has

been constructed, pT221, which contains, among others sequences from pBR322, a mDNA fragment of *C. acetomoni*, and an ARS region (autonomously replicating sequence) which permits replication in *S. cerevisiae*. Using the gene coding for hygromycin B phosphotransferase (HPT) of *E. coli*, it has been possible to transform protoplasts of *C. acetomoni* and demonstrate the presence of the HPT coding sequence. The principal goal of this work is to increase the yield of the enzyme.

Actinomycetes In the continuing search for suitable expression systems for mammalian proteins, research has turned to the organism *Dicystosium discoideum*, a eucaryote which has a life cycle in which a free-living amoeboid form alternates with a fruiting body in which resting cells are produced. A functioning expression system for this organism has been created; for a particular foreign protein, yields 300 times greater than that of an optimized *E. coli* system have been obtained. Since *Dicystosium* grows in the vegetative phase as an amoeba, diffusion is not a limiting factor in nutrient uptake so that continuous culture can be used to produce the desired protein in an unlimited fashion. However, whether correct protein processing and glycosylation occur has not been determined, and scale-up of this organism to a large-scale industrial process has not yet been accomplished.

Risks in genetic engineering

The creation of new types of viruses and the possible cloning of the DNA of tumor viruses has been perceived to be potentially dangerous. Since *Escherichia coli*, which exists in the human intestinal tract, has been used almost exclusively for the cloning of recombinant DNA, the possibility of human infection cannot be ruled out. Thus in 1974 numerous renowned American scientists called for a voluntary moratorium on certain cloning experiments until potential risks were resolved. The first guidelines for work involving recombinant DNA were established at an

international conference at Asilomar, California in 1975. They became the basis for the rules of the U.S. National Institute of Health (NIH). Since that time, however, the risks have been considered so slight that the NIH guidelines have been suspended. This has been because of:

- The development of suitable safety measures.
- The use of weakened *Escherichia coli* strains (e.g. x1776) with a probability of survival of 10^{-6} outside of the laboratory.
- Experimental evidence that there is only very slight danger of the transfer of cancer genes to humans.

Certain experiments are still prohibited or require authorization. Each country has its own specific guidelines for recombinant DNA research, and the operative regulations can be obtained by applying to the proper authorities.

3.7 USE OF GENETIC METHODS

Strain optimization

The primary goal of an industrial strain development program is the increase in the yield of the desired product. The genetic techniques for accomplishing this goal have been discussed in Sections 3.2-3.6. If the desired product, for example, an enzyme, is controlled by a single gene or a small group of genes, then high-yielding strains can be produced by:

- Isolation of mutants resistant to inhibitors of protein synthesis, which often overproduced proteins;
- Manipulation of regulatory signals to increase transcription or translation by cloning the gene on an expression vector or inserting the gene into a transposon which has a strong promoter;
- Modification of the gene by use of site-directed mutagenesis.

By increasing the gene dosage (gene amplification), the yield may be increased, provided the expression of the duplicated genes is not re-

stricted by regulation. Gene amplification can occur in the following ways:

- Increasing the number of DNA replication sites in growing bacterial cells causes amplification of the genes situated near the origin of replication.
- Diploidization of fungi increases gene dosage, although the strains are usually unstable.
- Isolation of hyperinduced strains, which have been cultivated under selective conditions over a long period. These strains are extremely unstable, however, and are usually not suitable for commercial processes.
- If the gene has been characterized, the greatest success is likely by use of genetic engineering methods, for example, cloning and amplification of the gene by means of a multicopy plasmid or a phage vector. For instance, by use of a cosmid system the formation of the enzyme penicillin acylase in *E. coli* has been markedly increased when compared to the wild type. A whole series of industrial enzymes have been optimized in this way.

Difficulties abound when the goal is the increase in yield of a multi-gene product such as a primary or secondary metabolite, although some successes have been achieved. Amino acid production has been increased by cloning the whole genome, first in *E. coli*, later in production strains such as *Corynebacterium*, *Brevibacterium*, or *Serratia* (see Chapter 9).

For secondary metabolites such as antibiotics, cloning and amplification of the rate-limiting enzyme of the biosynthetic pathway can be done. As a first step in this direction, the genes for a number of antibiotics have been isolated, cloned, and in a few cases expressed. These include actinorhodin, methylerythronycin, and undecylprodigiosin (*Streptomyces coelicolor*), cephalosporin (*Cephalosporium acetomoni*), erythromycin (*S. erythraeus*), oxytetracycline (*S. glaucus*) and tyrocin (*S. fradiae*).

In addition to increase in yield, genetic methods have been used to improve the fermentation

characteristics of the process. For instance, if the yeast *Saccharomyces cerevisiae* contains an α -amylase from *Bacillus subtilis*, ethanol fermentation can occur at 40°C and a stronger starch hydrolysis can take place. After transfer of the killer factor from a wild type to an industrial mutant of *S. cerevisiae*, wine, beer, sake, and alcohol production was protected from undesirable wild yeast contamination.

Qualitative change in the spectrum of antibiotics

Many producers of secondary metabolites also produce a range of chemically related byproducts. Mutants which produce only the desired antibiotic or only a small percentage of other substances can simplify the product recovery considerably and thus lower its cost. Table 3.12 shows examples which are industrially relevant.

Production of modified secondary metabolites

One way to produce therapeutically effective antibiotics is to use mutants which synthesize derivatives of known antibiotics. Such modified antibiotics may be synthesized either directly or through transformation of added precursors. With the use of the protoplast fusion or recombinant DNA techniques, hybrid antibiotics can be produced by bringing together in one strain the biosynthetic genes from several organisms.

Direct biosynthesis of modified antibiotics Strains with mutations in antibiotic synthesis produce substances whose structure is similar to the antibiotic of the parent strain (e.g. demethyl compounds from tetracycline, chlorotetracycline, rifamycin). One of the best examples is the production of adriamycin (14-hydroxydaunomycin) by construction of a mutant from *S. penicillatus* var. *caesi*. This derivative of daunomycin is more effective as an antitumor agent than daunomycin itself and is the most commonly used antitumor antibiotic today. Intermediate prod-

Table 3.12 Changes in byproducts of antibiotic-producing organisms through strain development

Fermentation product	By-products eliminated by strain development	Production strain
Polymyxin B	Other polymyxins Cephalosporin N and P Acetone (variation in amount produced) Mycellianamide	<i>Bacillus polymyxa</i> <i>Cephalosporium acremonium</i> <i>Clostridium</i> sp.
Griseofulvin		<i>Penicillium griseofulvum</i> <i>S. antibioticus</i>
Actinomycin D	Other actinomycins	<i>S. antibioticus</i> <i>S. aureofaciens</i>
Oleandomycin A	Oleandomycin B	
Tetracycline	Chlortetracycline	
Neomycin B	Neomycin C, Fractin	<i>S. fradiae</i>
Streptomycin	Mannosidostreptomycin, Cycloheximide, Vitamin B ₁₂	<i>S. griseus</i>
Cycloheximide	Streptomycin	<i>S. griseus</i>
Antimycin A ₁	Other antimycins	<i>Streptomyces</i> sp.
Tobramycin (Nebramycin factor 6)	Remaining factors of Nebramycin complex (1,1', 2,3,4,5,6 and 7)	<i>S. tenebrarius</i>
Apramycin (= Factor 7)		
Levorin	Levorisatin (reduced proportion)	<i>Actinomyces levis</i> No. 28
Leucomycin A ₁₀ A ₉	Leucomycin A ₁	<i>Streptoverticillium kitasatoensis</i>

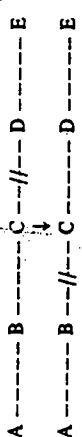
(Perlman, 1973; with additions)

ucts of biosynthesis, which in wild strains may be produced at concentrations too low to detect, may accumulate in blocked mutants, thus leading to the detection of new products. For instance, a new tylosin analog was found in this way in a tylosin-producing culture of *S. fradiae*.

Mutagenesis In mutagenesis (mutational biosynthesis), mutants are used which cannot synthesize a part of the antibiotic molecule. If

analog of this missing component are added, modified antibiotics may be produced, provided the analog penetrates the cell and is recognized by the respective enzymes. Table 3.13 shows several examples. Attempts have been made to use mutagenesis to produce modified macrolide and β -lactam antibiotics, as well as novobiocin analogs. Although several new compounds have been found, commercial use has not yet been possible due to the poor rate of precursor incorporation and the difficulty of obtaining proper mutants.

Cosynthesis In a mixed culture containing two mutants, the first mutant may accumulate an intermediate (C in the diagram below) which is transformed by the second mutant (blocked earlier in the synthetic pathway) into the end product.



This method, which has been primarily used in the study of biosynthetic pathways, has also been successful in the isolation of modified antibiotics. For example, cosynthesis between two mutants of the 6-demethyl tetracycline producer *S. psammodiscus* resulted in a new end product of tetracycline biosynthesis having a different spectrum of activity.

Hybrid antibiotics Genes for biosynthetic steps in different organisms can be combined in the same organism, thus leading to the production of new metabolites. By protoplast fusion, a blocked mutant of the streptomycin-producing *S. griseus* was combined with the istamycin-producing *S. tenimariensis*, leading to the production of the hybrid antibiotic indolizomycin. In another example, the genes for actinorhodin biosynthesis from *S. coelicolor* were cloned into a producing strain of the same antibiotic class and various new compounds were detected. Also, the gene for oleandomycin-producing *S. antibioticus* was transferred to a blocked mutant of eryth-

Table 3.13 Some examples of mutagenesis

Production strain (antibiotic produced)	Block in the biosynthesis of	Analog compound added	Mutagenesis product
<i>S. fradiae</i> (Neomycin)	2-Deoxystreptamine	Streptamine	Hybrimycin A ₁ , A ₂
<i>Bacillus circulans</i> (Butirosin)	2-Deoxystreptamine	2-Epistreptamine	Hybrimycin B ₁ , B ₂
	Neamine	2,5-Dideoxystreptamine	5-Deoxybutirosine
<i>Micromonospora inyoensis</i> (Sisomicin)	2-Deoxystreptamine	Streptamine	2-Hydroxybutirosin
	2-Deoxystreptamine	6'-N-Methylneamine	6'-N-Methylbutirosin
<i>Micromonospora sagamiensis</i> (sagamycin)	2-Deoxystreptamine	Streptamine	Mutamycin 1
<i>S. griseus</i> (Streptomycin)	Streptidine	2,5-Dideoxystreptamine	Mutamycin 2
	Streptidine	2-Epistreptamine	Mutamycin 4
	Streptidine	2-Deoxystreptamine	2-Hydroxysagamycin
	Streptidine	2-Deoxystreptidine	Streptomulin A

romycin-producing *S. erythraeus*. Although new compounds can undoubtedly be produced in this way, it appears unlikely that these more or less random approaches will lead to the formation of useful new compounds.

Another approach to the detection of new substances is the use of DNA sequences as gene probes for the detection of new antibiotics of the same chemical type.

Recombinant DNA products and specific applications of gene technology

Advances in the use of recombinant DNA technology to produce pharmacologically interesting proteins, enzymes, or amino acids is proceeding at a rapid pace. A whole series of recombinant DNA products are on the market, and more can be anticipated in the near future. Applications are in medical diagnostics, agriculture, and environmental protection as well as in basic research. The production of heterologous proteins by recombinant DNA technology has become widespread. However, the high development cost of these products is often not supported by a sufficiently large market. For some products, only kilograms are required on a world-wide basis. In general, the cost for developing a recombinant product may be as much as 100-fold higher than that for development of a conventional pharmaceutical product. A further handicap to rapid

development is the competitive situation, since many companies are often pursuing research on the same compound.

In the following, a brief summary of the current developments in use of recombinant DNA products is given.

Gene technology in basic research Genetic engineering has revolutionized research in basic biology. The clarification of control methods in both prokaryotes and eukaryotes, embryological development and differentiation, sequence determination of proteins has all made enormous advances. The first completely sequenced eukaryotic gene was the β -globin gene of rabbit. This led immediately to the discovery that eukaryotic genes were often split into coding regions (exons) and noncoding intervening regions (introns). Also discovered was the fact that the genome of an organism is not static but that genes exist whose role is in the regulation and expression of other genes (dynamic and hierarchical aspects of genetics).

As further examples from microbiology: the identification and characterization of antibiotic resistance genes, as well as regulatory and structural genes of biosynthetic pathways, the study of genetic instability and strain degeneration (primarily in streptomycetes), and genome analysis (for example, the demonstration that the archaeobacteria constitute a distinct group of organisms).

In medicine, the methods of gene technology have made important contributions in cancer research, immunology, circulatory system, as well as in the nerve and endocrine systems.

Protein engineering The use of X-ray crystallography has permitted the determination of the three-dimensional structure of proteins. By the use of computer-aided protein design, molecular models can be readily produced and their structure/function relationships determined. Changes in amino acid sequence lead to changes in protein structure, not always in a predictable way. By use of modelling, the intelligent selection of desirable mutations can be made and these mutations can then be created by recombinant DNA technology. The technique offers the possibility of modifying enzyme sensitivity to inhibitors, turnover rate, substrate specificity, stability, etc. As an example of the potentialities, changes in various domains of the protein tPA (tissue plasminogen activator) led to increase in the half-life of the enzyme in animal studies. In another example, changing a single amino acid (aspartic acid to serine at position 99) in subtilisin (*Bacillus amyloliquefaciens*) led to an alteration in the pH optimum.

Human therapy By the use of recombinant DNA technology, substantial quantities of proteins can now be produced that previously were only available in extremely small quantities, if they were available at all. Such proteins can be used therapeutically in those patients deficient in these proteins (for example, insulin for diabetics, human growth hormone for dwarfs, factor VIII for hemophiliacs). Another use is in the increase in an immune response, as for instance interferon. Over 100 proteins of human origin have been cloned and expressed in experimental cell systems. Several such products are now being marketed commercially (Table 3.14) or are in clinical trials (Table 3.15).

Recombinant DNA technology opens up further possibilities through:

- The development of vaccines, especially so-called "subunit vaccines" that represent the antibody-recognition sites of pathogens; for instance, surface proteins of viruses or pathogenic bacteria. Hepatitis B vaccine is already on the market in this area.
- The development of gene probes for use in diagnosis for virus, bacteria, and fungal diseases. Such probes are of special use for those disease agents for which culture methods are not possible, such as many of the viruses.
- Monoclonal antibodies are immunoglobulins that are very specific for their antigens. They are produced by the hybridoma technique, in which an antibody-producing lymphocyte is fused with a tumor cell (myeloma) to produce the so-called hybridoma. The value of the hybridoma is that it can be cultured indefinitely, permitting unlimited production of the particular immunoglobulin. Monoclonal antibodies have already been perfected for analytical and diagnostic purposes as well as for use in protein purification (for example, interferon). It can be anticipated that monoclonal antibodies will also play a role in cancer diagnosis, as well as in virus therapy.

Agriculture Advances in the production of cultivated plants, such as the production of disease-resistant plants, can be expected from genetic engineering. Genetic manipulation of plants has conventionally been a slow and difficult task, but has been greatly facilitated by the new genetic methods. It is possible to use plant cell culture procedures to select clones of plant cells that are genetically changed and then, with proper treatments, induce these cell cultures to make whole plants which can be propagated vegetatively or by seeds. One approach to the introduction of foreign genes into plants is the use of the bacterium *Agrobacterium tumefaciens*, which contains a large plasmid (the Ti plasmid) which can become integrated into the plant genome. The Ti plasmid can thus be used as a vehicle for the introduction of foreign genes into plant cells. In

Table 3.14 Recombinant DNA products on the commercial market

Substance	Trade name	Company	Application
Human insulin	Humulin	Eli Lilly/GeneTech	Diabetes
Interferon Alpha 2	Betrolor Intron A Wellferon	Boehringer Ingelheim Schering Plough/Bogen Wellcome Laboratories	Antiviral/Antitumor
Interferon Alpha (Nimelva)	Proferon	Serono/Interpharm	
Interferon Beta	Recombinax HB	Merck/Chiron	
Hepatitis B vaccine (HBsAg)	Protophin	GeneTech	Vaccine
Human growth hormone	Actilyse	Boehringer Ingelheim/ GeneTech	Thrombolytic activity (myocardial infarction)
Human tissue activator (tPA)			

Table 3.15 Recombinant DNA products in clinical trial

Substance	Application
Antithrombin III	Inhibition of blood coagulation
Atrial Natriuretic Factor	Diuretic
Blood coagulation factors VII and IX	Hemophilia
Epidermal growth factor	Healing of wounds
Erythropoietin (EPO)	Anemia in dialysis patients
Human gonadotropin (hCG, hMG)	Sterility
Human serum albumin	Blood substitute
Interferon Gamma	Antitumor therapy/Arthritis
Lymphokines (especially interleukin 2)	Stimulation of immune defenses (bacteria/virus infections; antitumor therapy)
Proteins as antigens (herpes, malaria, influenza, AIDS)	Vaccine production
Superoxide dismutase	Heart attacks
Tumor necrosis factor	Antitumor therapy

one example, the gene for the insect toxin produced by the bacterium *Bacillus thuringiensis* has been introduced into the tomato plant, rendering these plants resistant to insect infestation.

In animal agriculture, genetic engineering is being used for the production of proteins that modify the growth or productivity of farm animals. For instance, swine growth hormone and bovine growth hormone produced by recombinant DNA technology can be used to promote more rapid animal growth. Bovine growth hormone can also be used to increase milk production in dairy cattle by as much as 20%.

3.7 USE OF GENETIC METHODS / 5

Environmental protection Although still far in the future, there is the possibility of the use of engineered microorganisms for the elimination of toxic wastes. Many compounds produced industrially are highly persistent in the environment and, because of their real or potential toxicity, should be eliminated. Genetic engineering can be used to develop new strains of microorganisms capable of breaking down these persistent chemicals, opening up the possibility that these new microorganisms could play a role in environmental protection. However, much more information on the microbial ecology of these new microorganisms will be needed before they can be effectively and safely released into the environment.

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Media used in the cultivation of microorganisms must contain all elements in a form suitable for the synthesis of cell substances and for the production of metabolic products. In laboratory research with microorganisms, pure defined chemicals may be used in the production of culture media, but in industrial fermentations, complex, almost undefinable substrates are frequently used for economic reasons. Depending on the particular process, from 25 to 70% of the total cost of the fermentation may be due to the carbohydrate source. In many cases, media ingredients are byproducts of other industries and are extremely varied in composition. For strain development and fermentation control, the consequences are as follows:

- An optimally balanced culture medium is mandatory for maximal production. A supplement of critical elements must be used if necessary. Using statistical methods in computer-based programs, culture media may be optimized.

4 Substrates for industrial fermentation

- The composition of culture media must constantly be adapted to the fermentation process. New batches of substrate have to be carefully evaluated in trial fermentations before they can be used in production.
- In addition to product yield, product recovery must be examined in trial fermentations.
- If catabolite repression or phosphate repression cannot be eliminated by optimization of the nutrient medium or suitable fermentation management (e.g. feeding), deregulated mutants must be used as production strains.

Besides material cost and product yield, it must be considered whether materials used are readily available in sufficient supply without high transportation costs, and whether impurities will hinder product recovery or increase cost of product recovery.

We discuss below some of the frequently used substrates in industrial fermentation. More details can be found in references cited at the

end of the chapter and in chapters on the individual fermentations.

4.1 SUBSTRATES USED AS CARBON SOURCES

Carbohydrates are traditional energy sources in the fermentation industry. For economic reasons, pure glucose or sucrose can seldom be used as the sole carbon source, except in processes which demand exact fermentation control. Molasses, a byproduct of sugar production, is one of the cheapest sources of carbohydrate. Besides a large amount of sugar, molasses contains nitrogenous substances, vitamins, and trace elements. However, the composition of molasses varies depending on the raw material used for sugar production. Table 4.1 shows a comparison of the analysis of sugar beet and sugar cane molasses. Considerable variation in the quality of molasses occurs, depending on the location, the climatic conditions, and the production process of each

individual sugar factory. In addition to conventional molasses, the residue from starch saccharification which accumulates after the crystallization of glucose is also widely used as fermentation substrate. For example, "hydrolyzed molasses" is a byproduct of glucose production from corn.

Malt extract, an aqueous extract of malted barley, is an excellent substrate for many fungi, yeasts, and actinomycetes. Dry malt extract consists of about 90–92% carbohydrates, and is composed of hexoses (glucose, fructose), disaccharides (maltose, sucrose), trisaccharides (maltotriose), and dextrins, as shown in Table 4.2. Nitrogenous substances present in malt extract include proteins, peptides, amino acids, purines, pyrimidines, and vitamins. The amino acid composition of different malt extracts varies according to the grain used, but proline always makes up about 50% of the total amino acids present. Culture media containing malt extract must be carefully sterilized. When overheating occurs, the Maillard reaction shown in Figure 4.1 results, due to the low pH value and the high proportion of reducing sugar. In this conversion the amino groups of amines, amino acids (especially lysine), or proteins react with the carbonyl groups of reducing sugars: aldehydes, or ketones, which results in the formation of brown condensation products. These reaction products are not suitable substrates for microorganisms. The Maillard reaction is one of the main causes of damage to culture media during heat sterilization, resulting in considerably reduced yields.

Table 4.1 Composition of sugar beet and sugar cane molasses

Composition	Sugar beet molasses	Sugar cane molasses
Dry matter %	78 –85	77 –84
Sucrose	48.5	33.4
Raffinose	1.0	—
Invert sugar	1.0	21.2
Miscellaneous organic materials	20.7	19.6
N	0.2 – 2.8	0.4 – 1.5
P ₂ O ₅	0.02 – 0.07	0.6 – 2.0
CaO	0.15 – 0.7	0.1 – 1.1
MgO	0.01 – 0.1	0.03 – 0.1
K ₂ O	2.2 – 4.5	2.6 – 5.0
SiO ₂	0.1 – 0.5	—
Al ₂ O ₃	0.005 – 0.06	—
Fe ₂ O ₃	0.001 – 0.02	—
Ash	4 – 8	7 – 11
Thiamine	130	830
Riboflavin	41	250
Pyridoxine	540	650
Niacinamide	5100	2100
Pantothenic acid	130	2140
Folic acid	21	3.8
Biotin	5.3	120

(Rhodes and Fletcher, 1966; Imrie, 1969)

Table 4.2 Typical composition of malt extract

Component	% of Dry weight
Maltose	52.2
Hexoses (glucose, fructose)	19.1
Sucrose	1.8
Dextrin	15.0
Other carbohydrates	3.8
Nitrogenous materials	4.6
Ash	1.5
Water content	2.0

pH (10% solution) = 5.5

4.1 SUBSTRATES USED AS CARBON SOURCES / 61

fermentation, and the fermentative production of butanol, acetone, and isopropanol is also being considered. Work is in progress to develop one-step processes for direct conversion of cellulose to ethanol, using fermentative organisms which produce cellulases.

Whey, a byproduct of the dairy industry, is produced annually on a world-wide basis to the amount of 74 million tons (containing 1.2 million tons of lactose and 0.2 million tons of milk protein). Only about 56% of this product is used for human or animal feed. The lactose is used primarily for the production of ethanol or single-cell protein, but also in the production of xanthan gum, vitamin B₁₂, 2,3-butanediol, lactic acid, and gibberellic acid. Because of storage and transportation costs, whey is often not economical as a substrate.

Animal fats such as lard and animal and plant oils are readily utilized by some microorganisms, but are generally added as supplemental substrates rather than as the sole fermentable carbon source. For instance, in certain antibiotic fermentations, soy, palm, and olive oils are used.

With respect to its carbon content, methanol is the cheapest fermentation substrate, but it can be metabolized by only a few bacteria and yeasts. Methanol has commonly been used as a substrate for single cell protein production (see Chapter 16). Research has been carried out on processes for the production of glutamic acid, serine, and vitamin B₁₂ using methanol as the sole carbon source or as a cosubstrate.

Ethanol is available in ample supply from the fermentation of either saccharified starch or cellulose, and can be metabolized by many microorganisms as the sole carbon source or as a co-substrate. Acetic acid, for instance, is presently made by the oxidation of ethanol. At the present time, the cost of ethanol is too high to make it utilisable as a general industrial carbon source.

Alkanes with a chain length of C₁₂ to C₁₈ are readily metabolized by many microorganisms. The use of alkanes as an alternative to carbohydrates depends on the price of petroleum (see Section 16.2).

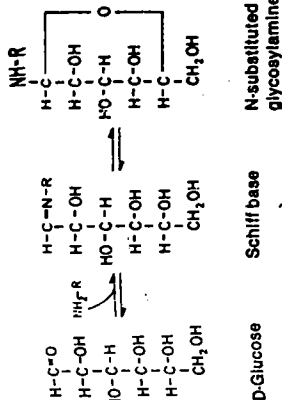


Figure 4.1 Maillard reaction

Starch and dextrins can be directly metabolized as carbon sources by amylase-producing organisms. In addition to glucose syrup, which is frequently used as a fermentation substrate, starch has become more important as a substrate for ethanol fermentation. The largest use of carbohydrate for ethanol fermentation has been the National Alcohol Program begun in 1975 by the Brazilian government. This program, which is based on starch from the cassava plant, produced about 9 million metric tons of ethanol in 1984.

Sulfite waste liquors, sugar-containing waste products of the paper industry, which have a dry weight of 9–13%, are primarily used in the cultivation of yeasts. Sulfite liquors from coniferous trees have a total sugar content of 2–3%, and 80% of the sugars are hexoses (glucose, mannose, galactose), the others being pentoses (xylose, arabinose). Sulfite liquors from deciduous trees contain mainly pentose sugars.

Because of its wide availability and low cost, cellulose is being extensively studied as a substrate for conversion to sugar or alcohol. The world's annual cellulose production is estimated at 10¹¹ metric tons; much of it exists as waste, in such forms as straw, corn cobs, wood wastes, peat, bagasse, and waste paper. It is usually not possible to use cellulose directly as a carbon source, so it must first be hydrolyzed chemically or enzymatically. The sugar syrup formed from cellulose hydrolysis has been used for ethanol

4.2 SUBSTRATES USED AS NITROGEN SOURCES

Many large-scale processes utilize ammonium salts, urea, or gaseous ammonia as nitrogen source. A nitrogen source which is efficiently metabolized is corn steep liquor, which is formed during starch production from corn. The concentrated extract (about 4% nitrogen) contains numerous amino acids, such as alanine, arginine, glutamic acid, isoleucine, threonine, valine, phenylalanine, methionine, and cystine. The sugar present in corn steep liquor becomes

Table 4.3 Composition of yeast extract

Composition (%)	Yeast extract produced by means of	
	Autolysis	Plasmolysis with NaCl
Dry matter	70	80
Total nitrogen	8.8	7.4
Protein (N X 6.25)	55	46
NaCl	< 1	18

Amino acids (% of total)	
Alanine	3.4
Aminobutyric acid	2.3
Arginine	0.1
Asparagine	2.1
Cystine	3.8
Glutamic acid	0.3
Glycine	7.2
Histidine	1.6
Isoleucine	0.9
Leucine	2.0
Lysine	2.9
Methionine	3.2
Ornithine	0.5
Phenylalanine	0.3
Proline	0.3
Serine	1.6
Threonine	1.6
Tyrosine	1.9
Valine	0.8
Vitamin content (ppm)	2.3
Thiamine	20-30
Riboflavin	10-15
Pyridoxine	50-70
Niacinamide	25-35
Pantoic acid	20-30
	600
	100
	350

(Data from Ohly Inc., Hamburg)

largely converted to lactic acid (9-20%) by lactic acid bacteria.

Yeast extracts are excellent substrates for many microorganisms. They are produced from baker's yeast through autolysis at 50-55°C or through plasmolysis in the presence of high concentrations of NaCl. Yeast extract contains amino acids and peptides, water-soluble vitamins, and carbohydrates. The glycogen and trehalose of yeast cells are hydrolyzed to glucose during yeast extract production—Table 4.3 shows some typical analyses. The composition of yeast extract varies, partly because the substrates used for yeast cultivation affect the quality of the yeast extract.

Peptones (protein hydrolysates) can be utilized by many microorganisms but they are relatively expensive for industrial application.

Table 4.4 Compositions of some typical peptones of different origins

Starting material	N-Z-Casein		HY-Casein		Soy	
	Autolysis	Enzymatic	Autolysis	Enzymatic	Autolysis	Enzymatic
Water content	4.8	3.5	4.0	4.0	4.0	4.0
Ash	5.8	39.7	14.0	14.0	14.0	14.0
Total nitrogen	12.8	8.3	9.2	9.2	9.2	9.2
Amino nitrogen	6.7	6.4	1.8	1.8	1.8	1.8
NaCl	3.0	38.0	5.9	5.9	5.9	5.9
Carbohydrate	—	—	29.0	29.0	29.0	29.0

Free amino acids (mg/g)	
Alanine	16.8
Arginine	14.1
Aspartic acid	3.0
Cysteine	8.7
Glutamic acid	33.2
Glycine	0.7
Histidine	38.6
Isoleucine	4.4
Leucine	13.3
Lysine	29.9
Methionine	10.7
Phenylalanine	6.1
Proline	22.7
Serine	33.0
Threonine	7.4
Tryptophan	28.7
Tyrosine	21.5
Valine	8.6
	13.1
	0.05
	1.6
	0.7
	3.8

(Data from Sheffield Chem. Co.)

Sources of peptones include meat, casein, gelatin, keratin, peanut seeds, soy meal, cotton seeds, and sunflower seeds. Peptone composition varies depending upon its origin. For instance, peptone from gelatin is rich in proline and hydroxyproline, but has almost no sulfur-containing amino acids. On the other hand, peptone from keratin has a large proportion of proline and cystine, but lacks lysine. Peptones of plant origin (soy peptone, cottonseed peptone) have large proportions of carbohydrates. The end product is also influenced by the type of hydrolysis, whether acid or enzymatic, especially in regard to its tryptophan content, as shown in Table 4.4.

Soy meal, the residue from soybeans after the extraction of soybean oil, is a complex substrate. Analysis shows a protein content of 50%, a carbohydrate content of 30% (sucrose, starch, yose, raffinose, arabinoglucan, arabinan, and acidic polysaccharides), 1% residual fat, and 1.8% lecithin. Soy meal is frequently used in antibiotic fermentations; catabolite regulation does not occur because of the slow catabolism of this complex mixture.

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Methods of fermentation

5.1 INTRODUCTION

The goal of biotechnology (bioengineering) is to obtain useful metabolic products from biological material. Biotechnology encompasses two distinct phases: fermentation and product recovery. Fermentation procedures must be developed for the cultivation of microorganisms under optimal conditions and for the production of desired metabolites or enzymes by the microorganisms. Product recovery or downstream processing involves the extraction and purification of biological products. Biochemical process recovery differs from chemical recovery primarily in that biological materials are frequently much more labile. Although many of the techniques used in biological or biochemical product recovery overlap with those used for strictly chemical processes (for instance, separation, distillation, heating, cooling, and drying), there is increasing use of methods specifically designed for biological products, such as chromatography and electrophoresis.

In fermentation processes, engineering is only an aid in the development and regulation of biological processes and the microorganism is the center of attention. All methods, such as genetic manipulation, regulation of metabolism by optimizing culture media, and provision of adequate oxygen supply under sterile conditions, are only ways of directing the process toward the desired product. The more precisely metabolic processes during growth and product formation are understood, the greater is the possibility of operating fermentation processes rationally rather than empirically. However, a fully automatic, computer-run fermentation process to obtain a metabolite such as an antibiotic has not yet been realized at the industrial scale.

5.2 GROWTH KINETICS OF MICROORGANISMS

A clear understanding of microbial growth kinetics is necessary if a large-scale process is to be properly managed. Growth kinetics are

nisms adapt to their new environment. Because of this transfer to a new medium, several parameters will probably be altered for the inoculum cells: change in pH value, increase in supply of nutrients, decrease of growth inhibitors. New transport systems for nutrients must be induced within the cells. Essential cofactors may diffuse out of a cell and enzymes of primary metabolism must be adjusted to the length of the lag phase. Moreover, the physiological condition of the inoculum is crucial to the length of the lag phase. If a culture for use as inoculum is still in the lag phase (see below), a lag may not occur and growth may begin immediately. However, inoculum taken from a culture in which growth has stopped due to substrate limitation needs more time to adapt to the new nutrient solution. The concentration of the inoculum also has an influence on the lag phase.

The proper cultivation and transfer of inoculum are essential for the later production of both primary and secondary metabolites. Although the regulation of product formation is not yet fully understood, other preculture media and culture conditions often have to be designed for optimal yields. However, the kinetics of product formation are not necessarily correlated with the length of the lag phase.

Log phase By the end of the lag phase, the cells have adapted to the new conditions of growth. Growth of the cell mass can now be described quantitatively as a doubling of cell number per unit time (yeasts and bacteria) or a doubling of biomass per unit time (filamentous organisms such as streptomycetes and fungi). By plotting the number of cells or biomass against time on a semilogarithmic graph, a straight line results, hence the term "log phase".

Although the cells alter the medium through uptake of substrates and excretion of metabolic products, the growth rate remains constant during the log phase. Growth rate is independent of substrate concentration as long as excess substrate is present. The rate of increase in biomass is correlated with the specific growth rate μ and

treated differently for conventional batch processes than for continuous processes.

Batch fermentation

A batch fermentation can be considered to be a "closed system." At the time $T = 0$ the sterilized nutrient solution in the fermenter is inoculated with microorganisms and incubation is allowed to proceed under optimal physiological conditions. In the course of the entire fermentation, nothing is added except oxygen (in the form of air), an antifoam agent, and acid or base to control the pH. The composition of the culture medium, the biomass concentration, and the metabolite concentration generally change constantly as a result of the metabolism of the cells. After the inoculation of a sterile nutrient solution with microorganisms and cultivation under physiological conditions, four typical phases of growth are observed: lag phase, log phase, stationary phase and death phase (Figure 5.1).

Lag phase When cells are transferred from one medium to another, there is initially no increase in the number of cells, although cell weight may change. During the lag phase, the microorganism

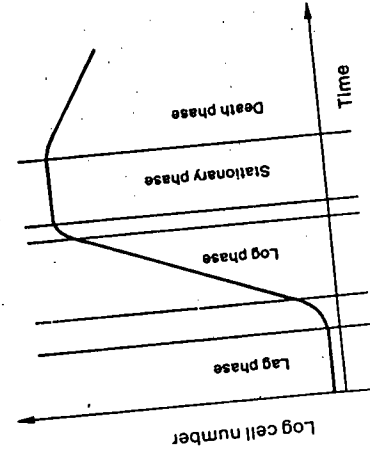


Figure 5.1 Growth curve of a bacterial culture

the biomass concentration X (g/l), whereas the rate of increase in cell number is correlated with μ and cell density N (1/l)

$$\frac{dX}{dt} = \mu \cdot X \quad (11)$$

or

$$\frac{dN}{dt} = \mu \cdot N \quad (12)$$

The specific growth rate, μ , is generally found to be a function of three parameters: the concentration of limiting substrate, S , the maximum growth rate μ_m , and a substrate-specific constant K_s .

$$\mu = \mu_m \frac{S}{K_s + S} \quad (13)$$

This equation is generally called the Monod equation since the relationship expressed was first described by Jacques Monod. The constant K_s is the substrate concentration at which half the maximum specific growth rate is obtained ($\mu = 0.5 \mu_m$). K_s is equivalent to the Michaelis constant in enzyme kinetics. If several limiting substrates are present, eq. (3) can be expanded.

$$\mu = \mu_m \left(\frac{k_1 S_1}{K_1 + S_1} + \frac{k_2 S_2}{K_2 + S_2} + \dots + \frac{k_n S_n}{K_n + S_n} \right) \left(\frac{1}{\sum_{j=1}^n \frac{K_j}{K_j + S_j}} \right) \quad (14)$$

If there is an excess of all substrates, then $\mu = \mu_m$, and the culture is in the log phase at its maximal growth rate. If one substrate has been exhausted, and another substrate is still present, there may be a second log phase during the metabolism of this second substrate with a second specific growth rate μ_1 .

The value for K_s is generally very low, in *Escherichia coli*, K_s values of 1.0 mg/l have been measured for glucose and 1.1 mg/l for tryptophan. Such low values also account for the fact that substrate levels equivalent to the K_s value are not obtained in normal batch fermentations during the log phase. Because of the large bio-

mass which is present by the end of the log phase, the substrate is quickly exhausted so that the period of time during which the substrate concentration is near that of the K_s is very short and the stationary phase is approached abruptly. Maximum specific growth rates are of considerable industrial importance. The μ_m is dependent on the organism and on the conditions of fermentation. The values for several fungi are listed in Table 5.1. The specific growth rates vary between 0.090–0.61 h⁻¹.

Since an organism needs extra energy to split long-chain substrates, the specific growth rate for simple substrates is greater than for long-chain molecules. One example in Table 5.2 shows the

Table 5.1 Maximal specific growth rates (μ_m) of some fungi on glucose

Organism	T°C	μ_m (h ⁻¹)	Doubling time (h)
<i>Aspergillus niger</i>	30	0.20	3.46
<i>Aspergillus nidulans</i>	20	0.090	7.72
	25	0.148	4.68
	30	0.215	3.23
	37	0.260	1.96
<i>Penicillium chrysogenum</i>	25	0.123	5.65
<i>Mucor hiemalis</i>	25	0.17	4.1
<i>Fusarium</i>	25	0.18	3.8
<i>Ascomycum</i>			
<i>Fusarium graminearum</i>	30	0.28	2.48
<i>Verticillium</i>			
<i>agaricinum</i>	25	0.24	2.9
<i>Geotrichum candidum</i>	25	0.41	1.7
<i>Geotrichum candidum</i>	30	0.61	1.1
<i>Neurospora sitophila</i>	30	0.40	1.73

(Anderson et al., 1975)

Table 5.2 Effect of glucose chain length on the maximal specific growth rate (μ_m) in *Fusarium graminearum* at 30°C

Substrate	Number of glucose units	μ_m	Doubling time (h)
Glucose	1	0.28	2.48
Maltose	2	0.22	3.15
Maltotriose	3	0.18	3.85

(Anderson et al., 1975)

growth of *Fusarium graminearum* with glucose, maltose (2 glucose units) and maltotriose (3 glucose units).

The Monod equation for the correlation of specific growth rate and substrate concentration does not hold true when the intracellular substrate concentration is reduced during fast growth, even though adequate substrate is still available in the medium. Additional mathematical models have been developed for handling such situations, in which the concentrations used are not those of the medium but of the intracellular milieu.

When complex nutrient solutions are used, two log phases frequently occur separated by a second lag phase. This process is called diauxy and arises because one of the substrates (usually glucose) is catabolized preferentially (see the discussion of catabolite regulation in Section 3.5). The presence of this one substrate represses the breakdown of the other substrates. The catabolic enzymes for the other substrates are induced (during the second lag phase) only after the first substrate has been completely metabolized.

Stationary phase As soon as the substrate is metabolized or toxic substances have been formed, growth slows down or is completely stopped. The biomass increases only gradually or remains constant during this stationary phase, although the composition of the cells may change. Due to lysis, new substrates (carbohydrates, proteins) are released, which then may serve as energy sources for the slow growth of the survivors. The various metabolites formed in the stationary phase are often of great biotechnological interest.

Death phase In this phase the energy reserves of the cells are exhausted. A straight line may be obtained when a semilogarithmic plot is made of survivors vs. time, indicating that the cells are dying at an exponential rate.

The length of time between the stationary phase and the death phase is dependent on the organism and the process used. In commercial processes, the fermentation is usually interrupted

at the end of the log phase or before the death phase begins.

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Fed-batch fermentation processes

In the conventional batch process just described, all of the substrate is added at the beginning of the fermentation. An enhancement of the closed batch process is the fed-batch fermentation, which is used in the production of substances such as penicillin. In the fed-batch process, substrate is added in increments as the fermentation progresses. The formation of many secondary metabolites is subject to catabolite repression by high concentrations of glucose, other carbohydrates, or nitrogen compounds (see Section 3.5). For this reason, in the fed-batch method the critical elements of the nutrient solution are added in small concentrations at the beginning of the fermentation and these substrates continue to be added in small doses during the production phase.

Since it is usually not possible to measure in situ substrate concentration directly and continuously during the fermentation, indirect parameters which are correlated with the metabolism of the critical substrate have to be measured in order to control the feeding process. For instance in the production of organic acids, the pH value may be used to determine the rate of glucose feeding. In fermentations with critical osmotic values, feeding can be regulated by monitoring the pO_2 -value or the CO_2 -content in the exhaust air.

Continuous fermentation

In continuous fermentation, an open system is set up. Sterile nutrient solution is added to the bioreactor continuously and an equivalent amount of converted nutrient solution with the microorganisms is simultaneously taken out of the system. Among the diverse kinds of continuous fermentation, two basic types can be distinguished:

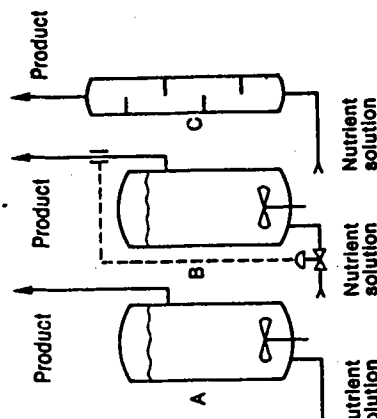


Figure 5.2 Continuous fermentation in the chemostat (A), and plug-flow reactor (B).

Homogeneously mixed bioreactor. This is run as either a chemostat or a turbidostat. In the chemostat in the steady state, cell growth is controlled by adjusting the concentration of one substrate (Figure 5.2 A). Any required substrate (carbohydrate, nitrogen compound, salts, O_2) can be used as a limiting factor. In the turbidostat, cell growth is kept constant by using turbidity to monitor the biomass concentration and the rate of feed of nutrient solution is appropriately adjusted (Figure 5.2 B). **Plug Flow Reactor.** In this type of continuous fermentation, the culture solution flows through a tubular reactor without back mixing. The composition of the nutrient solution, the number of cells, mass transfer (O_2 -supply) and productivity vary at different locations within the system (Figure 5.2 C). At the entrance to the reactor, cells must continuously be added along with nutrient solution (usually as a back flow in a bypass from the fermenter outlet or from a second continuous fermentation).

In a continuous process under steady state conditions, cell loss as a result of outflow must be balanced by growth of the organism.

$$D = \mu \quad [5]$$

$$\mu = 1/X \frac{dX}{dt} \quad [6]$$

Growth rate
Cell concentration in bioreactor

The rate of flow D is defined as the volumetric flow rate (in and out) through the fermenter volume.

By using the Monod equation describing dependence of the specific growth rate on the limiting substrate concentration (equation 3) and the yield constant (equation 7), the cell density X (equation 8) and substrate concentration S (equation 9) in the bioreactor can be determined.

$$YX = \frac{\text{Biomass (g)}}{\text{Substrate consumption (g)}} \quad [7]$$

$$X = YS \left(S_0 - \frac{D \cdot K_S}{\mu_m - D} \right) \quad [8]$$

$$S = \frac{D \cdot K_S}{\mu_m - D} \quad [9]$$

The substrate concentration in the incoming medium is designated as S_0 . Equations 8 and 9 show the dependence on flow rate. At a lower flow rate, the substrate is almost completely used up by the cells ($S \rightarrow 0$). The cell concentration is then $X = S_0/Y$. If D increases, X decreases slowly at first in a linear fashion and then at $D = \mu_m$ it drops sharply to 0. S increases slowly at first as D increases and approaches S_0 at $D = \mu_m$. When X is zero in equation 8, the wash-out point D_m has been reached.

$$D_m = \frac{\mu_m \cdot S_0}{K_S + S_0} \quad [10]$$

$$\frac{S_0}{K_S + S_0} = 1 \text{ therefore } D_m = \mu_m \quad [11]$$

Above D_m , a steady state is not possible. If the flow rate is only slightly lower than D_m , the system is very sensitive to external influences. Extreme changes in biomass may result from small deviations in feeding or cell removal. Figure 5.3 shows the interdependence of rate of flow, substrate concentration, cell concentration and rate

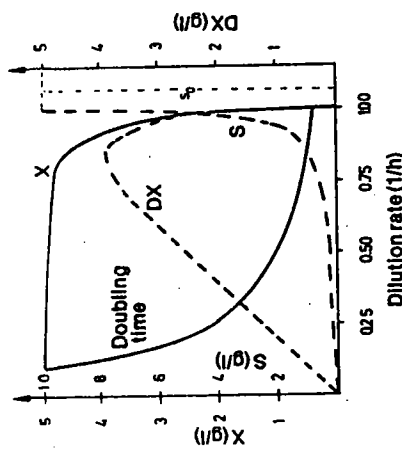


Figure 5.3 Effect of flow rate on substrate concentration (S), cell concentration (X), Doubling time t_d , and cell formation rate ($D \cdot X$).

of cell formation ($\mu_m = 1 \text{ h}^{-1}$, $K_S = 0.2 \text{ g/l}$, $Y = 0.5$).

The maximum specific growth rate is chosen in industrial processes so as to obtain the largest yield in the smallest fermenter volume and the shortest time.

The Monod model is not usually applicable to continuous fermentation at very low and very high dilution rates. At mean dilution rates the molar relationship between biomass produced and CO_2 given off is constant. At low dilution rates, the proportion of CO_2 becomes higher. This is because more of the energy source is used for maintenance of cell structure, for osmotic regulation, or for motility. On the other hand, at high flow rates, a part of the added carbon is incompletely oxidized (to products such as pyruvate, acetate or tricarboxylic acids) and productivity is thus reduced. With nitrogen as the limiting substrate, reserve materials such as polysaccharides are formed at low flow rates, which results in an increase in cell size and hence in biomass.

The production of metabolites is also not well described by the Monod model at very low or high flow rates. For example, with *Penicillium*

chrysogenum the production of penicillin ceases below a minimal growth rate, and conidiospores are formed.

In continuous fermentation at a constant flow rate, steady state conditions do exist and the substrate content, the biochemical reactions within the cells, and the rate of product formation do not change. However, there are differences in metabolism at a constant flow rate depending on the substrate limitation. Table 5.3 shows the influence of substrate limitation on the amino acid pool and glutamic acid dehydrogenase in *Saccharomyces cerevisiae*.

Furthermore the rate of product formation is dependent on the flow rate. This can be applied not only to individual enzymes, but to the excretion of primary and secondary metabolites as well (Figure 5.4). When the idiophase (product formation phase) is separate from the trophophase (growth phase), product formation cannot be carried out continuously in an optimally mixed bioreactor. Fermentation must then be done using a series of fermenters as a cascade or in a tower reactor using single reaction chambers separated by sieve plates.

Continuous fermentation processes have been developed for the production of single-cell protein, antibiotics, organic solvents, starter cultures and for cellulose decomposition (in laboratory studies). Pilot plants or production plants have been installed for continuous production of single-cell protein based on n-alkanes, C₁-compounds and starches (see Chapter 16).

Table 5.3 Effect of growth conditions on the amino acid metabolism of *Saccharomyces cerevisiae*
Dilution rate = 0.1 h^{-1} , 30°C

Limiting substrate	Amino acids (mM)		Glutamic acid dehydrogenase (mMol/min/mg protein)
	Total	Glutamic acid	
Ammonium ion	144	35.6	1730
Glutamic acid	180	49.2	510
Glucose	381	135.4	620
Phosphate	386	121.4	465

(Brown and Stanley, 1972)

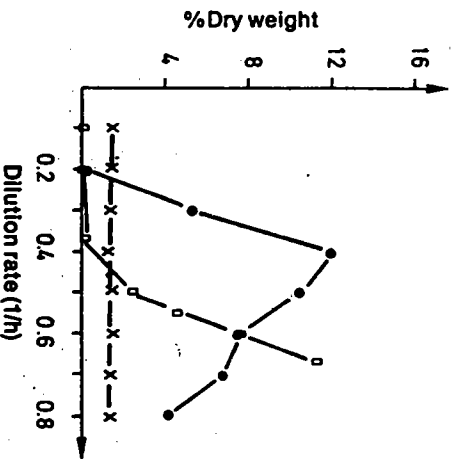


Figure 5.4 Effect of flow rate on the production of polyhydroxybutyric acid (PHB •—•) and glycogen (X—X) with limited carbon, and PHB with excess acetate (O—O), (Williamson and Munro, 1967)

Industrial processes using continuous fermentation include waste-water treatment (see Chapter 17) as well as the production of beer, glucose isomerase, and ethanol.

Although many fermentations for metabolite production work well as continuous processes at a laboratory scale, only a few processes have proved useful for practical application for several reasons:

- Many laboratory methods operate continuously for only 20 to 200 hours; for industrial use the system must be stable for at least 500 to 1000 hours.
- Maintaining sterile conditions on an industrial scale over a long period of time is difficult.
- The composition of substrates must be constant in order to obtain maximal production. The composition of industrial nutrient solutions varies (e.g. corn steep liquor, peptone, starch).
- When high-yielding strains are used, reverse mutants arise, which can overgrow the production strains in continuous culture.

Continuous cultures also find considerable use in research on ecological processes. Growth at low-substrate levels in a continuous culture is analogous to growth at low-substrate levels in nature, so that the continuous cultures provide excellent models for analysis of ecological processes.

Classification of fermentation processes

We have previously subdivided fermentation processes into three types: batch, fed-batch, and continuous. Another means of classification has been developed in which fermentations are classified according to the dependence of product formation on energy metabolism.

Type I In a type I process, the product is derived directly from the primary metabolism used for energy production. The product can also be biomass itself. Growth, carbohydrate catabolism, and product formation run almost in parallel (Figure 5.5 A) and trophophase and idio phase are not separated from each other. The reaction can be expressed as follows:



Processes for the production of single-cell protein, ethanol, and gluconic acid all belong to this category. Continuous fermentations in reactors without a cascade system are classified here as well.

Type II Here the product is also derived from the substrate used for primary energy metabolism, but production takes place in a secondary pathway which is separate from primary metabolism.



In batch fermentations of this type there are two maxima. At first, good growth occurs accom-

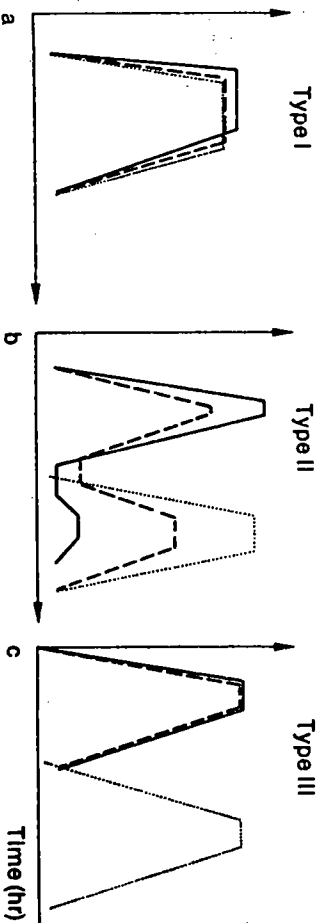


Figure 5.5 Fermentation categories defined by Gaden (1959). Specific growth rate (—), specific carbohydrate consumption (---), specific product formation rate (....)

panied by high substrate consumption and little or no product formation. Then growth slows down and product formation begins, accompanied by a high substrate consumption rate (Figure 5.5 B). In type II processes, the trophophase and idio phase are separated in time. Citric acid, lactic acid, and some amino acids are produced by this type of process.

Type III In the type III processes, primary metabolism and product formation occur at completely separate times. The product is not derived from catabolism, but from amphibolic pathways. In this type of fermentation, primary metabolism functions first, accompanied by substrate consumption and growth. Afterwards, the product is formed by the reactions of intermediary metabolism (Figure 5.5 C). Many antibiotics and vitamins are produced by this type of fermentation.

This classification cannot be applied to all fermentation processes. Depending on metabolite production, composition of culture medium, and regulation in the strain used, there may be intermediate forms. For example, the production of lactic acid falls between types I and II, and aminoglycoside antibiotic production falls between types II and III.

Production involving mycelium-producing microorganisms can not be classified among these simple fermentation types, since the cor-

relation of mycelium production and changes mycelium structure with antibiotic production complex. The stages in oxytetracycline formation with the mycelium-forming organism *Streptomyces rimosus* are shown in Table 5.4.

Table 5.4 Stages in the oxytetracycline fermentation with *Streptomyces rimosus*

Stage	Fermentation process
1 Lag phase	Duration 90 minutes for low inoculum. No metabolism evident.
2 Growth of primary mycelium	Duration 10–25 h, depending on inoculum. Hyphae dense. Respiration, nucleic acid synthesis and some other enzymatic reactions are high. Pyruvate content reaches the maximum. No antibiotic production.
3 Fragmentation of primary mycelium	Duration 10 h, no growth during this time. Respiration and nucleic acid synthesis decline. Pyruvate content falls to low value.
4 Growth of secondary mycelium	Duration 25 h, mycelium volume increases 4 times that of Stage 2, hyphae thin, good antibiotic production. Nucleic acid production resumes but respiration is low. Carbohydrate and ammonium content begin to increase again.
5 Stationary phase	Growth ceases, metabolism is low. Antibiotic production continues at a low rate.

(Bailey and Ollis, 1977)

Productivity and specific rate of production

The productivity of a fermentation is defined as

$$\text{Productivity (P)} = \frac{\text{Product concentration / l}}{\text{Fermentation time}} \\ = \text{Units/hr} \cdot \text{l}^{-1}$$

In evaluating the cost-effectiveness of a process, several factors must be considered: the production time of the fermentation, the time required to clean and set up the fermenter, the sterilization time, and the duration of the lag phase. In Figure 5.6, the slope of the tangent of the product formation curve is a measurement of maximal productivity and the slope of the straight line to the end of the product formation curve indicates total productivity.

Whether the fermentation process is terminated at the time of maximal productivity or later depends on the operating costs, which include energy, overhead, labor, and the capacity of the system. By using a graph such as that of Figure 5.6, an estimate can be made of how to optimize the process. In short-term fermentations (8–70 h) the set-up time is significant, whereas in long-term fermentations (> 3 days) the set-up time is less crucial.

The fermentation conditions directly affect productivity. Figure 5.7 shows the productivity of a single-cell protein process in relation to the substrate (hydrocarbon) content. There is a linear increase in productivity as the substrate content rises until the concentration is reached at which the petroleum hydrocarbon becomes the continuous phase (oil and water are not miscible). Above this concentration, productivity then decreases.

In continuous fermentations, productivity can be expressed as:

$$P = D \cdot X \quad (\text{g/l} \cdot \text{h}) \quad [12]$$

When the relationship of equation 8 is used, the following equation results:

$$P = D \cdot Y_S^X \left(S_0 - \frac{D \cdot K_S}{\mu_m - D} \right) \quad [13]$$

In continuous fermentations, maximal productivity is equal to total productivity since set-up time and lag phase can be disregarded. For instance, if the initial set-up time in a continuous fermentation requires 24 hours, and the continuous fermentation itself is run for 24 hours with a rate of flow of 0.1 h⁻¹, 24 hours with 0.15 h⁻¹ and at least 72 hours with 0.2 h⁻¹, the total productivity for product X is calculated as shown in Table 5.5.

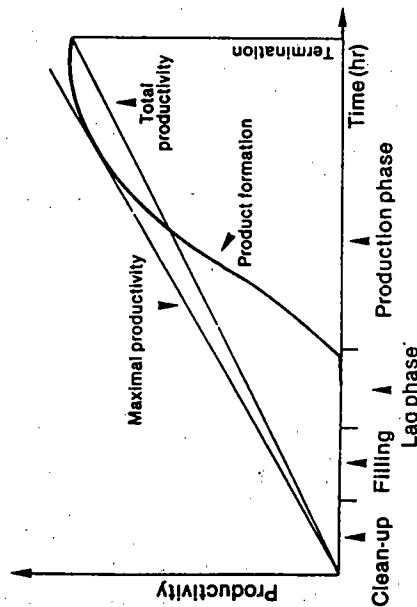


Figure 5.6 Total productivity and maximal productivity

Table 5.5 Total productivity of a continuous fermentation

Time period h	Flow rate h	Total productivity X(g/l)/D(h ⁻¹)
0–24	Batch	—
24–48	0.1	X·0.05
48–72	0.15	X·0.08
> 72	0.2	—
300	0.2	X·0.171
1000	0.2	X·0.191
5000	0.2	X·0.198

converted or to energy released or energy consumed. However, yield coefficients are not constants, since they are dependent on biological parameters (X, μ) and chemical parameters (pO₂, C/N ratio and P content of the medium).

Yield coefficients have been classified in three groups. The first group of coefficients (Y_g, Y_{ov}, Y_{kw}) gives information about the technical process and thus about cost-effectiveness. The second group (Y_C, Y_N, Y_W, Y_{am}) describes catabolism, anabolism, and anabolism. The third group includes Y_{am}, a coefficient which describes the energy relations of the cell. Some authors use other yield coefficients, which must be designated properly, as shown:

$$Y_{S^{CO_2}} = \frac{\text{Mol CO}_2 \text{ formed}}{\text{Mol substrate consumed}} \quad [17]$$

Heat production

In order to obtain optimal yields, fermentations must be carried out at constant temperature. We now discuss the parameters affecting the heat balance of a fermentation process.

The rate of heat production due to stirring and due to the metabolic activity of the microorganisms must be balanced by the heat loss resulting from evaporation and radiation plus heat removal by the cooling system (the jacket of the fermenter or cooling coils).

The heat production during stirring can be calculated from the energy consumption of the motor minus the losses through the seals, bearings, and drive shaft.

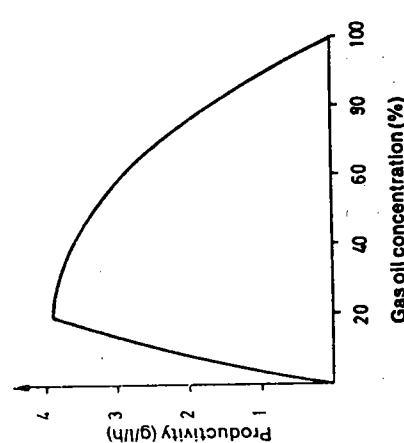


Figure 5.7 Productivity in relation to gas oil concentration in a single-cell-protein process

The specific rate of production q_p is derived from equation 1 in the following way:

$$\frac{dP}{dt} = q_p \cdot X \quad [14]$$

The value P is the concentration of product and the specific rate of production q_p is equivalent to the specific growth rate μ in equation 1. In type II or III processes, however, there is no direct correlation between μ and q_p .

Yield coefficients

Monod originally defined the yield coefficient, Y , as the ratio of cells produced to substrate consumed:

$$\frac{dX}{dt} = -Y \frac{dS}{dt} \quad [15]$$

or

$$Y_S = \frac{\text{Biomass (g)}}{\text{Substrate consumption (g)}} \quad [16]$$

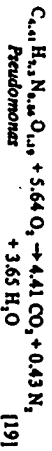
Today general yield coefficients are used to characterize fermentation processes, i.e. the relationship of cells produced to individual substrates

Heat production of the microorganism can be determined using the yield coefficient Y_{kcal} :

$$Y_{kcal} = \frac{Y_s}{H_s - Y_s \cdot H_c} \quad (18)$$

The yield coefficient Y_{kcal} is given in grams of cells/kcal energy released; Y_s indicates the ratio of grams of cells/gram of substrate utilized; the heats of combustion of the substrate H_s and of the biomass H_c are measured in kcal/g. Although there are tables giving the heats of combustion of different substrates, the heat of combustion of the cells (H_c) must be determined experimentally or calculated by using various assumptions.

One possibility for calculating H_c makes use of the electron theory of valence. In the transfer of an electron equivalent, 26.05 kcal are released. Since four valence electrons are shared by oxygen, 104.20 kcal/M O_2 are produced. One calculation using *Pseudomonas fluorescens* assumed that when the biomass is burned, CO_2 , N_2 and H_2O are produced.



The heat of combustion is calculated as:

$$H_c = \frac{5.64 \cdot 104.2}{4.41 \cdot 12 + 7.3 \cdot 1.0 + 0.86 \cdot 14 + 1.19 \cdot 16} = 6.44 \text{ kcal/g} \quad (20)$$

The value obtained must be corrected by about 10% due to the ash content of the cells.

$$H_c = 6.44 \cdot 0.90 = 5.8 \text{ kcal/g} \quad (21)$$

The heat production coefficients for some bacteria on different substrates are listed in Table 5.6. For the more highly oxidized substrates (e.g., glucose and acetate), less heat is produced per biomass than with the more reduced substrates (methane and methanol). In general, Y_{kcal} values increase linearly in relation to the specific growth rate μ of the cultures.

Table 5.6 Yield coefficients for bacteria with different substrates

Substrate	Y_{kcal} (g cells/kcal)
Malate	0.30
Acetate	0.21
Glucose	0.42
Methanol	0.12
Ethanol	0.18
Isopropanol	0.074
n-Paraffin	0.16
Methane	0.061

The rate of microbial heat production per unit time, which along with the heat produced by stirring must be removed from the bioreactor, is calculated from the rate of cell production (equation 1), the reactor volume and the heat coefficient:

$$\text{Heat production rate } Q_w = V \cdot \mu \cdot X \cdot \frac{1}{Y_{kcal}} \quad (22)$$

5.3 FERMENTER SYSTEMS

In a bioreactor, production of metabolites must be accomplished with maximum emphasis on reliability for the process and a minimum of capital investment and operating cost. Reliability is more difficult to achieve in a microbiological than in a chemical process so that bioreactors are more expensive to design and construct than chemical reaction vessels.

Microbiology must be the focus of all considerations concerning construction of a fermenter system. It must be decided in the planning stages whether the fermenter is to be used for a special process with one organism or for a variety of processes with different microorganisms. A number of different bioreactors for aerobic fermentation are diagrammed in Figure 5.8. They may be classified in four groups with respect to gas distribution.

1. Gas distribution by stirring. The stirred vessel (Figure 5.8, part 1) is the best understood

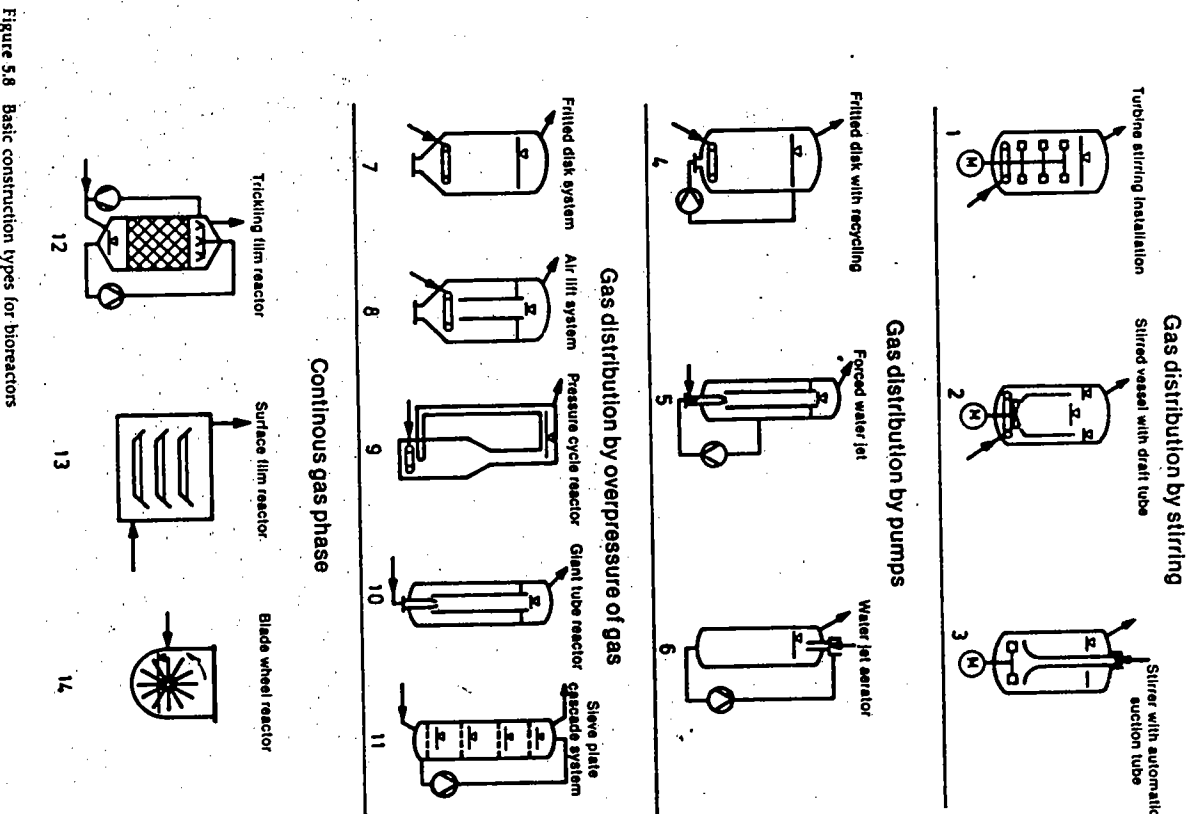


Figure 5.8 Basic construction types for bioreactors

and most widely used bioreactor and is quite flexible. Loop reactors with draft tubes (2) are also suitable for mass production. A system in which air is sucked through the tube by the low pressure on the lee side of the stirrer blades is used in vinegar production (3). The various surface aerators used in waste-water treatment also belong to this group.

2. **Gas distribution through pumps.** In a modified airlift system (4), the pump primarily transports liquid and the air is distributed more or less independently through the sparger. In the more efficient systems (5, 6), there is a direct mixing of air and fluid by means of a water jet pump.

3. **Gas distribution by means of pressurized air.** In most of these systems, no moving parts are present in the sterile area. The "pressure cycle fermenter" (9) was the first model of this kind used in biotechnology for single-cell protein production on an industrial scale.

4. **Continuous gas phase.** In these processes, air circulates over a film of microorganisms. In the trickle-film reactor (12), the organism grows on a solid, inert carrier material. In the surface reactor (13), it floats on or in the nutrient solution or grows in a semi-solid or solid culture medium. In the blade-wheel reactor (14), it grows on movable blades or drums and is dipped alternately in nutrient solution and in gas phase as the wheel turns. Reactors used in waste-water treatment, for the leaching of ores, and for vinegar or citric acid production, as well as the culture flasks on laboratory shaking machines, are included in this group.

Stirred bioreactors

For industrial use, especially in the pharmaceutical industry, the most versatile bioreactor is the simple stirred, aerated fermenter. However, no single system which adequately meets the needs of all biological systems can be constructed. Fermenters for laboratory experiments requiring volumes of up to 20 l are made of glass or stainless steel (Figure 5.9); for larger volumes con-

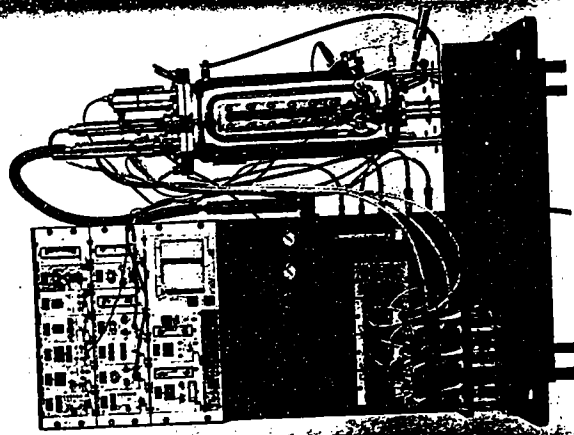


Figure 5.9 Laboratory fermenter

struction is of stainless steel. Figure 5.10 illustrates the design of a large scale fermenter.

Up to the 3000 l scale, fermenters can be standardized. However, the geometry of production fermenters changes as the volume and the process requirements change. The height and width relationship may vary from 2:1 to 6:1 and the stirrer may be top or bottom driven.

Baffles Baffles bring about the transfer of turbulence to the fermenter wall. Four baffles are commonly installed, with a width 1:10 or 1:12 of the reactor diameter. If heat dissipation is a problem, as it often is in large production fermenters (> 100 m³), up to 12 baffles can be used as heat exchangers.

Foam separators Foaming is frequently a problem in large-scale aerated systems. Antifoam chemical agents cannot always be used for the

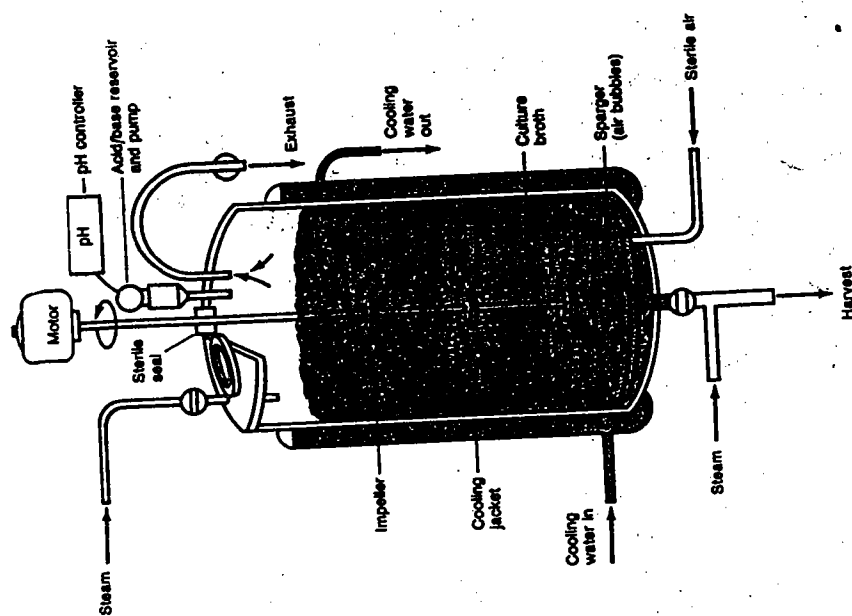


Figure 5.10 An industrial fermenter, illustrating the construction and facilities for aeration and process control.

reduction of foam, since they may have inhibitory effects on the fermentation. There are several methods by which the foam in the sterile area can be destroyed by mechanical methods. The simplest devices have rakes mounted on the stirrer. In the Frings system for vinegar production and the Fundafom system (Figure 5.11), both of which are currently used industrially, foam is destroyed by centrifugal force. The nutrient solution held in the foam flows back into the bioreactor and the air released from the foam leaves the sterile system.

Types of stirrers Only specific types of stirrers developed for chemical technology can actually be used for microbiological processes. Figure 5.12 shows the most important ones. The disc stirrer is the most common type; 4–8 radial blades project out beyond the edge of a disc of appropriate diameter. In the turbine-type stirrer, the blades are curved. Compared with the disc stirrer, the turbine type requires 50% less air for the same yield and energy consumption. Two other types consist of stirring arms with blades attached at an angle. Compared with both of the above-men-

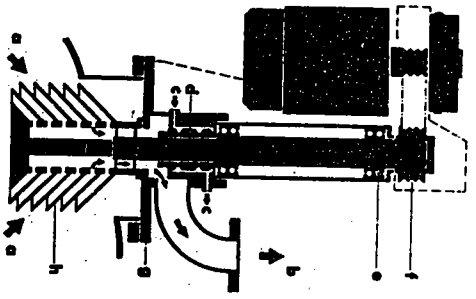


Figure 5.11 Mechanical foam separator (Fundatom, Chernap). a) Foam entrance, b) Gas exit, c) Lubrication, d) Double seal, e) Packing, f) Drive, g) Intermediate flange, h) Rotating plate.

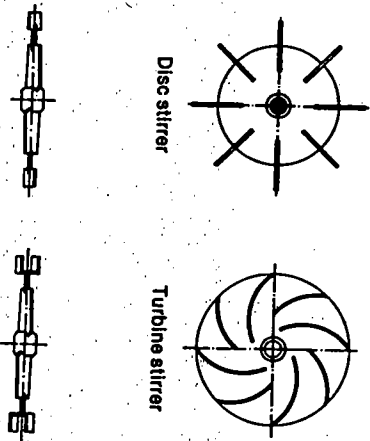


Figure 5.12 Impeller systems for fermenters

tioned stirrers, the MIG and INTERMIG stirrers (Figure 5.12) require 25% and 40% less energy, respectively for equivalent yields.

Maintenance of sterility There should be a minimum number of openings in the fermenter to favor maintenance of sterility. Small openings must be made leakproof with O-rings, larger openings with flat gaskets. Wherever a movable shaft penetrates the fermenter wall, special problems of sterility maintenance arise. Double mechanical seals on the agitator shaft are currently used and present noticeable advantages in comparison to the more conventional stuffing-box seal. If possible, the joints of all parts connected within the sterile area as well as all of the pipes both inside and outside the fermenter should be welded. There should not be any direct connection between the nonsterile and sterile area; that is, sampling devices and injection ports must be covered with steam-sterilizable closures. Sterile pipes must be slanted to collect the condensate and to drain it.

Although the fermenter vessel is the main focus of interest in the overall fermenter system, in an actual installation a large number of additional devices must be provided in order to obtain a functioning fermenter system. Figure 5.13 shows the layout for the installation of a fermenter system.

Reactors for immobilized enzymes or cells

An important type of large-scale biological reactions involves the attachment of the biocatalyst, enzyme or cell, to an inert support through which the substrate is passed. The product is then present in the effluent and can be directly purified. Bioreactors for use with immobilized enzymes or cells must be so constructed that the rate of movement between substrate and the biocatalyst is not rate-limiting. Internal mass transfer can only be influenced by the method by which the catalyst is immobilized. Figure 5.14 summarizes the various types of bioreactors for immobilized systems. Two basic types are used:

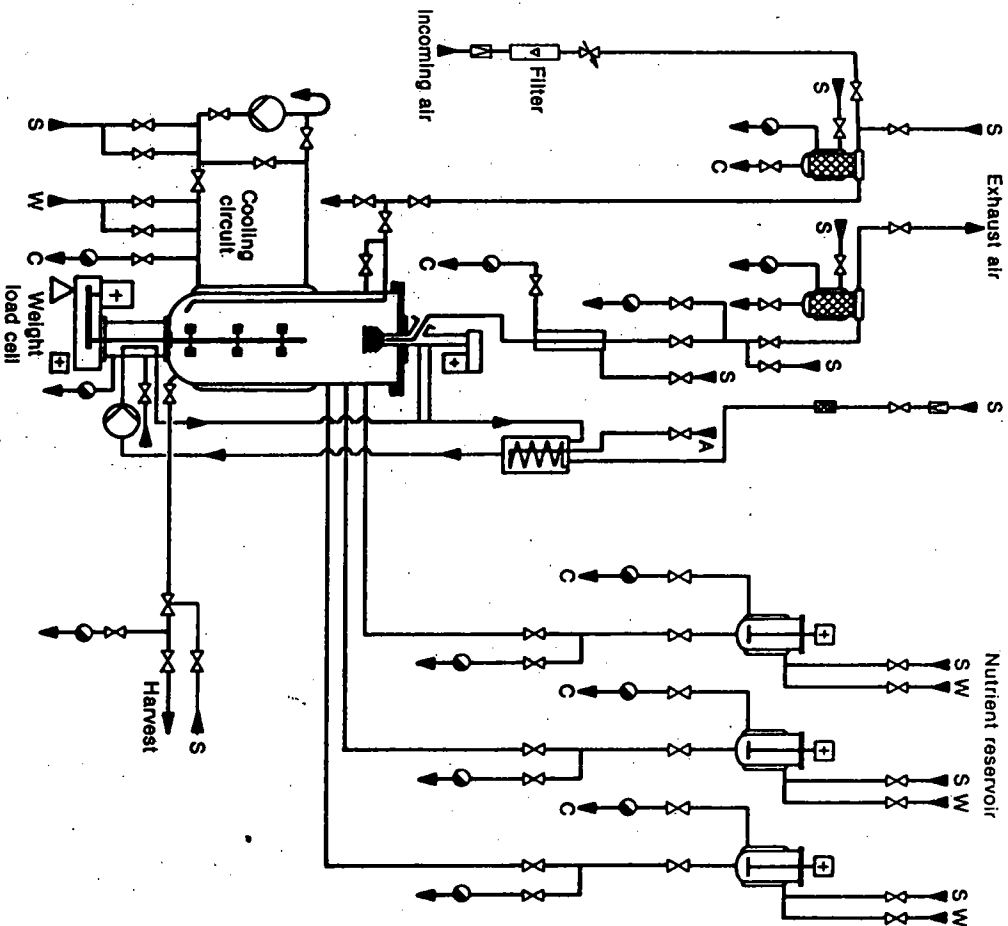


Figure 5.13 Installation of a fermenter. S, steam; C, condensate; W, water; A, air. The steam lines permit in-place sterilization of valves, pipes and seals. The input air is sterilized by both incineration and filtration.

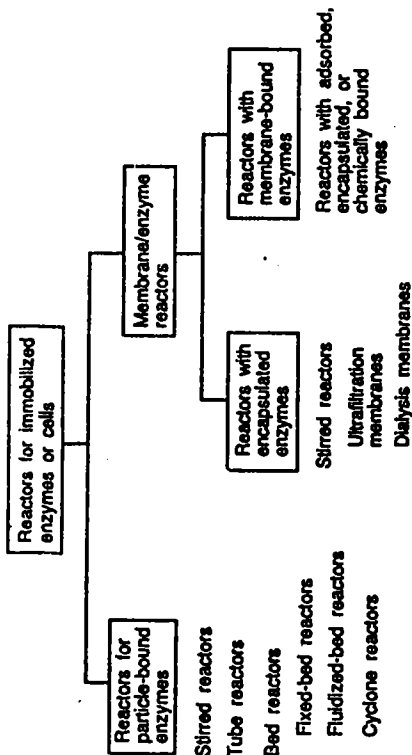


Figure 5.14 Types of systems for immobilized enzymes/cells

those in which the enzyme is incorporated into a membrane and those in which the enzyme is bound to an inert particle.

Particle-bound systems For systems that use particle-bound enzymes, stirred reactors or those mixed by pumps are used (see Figure 5.8, parts 1, 2, and 4) if high aeration or careful pH control is necessary, or if the fluid has a high viscosity. A disadvantage of the use of a stirred reactor is that the shearing force of the stirrer or pump is high, thus raising the possibility that the catalyst may be damaged.

Another approach is the use of a system in which the catalyst is immobilized in a bed, such as a fixed-bed, fluidized bed, or cyclonic reactor (Figure 5.15). The fixed-bed reactor adapts itself well to large-scale operation; because of the dense packing of the immobilized material a high rate of substrate conversion per unit time is obtained. Since the flow is from the bottom to the top, there are gradients of concentrations of substrate and product. Disadvantages of the up-flow fixed-bed reactor are: gas formation may reduce the contact surface between substrate and enzyme; channels in the fixed bed may develop, leading to the appearance in the product of un-

reacted substrate; regulation of temperature and pH is difficult. In the fluidized-bed reactor the height of the catalyst must be somewhat larger than that of the substrate solution. Because of the plug-flow nature of fluid movement, the catalyst is kept in suspension. Recycling of the fluid is not possible, the flow-rate cannot be varied, and scale-up is quite difficult.

Cyclone reactors can be well mixed, either by installation of a stirrer or by use of a gas under

Membrane/enzyme reactors In membrane/enzyme reactors the soluble enzyme and substrate are introduced on one side of an ultrafilter membrane. By means of a pump, the product is forced through the membrane under 0.5 to 5 atmospheres pressure; the membrane holds back the enzyme (Figure 5.16). Enzyme/membrane reactors have several advantages:

- The expense of immobilization is eliminated and the loss of enzyme is reduced.
- By periodic addition of enzyme to make up that lost by denaturation, the rate of reaction can be maintained at a constant rate.
- Multi-enzyme systems can be used.
- By recycling of the reaction mixture internal and external mass transfer can be optimized.
- Substrate and enzyme can be easily replaced.
- Scale-up presents no particular problems.

The main disadvantages of enzyme/membrane systems are the shearing stress from the fluid flow, which tends to denature the enzyme, and the relatively large surface area of membrane required.

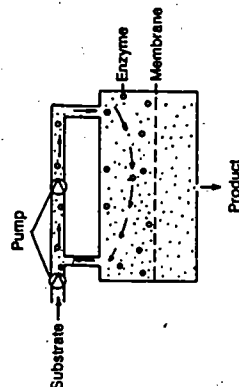


Figure 5.16 Use of ultrafilter membrane to separate enzyme and substrate from product

Modern ultrafiltration membranes made of polyamides (nylon) or polysulfones offer excellent molecular-weight separation in the range of 500–300,000 Daltons. They are also quite stable to temperature, pressure, and chemical attack and are also resistant to microbial degradation. In order to reduce the tendency for clogging, asymmetrical membranes can be used with smaller pores on the high-pressure side. Such membranes are also readily cleanable by backwashing. For industrial-scale operation, the separation is not carried out in the bioreactor itself but in a bypass in which the membrane is installed. Instead of ultrafiltration, it is also possible to separate product from enzyme and substrate by dialysis.

Bioreactors have also been developed in which the enzyme is adsorbed onto or incorporated within the membrane, either through physical or chemical means. In such systems the substrate is converted to product as it passes through the membrane. For instance, a cross-linked sulfonic acid cellulose has been developed which can be fabricated into paper sheets and wrapped into cartridges. In a radial reactor of this type, the enzyme mutase was used to quantitatively convert a solution of sucrose to isomaltulose. By passage through the cartridge the substrate was completely converted to product.

5.4 STIRRING AND MIXING

A microbial fermentation can be viewed as a three-phase system, involving liquid-solid, gas-solid, and gas-liquid reactions.

1. The **liquid phase** contains dissolved salts, substrates, and metabolites. A second liquid phase may occur in some cases if there is an insoluble substrate, e.g. alkane fermentations.
2. The **solid phase** consists of the individual cells, pellets, insoluble substrates, or precipitating metabolic products.
3. The **gaseous phase** provides a reservoir for oxygen supply, for CO₂ removal, or for the adjustment of pH with gaseous ammonia.

The transfer of energy, substrate, and metabolic within the bioreactor must be brought about by a suitable mixing device. The efficiency of the transport of any one substrate may be crucial to the efficiency of the whole fermentation.

For the three phases, the stirring of a bioreactor brings about the following:

- Dispersion of air in the nutrient solution.
- Homogenization to equalize the temperature and the concentration of nutrients throughout the fermenter.
- Suspension of microorganisms and solid nutrients.
- Dispersion of immiscible liquids.

Reynold's number

Stirring ensures the transport of nutrients within the culture liquid. In turbulent flow, two molecules of liquid move in relation to each other. The relative velocity between the nutrient solution and the individual cell should be about 0.5 m/sec. Turbulent flow can be characterized by the dimensionless Reynold's number, well discussed in texts on fluid mechanics. The Reynold's number is calculated as follows

$$\text{Reynold's number} \quad R_e = N_{Re} = \frac{D^2 \cdot N \cdot \rho}{\eta} \quad [23]$$

D = Stirrer diameter (cm)
 N = Stirrer speed (sec⁻¹)
 ρ = Density (g/cm³)
 η = Dynamic viscosity (g/cm · sec)

The following Reynold's numbers have been calculated for the penicillin fermentation with *Penicillium chrysogenum*. Note how much lower the values are than for pure water:

	Small fermenter	Production fermenter
Water		
$(\eta = 10^{-3} \text{ g/cm} \cdot \text{sec})$		
$(\eta = 1 \text{ centi Poise})$	$Re \ 4.0 \cdot 10^5$	$6.9 \cdot 10^4$
Culture medium		
$(\eta = 5 \text{ g/cm} \cdot \text{sec})$		
$(\eta = 500 \text{ centi Poise})$	$Re \ 8.0 \cdot 10^3$	$1.4 \cdot 10^4$

The Reynold's number describes the flow only at the periphery of the stirrer. To distribute the turbulence homogeneously within the entire reactor, an impeller of appropriate shape and diameter must be used. The flow rate is crucial to the distribution of turbulence. However, as turbulence may damage filamentous organisms, there are frequently limitations as to how fast a system can be stirred. The mixing time, θ_m , is the time needed for homogenization to the required degree of homogeneity in the entire reactor.

Power number

The power number (N_p = Newton's number = N_e) has been defined as a dimensionless parameter relating to the energy required by stirred reactors. It is calculated as follows:

$$\text{Power number} = N_p = \frac{\text{Imposed force}}{\text{Inertial force}} = \frac{P_o}{N^3 \cdot D^5 \cdot \rho} \quad [24]$$

P_o = Stirring power (KW)
 N = Stirring speed (sec⁻¹)
 D = Stirrer diameter (cm)
 ρ = Density of the medium (g/cm³)

The power number has been correlated with the Reynold's number for several types of stirrers. Figure 5.17 shows this relationship with six types of stirrer blades.

In the laminar flow range of mixing speed ($N_{Re} < 10$) the Reynold's number is correlated with the power number as follows:

$$N_p = K_1 \cdot (N_{Re})^{-m} \quad [25]$$

K_1 = Constant dependent upon the container geometry and the shape of the stirrer, but not dependent upon the reactor size
 $m = 1$

With laminar flow, the power required for stirring is not dependent on the density, but is correlated with fermentation parameters as follows:

$$P_o \sim N^2 \cdot D^3 \cdot \eta \quad [26]$$

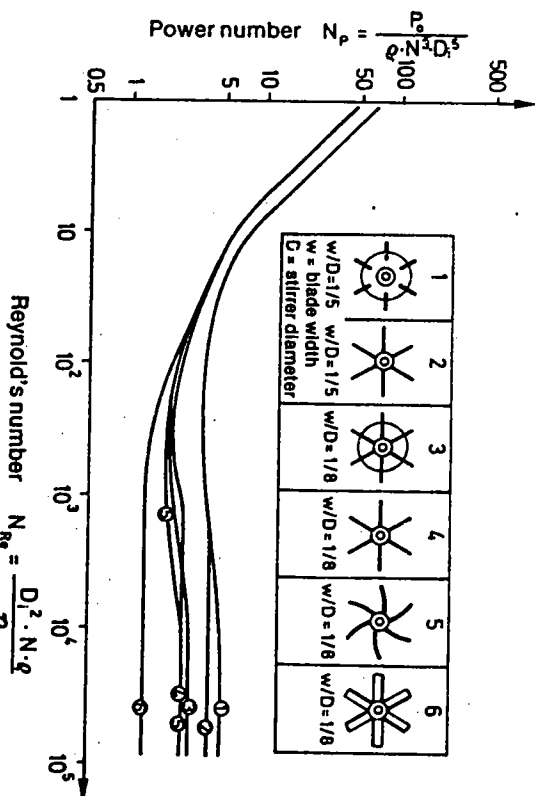


Figure 5.17 Correlation between power number and Reynold's number in Newtonian solutions for various impellers (Bates et al., 1963)

In the turbulent flow range of mixing speed ($N_{Re} > 10^3$), the power number is independent of the Reynold's number and is constant

$$N_p = K_2 = \text{Constant} \quad m = 0$$

Under these conditions the power number is independent of the viscosity

$$P_o = K_2 \cdot N^3 \cdot D^5 \cdot \rho \quad [27]$$

In the transient range of mixing speed ($N_{Re} = 10-10^3$), there is no simple correlation between the power number and the Reynold's number. Disc impellers have a higher power number than slanted blade or propeller stirrers.

In bioreactors where a vortex is formed, another dimensionless number, Froude's number, is useful. It is described as follows

$$N_{Fr} = \frac{N^3 \cdot D_1}{g} \quad [28]$$

g = Acceleration due to gravity

Power number, Reynold's number and Froude number are correlated with each other

$$N_p = K_3 \cdot (N_{Re})^m \cdot (N_{Fr})^y \quad [29]$$

No vortex is formed in bioreactors with baffle so that Froude's number is not relevant in the cases.

Effect of viscosity

Nutrient solutions can be subdivided into two groups according to the way they behave when stirred: viscous solutions with Newtonian and non-Newtonian properties; and viscoelastic solutions.

utions, in which normal liquid-state properties are not observed in stirred vessels.

There are only a few examples which fall into the second group (polysaccharides and certain antibiotic fermentations). During agitation of such solutions, the liquid does not flow tangentially, but rises up along the impeller, a behavior called the Weissenberg effect. With such solutions, scale-up calculations are difficult to make.

Most fermentation solutions fall into the first category. Uninoculated solutions and bacterial cultures often behave as simple Newtonian liquids. In such liquids, the dynamic viscosity (η) of a nutrient solution is a constant at a constant temperature, and it is dependent on the ratio of the shearing stress (T) to the rate of shear ($\dot{\gamma}$).

Newton's law of friction

$$\tau = \eta \cdot \dot{\gamma} \quad [30]$$

Newtonian fluid

$$\eta = \frac{T}{\dot{\gamma}} = \text{Constant} \quad [31]$$

T = Shear stress (kg/m^2)
 $\dot{\gamma}$ = Shear rate (sec^{-1})

With many mycelial organisms, changes occur during the fermentation not only in the amount of mycelium, but in the characteristics of the nutrient solution. Substrates are taken up during metabolism and the proportion of undissolved substrates is reduced. At the same time, metabolites are excreted, thus affecting the viscosity of the solution.

Figure 5.18 presents data on a fermentation system with non-Newtonian properties. In this figure, the dynamic viscosity of a *Micromonospora* culture is plotted against time (ν = dynamic viscosity/density; 1 Stokes = 1 St = $10^{-4} \text{m}^2/\text{sec}$). In this case, the viscosity increases by a factor of 100 during mycelium formation.

In non-Newtonian nutrient solutions, the dynamic viscosity is dependent not only on the temperature, but also on the rate of shear. The

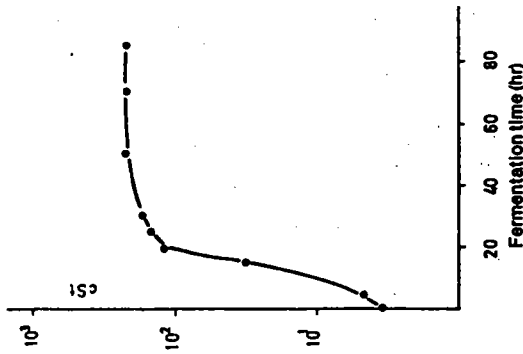


Figure 5.18 Progress of kinematic viscosity in *Micromonospora* fermentations ($1 \text{ cSt} = 10^{-2} \text{ m}^2/\text{sec}$, $D = 10 \text{ cm}$)

various types of non-Newtonian solutions are shown in Figure 5.19, which illustrates the dependence of shearing stress on rate of shear.

For solutions which exhibit pseudoplastic behavior, the apparent viscosity decreases as the rate of shear increases. With dilatant nutrient solutions, the apparent viscosity increases as the rate of shear increases. Bingham-plastic behavior is exhibited by nutrient solutions which will not flow unless a stress T_0 is imposed. The law is stated as follows

$$\frac{T - T_0}{\dot{\gamma}} = \eta = \text{Constant}$$

Table 5.7 lists the type of viscosity of some microbial mycelia.

It is more difficult to determine the amount of energy required for non-Newtonian solutions than for Newtonian solutions. The rate of shear changes with changes in the impeller speed, temperature, and time. The curves in Figure 5.19 are

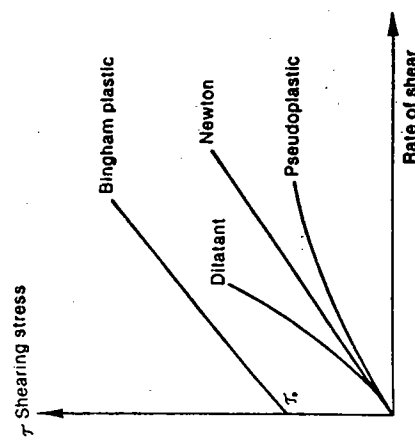


Figure 5.19 Correlation between shear rate and shear stress in nutrient solutions with Newtonian and non-Newtonian properties

Table 5.7 Viscosity of mycelium-producing microorganisms

Microorganism	Use	Viscosity
<i>Penicillium chrysogenum</i>	Penicillin	Pseudoplastic
<i>Coniothyrium helibori</i>	Steroid hydroxylation	Bingham
<i>Streptomyces noursei</i>	Nystatin	Newton
<i>Aspergillus niger</i>		Bingham
<i>Streptomyces nireus</i>	Novobiocin	Bingham
<i>Streptomyces griseus</i>	Streptomycin	Bingham
<i>Streptomyces</i> sp.		Newton and pseudo-plastic
<i>Endomyces</i> sp.	Glucosylase	Pseudoplastic

not linear and the Bingham-plastic line does not cross zero. Mycelial suspensions which are taken from a bioreactor to determine viscosity may change their behavior within minutes without constant gaseous conditions, stirring, and temperature.

Power requirement in aerated bioreactors

During aeration, as the effective density of the gas-liquid mixture is reduced, the power require-

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ment decreases. The correlation between power requirements of aerated and unaerated fermenters is given below:

$$P_g = K \cdot \left(\frac{P_1^2 \cdot N \cdot D^3}{Q^{0.56}} \right)^{0.45} \quad [32]$$

P_g = Power requirement of the aerated bioreactor (horsepower)

K = Constant (function of fermenter geometry)

P_1 = Power requirement of the unaerated fermenter (horsepower)

N = Stirring rate (rpm)

D_1 = Stirrer diameter (cm)

Q = Aeration rate (vvm)

This calculation describes both Newtonian and non-Newtonian fermentation systems. This equation has been shown to hold for a series of

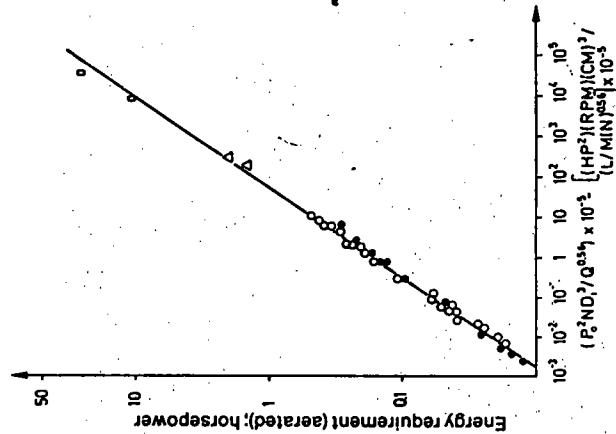


Figure 5.20 Michel and Miller correlation of energy required in geometrically similar aerated and unaerated bioreactors (Taguchi et al., 1968). • 30 l, ○ 100 l, ◻ 50,000 l fermenters

fermenters from 20 l to 30,000 l for the glucose/amine fermentation by *Endomyces* sp. (Figure 5.20).

Table 5.8 shows data for actual power requirements for some industrial fermenters in both pilot-plant and production-sized systems. Small fermenters are usually constructed so as to be oversized in power, in order to accommodate future needs. The power requirements of large fermenters are similar to those in the chemical industry.

5.5 GAS EXCHANGE AND MASS TRANSFER

One of the most critical factors in the operation of a large-scale fermenter is the provision of adequate gas exchange. Oxygen is the most important gaseous substrate for microbial metabolism, and carbon dioxide is the most important gaseous metabolic product. When oxygen is required as a microbial substrate, it is frequently a limiting factor in fermentation. Because of its low solubility, only 0.3 mM O_2 equivalent to 9 ppm, dissolves in one liter of water at 20°C in an air/water mixture. Due to the influence of the culture ingredients, the maximal oxygen content is actually lower than it would be in pure water. The solubility of gases follows Henry's Law in the gas pressure range over which fermenters are operated.

Henry's Law

Henry's law describes the solubility of O_2 in nutrient solution in relation to the O_2 partial pressure in the gas phase.

$$C^* = \frac{P_o}{H} \quad [33]$$

In this equation, C^* is the oxygen-saturation concentration of the nutrient solution, P_o is the partial pressure of the gas in the gas phase and H is Henry's constant, which is specific for the gas and the liquid phase. As the oxygen concentra-

Table 5.8 Energy demand in pilot plant and production fermenters

Fermenter size (m ³)	Power, kWh/d
0.1	11-13
1 - 5	4-7
15 - 120	1-3

(Einseler, 1978)

tion increases in the gas phase, the O_2 proportion of the nutrient solution increases. Consequently, the highest O_2 partial pressures are attained during aeration with pure oxygen. Compared to the value in air (9 mg O_2 /l), 43 mg O_2 /l dissolves in water when pure oxygen is considered. Henry's constants for various gases are summarized in chemical handbooks for standard conditions. As temperature rises, the O_2 solubility decreases.

$$C^* = \frac{468}{31.6 + T} \quad [34]$$

T = Temperature (°C)

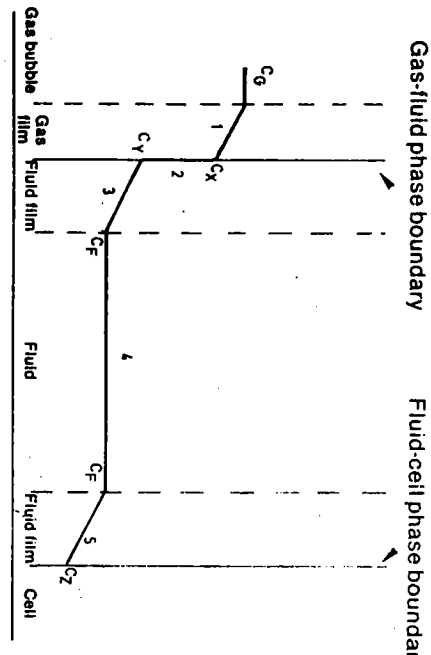
Oxygen transfer

For oxygen to be transferred from a gas bubble to an individual cell, several independent partial resistances must be overcome (Figure 5.21): 1) resistance within the gas film to the phase boundary; 2) penetration of the phase boundary between gas bubble and liquid; 3) transfer from the phase boundary to the liquid; 4) movement within the nutrient solution; 5) transfer to the surface of the cell.

For fermentations carried out with single-celled organisms such as bacteria or yeasts, the resistance in the phase boundary between gas bubble and liquid is the most important factor controlling the rate of transfer.

Microbial cells near gas bubbles may absorb oxygen directly through the phase boundary and the rate of gas transfer to such cells is increased. In cell agglomerates or pellets, the O_2 transfer

Figure 5.21 Resistances for oxygen transfer from the air bubble to the microbial cell



within the agglomerate can become the limiting factor.

Mass transfer of oxygen into the liquid can be characterized as follows:

$$N_A = k_L \cdot a \cdot (C^* - C_L) = OTR \quad [35]$$

N_A = Volume-dependent mass transfer (mmol O_2 /l · h)

k_L = Transfer coefficient at the phase boundary

a = Specific exchange surface

C^* = Saturation value of the dissolved gas in the phase boundary

C_L = Concentration of the dissolved gas (mmol/l)

OTR = Oxygen transfer rate (mmol O_2 /l · h)

Table 5.9 summarizes several oxygen transfer rates for three impeller systems.

The volumetric oxygen transfer coefficient has been thoroughly examined as a critical parameter for bioreactor function. The coefficient is dependent on the diameter, capacity, power, aeration system, and aeration rate of the bioreactor and on the density, viscosity, and composition of the nutrient solution, the structure of the microorganism, the antifoam agent used, and the temperature.

Table 5.9 Oxygen transfer rates (OTR) in bioreactors

Reactor volume m ³	Impeller	Method of assay	OTR mmol O_2 /h
0.1	Turbine	Sulfite	100-223
0.8	Turbine	Sulfite	94
1.2	Turbine	Sulfite	64
5.0	Turbine	Sulfite	45-72
47.7	Turbine	Sulfite	42
34.2	Walthof	Yeast	16-22
56.5	Vogelbusch	Yeast	26-43

(Hatch, 1975)

The volumetric oxygen transfer coefficient is dependent on the following fermentation conditions:

$$k_L a = k \cdot (P/V)^{0.66} \cdot (V_g)^{0.66} \cdot (N)^{0.66} \quad [36]$$

k = Constant

V = Bioreactor volume

V_g = Gas exit speed (cm/min)

The k_a value is frequently substituted for the $k_L a$ value in multi-level systems, as described in the following relationship:

$$k_a = (2.0 + 2.8 N_p) \cdot (P/V)^{0.66} \cdot V_g^{0.66} \cdot N^{0.66} \cdot 10^{-3} \quad [37]$$

N_p = Stirrer number

The value of the combined parameters $k_L a$ can be directly calculated, but k_L and a are difficult to calculate individually. The interfacial area a is usually unknown since it depends on bubble size. Table 5.10 shows some values for the interfacial area a .

Surface-active substances such as antifoam agents reduce the value of $k_L a$. In pure water, the bubble surface is constantly renewed through vibration and oscillation. As soon as surface-active substances are added, the renewal of the bubble surface by bubble movement ceases. Microorganisms themselves have an effect on the oxygen transfer by acting as a barrier, thus inhibiting the O_2 transfer. With filamentous organisms, there are variations depending on whether the mycelium is in loose form or in pellets. While the $k_L a$

value decreases gradually as the pellet increases in size, there is a much steeper decline with loose forms (Figure 5.22).

The gas bubbles are replenished in locations of the bioreactor where there is negative pressure, such as behind the agitator blades. As the aeration rate increases, various conditions can be characterized. At low aeration rates, large gas bubbles form behind individual turbine blades and smaller bubbles are spun off centrifugally into the nutrient solution. As the aeration rate is increased, gas bubbles collect behind all turbine blades and continue to accumulate. The energy input is one-third less than that used in un-aerated systems. In this intermediate stirring range, gas dispersion is the best. At very high aeration rates, many large gas bubbles adhere to each other and the impeller is flooded with gas, resulting in sharply lowered gas dispersion.

Oxygen as a substrate

As previously mentioned, growth in a microbial culture with limited substrate is calculated as follows:

Table 5.10 Values of the specific exchange coefficient a in some bioreactors

Bioreactor	Aeration type	Aeration rate vvm	Specific energy demand kW/m ³	Ex-coefficient a (m ⁻¹)
Stirred vessel				
			1	120
			1	300
			1	400
			2	600
			0.6	650
Bubble column bioreactor	Fitted disk	0.63		
			1.2	1000
Stirred vessel				
Tubular loop reactor	Ejector		4	1000
			10	1100
			0.9	1300
			0.9	1300
			1.8	1700
			1.8	1800
			7.2	2000
			0.9	2000
Bubble column bioreactor	Fitted disk	0.56		
			10	2200
			2.77	2315
Stirred vessel				
Tubular loop reactor	Ejector	0.48		
			7.2	2500
Bubble column bioreactor	Ejector	0.42	1.5	6000
			2.2	8000

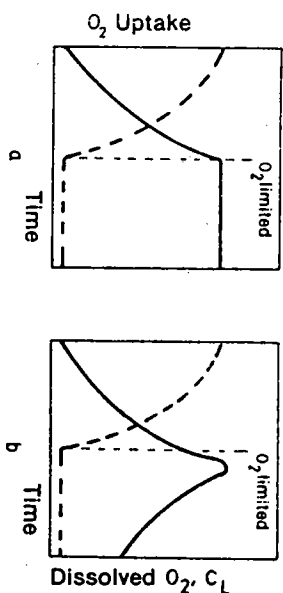


Figure 5.23 O_2 uptake rate and concentration of dissolved oxygen under O_2 limitation. (a) Bacterial culture, (b) Fungal fermentation. — O_2 absorption rate, --- C_L dissolved O_2 concentration

Table 5.12 Critical oxygen concentrations of some organisms

Organism	C_{Lm} (mg/l)
<i>Escherichia coli</i>	0.26
<i>Penicillium chrysogenum</i>	0.40
<i>Saccharomyces cerevisiae</i>	0.60
<i>Pseudomonas aeruginosa</i>	1.10
<i>Torulopsis utilis</i>	2.00

(Brown, 1970)

Table 5.13 Dependence of the critical oxygen concentration of *Sirpionomyces nitrovis* on fermentation parameters

Fermenter Volume (l)	Impeller D, (cm)	Impeller speed (rpm)	Critical O_2 Concentration (% saturation)
20	7.0	400	0
20	12.2	275	0
20	21.0	180	50
250	15.8	175	5
250	15.8	220	18
250	23.7	175	22
15,000	76.8	164	55
15,000	90.5	124	55
15,000	118	82	55

(Wang and Fewkes, 1977)

Dynamic method In batch fermentations, the actual dissolved O_2 concentration C_L can be determined by measuring the speed of the C_L decrease after interrupting the aeration, thus determining the volumetric oxygen transfer coefficient. The following expresses the change of the current O_2 content:

$$\frac{dC_L}{dt} = k_L a \cdot (C^* - C_L)_m - Q_{O_2} \cdot X \quad (40)$$

or

$$C_L = \frac{1}{k_L a} \cdot \left(\frac{dC_L}{dt} + Q_{O_2} \cdot X \right)_m + C^* \quad (41)$$

m = average value over the bioreactor

When the decline in O_2 concentration in the broth is plotted against time after interruption of aeration, a straight line results with a slope of $-Q_{O_2} \cdot X$. When the change in the O_2 concentration is plotted against time according to equation 1 (Figure 5.24), the slope of the straight line equals the reciprocal of the volumetric transfer coefficient $k_L a$.

The advantage of this procedure is that only one parameter must be calculated. The disadvantages are that the results are inaccurate due to the long response time of the oxygen electrodes and due to oxygen transfer from the headspace of the fermenter, thus requiring a gas exchange with nitrogen. In some cases the value of C_L is so low that an exact measurement in the fermenter is not possible. Also there may be a production deficit whenever the slightest interruption of aeration occurs (as is found in acetic acid production).

Sulfite method The sulfite method is frequently used to obtain a quantitative measure of the O_2

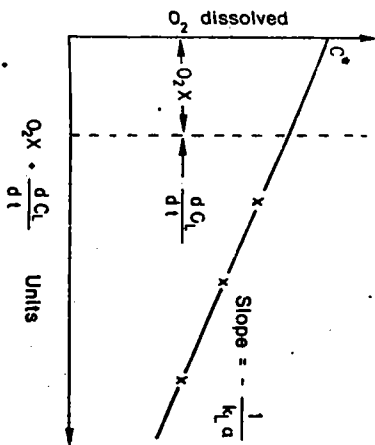
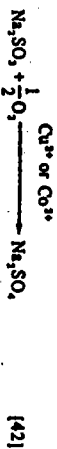
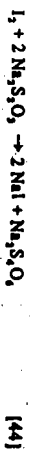


Figure 5.24 Determination of mass transfer coefficient using the dynamic method

transfer in bioreactors. In the presence of $10^{-3}M$ Cu^{2+} or Co^{2+} ions, sodium sulfite is oxidized with dissolved oxygen according to the formula



Since this reaction takes place in a very short time, if there is excess sodium sulfite, the O_2 concentration in the solution equals zero and the O_2 transfer from the gas phase into the solution is the limiting factor. Nonoxidized sulfite is determined by adding a sample to an excess iodine solution and by titrating the residual iodine with sodium thiosulfate.



If two samples are taken at intervals, the O_2 absorption per unit time can be calculated from the rate of sulfite decrease.

The sulfite procedure has a disadvantage: The process is performed as a chemical reaction in an aqueous solution without further additives. The

sulfite value (oxygen transfer rate, OTR) is a characteristic of the bioreactor and gives some information about the aeration device but gives no indication of the current oxygen energy supply in a fermentation broth. In addition, the cost of analysis materials needed for use of this method in a large bioreactor is considerable.

Direct measurement of the volumetric O_2 transfer rate A direct measurement of oxygen transfer is the most exact method, but it is time-consuming and requires precise analytical devices. In this method, the O_2 content in incoming and outgoing air is measured to determine the O_2 transfer rate. The exact oxygen absorption can be determined by multiplying the aeration rate by the difference in O_2 content (in and out) and taking the absolute temperature and pressure into consideration. The oxygen content in the air is measured specifically and very accurately by its paramagnetic properties or with a mass spectrometer, so that no other gases interfere.

$$Q(Y_{in} - Y_{out}) = N_A \quad (46)$$

Q = Aeration rate

Y_{in} = Oxygen content of the input

Y_{out} = Oxygen content of the exit

Determining the O_2 transfer rate via measurement of microbial growth The amount of oxygen required can be calculated from thermodynamic considerations, if it is assumed that a single carbon substrate is oxidized to carbon dioxide, water and ammonia by an organism. Under aerobic conditions the oxygen requirement C is calculated per gram of dry biomass produced:

$$C = \frac{A}{Y} - B \quad (47)$$

C = Oxygen demand/dry weight biomass (mm O_2 /g cell dry weight)

A = Oxygen required for oxidation of 1 g of substrate to CO_2 , H_2O , NH_3

B = Oxygen required for oxidation of 1 g biomass to CO_2 , H_2O , NH_3

Y = Cell yield/substrate (g/g)

Table 5.14 Cell yields obtained with various substrates

Substrate	Y_b Bacteria	Y_b Yeast
Glycerol	0.36	0.37
Glucose	0.51	0.50
Ethanol	0.40	
Bananol	0.68	
Iskanes	1.03	1.00
Ethane	0.62	

The bacterial data are from Abbott and Clamen (1973) and the yeast data from Bronn (1966).

The value of A can be calculated from the theoretical oxidation of the substrate. For B the biomass must be analyzed (in yeast $B = 934$ ml O_2 /dry weight).

Some examples of cell yields (Y_b) in yeast and bacteria are given for several substrates in Table 5.14.

Experimental values for the amount of O_2 needed are listed in Table 5.15. Since these are the total amounts needed, they do not provide any information about when during a fermentation the O_2 is needed. In order to determine the amount of O_2 required in a culture per hour, the specific growth rate μ (h^{-1}), the biomass X (g), and the Y_b value as grams of cells per gram of oxygen (g/g) must be known.

Table 5.15 Oxygen requirements of some microorganisms

Microorganism	Substrate	O_2 required (ml O_2 /g dry weight)
<i>Erubacter aerogenes</i>	Glucose- NH_3	515
<i>Scherichia coli</i>	Glucose- NH_3	400
<i>En. chrysogenum</i>	Glucose- NH_3	410
<i>Trich. cerevisiae</i>	Molasses	500
<i>Trich. cerevisiae</i>	Molasses	420-590
<i>Trich. cerevisiae</i>	Molasses	430-550
<i>Endonitella glutinis</i>	Glucose-peptone	510
<i>Enditella utilis</i>	Glucose-peptone	550
<i>Enditella utilis</i>	Glucose-urea	460
<i>Enditella utilis</i>	Glycerol-peptone	480

Bronn, 1966)

Hence

$$N_A = \mu \cdot X \cdot \frac{k}{Y_b} \quad [48]$$

$$k = \text{Correction factor} \quad 31.3 \frac{\text{mM } O_2}{\text{g } O_2}$$

For the cell yield per substrate, $Y_b = \text{g cells/g oxygen}$, the following equation is used:

$$\frac{1}{Y_b} = 16 \left[\frac{2C + H/2 - O}{Y_G \cdot M} + \frac{O'}{1600} + \frac{C'}{600} + \frac{N'}{933} - \frac{H'}{200} \right] \quad [49]$$

C, H, O = Number of carbon, hydrogen, and oxygen atoms in substrate

C', H', O', N' = Percent of carbon, hydrogen, oxygen, and nitrogen in biomass

M = Substrate molecular weight

Effects of carbon dioxide on the fermentation As mentioned earlier, carbon dioxide is the most important gaseous metabolite produced in fermentation. Although much study has been devoted to the effects of oxygen, too little attention has been given to the correlation between carbon dioxide and microbial metabolism. The negative influence of CO_2 on penicillin formation has been studied; erythronycin and rifamycin B formation rates are also reduced by excess CO_2 .

Figure 5.25 and Table 5.16 show the effect of CO_2 on production of the aminoglycoside antibiotic sisomicin. When 1% CO_2 is added to the incoming air in a 300 l fermenter, substrate is metabolized more gradually, mycelium is formed slowly, and the sisomicin yields are 33% less than in the control.

5.6 SCALE UP

Significance of scale up

The conversion of a laboratory procedure to an industrial process is termed scale up. It is well established in the field of industrial microbiology that a process which works well at the laboratory scale may work poorly or not at all when first attempted at large scale. It is generally not possible to take fermentation conditions that have

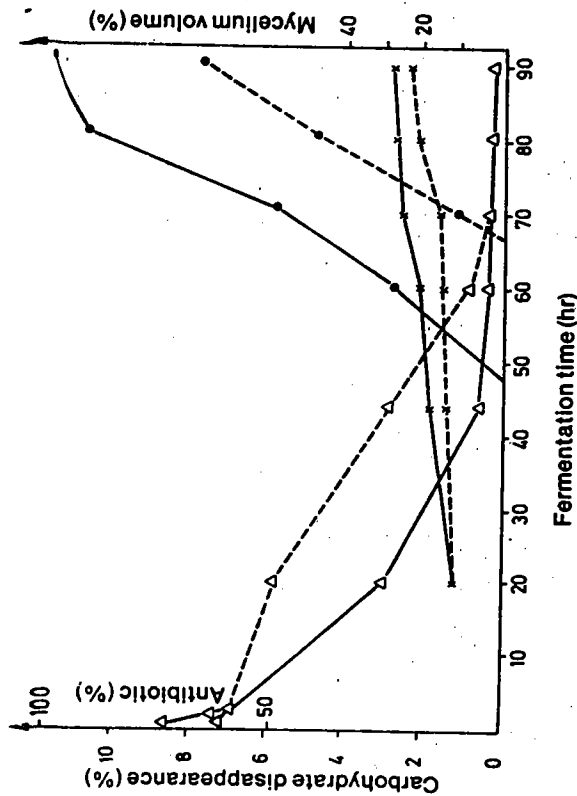


Figure 5.25 Sisomicin formation during aeration with a 1% CO_2 air mixture. --- CO_2 /air mixture, X control, • antibiotic, Δ carbohydrate.

Table 5.16 Influence of CO_2 on yield of the antibiotic sisomicin

CO_2 content of incoming air (%)	Relative sisomicin yield
0	100
1	66
2	15
3	0
4	0

worked in the laboratory and blindly apply them to industrial-scale equipment. All of the skills of the biotechnologist must be brought into play in order to develop a successful large-scale process.

Scale up is necessary in the following circumstances:

- A new process is implemented in the plant
- Mutants with 10-20% greater yield are to be introduced into large-scale production as soon as possible

- Construction of a completely new fermentation plant, a rare occurrence.

While laboratory microbiologists are primarily interested in yield based on weight of biomass, units per ml broth, or maximal yield per unit time, success in scale up is evaluated on the basis of maximal yield for minimal operating cost and time.

Comparing fermenters with similar geometries, Table 5.17 shows that at different fermenter sizes not all parameters can be kept constant. If the impeller increases in diameter by a factor of five, the fermenter volume increases by a factor of 125, from 80 liters to 10,000 liters. If one of several criteria is kept constant in scale up, e.g., energy consumed/volume or Reynold's number, the other parameters are quite different from the values obtained with the small fermenter.

Although many parameters have been tested for use as scale up criteria, there is no general

Table 5.17 Interdependence of scale-up parameters

Scale up criterion	Designation	Small fermenter 80 l	Production fermenter 10,000 l
Energy/ Input	P_g	1.0	125
Energy Input/Volume	P_g/V	1.0	1.0
Impeller rotation number	N	1.0	0.34
Impeller diameter	D_i	1.0	5.0
Pump rate of Impeller	F	1.0	42.5
Pump rate of Impeller/Volume	F/V	1.0	0.34
Maximum Impeller speed (max. shearing rate)	N/D_i	1.0	1.7
Reynolds number	ND_i^2/η	1.0	8.5
			25.0
			5.0
			1.0

(Oldshue, 1966)

formula because of the variation in fermentation processes. The most important methods are:

- Constant power consumption per unit of broth
- Constant volumetric oxygen transfer rate

Some metabolites (for example, phenylalanine, valine, leucine, and capreomycin) are formed best at oxygen concentrations below the critical level. Figure 5.26 shows the use of the critical O_2 concentration to scale up the formation of nikkomycin, an acaricide, from shake flasks to 40 m³ fermenters.

An additional criterion may be the impeller tip velocity, which is in the range of 5–7 m/sec

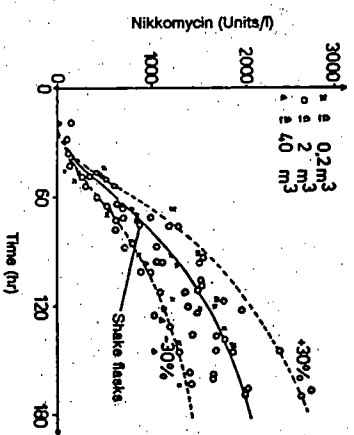


Figure 5.26 Scale up of nikkomycin production using the critical O_2 concentration as the scale-up criterion. (Cruiger et al., 1982).

In many production fermenters, mixing time is not suitable as a scale-up parameter because it increases as reactor size increases. It can be seen in Table 5.18 that mixing time is approximately related to the cube root of the fermenter volume. In reality, scale up is not usually done with geometrically similar fermenters in laboratory, pilot plant and production scale. There is usually no means nor any necessity of scaling up while keeping one parameter constant.

Scale up with constant power consumption per volume

In this simple method, one of the factors in the determination of energy input in aerated bioreactors is the previously mentioned correlation (see Section 5.4) between energy consumption (P_g) and fermenter parameters:

$$P_g = K(P_g)^{1/3} \cdot N \cdot D_i^5 / (Q^{1/3} \eta^{2/3})$$

Table 5.18 Mixing time in relation to fermenter volume

Fermenter volume	Impeller speed rpm	Mixing time sec
3	750	5
9	2,000	3
100	230	6.6
300	350	5
1,000	200	25
3,000	180	20
24,000	30	66

Studies have been done on the power input in a penicillin fermenter with an energy input of 1.5–3.0 horsepower/m³ and in a streptomycin fermenter with 2 horsepower/m³.

In Figure 5.27, novobiocin titers are plotted against energy input for three impeller types. The curves are parallel and maximal yields are obtained above 1.5 horsepower/m³. Below 1 horsepower/m³ considerably lower yields are observed. This graph also shows that at equivalent power inputs, antibiotic biosynthesis is dependent on the diameter of the impeller.

In another example, the flavonin process was transferred from a 4 m³ to a 40 m³ fermenter by keeping constant the P/V relationship.

In a comparative study of many production plants, no constant P/V ratio was found. Rather, the following relationship was observed:

$$P_g/V \sim (V)^{0.33} \quad (50)$$

Scale up with constant oxygen transfer rate

Scale up is most commonly performed on the basis of $k_L a$ data, which are calculated from sulfite

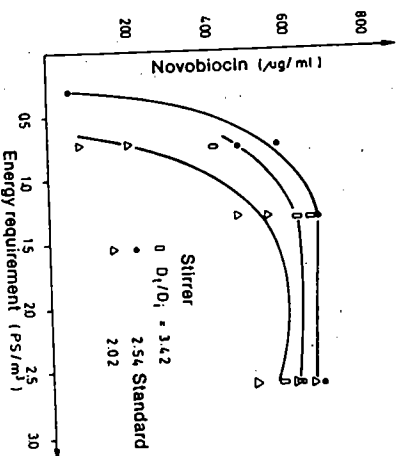


Figure 5.27 Effect of impeller geometry and energy input on novobiocin biosynthesis (Maxon, 1959)

fit values, as described earlier. But this calculation does not give any information on oxygen transfer in highly viscous non-Newtonian fermentation solutions (such as found in many antibiotic fermentations) or under conditions of high-speed stirring.

Figure 5.28 shows the production of baker's yeast in vessels ranging in volume from a small shake flask up to 114 m³, plotted as the sulfite value. In this study, neither the fermenter type nor the volume were crucial in determining yeast growth. Above 125 mM O_2 /h, optimal yields were attained. Scale up of penicillin and streptomycin processes has been successful in 15 l through 100 l and 3 m³ up to 63 m³ fermenters. Instead of the $k_L a$ value, a modified k_L value could be used as the constant parameter.

Figures 5.29 shows scale-up tests by Jazai (1977) on the value of $k_L a$ for the secondary metabolites (nystatin and fumagillin). For both these antibiotics (nystatin and fumagillin), there was a good agreement between $k_L a$ values and the yields in 6 l and 3000 l fermenters.

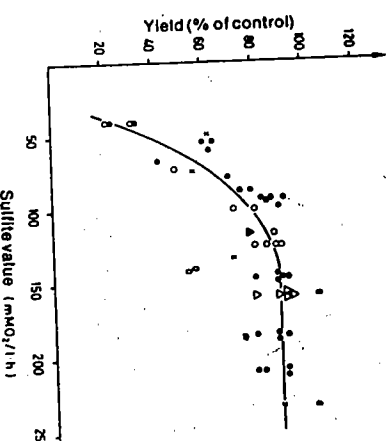


Figure 5.28 Results of the scale up of baker's yeast production (Strohm et al., 1959). Reactor volume: X 190 l shaker; □ 19 l, 800 l, impeller flooded; △ 19 l, 800 l, 255 l, 550 l, 114 m³, unstirred, fitted disk with stir holes; ◇ 265 l, unstirred, fitted disk with large holes; ○ 114 m³, unstirred.

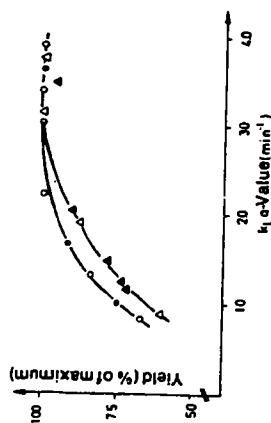


Figure 5.29 Nystatin and fumagillin titers in relation to K_a values during scale-up (Jara, 1972). —○—○— nystatin, —△—△— fumagillin; O, Δ 6 l, ●, Δ 3000 l

5.7 STERILIZATION OF GASES AND NUTRIENT SOLUTIONS

In virtually all fermentation processes, it is mandatory for a cost-effective operation to have contamination-free seed cultures at all stages, from the preliminary culture to the production fermenter. A bioreactor can be sterilized either by destroying the organisms with some lethal agent such as heat, radiation, or a chemical, or by removing the viable organisms by a physical procedure such as filtration.

The process of destroying a population generally follows first-order kinetics. Using the initial number of cells N_0 /ml, the number of destroyed organisms N' at time t (min), and the surviving cells N , the death rate can be calculated as follows

$$-\frac{dN}{dt} = k(N_0 - N') = kN \quad [51]$$

(k is the specific death constant/min)

When integrated between N_0 at time $t = 0$ and N at time $t = t$, the following equation is obtained

$$kt = \ln \frac{N_0}{N} \quad [52]$$

$$\ln \frac{N_0}{N} = -kt \quad [53]$$

The ratio of N_0/N is the inactivation factor, the ratio of N/N_0 is the survival factor and the $\ln$ of $N_0/N - \ln$ is the design criterion, a parameter which encompasses the contamination level of the medium to be sterilized, N_0 , and the desired sterility level, N .

The ideal curve expressing the exponential decline in survivors with time is plotted in Figure 5.30. This line has a negative slope and in Figure 5.31 part II corresponds to the death curve for vegetative cells of *Escherichia coli* at 54–60°C. This type of killing kinetics is referred to as a logarithmic death curve. In actuality, this curve is not always linear, as shown for the death curve of *Bacillus stearothermophilus* spores (Figure 5.31 part I). Mathematical models for such nonlogarithmic destruction curves have been developed.

In the above equations, k is a constant which expresses the specific death rate. It increases sharply with temperature and can be experimentally determined for an organism using equation 53. According to the Arrhenius equation the following connection between temperature and k value exists:

$$\frac{d \ln k}{dt} = \frac{E}{R \cdot T^2} \quad [54]$$

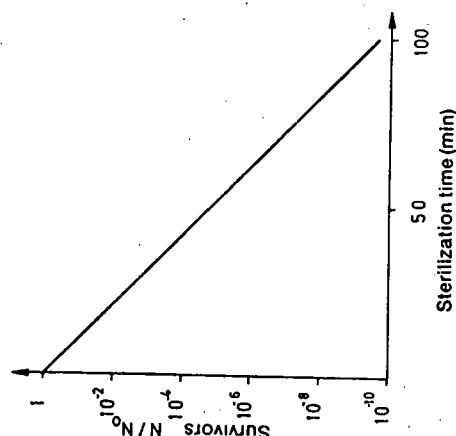


Figure 5.30 Theoretical death curve of a bacterial culture

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destruction rate is thus not a straight line, but a curve, as shown in Figure 5.32 for a culture consisting of two strains.

During fermentation the following points must be observed to ensure sterility:

- Sterility of the culture media
- Sterility of incoming and outgoing air
- Appropriate construction of the bioreactor for sterilization and for prevention of contamination during fermentation.

Sterilization of culture media

Nutrient media as initially prepared contain a variety of different vegetative cells and spores, derived from the constituents of the culture medium, the water, and the vessel. These must be eliminated by a suitable means before inoculation. A number of means are available for sterilization, but in practice for large-scale installations, heat is the main mechanism used.

Heat sterilization This is the most useful method for the sterilization of nutrient media. A number of factors influence the success of heat sterilization: the number and type of microorganisms

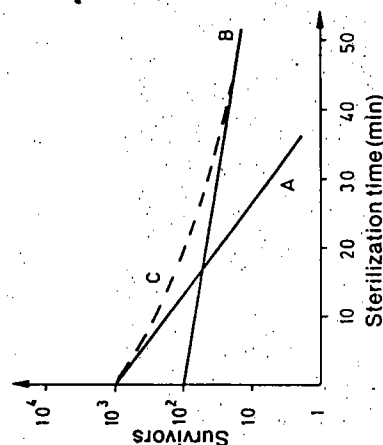


Figure 5.32 Death curve of a mixed culture (C). The straight lines (A) and (B) indicate the death rates of both pure cultures

present, the composition of the culture medium, the pH value, and the size of the suspended particles. Vegetative cells are rapidly eliminated at relatively low temperatures, as shown in Figure 5.31, but for destruction of spores, temperatures of 121°C are needed.

Spores of *Bacillus stearothermophilus* are the most heat resistant. Therefore they are used as assay organisms for testing the various procedures used to sterilize equipment. Table 5.19 provides data on the relationship between temperature, k value, and design criterion for this organism. A list of sterilization times and temperatures for various organisms is given in Table 5.20.

Radiation (UV, X Rays, or γ rays) Although occasionally used in the food industry, these agents are not used in industrial fermentation.

Chemical methods Although a number of chemical disinfectants are known, they cannot be used to sterilize nutrient media because there is a risk

that inhibition of the fermentation organism could occur from the residual chemical.

Mechanical removal of organisms Alternatives such as centrifugation, adsorption to ion exchangers, adsorption to activated carbon, or filtration are possible. Filtration is the only method in practical use. Filter sterilization is often used for all components of nutrient solutions which are heat sensitive and would thus be denatured through the steam sterilization process normally used in industrial fermentation. Vitamins, antibiotics or blood components are examples of heat-labile compounds which must be sterilized by filtration. Such materials must be completely dissolved before filtration, otherwise they would be filtered out of the mixture with the microorganisms. Deep filters (plate filters) are sometimes used to filter complex nutrient solutions. Two disadvantages of filtration are: 1) certain components of the nutrient solution may be adsorbed on the filter material, and 2) high pressures must be used (up to 5 bar), which are undesirable in industrial practice.

One approach which is cost-effective is the filtration of just the water which is to be used in the preparation of the culture medium. For instance, in steroid bioconversion processes, a concentrated nutrient solution is sterilized by heat in the fermenter and is then diluted to the normal concentration with water which has been filter-sterilized.

Batch sterilization

Most nutrient media are presently sterilized in batch volumes in the bioreactor at 121°C. Approximate sterilization times can be calculated from the nature of the medium and the size of the fermenter. Not only the nutrient media, but also the fittings, valves and electrodes of the fermenter itself must be sterilized. Therefore, actual sterilization times are significantly longer than calculated ones and must be empirically determined for the nutrient solutions in the fermenter. One method of sterilization is to inject

steam into the fermenter mantle or interior coils (indirect sterilization). Another method is to inject steam into the nutrient solution itself (direct procedure), in which case pure steam (free of chemical additives) is a prerequisite. Many industrial steam supplies contain potentially toxic chemicals derived from anti-corrosive additives used in the steam-manufacturing process. With direct steam injection, condensate accumulates within the fermenter and the volume of liquid thus increases during the sterilization process.

The drawbacks of the heat-sterilization process are shown for a typical sterilization of a 3000 l batch fermenter in Figure 5.33. It takes 2–3 hours to reach the sterilization temperature of 121°C, depending on the steam conduction and fermenter size. Once the proper temperature has been reached, another 20–60 minutes are required for the actual killing process, followed by cooling for about one hour. The energy required

for heating must subsequently be removed in order to cool the fermenter and if the hot water obtained during the cooling cannot be put to some use, batch sterilization becomes very costly.

Another disadvantage of heat sterilization (and from the standpoint of microbiology the most significant shortcoming) can be seen in Figure 5.33. The heating, sterilization and cooling phases not only kill microorganisms but also severely alter nutrient solutions. Discoloration and changes in the pH value result from caramelization and Maillard reactions (see Chapter 4). Vitamins are destroyed and the quality of the culture medium deteriorates. The extent to which the subsequent fermentation is affected depends on the organism and the process.

Continuous sterilization

The two main disadvantages of batch sterilization just mentioned, culture medium damage and

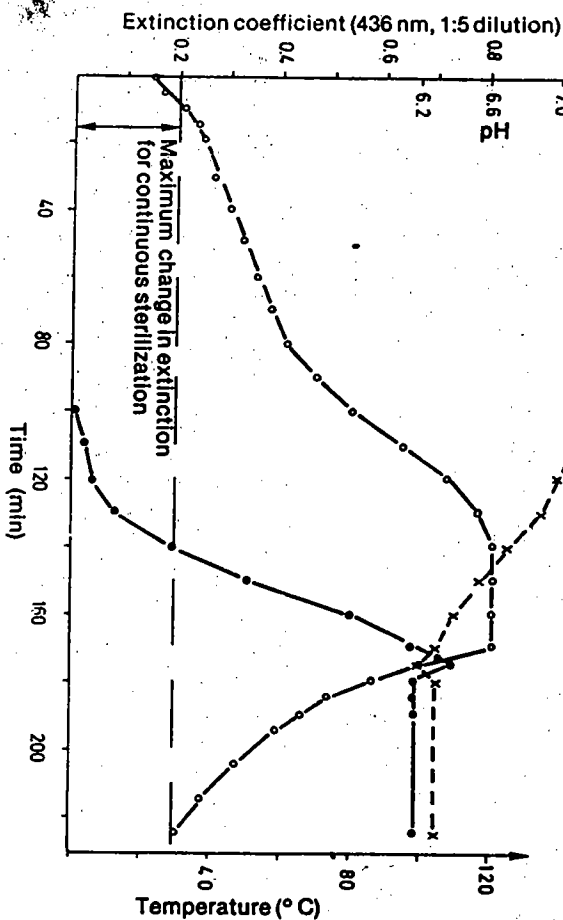


Figure 5.33 Temperature profile and culture medium change in a batch sterilization for a 3000 l fermenter. O — Temperature; X — pH value; — Extinction (1:5 diluted, 436 nm).

high energy consumption, can be largely avoided by use of a continuous sterilization procedure. Although continuous sterilization is the logical preliminary step for continuous fermentations in industrial scale, it is also of value in batch fermentations, making greater yields possible for the time and space allotted. The reason for this is because of the exponential relationship between death rate and temperature, making the time required for the complete elimination of life shorter if higher temperatures are used. While batch sterilization is carried out in 30–60 minutes at 121°C, continuous sterilization is normally accomplished in 30–120 seconds at 140°C.

The heating of culture media for continuous sterilization can be done either by injection of steam or by means of heat exchangers. Sterilization with steam injection is done by injecting steam into the nutrient solution. The temperature is raised quickly to 140°C and is maintained for 30–120 seconds. Due to the formation of condensate, the nutrient solution becomes diluted; to correct this, the hot solution is pumped through an expansion valve into a vaporizer and the condensate is removed via vacuum pumps so that the sterilized nutrient solution has the same concentration after the cooling process as before. The disadvantage of this process is the sensitivity it exhibits to changes in the viscosity of the medium and to pressure variations.

In the continuous process using heat exchangers (Figure 5.34), the nutrient solution in the first heat exchanger is preheated to 90–120°C within 3–30 seconds by the exiting previously sterilized nutrient solution. Then in the second heat exchanger, it is heated indirectly with steam to 100°C. This temperature is maintained for 30–120 seconds in a holding pipe before it is placed in the first exchanger for preliminary cooling and then in a third exchanger for cooling to the temperature of the fermenter. The cooling phase is only 20–30 seconds. Figure 5.35 shows the temperature profile of the nutrient solution during sterilization.

In the process using heat exchangers, 90% of the energy input is recovered. The disadvantage

of this method is that with some nutrient solutions, insoluble salts (e.g., calcium phosphate or calcium oxalate) are formed and crusts appear in the first heat exchanger, due to the extreme temperature differences between the sterilized nutrient solution and the cold incoming solution. The heat transfer coefficient is calculated as follows:

$$K = \frac{Q}{A \cdot \Delta T_m}$$

K = Heat transfer coefficient

Q = Heat consumed

A = Transfer surface

ΔT_m = Average temperature gradient

If precipitation occurs, the heat transfer coefficient decreases and the system must then be stopped, treated with cleaning agents (acid or base), and resterilized. By sterilizing the critical components of the nutrient solution separately, the value of k can be kept constant and the useful period can be extended for weeks.

Starch-containing solutions which become viscous when heated are difficult to use in continuous sterilization processes. Before the actual sterilization, a liquefaction and partial hydrolysis through acids or amylases must be carried out. Moreover, if there are suspended particles in the nutrient solutions, the short sterilization times in the continuous process may not be sufficient for the heat to permeate them thoroughly. The heating time for 1 mm particles is 1 second; for 1 cm particles it is 100 seconds. Therefore the particle size should be restricted to 1–2 mm in continuous sterilization processes.

Sterilization of fermentation air

Most industrial fermentations are operated under vigorous aeration and the air supplied to the fermenter must be sterilized. The number of particles and microorganisms in air varies greatly depending on location of the plant, air movement, and previous treatment of the air. On the

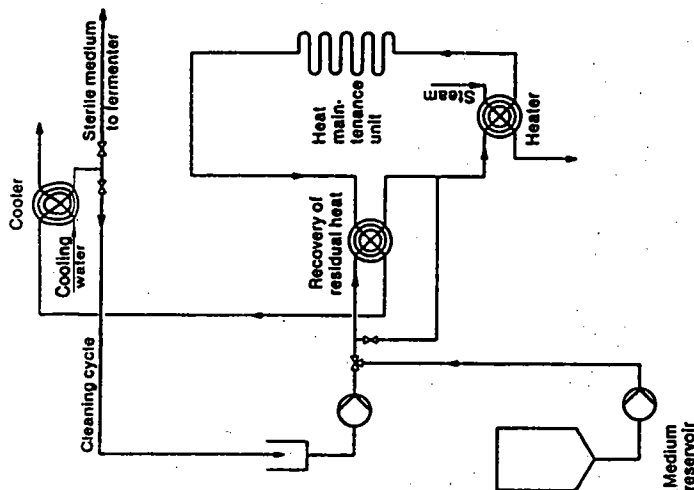


Figure 5.34 Diagram of continuous sterilization via spiral heat exchangers

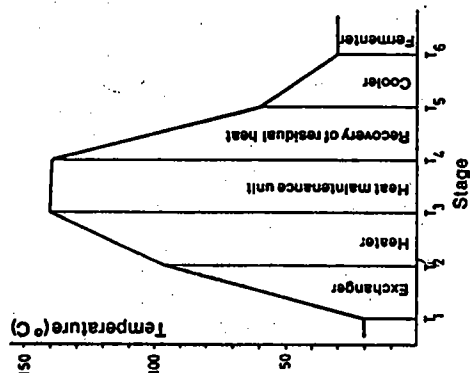


Figure 5.35 Temperature profile of nutrient solution during a continuous sterilization process

average, outdoor air has 10–100,000 particles per m^3 and 5–2,000 microorganisms/ m^3 . Of these, 50% are fungus spores and 40% are Gram-negative bacteria.

Fermenters generally work with aeration rates of 0.5–1.0 vvm (air volume/liquid volume·minute). A fermenter having a working volume of 50 m^3 with an aeration rate of 1 vvm needs 3000 m^3 sterile air per hour. The critical importance of air sterilization in industrial microbiology can be seen from these values.

The methods available for sterilizing gases include filtration, gas injection (ozone), gas scrubbing, radiation (UV), and heat. Of these, only filtration and heat are practical at an industrial scale. For many years, air was sterilized

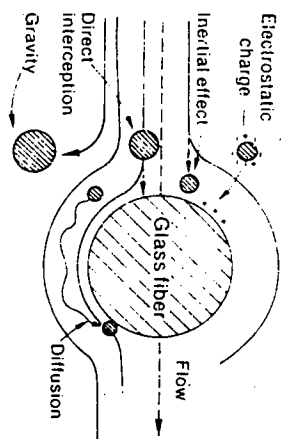


Figure 5.36 Mechanism by which particles are removed by a depth filter

by passing it over electrically heated elements, but due to the high cost of electricity, this process has been replaced by filtration.

In industrial systems, the air is sterile filtered. In older systems pure depth filters such as glass wool filters were installed, in which particles would be trapped by a combination of physical effects. As shown in Figure 5.36, particle

filtration involves inertial effects, blocking effects, diffusion, gravity separation, and electrostatic attraction. The last two mechanisms have a minimal effect on the removal of particles. The disadvantages of glass wool filters are shrinkage and solidification during steam sterilization. Glass fiber filter cartridges, which do not have these shortcomings, have replaced glass wool filters.

New cartridge filter systems using pleated membranes are now widely available. The advantage of these filters is that they are substantially smaller than glass wool or activated carbon towers. Operating the systems has become much simpler, because of the removable cartridge construction, replacing used filter elements is easy. Constructed of cellulose ester, polysulfone, or nylon, these membrane filters have the same structure as depth filters, but because they have a membranous structure they have an absolute filter effect. Figure 5.37 shows an example of the manner by which a filter cartridge is constructed. Figure 5.38 shows a fermenter installation with incoming and outgoing air filters. The dis-

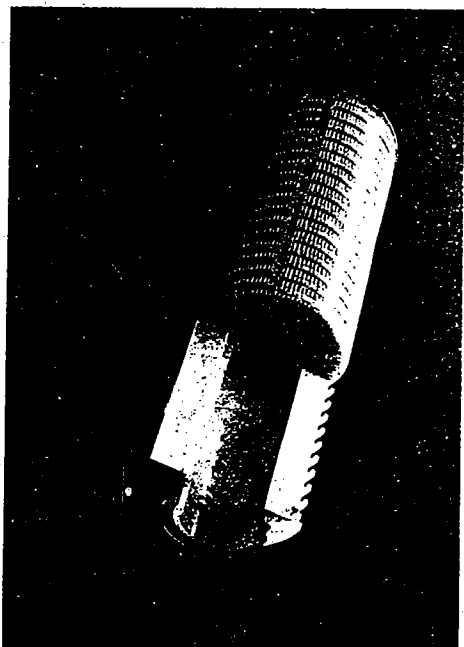


Figure 5.37 Structure of a filter cartridge

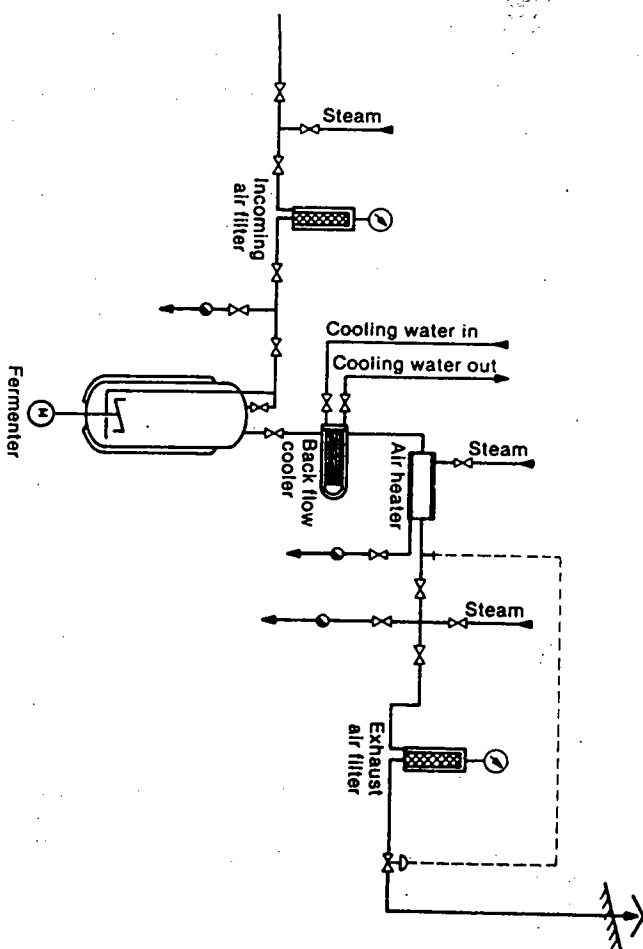


Figure 5.38 Installation of an air filter system (incoming and outgoing) in a fermenter

advantage of most systems installed today is that there is not yet any absolute filter for bacteriophages in industrial use. Bacteriophages can cause total failure in a system; for example, in the production of glutamic acid with *Corynebacterium glutamicum* or in the production of penicillin acylase with *Escherichia coli*.

5.8 FERMENTATION PROCESSES

An overall scheme of a fermentation process for the production of primary or secondary metabolites is given in Figure 5.39. The various stages of this scheme are discussed below.

Stage I: Inoculum preservation

The preservation of production strains over a long period is a basic requirement for a practical

Stage Course of the fermentation	
I Inoculum preservation	
II Inoculum build-up	a. 1-2 Shake flask cultures
III Prefe fermenter culture	b. Spore formation of solid medium
IV Production fermenter	a. 1-3 Prefe culture fermentations
	b. Continuous fermentation

Figure 5.39 General course of a fermentation in the production of primary and secondary metabolites

entation. The mere survival of strains is not main objective. Microorganisms can easily be viable through periodic transfer, but it is capability for product formation which must be reserved. High-yielding strains have often been damaged in primary metabolism during the fermentation process, and such strains frequently degenerate during successive transfers, probably as a result of spontaneous mutation. The objective of preservation is thus to maintain strains as long as possible without cell deterioration. "Master strains" should not be cultivated more than once in two years and activity levels should be checked with each usage. Depending on strain, selection procedures must be undertaken periodically. "Working strains" are derived from "master strains". Working strains should be selected for sterility and capability of product formation, and then stored until used.

The optimal method of preservation must be selected for each process, i.e., each strain. The following three techniques are most commonly

Storage at low temperatures (2–6°C) This method is the easiest, but also the least secure. Microorganisms are kept as stab cultures on agar or in culture in the refrigerator. There is a relatively high risk of contamination and reverse inoculation through frequent transfer (normally 8–16 weeks, at least once annually).

Freezing The most common method is to store at -18°C or -80°C in freezers or at -20°C under liquid nitrogen. Freezing down to -96°C must be done gradually ($1^{\circ}\text{C}/\text{min}$), and rapid freezing can be used if protective substances are added to prevent the formation of ice crystals. Frozen cultures may be kept for several years. The proportion of survivors is critical: between 90% and 95% of the microorganisms are generally killed during freezing and subsequent thawing.

Lyophilization The best method of strain preservation is freeze-drying (lyophilization) after

cultivation in special media. The addition of protective agents (such as skim milk or sucrose) reduces the lethality during the lyophilization process. Special equipment can be purchased which reduces the difficulties involved in freezing, drying, and aseptically sealing the ampoules. Lyophilization is the method of choice in large culture collections because the cultures can be satisfactorily maintained for an essentially unlimited amount of time.

Stage II: Growth of the Inoculum

The preserved culture is initially revived by growth in shaken liquid culture or on solid medium (if spore formation is needed). The conditions used in the initial culture (medium, temperature of incubation, etc.) will depend upon the specific process. Standard growth times can be expected as follows:

Lyophilized cultures		4–10 days
Frozen cultures		
Bacteria		4–48 hours
Actinomycetes		1–5 days
Fungi		1–7 days
Refrigerated cultures		
Bacteria		4–24 hours
Actinomycetes		1–3 days
Fungi		1–5 days

In order to obtain sufficient inoculum for small fermenters, a second series of shake cultures is usually made in more flasks. In some fermentations, the large-scale inoculum must consist of spores. To obtain a spore crop, the preserved culture is cultivated on a solid substrate in 2–10 liter glass vessels under conditions of constant temperature and sterile aeration for 8–24 days. The substrate for the production of large amounts of spores is generally a granular material such as bran, peat, rice, or barley. In order to ensure continued aeration, the substrate must be shaken daily, which makes maintenance of aseptic conditions difficult. Further, many of the spores produced may be incapable of germinating.

For inoculation, the total culture (spores plus culture medium) is suspended with the aid of a surface-active agent (e.g., Tween 80) and transferred into the fermenter.

The proper cultivation of inoculum is vital for optimal titers in the later production-scale process. For optimal yields, not only the number of cells and spores have an influence but also the nutrient medium used for the inoculum, the temperature of growth, and the inoculum age. Induction or repression phenomena in the culture used for inoculum may affect the rate of production.

Stage III: Fermenter preculture

Fermenter precultures must be made in order to have enough inoculum for a large fermenter. If a production fermenter is started with too little inoculum, growth is delayed and the product formation rate can be unsatisfactory (see Section 5.2). The optimal inoculum concentration for the production fermenter determines the number of stages of fermenter preculture that are needed. In general the following inoculum concentrations are required:

Bacteria	0.1–3.0%
Actinomycetes	5–10%
Fungi	5–10%
Spore suspension	$1\text{--}5 \cdot 10^3/\text{l}$ culture solution

At times, the production culture medium is used for the last stage of inoculum build-up in order to induce product formation.

Stage IV: Production fermentation

Depending on the fermentation, reactors of various sizes are used and no general scheme for the inoculation of a production fermenter can be given. Table 5.21 gives the sizes of various production fermenters in actual use.

Nutrient media for production must be optimized not only in the ingredients used but in

Table 5.21 Fermenter sizes for various processes

Size of fermenter m ³	Product
1–20	Diagnostic enzymes, substances for molecular biology, recombinant organisms
40–80	Some enzymes, antibiotics
100–150	Penicillin, aminoglycoside antibiotics, proteases, amylases, steroid transformations, amino acids
–450	Amino acids (glutamic acid), single-cell protein

how the nutrient medium is prepared. Several parameters which must be optimized are:

- Composition of ingredients, quality, carbon/nitrogen relationship, impurities, variability from batch to batch.
- Order of solution or suspension of ingredients, pH value before and after sterilization, effect of sterilization on the entire nutrient solution or on individual components.
- Changes in the sterilized nutrient solution before inoculation due to increase in temperature and aeration.

The most important parameters during the fermentation are:

Temperature Fermentations are run either in the mesophilic range (temperature optimum 20–45°C) or thermophilic ($>45^{\circ}\text{C}$) range. The appropriate temperature must be chosen to achieve maximum growth on the one hand and optimal product formation on the other hand. In some fermentations, higher temperatures are used to obtain increased growth of the culture and then the temperature is decreased at the onset of the idiophase. As an example of how important temperature control is, an increase of 1°C above the optimum produces a 20% lower yield in the penicillin acylase fermentation!

Aeration The aeration rate is 0.25–1.0 vvm (air volume/liquid volume-minute). The aeration rate must be adjusted to the amount of O_2 required.

Pressure. In order to minimize the risk of contamination, an overpressure of 0.2–0.5 bar is used. The hydrostatic pressure also has to be taken into account in large fermenters, since this influences the O_2 and CO_2 solubility in the nutrient solution.

Stirring. Depending on construction, the following stirring rates are used with disc impellers.

Fermenter size (m ³)	Impeller speed (rpm)
0.02	250–450
0.2	250–350
1 – 20	120–180
40 – 150	120–150
450	60–120

The installation of a continuous drive system is desirable in industrial fermenters in order to be able to precisely adjust the stirring rate to the process.

5.9 INSTRUMENTATION

To carry out measurements during fermentation for data analysis and control of the process, **special sensors**, which differ somewhat from those in the chemical industry, have been developed for bioreactors. 1. All sensors located in the sterile area must be sterilizable. 2. Some sensors must be specifically adapted to biochemical needs. The physical and chemical parameters listed in Table 5.22 can either be measured directly at many pilot plant or production fermenters or can be measured off-line in the laboratory.

The biological parameters listed must all be measured outside of the fermenter, with the exception of the NADH, measurement, which can be done on-line using a fluorescent method. There are interesting developments in the field of enzymatic electrodes, so-called **biosensors** (see Section 11.11). Such sensors, however, cannot be sterilized.

A standard procedure is to determine O_2 and CO_2 in the ingoing and outgoing air separately through the paramagnetic property of O_2 , and the infrared absorption of CO_2 . Sensors for measur-

Table 5.22 Parameters that can be measured in fermentation processes

Physical parameters	Chemical parameters	Biological parameters
Temperature	pH	Biologically active product
Pressure	Dissolved O_2 , O_2 and CO_2 in waste gas	Enzyme activity DNA and RNA content
Power consumption	Redox potential	NADH, and ATP
Viscosity	Flow rates (air and liquid)	Substrate concentration
Turbidity	Product concentration	Protein content
Weight of fermenter	Ionic strength	

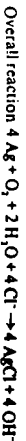
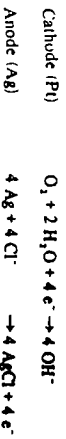
ing these gases are well-developed and function with few interruptions. But the **mass spectrometer** is more versatile, since it can also measure N_2 , NH_3 , methanol, and ethanol simultaneously as well as give qualitative and quantitative information on exchange of O_2 and CO_2 . By the use of gas-permeable membranes, it is also possible to measure dissolved gases in nutrient media. Devices have been developed which analyze up to eight gases in the fermentation simultaneously.

Equipment for making accurate pH measurements is readily available. Combination electrodes (glass electrode, reference electrode, and temperature compensator in a single unit) are available which are able to withstand sterilization temperatures, pressures, and mechanical stresses. Response time and sensitivity of these electrodes is satisfactory for the usual fermentation requirements. However, although electrodes are available for measuring many other inorganic ions, they are not as sensitive as those for the hydrogen ion (see the paper by Fiechter et al., 1987).

CO_2 electrodes and oxygen electrodes have been used in commercial operations with varying success. The electrodes are of the amperometric type (galvanic or polarographic).

The current and not the voltage is altered by the oxygen concentration. In the polarographic

electrode, oxygen is reduced at the cathode and silver is oxidized at the anode:



The current produced in this reaction is proportional to the oxygen partial pressure.

Commercial oxygen electrodes suitable for fermentation processes are widely available. However, the durability of these electrodes varies. Failure rates after sterilization lie above 50%, depending on the nutrient medium.

5.10 USE OF COMPUTERS

Computers can serve a variety of functions in fermentation process control and analysis:

1. **Optimization via computer.** Computers are used in scale up to store and evaluate fermentation parameters and to measure the effects of individual parameters on the metabolic behavior of cultures.

2. **Control via computer.** Computers can control fermentation processes. On-line fermentation control is widely used in the production scale in many companies.

Computer applications in biotechnology are not yet as widespread as in the chemical industry for several reasons: sensors suitable for use in sterile systems are not yet reliable enough to take advantage of computer capacity; biosynthesis and regulation of metabolite formation are not yet fully understood; fermentation cost reduction by using computers is difficult to calculate.

Thus, in biotechnology, computers are used primarily for data acquisition, data analysis, and development of fermentation models.

Data acquisition. Data can be acquired directly at the fermenter with on-line sensors. The information acquired can be data such as pH, PO_2 ,

temperature, pressure, viscosity, ferment weight, power uptake, aeration rate, and O_2 at CO_2 content in the gas stream. Other data can be obtained from laboratory measurements at fed into the computer off-line, e.g., biomass concentration, nutrient content, metabolite formation. This information can be entered as raw data and can be converted by the computer to standard units; for example, to adjust volumes for standard temperature, temperature-corrected data can be used to calculate the true aeration rate for a production system.

An alarm system can be hooked up to it data-acquisition system to inform the attendant when deviations from standard values occur. Data about the course of fermentation can be stored, retrieved, and printed out and production calculations can be documented.

Data analysis. The data entered or measured used in calculations such as CO_2 formation rate, O_2 uptake rate, respiratory quotient, specific substrate uptake rate, yield coefficient, heat balance, productivity, volume-specific energy uptake, an Reynold's number. When biomass is not contin-

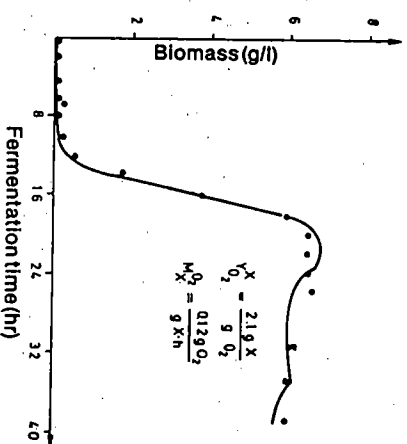


Figure 5.40 Comparison of experimentally determined *Thermotrichomyces* sp. concentrations (g) with growth curve generated from model calculations (Zabritskie, 1976)

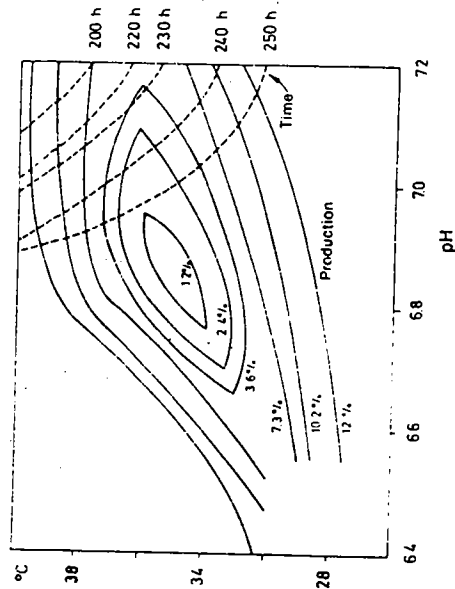


Figure 5.41 Isoproduct and isotherm curves for the erythromycin production process in relation to temperature and pH. The percentages are based on the lowest yields obtained (Cherny and Durand, 1979)

uously measured, the biomass concentration can be calculated through the O_2 uptake rate. It is assumed that the yield constant and the proportion of O_2 needed for maintenance metabolism are known. The calculations must be adjusted if, for example, secondary metabolites are formed or the yield constant changes during the fermentation. Figure 5.40 shows the agreement between the measured biomass of *Thermotoga* and the calculated growth curve. The value M is the amount for maintenance metabolism.

As a further example of data analysis, the optimization of erythromycin biosynthesis is illustrated in Figure 5.41. After fermentation at different pH and temperature levels, "isoproduct" and "isotherm curves" are computed. The fields are given as percentage of the minor production compared with the maximum erythromycin titer. Optimal productivity (production/fermentation time) for a given set of operating conditions can be ascertained from this graph.

Development of fermentation models By using mathematical models, it is possible to better understand the fermentation process and to calculate the effects of process variables on the fermentation results, and thus optimize a process

faster and more cost-effectively. The use of models can also aid the development of better control strategies for fermentations. A large number of models exists for batch and continuous fermentations. However, each model is only applicable to a specific process and cannot be used for other processes.

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6

Product recovery

specific purification processes had to be perfected for biological materials. The bioreactor is no longer considered in an isolated manner from the apparatus used in purification. Today, the bioreactor and the purification techniques are considered as an integrated system, all components of which must be optimized together.

Further, the work of the biochemical engineer has been considerably expanded by the development of processes for expressing foreign genes of eucaryotes in procaryotes. As discussed in Chapter 3, eucaryotic gene products are often formed in procaryotes in the form of highly insoluble inclusion bodies, frequently in inactive granules held together by disulfide bridges. Such inclusion bodies must be released and purified and eventually converted into active proteins.

6.2 UNIT OPERATIONS IN PRODUCT RECOVERY

The microorganism itself can be the desired product, as, for example, the ICI Pruteen process.

6.1 INTRODUCTION

One of the most critical aspects of an industrial fermentation process is the recovery and purification of the product. The selection of the appropriate purification steps depends on the nature of the end product, its concentration, the side products present, the stability of the biological material, and the necessary degree of purification.

When discussing product recovery, one must distinguish between the concepts of purification and concentration. A recovery step changes either the purity or the concentration of a metabolite. In the ideal recovery process, both of these parameters are optimized. In the early days of industrial microbiology, the techniques used in product recovery (for instance, extraction, distillation, dialysis, crystallization, precipitation, drying) were taken over more or less directly from chemical engineering without more than an approximate attempt to adapt them to biological materials. Gradually it came to be realized that

for the production of biomass as single-cell protein (SCP). In this case, if the cells are heated they aggregate into large clumps which can be readily separated from the fermentation broth by sedimentation. However, in most large-scale processes, the desired product is a metabolite, which is present either intracellularly or extracellularly. Examples of intracellular metabolites include nucleic acids, vitamins, enzymes, and certain antibiotics, such as streptomycin and griseofulvin. Examples of extracellular metabolites include amino acids, citric acid, alcohol, some enzymes (for example, amylases and proteases) and most antibiotics (for example, penicillin and streptomycin). In a few cases, metabolites are found both in the cells and the culture filtrate (for example, flavonoids, vitamin B₁₂).

The first step in product recovery is therefore the separation of cell biomass and insoluble nutrient ingredients from the supernatant. For this purpose, several methods are available, including flocculation, flotation, filtration, or centrifugation. If an intracellular metabolite is to be isolated, it must be liberated from the cells.

Once the metabolite has been separated from the cells, the selection of further purification steps will depend upon the desired product. Figure 6.1 summarizes various purification processes, arranged according to the separation principle for particles of various sizes. The last stages in process recovery involve precipitation, crystallization, and/or drying.

Fermentation products are present in culture solutions primarily at quite low concentrations (Table 6.1). In order to keep costs down, it is desirable in the very first stages of purification to reduce the volume to the smallest possible.

Table 6.1 Concentrations of some metabolites at the end of fermentation

Metabolite	Yield (g/l)
Vitamin B ₁₂	0.06
Riboflavin	0.1 - 7
Antibiotics	0.2 - 60
Amino acids	2 - 100

Flocculation and flotation

Single cells in the size range of 1 to 10 μm settle only very slowly and are difficult to bring down even with the centrifuge. In some cases (for example, SCP production and sewage treatment), flocculation can be used to produce large aggregates which will settle more readily. In most cases, a flocculating agent is added, such as an inorganic salt, an organic polyelectrolyte, or a mineral hydrocolloid. Depending on the agent used, the flocculation process can be either reversible or irreversible. The flocculation process is also influenced by the nature of the cells and the ionic constituents and their concentrations.

The reverse of flocculation is flotation, which is most readily accomplished by introducing gas into the liquid. The cells become adsorbed to the gas bubbles and rise to the foam layer at the top of the vessel, where they can be collected and removed from the bioreactor.

Filter systems

Most commonly, the first step in process recovery, separation of the biomass and the culture filtrate, is carried out by some sort of filtration process. Two major types of filters are so-called depth filters and absolute filters. A depth filter is constructed of a filamentous matrix, such as glass wool, filter paper, or asbestos, the filtration process occurring because the particles to be removed are trapped within the matrix. The particles being filtered are often smaller than the spaces in the filter, but are removed anyway as they pass through the contorted interstices of the filter. Absolute filters, on the other hand, are membranes, in which the pore sizes are smaller than the particles being filtered. The particles are therefore removed on the surfaces of the filters. Filamentous fungi are most commonly filtered through depth filters such as cloth, sometimes in the presence of a filter aid. Bacterial cultures, on the other hand, must be filtered with absolute filters. In either case, the efficiency of filtration is influenced by numerous factors, such as the

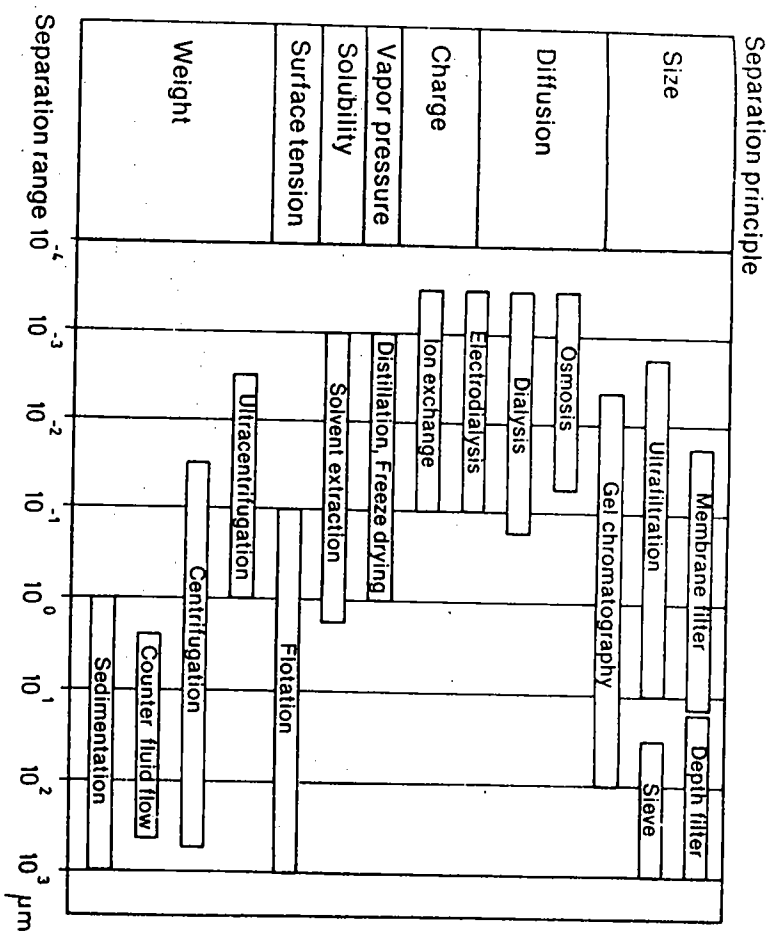


Figure 6.1 Concentration processes using different chemical and physical properties of particles and molecules

size of the organism, its morphology, the pH, viscosity, presence of slimes, temperature, and presence of other organisms as possible contaminants. As the cell material piles up on the filter, a filter cake is formed which reduces the filtration rate.

In the classical purification process used in the antibiotic industry, biomass is separated from the culture filtrate on a vacuum rotating drum filter. A cloth or rope is wound around the outside of the apparatus and a vacuum is pulled from the inside of the drum. To increase the efficiency of the filtration process, a filter aid such as diatomaceous earth is generally used. To

maintain the efficiency of filtration, an automatic knife is used to continuously scrape the biomass plus a thin layer of the filter aid from the filter. By arranging the drum in segments, the filter aid can be washed as the drum turns. Typical filtration efficiencies for arrangements of this type are shown in Table 6.2. Vacuum drum filters are especially good for suspensions which contain a high concentration of solids (20–60% mycelial volume).

Another type of filter, better suited for solutions in which the solid concentration is lower, is the filter press, in which a stack of flat porous plates are used as supports for a filter, either cloth

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The reverse of flocculation is flotation, which is most readily accomplished by introducing gas into the liquid. The cells become adsorbed to the gas bubbles and rise to the foam layer at the top of the vessel, where they can be collected and removed from the bioreactor.

Filter systems

Most commonly, the first step in process recovery, separation of the biomass and the culture filtrate, is carried out by some sort of filtration process. Two major types of filters are so-called depth filters and absolute filters. A depth filter is constructed of a filamentous matrix, such as glass wool, filter paper, or asbestos, the filtration process occurring because the particles to be removed are trapped within the matrix. The particles being filtered are often smaller than the spaces in the filter, but are removed anyway as they pass through the constricted interstices of the filter. Absolute filters, on the other hand, are membranes, in which the pore sizes are smaller than the particles being filtered. The particles are therefore removed on the surfaces of the filters. Filamentous fungi are most commonly filtered through depth filters such as cloth, sometimes in the presence of a filter aid. Bacterial cultures, on the other hand, must be filtered with absolute filters. In either case, the efficiency of filtration is influenced by numerous factors, such as the

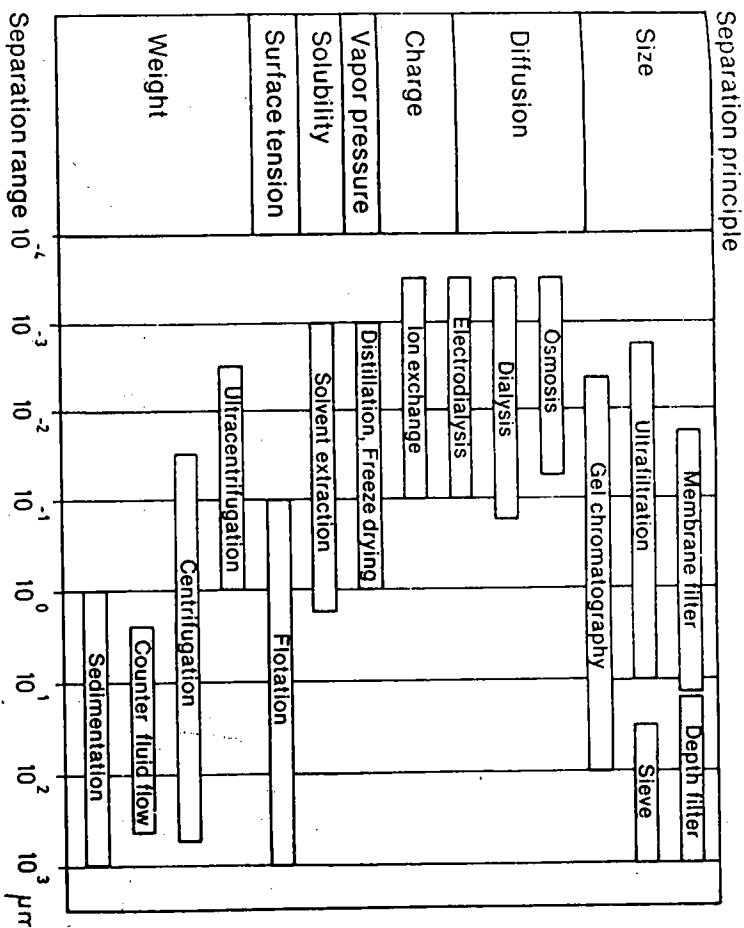


Figure 6.1 Concentration processes using different chemical and physical properties of particles and molecules

size of the organism, its morphology, the pH, viscosity, presence of slimes, temperature, and presence of other organisms as possible contaminants. As the cell material piles up on the filter, a filter cake is formed which reduces the filtration rate.

In the classical purification process used in the antibiotic industry, biomass is separated from the culture filtrate on a vacuum rotating drum filter. A cloth or rope is wound around the outside of the apparatus and a vacuum is pulled from the inside of the drum. To increase the efficiency of the filtration process, a filter aid such as diatomaceous earth is generally used. To

maintain the efficiency of filtration, an automatic knife is used to continuously scrape the biomass plus a thin layer of the filter aid from the filter. By arranging the drum in segments, the filter aid can be washed as the drum turns. Typical filtration efficiencies for arrangements of this type are shown in Table 6.2. Vacuum drum filters are especially good for suspensions which contain high concentration of solids (20-60% mycelial volume).

Another type of filter, better suited for solutions in which the solid concentration is low is the filter press, in which a stack of flat porous plates are used as supports for a filter, either clot

Table 6.2 Typical filtration rates for antibiotic fermentations

Antibiotic	Organism	Filtration rate l/h-ft ²
Penicillin G	<i>P. chrysogenum</i>	130-170
Kanamycin	<i>S. kanamyceticus</i>	8
Lincomycin	<i>S. lincolnensis</i>	35
Neomycin	<i>S. fradiae</i>	12

(Beller, 1985)

or membrane. The volume of the chamber within the plates determines the biomass capacity of this type of filter.

Although mycelium (from either fungi or actinomycetes) can be removed with a drum filter, single-celled bacteria are generally separated from the medium by use of membrane filters. Two types of membrane filtration processes, static and cross-flow, are compared in Figure 6.2. Because the filtration process is strictly dependent on the size of the pores and the size of the particles, clogging of filters is a problem. The cross-flow method was developed in order to reduce the tendency for clogging. In this method, the solution is pumped in a crosswise fashion across the membrane. The filtrate passes through the membrane and the biomass is washed off the filter and carried out with the retentate. With the cross-flow method, an increase in filtration rate of 100-fold can be obtained in comparison to static filtration.

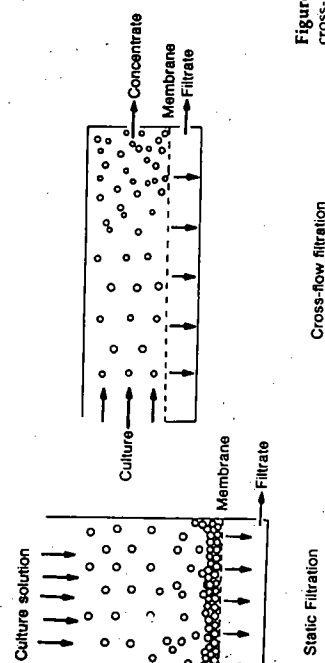


Figure 6.2 Comparison of static and cross-flow filtration

Depending on the sizes of particles being filtered, three major types of filtration process are recognized: reverse osmosis, for particles of 0.0001-0.001 μm ; ultrafiltration, for particles of 0.001-0.1 μm ; and microfiltration, for particles of 0.1-10 μm . The capabilities of the ultrafiltration and reverse osmosis processes are generally given in terms of the nominal molecular weights of substances separated (molecular-weight cut-off). Reverse osmosis generally involves substances of molecular weight less than 1000, ultrafiltration molecular weights greater than 1000.

The microfiltration process concerns mainly the separation of cells or cell fractions. Membranes made of cellulose esters, polyvinylfluorides, polycarbonates, polysulfones, and cellulose are used. Membranes are arranged either in cassettes, as spiral-wound modules, as bundles of tubes of 1-2 cm diameter, or as capillary bundles (Figure 6.3). Numerous parameters must be considered when selecting a filtration system. Among these are the pore size and particle selectivity, the rate at which fluid can be passed through when viscous solutions are being filtered, the ease with which the filter can be cleaned, and the number of times the filter can be reused. Other factors to consider are the cost of the filter system, the membrane surface area presented, the dead volume within the filter (which determines how much of the product will be lost), and the sterilizability of the system. Another factor that often reduces the efficiency of

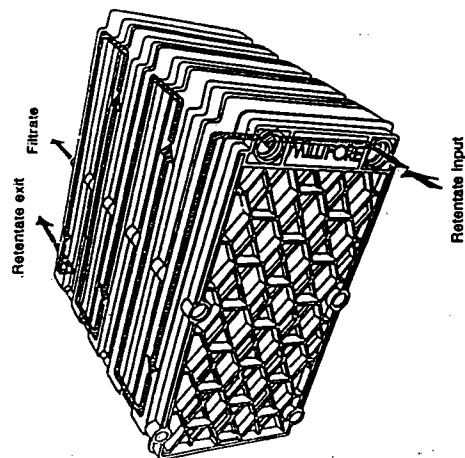


Figure 6.3 Comparison of a cassette system and a cross-flow cartridge module

a filter is the presence of antifoam agents, which are often used in large-scale fermentations.

The following data are typical for an industrial process for the concentration of *Escherichia coli* from a culture broth: fermenter volume, 6 m³; working volume, 4 m³; filtration rate, 25-80 l/h-m²; process time, 12 h; concentration factor, 1:10. Table 6.3 presents data for filtration processes with several organisms with various types of filters.

The following are typical costs that must be incurred for filtration in a large-scale industrial process: Production of 1 m³ retentate: membranes (\$420), capital investment (\$1600), labor (\$70), energy (\$230). Production of 1 m³ filtrate: membranes (\$14), capital investment (\$60), labor (\$3), energy (\$8).

In addition to their use in filtration, membranes also play a role in the fermentation pro-

cess itself. They can function in providing sterile air, and they can be used in other ways in the fermenter (Figure 6.4).

Centrifugation

The sedimentation rate of a particle in a gravitational field can be represented by Stoke's law:

$$v = \frac{d^2(\delta - \delta_l)}{18\eta} \cdot r \cdot \omega^2$$

where v = the sedimentation rate (m/s), d = the diameter of the particle (m), δ = the density of the particle (kg/m³), δ_l = the density of the liquid (kg/m³), r = the radius of the centrifuge head (m), ω = the angular velocity (rad/s), and η = the dynamic viscosity (Pa·s).

Centrifugation is used not only for separating solid particles from the liquid phase (fluid/part-

Table 6.3 Concentration of microorganisms by use of cross-flow filtration

Organism	Type of membrane	Concentration	Average flow rate
		% cell wet volume	l/h·m ²
<i>Bacillus cereus</i>	Capillary tubes polypropylene	10 — 30	124
<i>B. cereus</i>	Capillary tubes polysulfone	0.8 — 15	47
<i>Brevibacterium</i> sp.	Cassette polyacryl	3.5 — 32	37
<i>Escherichia coli</i>	Capillary tubes polysulfone	4.0 — 40	15
<i>E. coli</i>	Cassette PVDF	4.2 — 48	16
<i>Candida boydii</i>	Capillary tubes polypropylene	10 — 40	56
<i>Klebsiella pneumoniae</i>	Capillary tubes polycarbonate	2 — 58	73
<i>Lactobacillus casei</i>	Cassette PVDF	1.5 — 28	28
		0.45 µm	

Kroner et al. (1984)

title separation) but also for fluid/fluid and fluid/fluid/particle separation. Fluid/particle separation is of most significance, although fluid/fluid separation is used in penicillin production (for the separation of the antibiotic extracting solvent from the aqueous phase by means of a two-stage continuous countercurrent extractor; see Section 13.2).

Two distinct types of centrifuges are used, filter- and sieve centrifuges, and baffle centrifuges. In the filter- or sieve centrifuge the separation occurs as the particles are forced against a filter material. In the baffle centrifuge the separation occurs because of density difference between the particles and the liquid. The product can be removed either continuously or batchwise. A wide variety of centrifuges are marketed for large-scale centrifugation processes, and the capabilities of the major types are given in Table 6.4. In the case of yeast, machines capable of handling volumes as large as 300 m³/h have been developed.

Figure 6.5 shows cross-sectional diagrams of two major types of baffle centrifuges, a plate separator and a decanter. The major considerations

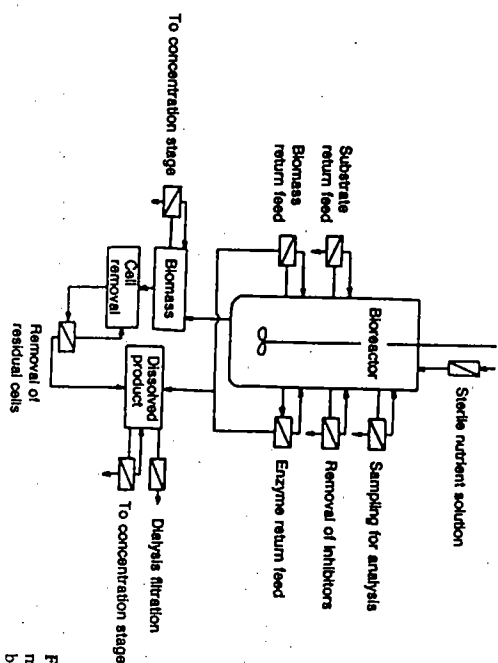


Figure 6.4 Various locations where membrane systems can be used in a bioreactor unit

in the selection of a centrifuge for biotechnological processes depend on the task at hand. For fluid/particle separation, factors to consider are: 1) the needed purity of the fluid phase; 2) the needed recovery of the fluid phase; 3) the needed recovery of the particle phase; 4) the needed permissible moisture content of the particles; 5) the specific density of the particles. For fluid/fluid separation, the major factors to consider are: 1) the needed purity of the lighter or heavier liquid; and 2) the needed recovery of the lighter or heavier phase.

Table 6.4 Comparison of a separator, a decanter, and a tube centrifuge

Parameter	Separator	Decanter	Tube centrifuge
Solids, %	1-30	5-80	1-5
Maximal centrifugal force (g)	5000-15,000	1500-4500	13,000-17,000
Dewatering capacity	Average	Average	Good
Removal of small particles	Good	Moderate	Very good
Cleanability	Good	Good	Very good

For use in the pharmaceutical industry, separators are available that are capable of withstanding temperatures of 121°C and hence can be completely sterilized. An example of such a machine is shown in Figure 6.6.

Disintegration of microorganisms

In some cases, the recovery of product requires that the microorganisms be fragmented, either by chemical, physical, or biological means. A summary of methods for disintegrating microorganisms is given in Figure 6.7. The selection of a method depends principally on the nature of the cells. Although the cell membrane offers no special resistance to breakage, cell walls vary widely in how readily they are broken. Gram-positive bacteria and yeasts are much more difficult to break than gram-negative bacteria or filamentous fungi. Even within a given organism, cells vary significantly in sensitivity to breakage depending upon their physiological state. It is also important, when selecting a disintegration method, that the target biomolecule not be destroyed. For instance, acid can be used to break many orga-

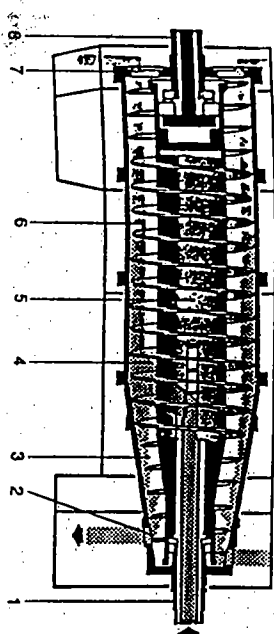
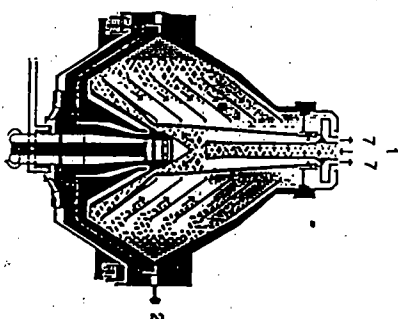


Figure 6.5 Construction details for a decanter (left) and a separator. 1 = inflowing stream; 2 = removal of particles; 3 = zone of low water content; 4 = particle sedimentation and drum cover; 5 = decanter screw; 6 = fluid level; 7 = clear liquid effluent; 8 = drive shaft



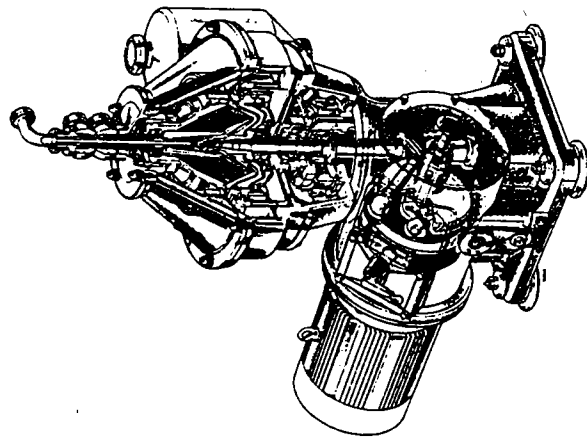


Figure 6.6 Sterilizable separator. This is type BTPX 20s manufactured by Alfa Laval. It contains 80 plates, an interior volume of 3.1 liter, a maximal centrifugal force of 12,800 X gravity, a maximal rotation of 9,650 revolutions per minute, a throughput for biotechnological processes of around 400–500 liters/hour.

isms but cannot be used if the product is acid labile.

For the extraction of yeast cells, autolysis is often used. Endogenous autolytic enzymes of the yeast cells themselves bring about the destruction of the cell wall. Addition of a sodium chloride, ethyl acetate, or chloroform, and incubation or 24 h at 45°C increases the rate of the autolytic process. Another technique is plasmolysis, following the addition of a high concentration of NaCl. Ultrasonic disintegration is widely used in the laboratory, but because of the high cost it is not suitable for large-scale industrial processes.

Industrially the most widely used methods of cell disintegration involve mechanical action. Ball mills bring about the breakage of cells

through the action of glass beads. The cells and the glass beads are mixed together and subjected to high-speed mixing in a reaction vessel. Breakage occurs when a cell is forced against the wall of the vessel by a glass bead. The efficiency of the process depends on the following parameters: 1) the cell concentration (optimal at about 40% cell wet volume); 2) rate of flow through the mill; 3) orientation of the apparatus (horizontal or vertical); 4) the cell/glass bead ratio; 5) the size of the glass beads (generally 0.5–0.8 mm). The procedure must, of course, be optimized for the particular process. One can expect a maximal breakage rate of about 80% of the cells.

Another approach is the use of homogenizers. In such devices, the biological material is placed under a high hydrostatic pressure (around 500 atmospheres) and the pressure suddenly released by allowing the liquid to exit through a valve. A cross section of the valve arrangement is shown in Figure 6.8. The operating conditions must be carefully optimized for each situation. Depending on the organism and the desired metabolite, from 10–60% of the cells can be broken. In order to achieve 90% breakage, two or three passes of the material through the homogenizer must be carried out. As seen in Figure 6.9, the efficiency of the process depends on the applied pressure. In general, the best pressure to use is between 450 and 600 atmospheres.

Chromatography

Some of the most expensive stages in product recovery involve the use of chromatographic methods. Such methods are especially valuable for the purification of biologically sensitive materials and find widespread use in the preparation of pharmaceuticals, diagnostic reagents, and research materials. In many cases, several separate types of chromatographic methods must be used, one following the other (Figure 6.10). Separation generally occurs in a column: a stationary phase (generally a resin) is used to adsorb the product which is then eluted with a mobile (liquid) phase. The elution liquid can be at a single density or

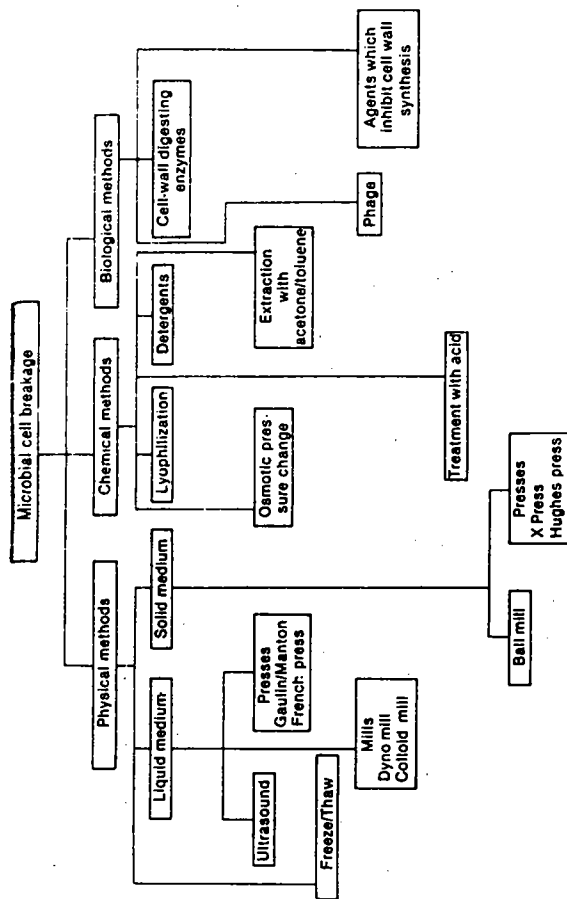


Figure 6.7 Methods of opening cells to isolate microbial metabolites

a gradually changing density (density gradient) can be used. By use of a suitable detector at the exit of the column, the eluted fractions which contain the desired product can be determined. Suitable detectors use ultraviolet or visible light absorption or conductivity. Summarized below

are various chromatographic methods that are widely used.

Gel filtration (molecular sieve chromatography)
Molecules of different sizes can be separated by passage through gels with different pore sizes. Large molecules do not penetrate the gel and pass directly through the column, exiting with the elution front. Small molecules penetrate more or less deeply into the gel and their mobility is hence retarded from the solvent front. Standard curves can be developed which relate molecular weight to the elution position. At the industrial scale, gel filtration is used mainly to remove salts and to separate low-molecular-weight impurities. At a smaller scale, gel filtration is used to fractionate and purify protein molecules (for instance, insulin or interferon). The most widely used gels for gel filtration are the Sephadex types P and A from BioRad and the Sephadex series S, G, and LH from Pharmacia.

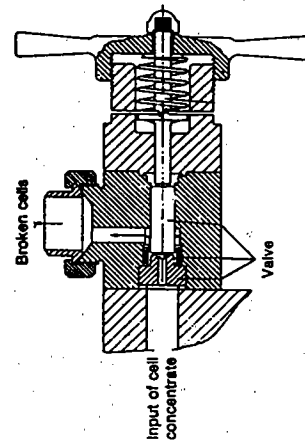


Figure 6.8 Outlet valve for a homogenizer

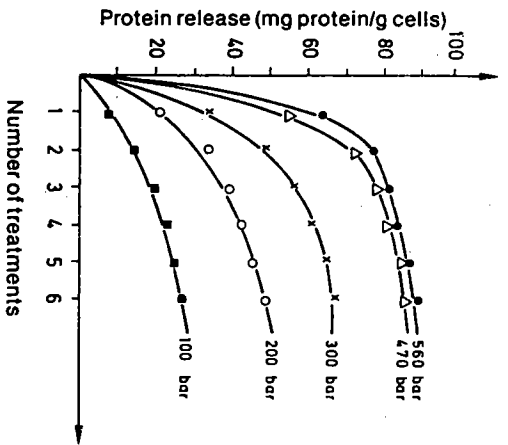


Figure 6.9 Protein release from yeast cells with a pressure homogenizer (Whitworth, 1974)

Adsorption chromatography In adsorption chromatography, separation involves hydrophilic or hydrophobic interactions (van der Waals forces) between the carrier and the biological material. Elution and fractionation are accomplished by means of solutions of higher or lower polarity or ionic strength. Adsorption materials include silicates, alumina, activated carbon, cross-linked dextrans, and hydroxylapatite.

Ion exchange chromatography Macromolecules with ionic groups can be separated by ion exchange chromatography. The macromolecule adsorbs to the carrier and is eluted by a solution of defined ionic strength. Depending on the ionic nature of the molecule and the ionic strength of the eluting solution, the separation can be quite specific. Ion exchange chromatography finds wide use for purification of antibiotics from fermentation broths. It also is widely used in large-scale purification of proteins. The most widely used ion materials for cation exchange are Dowex HCR and OCR, Amberlite IR and IRC, and Lewatit S. Anion exchangers used are Dowex SAR and MSA, Amberlite IRA, and Lewatit M.

Affinity chromatography Affinity chromatography is an absolutely specific method for purification of biological materials. The desired material binds specifically and reversibly to a ligand which has been fixed to an inert carrier. For example, nucleotide adenine dinucleotide (NAD) can be purified by allowing it to bind to a carrier containing a dehydrogenase enzyme. The antibiotic bacitracin can be used as a ligand to isolate Asp-, Ser-, Cys- and metalloproteases from a crude mixture.

Isoelectric focussing By means of isoelectric focussing, proteins can be purified by use of pH gradients in association with ion exchange gels. A linear pH gradient is constructed and the proteins are separated according to their isoelectric

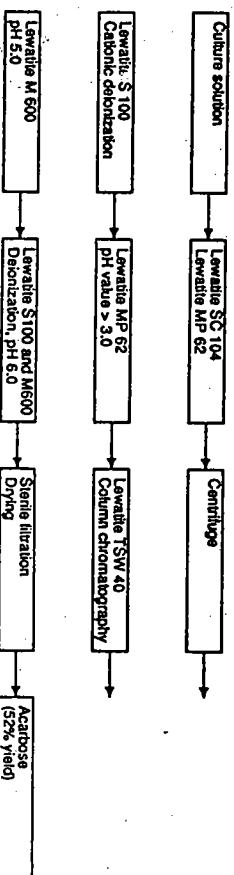


Figure 6.10 Flow diagram for the purification of a particular product, the invertase inhibitor Acarbose[®] (from Raunbush and Schmidt, 1978).

points. The separation of proteins by this method is very good, but only small volumes can be handled.

Extraction

Many products can be purified by use of solvent extraction. A two-phase system is set up, using a solvent which is immiscible with the aqueous fermentation broth. After the desired product is concentrated in the solvent phase, it can be further purified. Ideally, the product will transfer quantitatively to the solvent phase. Solvent extraction has been widely used in the antibiotic industry, for antibiotics such as penicillin, using organic solvents such as amyl or butyl acetate.

For the purification of enzymes in the active state, organic solvents cannot be used, but two-phase aqueous systems can be used, prepared in a salt solution with hydrophilic polymers. In both phases, water is the main component, but the two phases are not miscible. Cells remain in one of the phases and the enzyme is transferred to the other phase without any loss of activity. Human fibroblast interferon can be concentrated 70-fold with 100% yield using this method, at the same time that a four-fold reduction in volume is achieved.

An interesting variant of this procedure is extraction with supercritical solutions. This approach is used in the extraction of caffeine from coffee beans and hops for brewing, but can also be used for the extraction of pigments and flavor ingredients from biological materials.

Crystallization and precipitation

Once the metabolite has been extracted, it can be further concentrated and purified by crystallization, either by evaporation or by transfer to low temperature. Low-temperature crystallization is a very gentle way of purification. In the case of precipitation, a chemical agent is sometimes added to promote the concentration reaction.

Drying

For biological materials, it is essential to dry the final product in such a way that its activity is not lost. Drying essentially involves transfer of heat to the wet product, and removal of the moisture in a stream of gas. Heat transfer can be either by direct contact, by convection, or by radiation. Numerous types of apparatus are commercially available for carrying out the drying process. They include convection dryers (such as pulverizing, rotating, and band dryers) and contact dryers (such as thin layer and chamber dryers).

In some industries, the final product is a living microbial culture (for instance, starter cultures in the dairy industry). To prepare dry cultures, the freeze-dry technique must be used, since in this way the living organism is never subjected to heating. Freeze drying is also used in the preparation of some pharmaceutical products (for instance, penicillin is freeze dried directly in amples).

6.3 YIELD

Recovery losses depend on the sensitivity of the substance to the process and on the number of purification stages. Table 6.5 shows typical yield losses. Further details will be given in the discussions of particular processes later in this book.

The fraction of the total cost attributed to purification is on the average around 20%, but in extreme cases with certain kinds of intracellular metabolites can be as high as 90%.

In Table 6.6, the data for glutamic acid purification demonstrate the extent to which recovery methods must be combined and optimized.

Table 6.5 Recovery losses in typical fermentation processes

Fermentation product	Recovery loss %
Single cell protein	5
Antibiotics	20-50
Extracellular enzymes	10
Intracellular enzymes	90

Table 6.6 Glutamic acid purification process

Process	Direct crystallization	Yield %	Anion exchange process	Yield %	Cation exchange process	Yield %
Stages of process	Culture solution Filtration pH 7.5-8.0 Filtrate evaporation	100 90 98	Culture solution Filtration pH 7.5-8.0 Adsorption to anion exchanger Elution with acid Crystalline acid	100 90 95 98 90	Culture solution Filtration pH 7.5-8.0 Adsorption to cation exchanger Elution with base Crystalline Na salt	100 90 98 95 93
Chemicals used	3 Equivalents of acid 1 Equivalent of NaOH		3 Equivalents of acid 3 Equivalents of NaOH		3 Equivalents of acid 2 Equivalents of NaOH	
Total yield		56.6		70.0		78

(Wolf, 1974)

The final yield is only one factor, since the cost of chemicals, equipment, and wages must also be considered in the overall calculation.

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7

Organic feedstocks produced by fermentation

7.1 GENERAL

The chemical industry makes use of a variety of simple organic compounds as feedstocks, that is, raw materials, for synthetic processes. Due to the low cost of petrochemical products in the post-World War II era, little attention was given to studies involving the microbial production of organic feedstocks from plant substances. Since the mid-1970's, due to political and economic factors, petroleum and natural gas have become scarce and more attention has turned to microbial production of feedstocks. Of the 180×10^9 tons of plant material annually produced on earth, only 1-2% is utilized for animal and human nutrition and 1% for energy and fiber production.

The requirements for a cost-effective fermentation of organic feedstocks from plant carbohydrates are as follows:

- Low transportation costs of raw materials.
- Low costs to convert polymers (wood, cellulose, hemicellulose, starch) to usable monomers and disaccharides.

- Use of mixed cultures in order to catabolize different substrates and to convert them into the desired metabolite.
- Use of thermophilic strains to save costs for cooling, to bring about higher conversion rates, and to reduce contamination risks.
- Due to the high energy demand for aeration, anaerobic processes are preferable.
- Amenable to a continuous process.
- Low recovery and concentration costs.

7.2 ETHANOL

For thousands of years, ethanol has been produced for human consumption, and for at least a thousand years it has been possible to make concentrated alcoholic drinks by means of distillation. Ethanol for use as a chemical feedstock was produced by fermentation in the early days of industrial microbiology; however, for many years it has been obtained by chemical means instead, primarily through the catalytic hydration of ethylene. In recent years, attention has turned again to the production of ethanol for chemical

and fuel purposes by fermentation. For example, in three major countries the following amounts of industrial ethanol were produced by fermentation in 1986: United States, 2.5×10^9 liters; Federal Republic of Germany, 1×10^9 liters; Brazil, 1.1×10^9 liters. In countries with large agricultural areas, such as Brazil, South Africa, and the United States, intensive studies are being conducted on the production of ethanol from carbohydrates such as sucrose and starch. The objective of such research is to use ethanol for automobile fuel (generally mixed with gasoline). In some countries, fermentation ethanol is also used to produce ethylene and other petrochemicals.

The efficiency of energy conversion by ethanol fermentation varies considerably depending on the starting material. For instance, under optimal conditions the efficiency of energy yield (ratio of energy demand to energy produced) is as follows: sugar beet, 86%; potatoes, 59%; corn, 25%; cassava, 50%; sugar cane, 66%. Ethanol is more expensive than petroleum when used as a motor fuel but some countries, for political or internal economic reasons, subsidize ethanol production.

In Brazil, as early as 1982, about 30% of petroleum imports were replaced by ethanol production from sugar cane (5.2×10^6 m³), involving 60-80 separate fermentation plants. In 1986 in the United States, 65 separate fermentation plants were in operation, with a total capacity of 2.6×10^6 m³ per year, using corn starch as the starting material. The capacity of these plants varied from 20,000 to 570,000 m³. However, the total capacity was only about 25% of that which had been planned and some of the plants which had opened were unsuccessful and closed. The main problems were lack of technical and microbiological expertise and the high price of corn starch (which constituted 60-70% of the total production cost).

Biosynthesis of ethanol

Both yeasts and bacteria have been used for the production of ethanol. Among the bacteria, the

most widely used organism is *Zymomonas mobilis*. *Saccharomyces cerevisiae* is the most commonly used yeast but *Kluveromyces fragilis* has also been employed.

Under aerobic conditions and in the presence of high glucose concentrations, *Saccharomyces cerevisiae* grows well, but produces no alcohol. Under anaerobic conditions, however, growth slows and pyruvate from the glycolytic pathway is split with pyruvate decarboxylase into acetaldehyde and CO₂ (Figure 7.1). Ethanol is then produced from the acetaldehyde by reduction with alcohol dehydrogenase.)

Batch systems for ethanol production are started aerobically to obtain maximum biomass since if anaerobic conditions begin too early, the population density is not high enough to obtain a good conversion rate. Forced aeration may even be necessary for a short time in order to avoid yield losses. In continuous processes, optimal yeast growth and ethanol production are carried out under sugar limitation (≤ 1 g/l) and in a microaerobic environment (0.2-5 mg O₂/g dry substance/h). Theoretically, from one gram of glucose, 0.511 grams of ethanol can be obtained. When pure substrates are fermented, the yield is 95% and reduces to 91% when industrial-grade starting materials are used. One hundred grams of pure glucose will yield 48.4 grams of ethanol and 46.6 grams of CO₂, 3.3 grams of glycerol, and

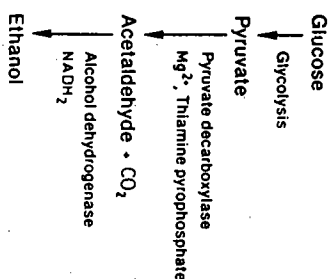


Figure 7.1 Biosynthesis of ethanol

2 grams of biomass (yeast cells). If corn starch is used, 100 kg starch (corresponding to 180 kg of corn) ferments to 51.5 grams of ethanol.

Ethanol is inhibitory at high concentrations and the alcohol tolerance of the yeast is critical or high yields. The ethanol tolerance of different strains varies considerably. As the concentration of ethanol increases, the growth rate is first reduced, whereas at higher concentrations the biosynthesis of ethanol itself is inhibited. However, yeast is more sensitive to endogenously produced ethanol than to that added from external sources to the fermentation system. Growth generally ceases at 5% ethanol (volume ethanol in volume water = vv) and the production rate is reduced to zero at 6–10% (vv). Using molasses, 2% sucrose, weight per volume water = ww, glucose is normally converted to 6% (vv) ethanol within 36 hours.

With pure sugar solutions, the ethanol concentration can reach as high as 10%, although ethanol-tolerant mutants have been isolated that in produce 12–13% ethanol and are currently being scaled up to full production. The bacterium *Imomonas mobilis* has in recent years come under increasing study because it has a number of potential advantages:

Osmotic tolerance to higher sugar concentrations (up to 400 g/l).

Relatively higher ethanol tolerance (up to 130 g/l).

Higher specific growth rate than yeast (growth rate, μ of 0.27 compared to 0.13 for yeast; laboratory culture studies).

Anaerobic carbohydrate metabolism is carried out through the Entner-Doudoroff pathway, where only one mole of ATP is produced per mole of glucose used, thus reducing the amount of glucose that is converted to biomass rather than ethanol.

The pH optimum of the bacterial ethanol fermentation is considerably broader (pH 5–7) and temperature optimum is higher (30°C). Even at 17°C the yield of ethanol from glucose is 97% theoretical. A comparison of the kinetic pa-

rameters for *Zymomonas* and *Saccharomyces* shows the following: ethanol formation rate (g/g-h) 2.9 times higher, growth rate (μ) 2.4 times higher, and glucose uptake rate 2.6 times higher.

In order to increase even more the productivity, both the bacteria and yeast processes are being carried out in large scale by continuous fermentation. The optimal amount of cell recycling is under study. Maximal productivity with glucose as carbon source that has been reached is about 82 g/l-h for yeast and 120 g/l-h for *Zymomonas*. A further possibility is the use of immobilized cells. In a direct comparison of yeast using molasses as substrate, the following results have been obtained: batch process, 2.0 g/l-h ethanol; continuous process, 3.35 g/l-h; immobilized cells under continuous conditions, 28.6 g/l-h. The immobilized cell process has already been placed in large-scale production in Japan.

Ethanol production process

Ethanol is produced in three steps, each of which must be optimized: 1) Preparation of the nutrient solution; 2) fermentation; 3) distillation of ethanol. Figure 7.2 shows the stages in the production of ethanol from corn meal.

Preparation of the nutrient solution Three types of substrates are used in the ethanol fermentation: 1) starch-containing roots, tubers, or grains; 2) molasses or juice from sugar cane or sugar beet; 3) wood or waste products from processed wood. The production of alcohol from milk whey has been tested but is not yet in use commercially.

The most important root starch is derived from the tropical plant *Manihot esculenta*, from which cassava, manioc or tapioca flour is obtained. In Brazil, 28 × 10⁶ tons of this plant are harvested annually.

Since *Saccharomyces cerevisiae* has no amylases, the starch must be hydrolysed. The roots (containing 20–35% starch and 1–2% protein) are first ground, squeezed, and dried. The starch is liquefied by boiling under pressures and then

exol pressure

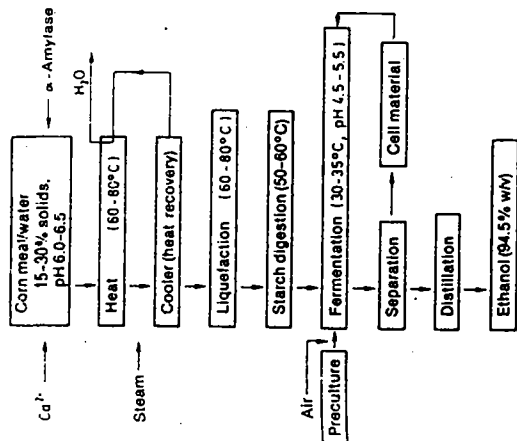


Figure 7.2 Steps in ethanol production from a substrate containing starch

hydrolysed enzymatically. By adding cellulose-splitting enzymes of different microbial origin, the proportion of reducing sugars can be increased in the cassava paste.

The starch-containing grain used in Asia is rice, in the United States mainly corn but also millet in some areas, in Europe potatoes in addition to other grains. Grain can be used whole or, in the case of corn, coarsely ground; the kernels are soaked at 40–50°C for several hours, then steeped and liquefied.

Continuous liquefaction and saccharification processes using first steam injection (3 min at 150°C) and then vacuum cooling are used in modern plants. Some α -amylase is added before heating in order to decrease the viscosity which develops after the steeping process. For glucose production, glucoamylase and α -amylases are added after the heating and cooling process. Approximately 1 l commercial α -amylase and 3.5 l glucoamylase are added for 1 ton of starch.

Molasses, a byproduct of sugar crystallization, is commonly used today as an ethanol source, but may become less important in the future since it has better uses as animal feed or in other fermentations.

When molasses is made from sugar beets, the juice is extracted from the beet chips with hot water. With sugar cane, the cane juice is released by means of presses. The residue from the pressing of sugar from sugar cane stems is called bagasse. Eighty percent of the bagasse formed can be burned as an energy source in the distillation process and 20% more can be fermented after chemical hydrolysis. In the Federal Republic of Germany a hybrid plant which is a cross between sugar beet and turnip is being developed as a source of carbohydrate for ethanol production.

Wood has not yet been used in the commercial production of ethanol. Because of the large amount of waste wood available, the direct fermentation of wood which has been hydrolyzed with cellulases would be of considerable importance.

Sulfite waste liquor is another potential source of sugar. This is formed during production of paper from conifers and contains fermentable hexoses. Sulfite waste liquor from deciduous trees cannot be used for commercial processes because of the high fraction of the sugar which is in the form of pentoses rather than hexoses.

Fermentation Continuous fermentations are only slowly being brought into large-scale operation. In the United States, among 10 large-scale plants using yeast, four are operating continuously, one of which also employs cell recycling. A few continuous plants have also been started in Brazil. A Danish system has been described in detail: The nutrient solution was molasses with diammonium phosphate additive. The pH value was adjusted to 5 with H₂SO₄ and pasteurization was then carried out. The fermentation temperature was 35°C and the yeast production was 10 g/l. After the fermentation, the cells were separated by centrifugation and channeled back into the first fermenter. After 10.5

hours, the yield in the first fermenter was 6.1% (vv) alcohol with 1% residual sugar, whereas in the second fermenter the yield was 8.4% (vv) with 0.1% residual sugar.

In the Federal Republic of Germany, ethanol production with *Zymomonas mobilis* is being carried out in two parallel bioreactors, each of which has a volume of 70 m³. The fermentation operates continuously but without cell recycling, using as a starting material a starch fraction which was a byproduct of enzymatic glucose production. The ethanol concentration ranged from 7-8%. The process could be operated for up to three weeks before a *Lactobacillus* contaminant became established and the process had to be restarted from scratch.

Batch fermentation is more commonly employed for ethanol production. Production is carried out in batch processes (fermenter volume of 600 m³) with either starch hydrolysate (up to 20% dry weight) or molasses, using a 3% inoculum (cell density 3×10^6 /ml). Within 12 hours, *Saccharomyces* produces 10% (vv) ethanol with 10-20 g cell dry weight/l when the process is carried out at 35-38°C, pH 4.0-4.5, the maximal productivity was 1.9 g/l/h. The short fermentation time was accomplished by use of cell recirculation; 80% of the cells were removed in a separator and brought back again into the fermenter. When high quality molasses is used, the maximal yield is 95% of the theoretical. Table 7.1 shows a comparison of the operating parameters of three different types of fermentation systems, batch, continuous, and continuous with cell recycling.

Before distillation, the cell mass is separated by centrifugation or sedimentation. Figure 7.3 shows the layout of a modern system.

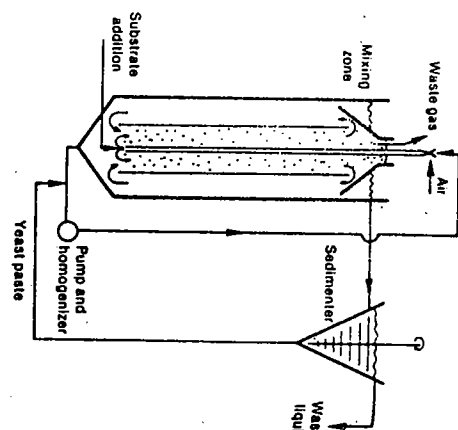


Figure 7.3 ALCO-FLOC system using air suction and cell flocculation (Meyraht, 1979)

Energy balance of the ethanol fermentation Since ethanol is produced primarily as a fuel and energy source, the energy balance of the whole process determines its economic viability. A summary of the energy requirements of the different stages in the process for ethanol production from various substrates is given in Table 7.2.

It can be seen that the product-recovery stage, the distillation of ethanol, is the single most energy-demanding step of the whole process. Because of this, improvements in the distillation process will have more impact on the success of the process than will improvements in the fermentation itself. If the energy yield of the ethanol produced is related to the total energy inputs from the various stages of the process, it is found

Table 7.1 Comparison of several kinds of fermentation systems for ethanol production

Fermentation process	Sugar concentration %	Cell concentration g/l	Ethanol concentration g/l	Specific productivity g/g cells-hr	Volume-specific productivity g/l-h
Batch	16.7	21.3	85.6	0.42	11.8
Continuous	15.6	19.7	79.1	0.72	14.1
Continuous with cell recycling	16.7	100	85.6	0.42	42.5

(Majorella et al., 1984)

Table 7.2 Energy (MJ/l of pure ethanol) required to produce absolute alcohol

Process Stage	Beets	Cane	Starchy raw materials
Digestion/hydrolysis, batch	4 - 5	-	7 - 8
continuous	-	1.1 - 1.5	2
Cane mill Extraction	0.8 - 1	2 - 3	-
Fermentation, batch	0.06	0.06	0.06
continuous	0.1	0.1	0.1
Distillation, single-stage optimized	10 - 13	10 - 13	10 - 13
Process, conventional optimized	16	13	19
	7	7	8

(Misselhorn, 1979)

that there is actually either an approximate balance or a net energy loss. This emphasizes the importance in optimizing all steps in the process to the utmost. Because the profit margin is so low, such optimization is economically much more important for the production of an energy source than it is for the production of a fine chemical or antibiotic. Ethanol with a purity of 92.4% is used as a solvent in the cosmetic, pharmaceutical, and chemical industry and at 99.2% purity as a motor fuel. In order to produce the virtually water-free product, better distillation methods would be desirable.

7.3 ACETONE/BUTANOL FERMENTATION

Pasteur first observed the production of butanol by bacteria in the 19th century. Before World War I, microbial processes were developed for the purpose of obtaining butadiene for synthetic rubber, at which time Chaim Weizmann performed fundamental research on the fermentation of *Clostridium acetobutylicum* for the production of acetone, butanol, and ethanol. During

World War I, the product of interest was acetone used for the production of the explosive trinitrotoluene (TNT), but after the war butanol became more important, used primarily for the production of nitrocellulose lacquers. After World War II, petroleum-based processes replaced biological fermentation processes, and the majority of plants with fermenters less than 200 m³ in volume were closed. However, some plants of this size are still functioning today in countries where economic, political, geographical, or climatic factors are favorable, such as in Taiwan and South Africa.

Butyric acid, butanol, acetone, and isopropanol are obtained through *Clostridia* fermentation of starch, molasses, sucrose, wood hydrolysates, and pentoses. The relative proportions of each of these products in the fermentation depends on the bacterial strain used and on the fermentation conditions. Three fermentation types can be categorized, according to their fermentation products:)

- Acetone-butanol fermentation with *Clostridium acetobutylicum*
- Butanol-isopropanol fermentation with *Clostridium butyricum*
- Butyric acid-acetic acid fermentation with *Clostridium butyricum*.

Biosynthesis

Figure 7.4 shows the biosynthesis of the main fermentation products: acetone, butanol, butyric acid, isopropanol, and acetic acid. The butanol referred to here is entirely n-butanol. Hydrogen and carbon dioxide are also produced. In addition, ethanol can be produced through the reduction of acetaldehyde. The organism forms a net yield of 2 ATP in the steps from glucose to pyruvate and this is the total energy released when no acetic acid is produced.

Only the acetone-butanol fermentation is of current economic interest. Butanol at a concentration of less than 0.5% has no influence on the cells, whereas at higher concentrations it causes

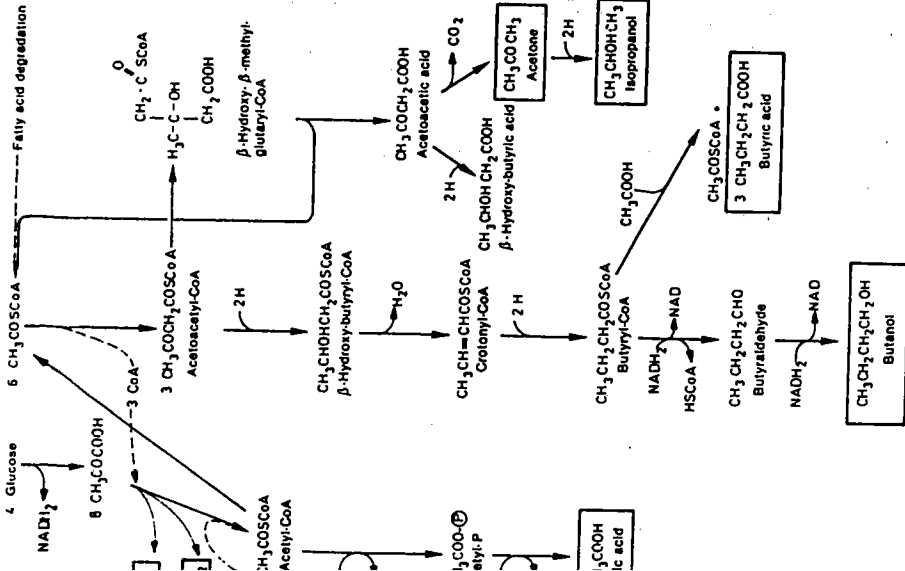


Figure 7.4 Pathway of biosynthesis of butanol, butyric acid, acetic acid, acetone, and isopropanol

To the phospholipids of the cell membrane concentrations above 1.3% butanol production ceases. In addition, depending on the fermentation conditions, autolytic products are produced that cause the cells to lyse. Laboratory studies are underway to develop various fermentation processes for butanol, either cell suspensions or immobilized

Production process

One of the few remaining systems for acetone-butanol production still in use is in South Africa. Stock cultures of *Clostridium acetobutylicum* are stored as spores in sand for up to 30 years. Both the production of the spores and the preparation of the inoculum are critical. Although production fermenters must be inoculated only in the low

inoculum ratio of 1:3000, build-up of the inoculum over several stages has the following advantages: greater resistance to contamination, shorter fermentation times, greater yields, greater fermentable sugar concentrations.

After sterilization, the fermenters (12 X 90 m³ fermenters) are gassed with CO₂ and before and after inoculation the fermenter contents are stirred with CO₂. The base for the fermentation is molasses plus corn steep liquor with a beginning pH of 5.8-6.0 at 34°C.

The 36-hour fermentation has three phases (Figure 7.5):

1. During the first 18 hours, the pH value decreases to 5.2 due to the formation of acetic acid and butyric acid.
2. During the next 18 hours, the pH value increases through the metabolism of these acids to acetone and butanol. When excess NADH₂ is present, the cells take up butyric acid and reduce it to butanol.
3. In the next stage, growth and solvent production stop. The pH value (5.8) remains constant. The products can now be recovered.

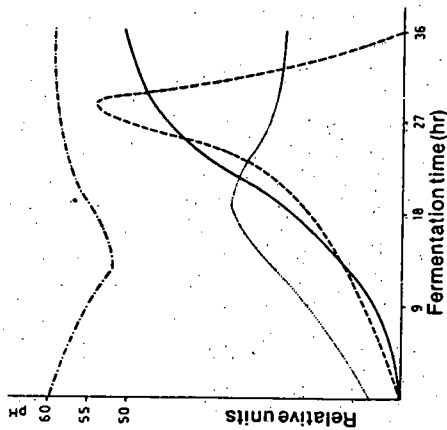


Figure 7.5 The kinetics of the acetone-butanol fermentation (Spivey, 1978). — solvent, --- titratable acids, ---- gas production, pH value

Table 7.3 Yields from a 90m³ fermenter

Product	% of converted sugar	Amount produced
Butanol	30 (ratio 6:3:1)	1053 kg
Acetone		526 kg
Ethanol		175 kg
CO ₂	50	2900 kg
H ₂	2	117 kg

(Spivey, 1978)

Contamination due to bacteriophages and lactobacilli is a common problem and absolute sterility is therefore a necessity. Table 7.3 shows typical yields of a 90 m³ fermenter containing 5.85 tons of fermentable sugar.

CO₂ produced during the fermentation is recovered and converted to liquid CO₂ or dry ice. Acetone, butanol, and ethanol are recovered through continuous distillation and fractionation. The residue left after the distillation may be dried and utilized as animal feed.

7.4 GLYCEROL

Glycerol has wide uses in commerce and is also a starting material for explosives manufacture. During both world wars, glycerol was produced by microbial processes, but today these processes are no longer of any commercial value. However, the glycerol fermentation is of theoretical interest in demonstrating how modification of fermentation conditions can lead to modification of product type.)

Glycerol is formed by yeast along with ethanol during the alcoholic fermentation (Figure 7.6). Normally, the amount of glycerol formed is tiny, but by modifying the fermentation balance the amount of glycerol produced can be greatly increased. This is done in the following way: An intermediate in the ethanol fermentation is acetaldehyde, and if sodium bisulfite is added, an acetaldehyde-sulfite complex is formed. Normally, the NADH₂ formed during the first part of glycolysis is reoxidized by acetaldehyde to form ethanol, but if the acetaldehyde is removed in this sulfite complex, the NADH₂ is available

Organic acids

8.1 INTRODUCTION

Organic acids find wide use both as additives in the food industry and also as chemical feedstocks. Fermentation processes play a major role in the production of most organic acids. All acids of the tricarboxylic acid cycle can be produced microbiologically in high yields. Some acids derived indirectly from the Krebs cycle, such as itaconic acid, can likewise be produced. Other organic acids are derived directly from glucose (for example, gluconic acid), or they are formed as end products from pyruvate or ethanol (for example, lactic acid and acetic acid).

Except for the production of citric acid which is manufactured entirely by fermentation, there is frequently great competition between microbiological and chemical processes for production of the various organic acids. For some acids (lactic acid, acetic acid) microbiological and chemical methods for preparation are simultaneously in use. For others (α -ketoglutaric acid, malic acid), fermentation processes have been developed

which are not used commercially, either due to lack of sufficient need for the acid or for economic reasons. Table 8.1 lists several acids that have been produced microbially or for which processes have been extensively developed. In this chapter, we will discuss the most important processes of commercial interest.

8.2 CITRIC ACID

Citric acid has been known as a natural plant substance since the end of the nineteenth century, and since 1893 scientists have known that it is produced by filamentous fungi. In 1923, the first practical fermentations for the production of this organic acid were started, using microorganisms growing in surface culture. Fermentation using deep-vat fermenters began in the 1930's. Although in South America and Mexico there are some small factories where citric acid is isolated from unripe citrus fruits (lemons contain 7-9% citric acid), today over 99% of the world's output of citric acid is produced microbially.

Table 8.1 Practical microbiological processes for organic acid production

Organic acid	Organism	Commercially produced	Substrate	Process	Yield %	Use of product	References
Fumaric acid	<i>Rhizopus</i> sp., <i>Candida</i>	Yes	Glucose	3 days, 33°C	65	Resin	Buchta, 1983b
Propionic acid	<i>Propionibacterium</i>	Yes	Lactose Glucose Starch	7 days, 30°C 8-12 days, 30°C	84 60	Perfume Fungicide	Buchta, 1983b
Malic acid	<i>Leuconostoc brevis</i> , <i>Candida</i>	No	Fumaric acid n-Alkane	24 h	99	Food	Buchta, 1983b
α -Keto-glutaric acid	<i>Candida</i> , <i>Aerobacter</i>	No	n-Alkane Glutamic acid	40h	67 46		Rose, 1978
5-Keto-gluconic acid	<i>Acetobacter suboxydans</i>	Yes	Glucose	5-6 days	85	L-Tartaric acid	Buchta, 1983b
2-Keto-gluconic acid	<i>Serratia marcescens</i>	Yes	Glucose	16 h	95-100	Isoscorbic acid	Buchta, 1983b

The following amounts of citric acid have been produced microbially:

1929	5,000 tons
1953	50,000 tons
1976	200,000 tons
1979	220,000 tons
1982	350,000 tons

The fermentation processes are carried out either in surface culture or in fermenters up to 220 m³ in volume. Citric acid is marketed as citric acid-1-hydrate or as anhydrous citric acid. Most citric acid (60% of the total) is used in the food and beverage industry. The flavor of fruit juices, fruit juice extracts, candy, ice cream, and marmalade is enhanced or preserved by the addition of citric acid.

The pharmaceutical industry (10% of total usage) uses iron citrate as a source of iron and citric acid as a preservative for stored blood, tablets, ointments, and in cosmetic preparations. In the chemical industry (25% of total usage), citric acid is used as an antifoam agent, as a softener, and for the treatment of textiles. In the metal

industry, pure metals are produced as metal citrates.

Citric acid is being used more and more in the detergent industry because it can replace polyphosphates. The higher cost of citric acid as compared to polyphosphates formerly restricted its use; however, detergents containing polyphosphates have been prohibited in some locations and the polyphosphates have been completely replaced with citric acid.

Strains for citric acid production

Many strains excrete traces of citric acid as a metabolite of primary metabolism. Examples are *Aspergillus niger*, *A. wentii*, *A. clavatus*, *Penicillium luteum*, *P. citrinum*, *Mucor piriformis*, *Pectinomyces divaricatus*, *Citromyces plefferianus*, *Candida guilliermondii*, *Saccharomycopsis lipolytica*, *Trichoderma viride*, *Athrobacter paraffinicus*, and *Corynebacterium* sp. However, only mutants of *Aspergillus niger* are used for commercial production. Compared to *Penicillium* strains, the *Aspergillus* produce more citric acid per unit time. Moreover, the production of undesirable side

products, such as oxalic acid, isocitric acid, and gluconic acid, can be more efficiently suppressed in these mutants.

Biosynthesis

Citric acid (2-hydroxypropene-1,2,3-tricarboxylic acid) is a primary metabolic product and is formed in the tricarboxylic acid cycle. Glucose is the main carbon source used for citric acid production. In many organisms, 80% of the glucose used for citric acid biosynthesis is broken down by reactions of the Embden-Meyerhof-Parnas (EMP) pathway and 20% by reactions of the pentose phosphate cycle. During the growth phase, the relationship between these two pathways is 2:1. In citric acid producers, the enzymes of the EMP pathway are present throughout the fermentation process and the activity of the pathway is regulated in a positive manner with phosphofructokinase and in a negative manner by pyruvate kinase. When pyruvate is decarboxylated with the formation of acetyl-CoA, the acetate residue is channeled into the tricarboxylic acid cycle. During the idliophase, all enzymes of the Krebs cycle are expressed except α -ketoglutarate dehydrogenase. The citrate synthase activity (condensing enzyme) is increased by a factor of 10 during the production of citric acid, while the activities of the enzymes which catabolize citric acid, such as aconitase and isocitrate dehydrogenase, are sharply reduced as compared to their activity in the trophophase. One of the three isocitrate dehydrogenase isozymes, a mitochondrial enzyme that is specific for NADP, is inhibited by glycerol that accumulates during the spore germination process. In addition, citric acid production is inhibited by high intracellular concentrations of ammonium ion.

Citrate synthase cannot be solely responsible for maintaining the activity of the tricarboxylic acid cycle, since the cycle would cease if acetate were removed. Distinct sequences which replenish the tricarboxylic acid cycle intermediates (anaplerotic sequences) must thus exist in the production phase, as outlined in Figure 8.1. The

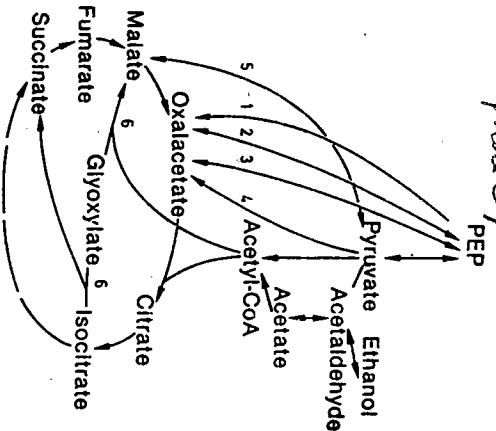


Figure 8.1. Anaplerotic reactions which connect with the tricarboxylic acid cycle

first anaplerotic enzyme present in *Aspergillus* is a pyruvate carboxylase (reaction 4, Figure 8.1), which converts pyruvate and CO₂ into oxalacetate, inorganic phosphate, and ADP (while consuming ATP). The reaction is dependent on Mg²⁺ and K⁺ ions; acetyl-CoA is not needed for the reaction, in contrast to its requirement for the metabolic reactions of other microorganisms. Pyruvate carboxylase is the key enzyme for citric acid production.

The second anaplerotic sequence involves a phosphoenolpyruvate carboxykinase (reaction 2, Figure 8.1), which converts PEP and CO₂ into oxalacetate and ATP in the presence of ADP. The system requires Mg²⁺ or Mn²⁺ and K⁺ or NH₄⁺. If acetate or higher aliphatic compounds such as n-alkanes (C₆-C₁₂) are used as carbon source,

a third anaplerotic sequence is present in *Aspergillus niger*. In the absence of glucose, the glyoxylate cycle (reaction 6, Figure 8.1) is operable: isocitrate lyase is induced and malate synthase is present. If glucose is added, the glyoxylate cycle is then repressed, although isocitrate lyase is still partially active.

Yields

In the trophophase, part of the added glucose is used for the production of mycelium and is converted through respiration into CO₂. In the idliophase, the rest of the glucose is converted into the organic acids and during this phase there is a minimal loss through respiration. Table 8.2 gives data on citric acid yields in various processes. The theoretical yield is 123 g citric acid-1-hydrate or 112 g anhydrous citric acid per 100 g sucrose. However, such yields are not obtained in practice, due to the loss during the trophophase.

Nutrient media

The media used for citric acid production have been highly perfected over the many years that the commercial process has been underway.

Carbohydrate source A 15-25% sugar solution is converted during fermentation. A variety of starting materials can be used as carbohydrate sources: Starch from potatoes, starch hydroly-

sales, glucose syrup from saccharified starch, sucrose of different levels of purity, sugar cane syrup with two-thirds of the sucrose converted into invert sugar, sugar cane molasses, sugar beet molasses.

If starch is used, amylases formed by the producing fungus or added to the fermentation broth hydrolyze the starch to sugars. If hydrolysates or syrup are used, a preliminary treatment with either precipitants or cation exchangers must be carried out to remove cations. For elimination of metals, molasses is generally treated before sterilization with calcium hexacyanoferrate or is purified by cation exchangers. Although the effect of the hexacyanoferrate is primarily to bring about metal precipitation, there are also studies which show a direct influence on the fermentation, since an excess of hexacyanoferrate over that needed to complex metals increases the citric acid yield.

Production strains are optimized based on the carbon sources used. A strain which produces well with one carbohydrate source generally cannot be used with another starting material without a substantial yield reduction.

There are no analytical methods that permit an evaluation of a molasses source for its use in citric acid production; each batch of molasses must be given a preliminary fermentation test. Firms which use molasses as starting material generally optimize molasses treatment and the make-up of the culture media in model fermentations.

Table 8.2 Citric acid yields in different fermentation processes

Microorganism	Raw material (Sucrose, %)	Process	Actual yield		Theoretical maximum yield		% of Theoretical maximum yield
			kg %	kg %	kg %	kg %	
<i>A. niger</i>	Beet molasses (50%)	Surface	42	84	61.4	68.4	68.4
<i>A. niger</i>	Beet molasses (50%)	Submerged	41	82	61.4	66.8	66.8
<i>A. niger</i>	Beet molasses (54%)	Submerged	46	85	66.3	69.4	69.4
Yeast	n-Alkane	Submerged	165	82	247.0	66.8	66.8
Yeast	Methanol	Submerged	40	45	109.4	36.6	36.6
Yeast	Ethanol	Submerged	60	48.4	145.8	41.1	41.1

(Siebert and Schulz, 1979)

log = exponential phase
idliophase = stationary phase
trophophase

tations of up to 30 m³, before the molasses is used in production.

Trace elements: Extensive research on the role of trace elements in the citric acid fermentation was begun in the 1940's and has continued as analytical methods have improved. Copper, manganese, magnesium, iron, zinc, and molybdenum are necessary in the ppm range for optimal growth. However, if optimal concentrations are exceeded, there may be a toxic effect.

The role of iron is especially interesting. While optimal growth requires higher iron concentrations (among other things as a cofactor for aconitase), only 0.05–0.5 ppm is needed for maximal production of citric acid. Optimal iron concentration is dependent on the starting material used. For instance, with pure sucrose, 2.0 ppm of iron is optimal for growth, whereas when raw materials such as invert sugar or starch hydrolyzates are used, growth without acid production is obtained by adding 0.2 ppm iron, due to the fact that the iron content of the raw material solution is so high. The sensitivity of strains to heavy metals such as zinc, iron, manganese, and copper decreases with a decrease in the temperature. In addition, copper reverses the inhibitory effect of iron (Table 8.3).

Besides the composition of the nutrient solution (sugar content, metal-salt concentration, and phosphate and nitrogen contents) the pH of the medium influences the yield. During the idio phase, the pH must be below 3 in order to suppress oxalic and gluconic acid formation, but dur-

Table 8.3 Reversal by copper of the inhibition of citric acid formation by 10 ppm Fe³⁺

Copper (ppm)	Citric acid % Yield
0	—
5.0	77.8
10.0	77.4
20.0	75.0
30.0	75.6
50.0	74.0

(Noyes, 1969)

ing the trophophase, the pH is generally started at 5. In the first 48 hours during the trophophase, the pH falls to below 3.0 as a result of the metabolism of ammonium ions. For better control of yield in submerged cultures, the trophophase and idio phase can be separated by reducing the pH to under 2 once growth has ceased.

Another benefit of the low pH is the decreased risk of contamination, since at these low pH values only *Penicillium* and yeast cause contamination. Especially in surface fermentations, the frequency of contamination is a significant economic factor.

Production processes

Citric acid is produced by both surface and submerged processes. Surface processes can be further subdivided according to the state of the culture medium used: solid or liquid. Two types of submerged processes are used, stirred fermenters or airlift fermenters (Figure 8.2). The surface process employing solid culture medium is in use solely in small plants and a maximum of 500 tons per year are produced, but the other three processes are widely used. Several factors affect the choice of production type: availability of investment capital, energy availability, cost of labor, and training, and availability of techniques for measurement and regulation.

Production of inoculum The material used as inoculum for citric acid production is a spore suspension. Spores are produced in glass bottles on solid substrates at 25°C with incubation times of 10–14 days. Besides the total numbers, the viability of the spore crops is critical. When a submerged fermenter is to be inoculated, the spores are induced to germinate in a preliminary fermentation. A nutrient solution containing 15% sugar (from molasses) is used in this seed fermenter, and to induce the formation of mycelium in the form of pellets, cyanide ions are added. If too little cyanide is added, growth proceeds well but the citric acid yields in the later production-scale fermenter are low. The spores germinate at

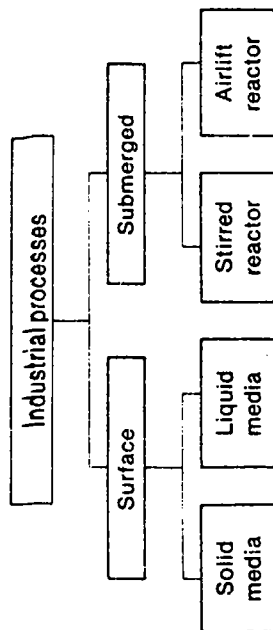


Figure 8.2 Various processes for citric acid manufacture

32°C and form pellets 0.2–0.5 mm in diameter within 24 hours. During this period, the pH falls to 4.3. These pellets are then used as the inoculum for production fermenters. Conversion rate and efficiency in the production fermenter are extremely dependent on the manner in which the spores and pellets are produced.

Surface processes

Surface processes employing solid substrates may use either wheat bran or pulp from sweet potato starch production as a culture medium. In this process, the *A. niger* strains are not as sensitive to trace elements as in the other processes. The pH of the bran is reduced to 4–5 before sterilization; after sterilization the material is inoculated with spores, spread on trays in layers 3–5 cm thick and incubated at 28°C. Growth can be accelerated by adding α -amylases, although the fungus can hydrolyze starch with its own amylase. The solid surface process takes 90 hours, at the end of which the entire solution is extracted with hot water to isolate the citric acid.

Surface processes using liquid nutrient solutions are the oldest production methods and account for 20% of the world's supply of citric acid. These processes are still in use because of the low investment (for the bioreactor), the low energy cost for the cooling system, and the simple technology. The labor costs, however, are substantially greater than for the deep vat processes because of the manpower needed to clean the pipes, trays, and walls of the system.

Table 8.4 shows the composition of a typical nutrient solution; in surface processes, the sucrose is supplied as beet molasses instead of sugar cane molasses. In modern systems, continuous sterilization of the medium is used.

Figure 8.3 shows the layout of a typical fermentation system. The sterilized nutrient solution automatically flows over a distribution system onto the trays. Inoculation is done in the incubation chamber at 30–40°C by blowing dry spores ($2\text{--}5 \times 10^7$ spores/m²) or by spraying spore suspensions.

The temperature is kept constant at 30°C during the fermentation by means of an air current. Ventilation is also important for gas exchange because the rate of citric acid production falls if CO₂ in the atmosphere increases to > 10%. Within 24 hours after inoculation, the germinating spores form a thin cover of mycelium on the

Table 8.4 Composition of nutrient solution for a surface culture

Substrate	g/l Substrate
Sucrose	160–200, ca. 320–400 g molasses
NH ₄ NO ₃	1.6–3.2
	Not needed with molasses
CaH ₂ PO ₄	0.3–1.0
MgSO ₄ ·7 H ₂ O	0.2–0.5
	Not needed with molasses
ZnSO ₄	0.01–0.10
Calcium hexacyanoferrate	0.4–2.0

(Schultz and Rauch, 1975)

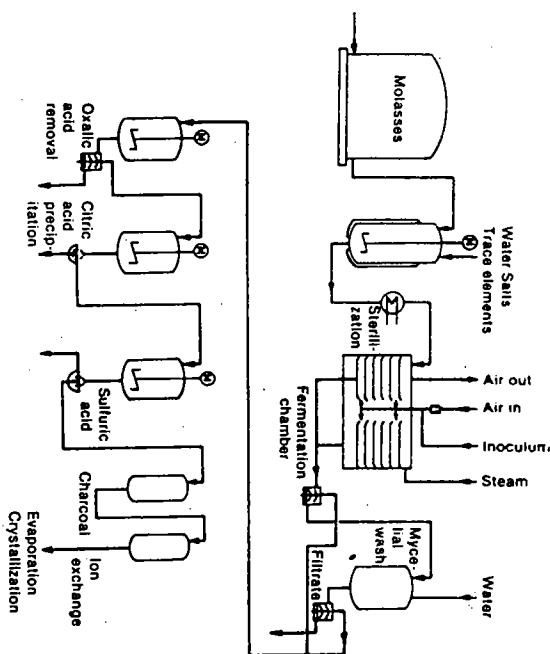


Figure 8.3 Block diagram of a citric acid process using the surface method

surface of the nutrient solution. As a result of the uptake of ammonium ions, the pH in the culture liquids falls to 1.5–2.0. Figure 8.4 shows the course of the fermentation during the first 60 hours. After 30 hours, the idiophase begins. If too much iron is present, oxalic acid is produced and a yellowish pigment is formed which later hinders the recovery process. The fully developed mycelium floats as a thick convoluted white layer on the nutrient solution. Through evaporation, the temperature can be maintained constant, but the culture then loses 30–40% of its original volume. The fermentation is stopped after 8–14 days.

Yields from the surface process using liquid nutrient solution amount to 1.2 to 1.5 kg citric acid monohydrate/m² of fermentation surface per hour. The heat production is 12,500 kJ/kg of citric acid produced.

During recovery, the mycelium and nutrient solution are removed from the chambers. Due to its volume, the mycelium must be carefully washed in sections. In some cases, mechanical

Submerged processes

Eighty percent of the world's supply of citric acid is produced by submerged processes. The submerged process, although taking a longer time (8 days), has several advantages: lower investment for construction by a factor of 2.5, 25% lower total investment, and lower labor costs. The disadvantages are the greater energy costs and the more sophisticated control technology which requires more highly trained personnel. Three factors are especially important for production in submerged processes: a) quality of the material used to construct the fermenter, b) mycelium structure, c) oxygen supply.

Fermenters for citric acid production must be either protected from acids or constructed of stainless steel. At pH values between 1–2, the heavy metals leached from normal steel fermenter walls can inhibit the formation of citric acid.

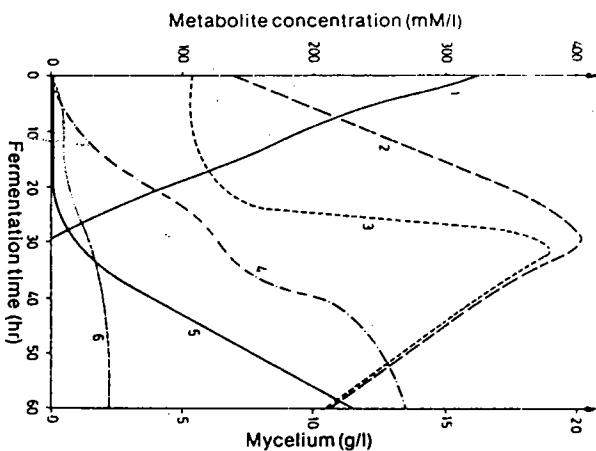


Figure 8.4 Kinetics of the citric acid fermentation (Siebert and Schulz, 1979)
1, Sucrose; 2, Glucose; 3, Fructose; 4, Mycelium; 5, Citric acid; 6, Gluconic acid

With small fermenters (up to 1000 l), even stainless steel chambers must have plastic liners because of the large surface/volume ratio. In large fermenters, the lining is not necessary if stainless steel is used.

The structure of the mycelium that forms in the submerged culture during the trophophase is vital to a successful production process. If the mycelium is loose and filamentous, with limited branches and no chlamyospores, little citric acid is produced in the idiophase. Mycelium for optimal production rates consists of very small solid pellets. The ratio of iron to copper in the medium determines the mycelium structure. Since *Aspergillus* is genetically unstable, only a minimum number of preliminary culture steps should be performed. In some cases, production fermenters are inoculated directly with spores.

Although *Aspergillus niger* requires relatively little oxygen, it is sensitive to oxygen deficiency. Throughout the entire fermentation period, there must be a minimum oxygen concentration of 20–25% of the saturation value. Short interruptions in the oxygen supply cause production to cease irreversibly. The aeration rates in deep vat fermenters should be 0.2–1 volume/volume/minute (vvm) during the acid production phase. Due to the low viscosity, stirring is not necessary. Thus, although some plants use stirred fermenters, air-lift bioreactors can also be used.

Most fermenters for citric acid production are constructed in the range of 120–220 m³. Submerged systems are somewhat less sensitive than surface systems to variability in the medium composition and in molasses composition (which is notoriously variable), so that they can be operated much more reproducibly than surface systems.

Foaming is a problem in submerged fermentations. A foam chamber 1/3 the size of the fermenter volume is needed in both air-lift and stirred bioreactors. Antifoam agents, such as land oils, can be added at frequent intervals, and mechanical anti-foam devices can also be used.

Continuous fermentation for citric acid production has not yet been accomplished. The complexity of pellet formation and the control requirements of the idiophase could only be satisfied by a multistage system with fermenters connected in a series; for economic reasons, however, such a system would not be sensible, even if the microbiological and sterility problems could be solved.

Citric acid production with alkane-utilizing microorganisms Several processes for the production of citric acid from C₂–C₄ hydrocarbons have been described, using both yeast and bacteria. The best producer is *Candida lipolytica*, which can be employed in batch, semicontinuous, or continuous fermentations. Compared to carbohydrate fermentations, the specific yields are measured as higher with hydrocarbons (145% g citric acid/g paraffin), but there are difficulties

the low solubility of the alkanes and the small amount of isocitric acid produced, it must later be separated from the citric acid (8.5). Studies by Ermakova et al. (1986) and Inogenova et al. (1986) have focussed on regulation of anaplerotic pathways and the action of citric acid versus isocitric acid in *lipolytica* when different substrates are used. In some cases, the proportion of isocitric acid can be up to 50% of the citric acid content. Figure 8.5 shows citric acid production from hydrocarbons by both batch and semicontinuous fermentations.

Product Recovery

Citric acid should be formed as a side product of suboptimal fermentation control; it is produced as calcium oxalate at a low pH, leaving the citric acid in solution as the monocalcium citrate. Rotating filters or centrifuges are used to separate the mycelium and the precipitated calcium oxalate from the liquid. At pH 0.2 and 70–90°C, citric acid is precipitated, can in turn be separated by means of rotating filters, and dried. For some uses the citric acid is further purified by adding sulfuric acid to the citric acid, with a calcium sulfate precipitate forming. The subsequent recovery includes treatment with activated carbon, ion exchange with cation and anion exchangers, and crystallization as citric acid or citric acid monohydrate. Above 40°C citric acid crystallizes as the monohydrate, and below 36.5°C as the monohydrate.

The required purity of the product depends on the intended use. Citric acid for addition to

Table 8.5 Citric acid and isocitric acid production in batch and semicontinuous fermentations

	Citric acid g/l	Isocitric acid g/l	Isocitric acid %, of total
1. Batch	125.5	11.4	8.3
2. Semicontinuous	89.4	7.0	7.3
3. Continuous	92.0	49.0	34.8

(Ermakova et al., 1978)

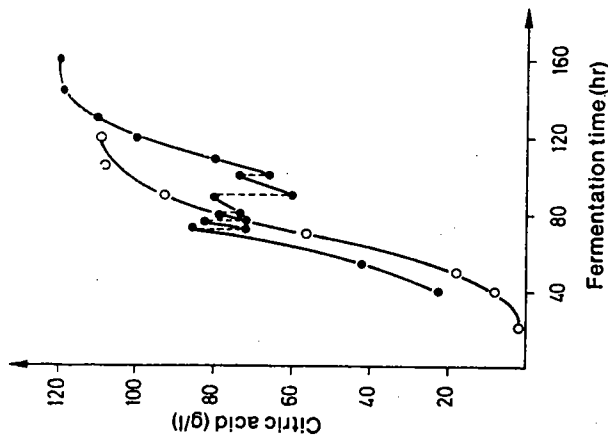
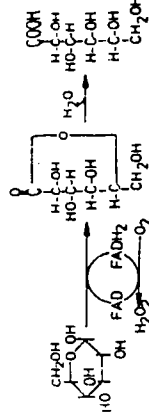


Figure 8.5 Citric acid production with *Candida lipolytica* ATCC 8661 in batch (O—O) and semicontinuous (●—●) fermentation (Gledhill et al., 1973)

foods must of course be purer than that needed for industrial purposes.

8.3 GLUCONIC ACID, GLUCONOLACTONE, AND GLUCOSE OXIDASE

Gluconic acid is used in the manufacture of metal, leather, and food; sodium gluconate is used as a sequestering agent in many detergents; δ -gluconolactone functions as a baking powder additive; and calcium gluconate is used in medicine. The production of gluconic acid from glucose by means of glucose oxidase is a simple reaction which can be carried out by many microorganisms (Figure 8.6). The gluconolactone formed in the first step hydrolyzes either spontaneously or enzymatically to gluconic acid. Dur-



D-Glucose δ -D-Gluconolactone Gluconic acid
Figure 8.6 Production of gluconic acid from glucose

ing the transfer of hydrogen from FADH₂ to oxygen, hydrogen peroxide (H₂O₂) is produced and is immediately split into water by the enzyme catalase. Because the activity of the enzyme leads to the formation of hydrogen peroxide, which has antimicrobial activity, glucose oxidase has also been identified as an antibiotic in fermentation broths and has thus been known under the names notatin and penicillin B.

Gluconic acid has had a long history in industrial microbiology. In 1911 Alsberg described the production of gluconic acid with *Pseudomonas*. The first process with a fungus, a surface process using *Penicillium luteum-purpureum*, was begun in 1928. In this process the yields amounted to 80–87% of theoretical.

Today only submerged processes are used, with either the fungus *Aspergillus niger* or the bacterium *Acetobacter suboxydans*. In these processes, gluconic acid, sodium and calcium gluconate, and glucose oxidase are manufactured. Other organisms which have been optimized to produce gluconic acid but which have not been used commercially include the fungi *Penicillium*, *Scopulariopsis*, *Gonobotrys*, *Endomycopsis*, and *Pullularia* and the bacteria *Vibrio* and *Pseudomonas*.

A process for immobilizing cells or glucose oxidase has been described which leads to yields of 93% when pure oxygen is used. There is considerable competition between the microbiological process and chemical methods, which also produce high yields.

Table 8.6 Relationship between gluconic acid production and pressure

Pressure (bar)	Yield by weight
1	42.5
3	80.4
4	82.4
5	81.3
6	86.1

(May et al., 1934)

Production

The fermentation, carried out at pH 4.5–6.5, requires a growth medium in which both phosphorus and nitrogen are limiting. If calcium gluconate is to be produced, no more than 13–15% glucose can be added as substrate because of the low solubility of calcium gluconate (4 g/l at 30°C); higher substrate concentrations would allow higher calcium gluconate levels, which would spontaneously crystallize, making purification difficult. In the production of sodium gluconate (solubility 396 g/l), a glucose concentration as high as 28–30% can be used.

The fermentation runs 20 hours at 28–30°C with a high aeration rate (1–1.5 vvm). By raising the pressure in the system, the oxygen solubility and therefore the gluconic acid yields can be increased (Table 8.6). Commercial yields today are 90–95%.

8.4 ACETIC ACID

The production of acetic acid from alcoholic liquids has been known as long as the production of wine (about 10,000 years). The Romans and Greeks, who used diluted vinegar as a refreshing drink, produced vinegar by leaving wine open to air. Vinegar was produced only for local consumption until the Middle Ages. The first industrially manufactured vinegars were produced in flat open vats. These were slow processes, in which a film of bacteria floated on the surface of the wine. In the nineteenth century, surface fermentations were developed into more rapid processes. One of these, the trickling generator pro-

CSS is still used today. Beginning in 1949 submerged processes were developed. Both types of process, the trickling generator and the submerged fermenter, are used worldwide; the older methods still being used because of the better flavor of the product. The submerged process has been extensively improved by the Kring Company in Germany. As an example of the scale of the process, in 1980 production of vinegar with 10% acetic acid amounted to 356 X 10⁶ liters in the European Economic Community, 469 X 10⁶ liters in the United States and 1600 X 10⁶ liters worldwide (excluding China and the Soviet Union).

Although acetic acid is produced by many fermentative bacteria, only members of a special group, the acetic acid bacteria, are used in commercial production. The acetic acid bacteria can be divided into two genera, *Gluconobacter* and *Acetobacter*, the first group oxidizing ethanol solely to acetic acid and the second group (the overoxidizers) able to oxidize ethanol first to acetic acid and then further to CO₂ and H₂O. Members of the genus *Acetobacter* are Gram-negative and acid tolerant. Strains used commercially belong to the species *Acetobacter aceti*, *A. pasteurianus* or *A. peroxidans*. Members of the genus *Gluconobacter* are not overoxidizers. The species *Gluconobacter oxydans* (formerly *Acetomonas oxydans*) and several subspecies of this species are used commercially. Mixed cultures appear during production, even when the inoculum is assumed to be pure, particularly in surface processes. Production strains seem to lose their high-yielding character if they are transported to a new location on agar. To avoid this problem, production strains are shipped to new vinegar plants in transportable small fermenters.

Biosynthesis

Acetic acid production is an incomplete oxidation rather than a true fermentation, because the reducing power which is produced is transferred to oxygen. The first oxidation step from ethanol leads to acetaldehyde with a NAD- or NADP-

specific alcohol dehydrogenase. Then there is a hydration to acetaldehyde hydrate and a second oxidation with acetaldehyde dehydrogenase to acetic acid (Figure 8.7). During the oxidation, 1 mole of acetic acid is produced from 1 mole of ethanol. From 1 liter of 12% (v/v) alcohol, 1 liter of 12.4% (w/v) acetic acid is produced.

For optimal production, sufficient oxygen is required, which is reduced by means of the respiratory chain. Six ATP are produced per mole of acetic acid produced. If sufficient oxygen is not available, at high acetic acid and ethanol concentrations the cells die. At a concentration of 5% acetic acid plus ethanol, 34% of the bacteria die after a 2-minute interruption of aeration, whereas at a total concentration of 12% acetic acid plus ethanol, the same killing occurs after only 10–20 seconds.

Both acetic acid and ethanol must be present for optimal growth of *Acetobacter*. Ethanol supply is critical and with less than 0.2% (v/v) in solution the death rate increases. However, the maximal ethanol content should not exceed 5% in conventional processes. Today high-yielding strains produce 13–14% acetic acid.

Production of vinegar

Starting materials with low alcohol content, such as wine, whey, malt, or cider, do not require any additional components to constitute a complete nutrient solution. However, if potato or grain spirits or technical alcohol is used, nutrients must be added in many cases to obtain optimal growth and acetic acid production (Table 8.7). In submerged fermentation, concentrations of nutrients must be five times higher due to the increased biomass which is taken out of the reactor with the vinegar. The inoculum can be either pure cultures or vinegar from previous batches.

Surface processes Despite the success of the submerged process, the trickling generator is still widely used in vinegar production (Figure 8.8) especially for making household vinegar. The wooden bioreactor has a total volume of up to

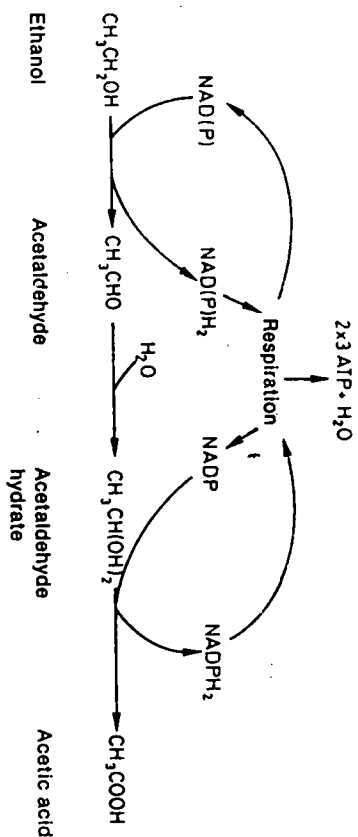


Figure 8.7 Oxidation of ethanol to acetic acid

60 m³ and is filled with beechwood shavings. The starting material is sprayed over the surface and trickles through the shavings containing bacteria into a basin in the bottom, where the partially converted solution is cooled and pumped back up to the top.

Of the alcohol added, 88–90% is converted to acetic acid in the trickling generator process. The rest of the alcohol is used for primary metabolism or it escapes with the waste air. The temperature in the upper part of the system is about 29°C and is about 35°C in the lower part. The time needed to produce 12% acetic acid in this process is about three days.

Submerged processes Fruit wines and special mashes with low total concentrations of alcohol were first used in submerged processes. With

Table 8.7 Additives for acetic acid fermentation

	Gram/m ³ nutrient solution	
	Surface process	Submerged process
Sugar solution from grain hydrolysate	200	1000
Ammonium phosphate	96	480
Magnesium sulfate	24	120
Calcium citrate	24	120
Calcium pantothenate	0.24	1.2

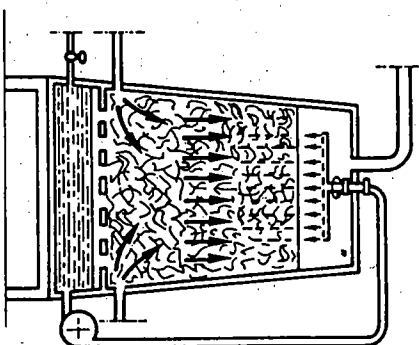
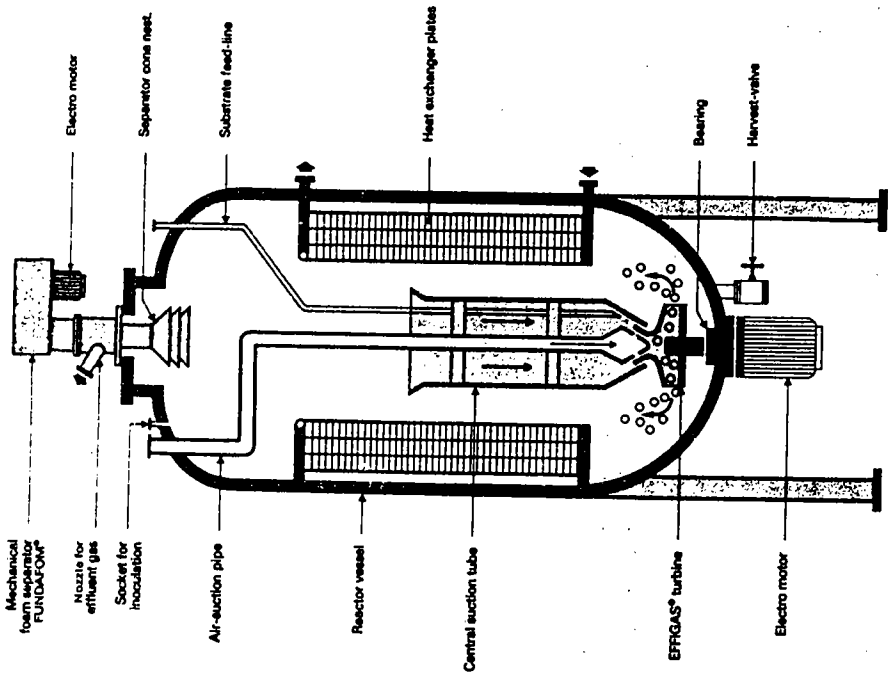


Figure 8.8 Diagram of a trickling generator for vinegar production



submerged fermenter for acetic acid production (Chemap)

pipe from the top of the fermenter. ingers control the temperature and foam eliminators must usually be in- sehold vinegar (13% acetic acid) is a semicontinuous, fully automatic der continuous stirring and aeration, with a starting material that contains ic acid/100 ml and 5% ethanol. The centration is measured continuously

during the process and when the concentration sinks to 0.05–0.3% (about 35 hours), 50–60% of the solution is removed and replaced with a new mash containing 0.2 g acetic acid/100 ml and 10–15% ethanol. Fully continuous processes with yields of 98% at 40°C have also been described.

With the submerged process, the production rate per m³ is 10 times higher than with the sur-

face fermentation and about 5% higher than with the trickling generator process. Other advantages are: lower capital investment per production amount, only 20% of the plant area required for the installation, capability of conversion to other mashes in a short time, and low personnel cost due to fully automatic control.

Recovery The acetic acid obtained in the submerged process is turbid due to the presence of bacteria and the product must be clarified by filtration. Plate filters and filter aids are generally used. Once the filtrate is obtained, K₂Fe(CN)₆ is used to decolorize the final product when desirable.

8.5 LACTIC ACID

The first microbial production of an organic acid was the production of lactic acid, carried out in 1880. Today chemical methods for the production of lactic acid are very competitive at the same cost as biological processes. Two kinds of lactic acid bacteria are recognized, heterofermentative and homofermentative. The heterofermentative organisms produce a great number of byproducts and are not suitable for commercial purposes.

With homofermentative organisms, as little substrate as possible is converted into cell material plus byproducts and as much as possible is metabolized into lactic acid. The biosynthesis of lactic acid from glucose proceeds via glyceraldehyde-3-P, 1,3-di-P-glycerate and pyruvate. The reducing power produced during the oxidation of glyceraldehyde phosphate is transferred with

an NAD-dependent lactate dehydrogenase to pyruvate, which is reduced stereospecifically to L(+) or (D–) lactic acid (Figure 8.10).

Theoretically, two moles of lactate are produced from 1 mole of glucose. Over 90% of this theoretical yield is actually attained in practice.

The organisms used are *Lactobacillus delbrueckii* and *L. leichmannii* when glucose is used as substrate, *Lactobacillus bulgaricus* with whey, and *Lactobacillus pentosus* with sulfate waste liquor. These organisms are facultative rather than obligate anaerobes, and therefore the bioreactors need not be run with complete oxygen exclusion.

In addition to 12–13% glucose and (NH₄)₂HPO₄ (0.25%), the fermentation medium must contain B vitamins. Production is carried out in 25–120 m³ fermenters at 45–50°C with an excess of CaCO₃ added to maintain the pH between 5.5 and 6.5. The fermentation time is about 72 hours. Since free lactic acid is toxic to the organism, procedures for removing the product continuously have been studied. In one approach, electrodialysis has been used. In another procedure, continuous culture was done in a membrane reactor, leading to the production of 80 g/l-h.

Only L+ lactic acid is produced commercially. When the fermentation has completed, the broth is heated to dissolve the calcium lactate. The heated broth is filtered and the calcium precipitated by addition of sulfuric acid. After concentration of the lactic acid, it is further purified. Total microbial production per year amounts to about 20,000 metric tons.

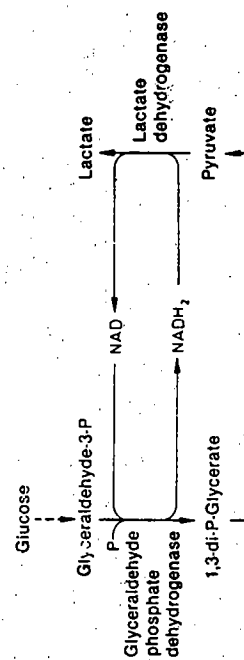


Figure 8.10 Lactic acid production from glucose

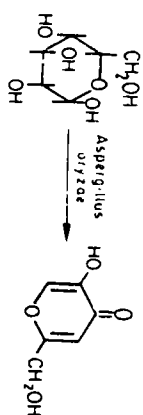


Figure 8.11 Biosynthesis of kojic acid

8.6 KOJIC ACID

Kojic acid (5-hydroxy-2-hydroxymethyl-4-pyrone) has found minimal commercial use as a feedstock in the plastics industry. For a long time, it has been produced by the direct fermentation of glucose. Yields of 70–90% can be obtained from glucose with *Aspergillus flavus* and *A. oryzae* (Figure 8.11). This acid can also be produced from pyruvate, glycerol, acetate, or ethanol. The production using glucose is very similar to that of the citric acid and itaconic acid processes. By changing the fermentation parameters, such as stirring or aeration rate, some strains can be converted from citric acid or itaconic acid production to kojic acid production.

8.7 ITACONIC ACID

In 1931, itaconic acid was first shown to be a metabolic product of *Aspergillus itaconicus*. In the same decade, it was discovered that some strains of *A. terreus* also excrete itaconic acid. Mutants of both strains are used today for commercial production.

At present there are only a few commercial systems which produce itaconic acid: 2 in Western Europe, one each in America, USSR and Japan. The main use of itaconic acid is in the plastic industry. Itaconic acid forms copolymers with its esters and other monomers; these are used in the paper industry for wall paper and other paper products and in the production of adhesives. An

itaconic acid acrylonitrile copolymer is more readily dyed than some other polymers.

Itaconic acid is produced by way of the tricarboxylic cycle from cis-aconitic acid via decarboxylation. The enzyme cis-aconitic acid decarboxylase has been purified and characterized (Figure 8.12). Another biosynthetic pathway from pyruvate through citramalic acid, citraconic acid, and itatartaric acid also results in itaconic acid.

The undesirable by-products succinic acid and itatartaric acid are produced from itaconic acid (Figure 8.13). Calcium inhibits the enzyme itaconic acid oxidase and hence calcium additions increase the yield of the desired product. By the

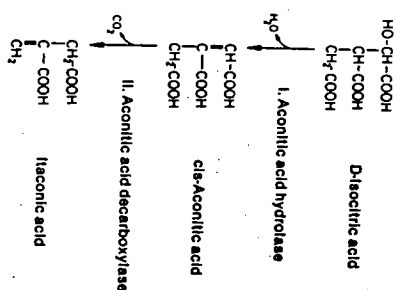


Figure 8.12 Biosynthesis of itaconic acid out of the tricarboxylic acid cycle

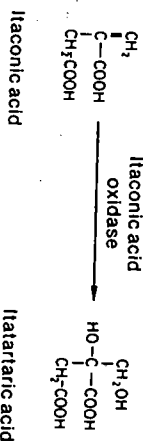


Figure 8.13 Destruction of itaconic acid by conversion to itatartaric acid

use of immobilized cells, production has been increased to about 0.73 g/l/h.

A. terreus is used only in batch submerged fermentation at present. With a 15% sucrose solution, conversion into itaconic acid can be achieved at 78% of the theoretical yield.

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Amino acids

9.1 INTRODUCTION

The taste-enhancing properties of glutamic acid were discovered in 1908 in Japan, and commercial production of sodium glutamate from acid hydrolysates of wheat and soy protein began soon after. In 1957, L-glutamic acid was discovered as a product in the spent medium of *Corynebacterium glutamicum*, and this organism subsequently became the major source of sodium glutamate. This development was an enormous boost to the fermentation industry in Japan, and fermentative processes for the production of amino acids as well as nucleotides (see Chapter 10) have been almost exclusively Japanese developments.

Thirteen of the 17 major manufacturers are situated in Southeast Asia (Japan, Korea, Taiwan), the foremost Japanese firms being Ajinomoto Co., Kyowa Hakko Kogyo Co. and Tanabe Seiyaku Co. The most important western manufacturers of sodium glutamate are Stauffer Chemical Co. (USA) and Orsan S. A. (France).

The most well-known manufacturer of L-lysine is Eurolysine Co. (France) and of L-aspartic acid the two French firms Orsan S. A. and Rhône-Poulenc S. A.

In 1985 the world production of biochemically produced amino acids was 460,000 tons, of which 370,000 tons per year were sodium glutamate and 70,000 were L-lysine. It is anticipated that increased demand will result in yearly increases of around 10%.

Further increases in the production of amino acids are expected because of the following developments:

- The isolation of improved production cultures through the addition of recombinant DNA and protoplast fusion techniques to the currently used mutation methods to isolate derepressed strains.
- Combining chemical synthesis with microbial and enzymatic processes.
- Adaptation of existing processes to the use of less expensive starting materials.

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9.3 METHODS OF PRODUCTION / 151

- Optimization of scale-up conditions.

9.2 COMMERCIAL USES OF AMINO ACIDS

Amino acids have extensive industrial applications. About 66% of the amino acids produced are used in the food industry, 31% as feed additives, in 4% in medicine and cosmetics and as starting materials in the chemical industry. In the food industry amino acids are used alone or in combination to enhance flavors. The flavor-enhancing effect of sodium glutamate has been mentioned; sodium aspartate and D,L-alanine are added to fruit juices to round off the taste, and glycine is added to foods containing sweeteners. L-Cysteine improves the quality of bread during the baking process and acts as an antioxidant in fruit juices. L-Tryptophan combined with L-histidine also acts as an antioxidant and is used to keep powdered milk from getting rancid. Aspartame® (L-aspartyl-L-phenylalanine methyl ester), which is made from L-phenylalanine and L-aspartic acid, is used as a low-calorie sweetener in soft drinks.

Plant proteins are often deficient in essential amino acids such as L-lysine, L-methionine, L-threonine, or L-tryptophan. The lysine contents of some foods and feeds are given in Table 9.1. Lysine is added to bread in Japan, and in some countries soy products are enhanced by the addition of methionine. L-Lysine and DL-methionine are used to upgrade animal feeds, and the addition to foods of L-threonine and L-tryptophan is being developed. In view of the scarcity of nutritious food in the Third World, there will be an ever-increasing need to supplement plant

Table 9.1 L-lysine content of several foods and feeds

Food/feed	Lysine content (%)	Food/feed	Lysine content (%)
Corn	0.21	Soy meal	2.90
Oats	0.50	Dry yeast	3.40
Barley	0.40	Skim milk powder	2.50
Wheat	0.60	Meat-meal	2.60

proteins with those essential amino acids which occur at suboptimal levels. Aspartame® (L- α -aspartyl-L-phenylalanine methyl ester) is a major product in the soft-drink industry, serving as an artificial sweetener in so-called "diet" soda. The sweetness level of aspartame in aqueous solution is 200-times higher than a 3-4% sucrose solution. As the starting material for the industrial production of aspartame, L-phenylalanine and L-aspartic acid are used, both of which are products of industrial microbiology. The significance of aspartame can be seen from the fact that during the period from 1981 to 1985 L-phenylalanine production rose from 50 to over 3000 metric tons, and similar rates of increase occurred for aspartic acid. By means of genetic engineering it has been possible to obtain an organism capable of producing an alternating copolymer of phenylalanine and aspartic acid, which can be converted into aspartame dipeptide by enzymatic splitting. The latter is esterified to produce aspartame.

Many amino acids are used in medicine, particularly as ingredients of infusion solutions in post-operative treatment.

In the chemical industry, amino acids are used as starting materials for the manufacture of polymers, such as polyalanine fibers and lysine isocyanate resins. Poly- γ -methylglutamate is used as a surface layer in the manufacture of synthetic leather, and the N-acyl derivatives of some amino acids are used in the manufacture of cosmetics and as surface-active substances. Urocanic acid, used as a suntanning agent, is produced by the biotransformation of histidine. Another amino acid, glycine, is used as a starting material for the production of the herbicide glyphosate, and threonine serves a similar purpose for aztreonam (see Section 13.2).

9.3 METHODS OF PRODUCTION

Amino acids that are currently in commercial production are summarized in Table 9.2. Fermentation processes have been developed for all amino acids except glycine, L-cysteine and L-cystine, but not all are in commercial use. In the

Table 9.2 World production of amino acids: production processes and applications

Amino acid	Annual production (metric tons)	Production methods	Application
L-Alanine	130	1, 3c	Flavor enhancer
D,L-Alanine	700	2	Flavor enhancer (beverage production)
L-Arginine	1000	1, 3a	Infusions Therapy (liver diseases) Cosmetics
L-Aspartic acid	4000	1, 3c	Therapy Flavor enhancer Aspartame® production
L-Asparagine	50	1, 2	Therapy
L-Cysteine	700	1	Bread production Antioxidant Therapy (bronchitis)
L-Glutamic acid	370,000	3a	Flavor enhancer
L-Glutamine	500	3a	Therapy (ulcer)
Glycine	5000-6000	2	Component of sweeteners
L-Histidine	200	3a, 1	Therapy (ulcer)
L-Isoleucine	150	3a	Infusions
L-Leucine	150	1, 3a	Infusions
L-Lysine	70,000	3a, 3c	Feed additive Infusion solution
DL-Methionine	70,000	2	Feed additive
L-Methionine	150	3c	Therapy
L-Ornithine	50	3a, 3c	Liver therapy
L-Phenylalanine	3000	3a, 3c	Infusions Therapy Aspartame® production
L-Proline	100	3a	Infusions
L-Serine	50	3a, 3b	Cosmetics
L-Threonine	160	3a	Feed additive
L-Tryptophan	200	3a, 3c	Infusions
L-Tyrosine	100	1, 3c	Infusions Precursor for L-DOPA synthesis
L-Valine	150	3a, 3c	Infusions

Production methods: 1. Extraction of protein hydrolysates; 2. Chemical synthesis; 3a. Direct fermentation; 3b. Microbial transformation of precursors; 3c. Use of enzymes or immobilized cells (Soda et al., 1983; Kinoshita, 1987)

commercial manufacture of amino acids there are three rival processes:

1. The extraction of amino acids from protein hydrolysates. This method is used to obtain L-cysteine, L-cystine, L-leucine, L-asparagine and L-tyrosine.
2. Chemical synthesis. The production of glycine, D,L-alanine, D,L-methionine and D,L-tryptophan always involves chemosynthesis. Chemical synthesis is cheaper than microbial production, but the chemical product is the optically inactive mixture of the D- and L-isomers.
3. Microbiological production, discussed below.

There are three approaches to microbiological production. The first, outlined in Table 9.3, is by direct fermentation of amino acids using different carbon sources, such as glucose, fructose, molasses, starch hydrolysate, n-alkanes, ethanol, glycerol, acetate, propionate, etc. Fermentations using methanol as a starting material have been studied extensively because of the low cost of this substrate, but have not yet been used commercially due to the low yields.

The second approach is by converting inexpensive intermediate products via biosynthesis. For example glycine, which is inexpensive, can be converted to L-serine (see Table 9.4). The third approach is by the use of enzymes or immobilized cells, sometimes in continuous processes involving enzyme-membrane reactors (see Section 5.3). The most common reactions of this type are:

- Use of a stereospecific amino acylase from *Aspergillus oryzae* to selectively split a chemically synthesized D,L-amino acid that had been chemically acetylated. In this way, the L form of the amino acid is selectively liberated (see Figure 9.1). A German company, Degussa AG, has been using an enzyme-membrane reaction technique of this type for many years to produce L-alanine, L-methionine, L-phenylalanine, L-tryptophan, and

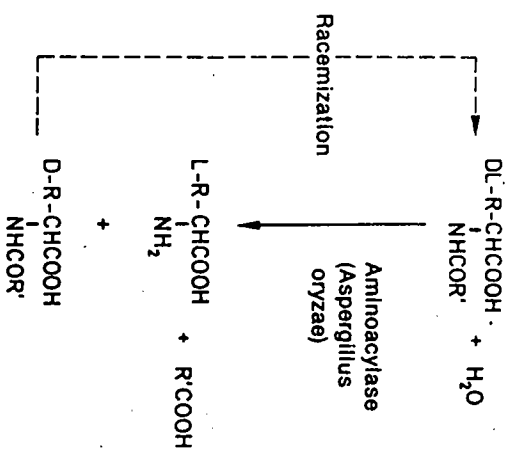


Figure 9.1 Use of an enzyme for the separation of a racemic mixture in the manufacture of L-amino acids (R = side chain of an amino acid; R' = e.g., -CH₃)

L-valine (monthly production of 5-20 metric tons).

- L-α-amino acids can be produced by reductive amination of α-ketoacids using amino acid dehydrogenases. The most commonly used dehydrogenase is an enzyme from *Bacillus* strains such as *B. megaterium*, which requires NADH as coenzyme. L-alanine, L-leucine, and L-phenylalanine are produced in this way either with immobilized cells or in enzyme-membrane reactors in combination with a NADH regeneration system.
- D,L-5-monosubstituted hydantoins, which are easily produced chemically, can be converted by means of stereospecific enzymatic cleavage to optically active D- or L-amino acids. The hydantoin cleavage by either D- or L-hydantoinase leads to the production of D- or L-N-carbamoyl-amino acids. The carbamoyl residue is chemically or enzymatically removed with the corresponding car-

9.3 Production of amino acids by fermentation

no acid	Strain used	Genetic characteristics	Yield (g/l)	Carbon source
Alanine	<i>Microbacterium ammoniophilum</i>	ArgHx ⁺	60	Glucose
alanine	<i>Pseudomonas</i> No. 483	Wild type	17.5	Glucose
Arginine	<i>Serratia marcescens</i> AT 428 (<i>aru argR2 argA2</i>)	Transduction Canavanine ⁺	50	Glucose
aspartic acid	<i>Escherichia coli</i> K1-1023 ⁺		56	Fumaric acid
lactic acid	<i>Corynebacterium glutamicum</i>	Wild type	100	Glucose
	<i>Brevibacterium flavum</i>		98	Acetate
	<i>Arthrobacter parviflavus</i>		82	n-Alkanes
lutamine	<i>Corynebacterium glutamicum</i>	Wild type of glutamic acid producer	58	Glucose, with high biotin and NH ₄ Cl content
histidine	<i>Serratia marcescens</i> L 120 (pSH 368)	rDNA	40	Sucrose
isoleucine	<i>B. flavum</i>	Ehr ⁺	30	Acetate
eucine	<i>Brevibacterium lactofermentum</i>	Ile ⁺ Met ⁺ TA ⁺	28	Glucose
lysine	<i>B. lactofermentum</i> A1 11204	AEC-Ala ⁺ CCL-ML-FPs	70	Glucose
	<i>B. flavum</i>	Hom ⁺ Thr ⁺	75	Acetate
Methionine	<i>C. glutamicum</i> KY 9276	Thr ⁺ Ehr ⁺ MetHx ⁺	2	Glucose
Ornithine	<i>C. glutamicum</i>	Arg ⁺	26	Glucose
Phenylalanine	<i>B. lactofermentum</i>	SMT-PFP-PFP ⁺ Dec ⁺ TyrMet ⁺	25	Glucose
Proline	<i>C. acetosidophilum</i>	Uncharacterized mutant	108	Glucose + Glutamic acid
Serine	<i>B. lactofermentum</i>	SC ⁺	4.5	Glucose
Threonine	<i>E. coli</i> VL 344 (pYN7)	rDNA	55	Sucrose
Tryptophan	<i>E. coli</i> JP 4114	rDNA	23.5	Glucose
Tyrosine	<i>C. glutamicum</i> Pr-20	Phe ⁺ PFP ⁺ PAP ⁺ PAT ⁺ TyrHx ⁺	18	Glucose
Valine	<i>Brevibacterium lactofermentum</i> No. 487	TA ⁺	31	Glucose

technique used for the production of L-aspartic acid between 1960 and 1973. Resistance: AEC 5-(β -Aminoethyl)-L-serine; ArgHx Arginine hydroxamate; CCL, α -Chloroacetyl; Dec, Decoyinine; Ehr Ethionine; FP, Fluoropyruvate; MetHx Methionine hydroxamate; ML, γ -Methyl-L-lysine; 5 MT 5-Methyltryptophan; PAP p-nitro-phenylalanine; PAT p-Aminotyrosine; PFP p-Fluorophenylalanine; SC Sulfaguanidine; TA 2-Thiazolalanine; rHx Tyrosine hydroxamate. Auxotrophs: Ala, alanine; arg, arginine; hom, homoserine; ile, isoleucine; met, methionine; phe, phenylalanine; thr, threonine; tyr, tyrosine. *aru*: block in arginine degradation; *argR*: biosynthetic enzyme is derepressed; *argA*: N-acetylglutamate synthesis is insensitive to feedback inhibition by arginine.

Table 9.4 Enzymatic production of amino acids

Amino acid	Enzyme	Enzyme-producing microorganism	Reaction	Yield (g/l)	Efficiency (%)
L-Alanine	Aspartate decarboxylase (RC)	<i>Pseudomonas dacunhae</i>	L-Aspartate $\rightarrow$ L-Alanine + CO ₂	600	100
L-Aspartic acid	Aspartase (IC)	<i>Escherichia coli</i>	Fumarate + NH ₄ ⁺ $\rightarrow$ L-Aspartic acid	560	99
L-Cysteine	Cysteine desulfhydrase (RC)	<i>Bacillus sphaericus</i>	β -Chloro-L-alanine + Na ₂ S $\rightarrow$ L-Cysteine + NaCl + NaOH	70	80-85
L-DOPA (L-3,4-Dihydroxyphenyl-alanine)	Tyrosinase (RC)	<i>Erwinia herbicola</i>	Pyrocatechol + Pyruvate + NH ₄ ⁺ $\rightarrow$ L-DOPA + H ₂ O	59	95
L-Leucine	L-Leucine dehydrogenase (EMR)	<i>Bacillus sphaericus</i>	α -Ketoglutaric acid + NH ₄ ⁺ + NADH $\rightarrow$ L-Leucine + H ₂ O + NAD ⁺	42	max. 99.7
L-Lysine	D-Amino-caprolactam racemase, L-Amino-caprolactam hydrolase (RC)	<i>Achromobacter obae</i> , <i>Cryptococcus laurentii</i>	DL-Aminocaprolactam + H ₂ O $\rightarrow$ L-Lysine	100	100
L-Phenylalanine	L-Phenylalanine dehydrogenase (EMR)	<i>Brevibacterium</i> sp.	Phenylpyruvate + NH ₄ ⁺ $\rightarrow$ L-Phenylalanine + H ₂ O	37	81-d-1
	Acetamidocinnamate amidohydrolase (RC)	<i>Bacillus sphaericus</i>	Acetamido cinnamic acid $\rightarrow$ L-Phenylalanine	8	94
	Phenylalanine ammonia lyase (RC)	<i>Rhodotorula glutinis</i>	trans-Cinnamic acid + NH ₄ ⁺ $\rightarrow$ L-Phenylalanine	18	70
	Aspartate phenylalanine transaminase (EMR)	<i>E. coli</i>	L-Aspartate + Phenylpyruvate $\rightarrow$ L-Phenylalanine + Oxalacetate	30	98
	D-Hydroxyisocaproate dehydrogenase	<i>Lactobacillus casei</i>	D,L-Phenyllactate $\rightarrow$ Phenylpyruvate		
	L-Hydroxyisocaproate dehydrogenase	<i>Lactobacillus confusus</i>	Phenylpyruvate + NH ₄ ⁺ $\rightarrow$ L-Phenylalanine + H ₂ O	28	81-d-1
L-Serine	L-Phenylalanine dehydrogenase (EMR)	<i>Rhodococcus</i> sp.	Glycine + HCHO $\rightarrow$ L-Serine	450	88
	Serine hydroxymethyl transferase (CCE)	<i>Klebsiella aerogenes</i>	Glycine + HCHO $\rightarrow$ L-Serine		
L-Tryptophan	Serine hydroxymethyl transferase	<i>Klebsiella aerogenes</i>	L-Serine + Indole $\rightarrow$ L-Tryptophan + H ₂ O	200	95 based on indole
	Tryptophanase (EMR)	<i>E. coli</i>	Glycine + HCHO $\rightarrow$ L-Serine		
L-Tyrosine	Serine hydroxymethyl transferase	<i>Klebsiella aerogenes</i>	L-Serine + Phenol $\rightarrow$ L-Tyrosine + H ₂ O	26	61 based on glycine

Aida et al. (1986); Groeger and Sahm, 1987; Schmidt et al. (1987)

EMR, enzyme/membrane reactor; IC, immobilized cells; RC, resting cells; CCE, crude cell extract.

bamolyase. L-hydantoins have been found in only a few bacteria, for example *Flavobacterium ammoniogenes*, whereas D-hydantoins are found in numerous bacteria, actinomycetes, yeasts, and some strains of *Aspergillus*. In addition to the use of this approach for the production of L amino acids, it can also be used for the manufacture of D-N-carbamoyl amino acids and D amino acids, which find a role as side-chain precursors for the production of semi-synthetic penicillins and cephalosporins.

In some manufacturing processes, several enzymatic reactions are coupled. For example, L-alanine is produced by sequential conversion of fumarate to L-aspartic acid with immobilized cells of *E. coli* and *Pseudomonas dactinifera*. This process, which has been in commercial operation by Tanabe Seiyaku in Japan since 1982, involves the connection of two reactors in series.

Enzymes or immobilized cells are used in the production of L-alanine, L-aspartic acid, L-dihydroxy phenylalanine (DOPA), L-lysine, L-methionine, L-phenylalanine, L-tryptophan, L-tyrosine, and L-valine. The most important approaches used are summarized in Table 9.4. The current interest in the production of L-phenylalanine as a precursor for aspartame production has led to the development of numerous competing biotransformation reactions, which have not as yet replaced the direct fermentation process.

9.4 STRAINS FOR AMINO ACID PRODUCTION

Although microorganisms which excrete glutamic acid are easily obtained from nature, it has been more difficult to obtain natural isolates which excrete other amino acids for large-scale production. Extensive screening has led to the isolation of only a few strains which excrete D,L-alanine and L-valine. Figure 9.2 shows the overall biosynthetic pathways for the other amino acids. Natural isolates rarely excrete these amino

acids, since cell metabolism regulates production. In strain development, several methods of eliminating the regulation control are available.

- The use of **auxotrophic mutants** which can no longer form the regulatory effector or co-repressor (usually the end product), and excrete the intermediate of the biosynthetic pathway which is just in front of the block.
- The use of **regulatory mutants**. For instance, if an excess of the end product of an unbranched biosynthetic pathway is to be produced, mutants with a feedback-insensitive key enzyme are selected either from antimetabolite-resistant strains or from a population of revertants from auxotrophs.
- Use of **genetic recombination** to combine auxotrophy or regulatory mutants from different strains into a single hybrid strain. In *Serratia marcescens* transduction with phage P520 has been used for creating suitable hybrids. Protoplast fusion has been used in *Corynebacterium glutamicum*, *Brevibacterium lactofermentum*, and *B. flavum*.
- Use of **recombinant DNA technology** to amplify the gene dosage of the rate-limiting enzyme. This procedure leads to yield increases if the amount of enzyme synthesis rises in parallel with the gene copy number. This approach has been used with *E. coli*, albeit without significant advantages. In this case, the cloning vector pBR322 was used to increase the gene dosage of genes for the production of the following amino acids: L-glutamic acid, L-histidine, L-lysine, L-phenylalanine, L-proline, L-threonine, and L-valine. The yields were in most cases significantly lower than the yields of commercial production strains of *coryneform* bacteria. However, cloning systems developed for these commercial strains have permitted more success for several Japanese companies. Kyowa Hako has used recombinant DNA technology to create hybrid strains of *Corynebacterium glutamicum* for the production of L-cysteine, L-histidine, L-isoleucine, L-phenylalanine, L-serine, and

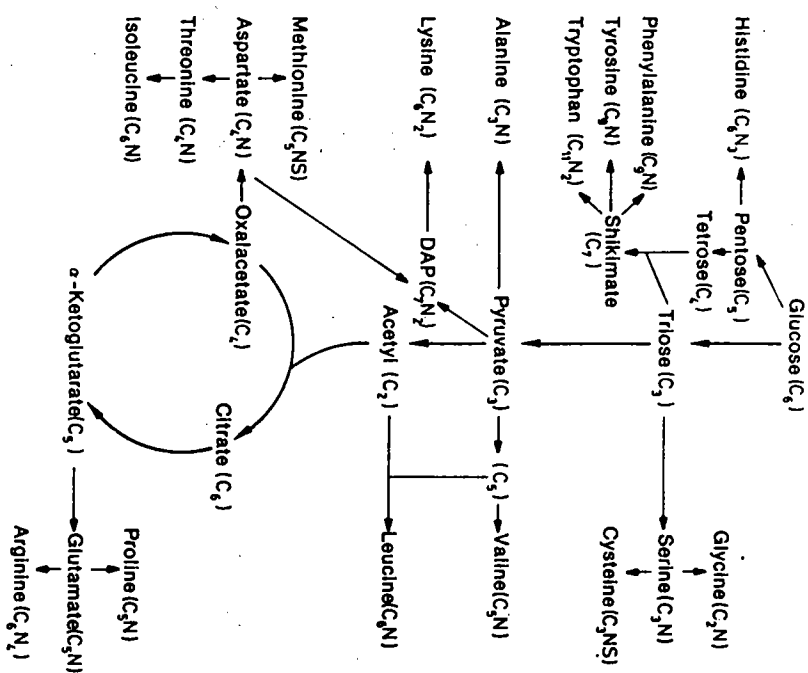


Figure 9.2 Biosynthesis of amino acids using glucose as the carbon source

L-threonine; Ajinomoto has modified *Brevibacterium lactofermentum* for the production of L-histidine, L-phenylalanine, L-proline, L-threonine, and L-tyrosine; Tanabe has altered *Serratia marcescens* for the production of L-histidine, L-proline, and L-threonine.

Mutants producing excess amounts of amino acids have also been isolated by combining several of the above methods.

Table 9.3 lists the strains available for the microbial production of amino acids, their ge-

netic characteristics, and the yields obtained. Processes using various carbon sources have been developed for different production strains. In this list, only the processes with the highest published yields have been given and the actual industrial yields are assumed to be significantly higher.

9.5 PROCESS CONTROL

Amino acid production runs for 2-4 days in batch processes in vessels containing up to 450 m³.

Methods using continuous processes have been developed, but have not yet been implemented in commercial plants.

Infections due to bacteriophages may cause considerable losses in production. Phage-resistant mutants and chemical agents which inhibit phage reproduction may be used to avoid phage infection. Intensive research on the maintenance of pure cultures (sterilization of equipment, media, and air) is underway; the sterilization of nutrient solutions usually takes place continuously.

Because of the high rate of sugar breakdown and the high respiratory activity during the fermentation, a high oxygen requirement is exhibited. The optimal aeration rates for individual processes depend on the strain, the substrate, and the biosynthetic pathway. Excess heat must be simultaneously dissipated, since the processes are carried out at temperatures between 28–38°C. The pH is kept constant between 6.8–8.0, depending on the process; gaseous NH_3 is frequently used for pH control and is simultaneously metabolized as a nitrogen source. During the fermentation of L-glutamic acid, such critical parameters as aeration rate, temperature, pH, and antifoam dosage are all automatically regulated and in a few cases are under computer control. Biosensors have been developed which permit continuous measurement of the amino acid concentration in the fermenter. For instance, a glutamic acid sensor has been prepared using the glutamate synthase enzyme obtained from a thermophilic organism, *Bacillus stearothermophilus*.

9.6 PRODUCT RECOVERY

Centrifugation is used at the end of the fermentation process to separate the cell material. The amino acids are obtained after acidification through precipitation at the isoelectric point, ion exchange chromatography, electrodialysis, or extraction with organic solvents.

9.7 PRODUCTION OF INDIVIDUAL AMINO ACIDS

The demand for amino acids is increasing. While L-lysine and D,L-methionine can be manufac-

tured economically for use as feed additives, the demand for L-threonine and L-tryptophan cannot yet be filled due to the low yields of existing processes.

In order of their importance, the processes for the production of L-glutamic acid, L-lysine and L-tryptophan are discussed below. Further details for these and other amino acids are available in the references at the end of this chapter.

9.8 L-GLUTAMIC ACID

Production strains

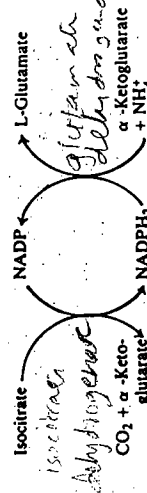
L-glutamic acid is manufactured predominantly by microbial means, although it is also manufactured chemically. Japanese researchers began developing a direct fermentation process because the D,L-glutamic acid which is formed by chemical synthesis is the racemic mixture. In screening about 2,000 microorganisms on different media, L-glutamic acid production was found to occur in a wide variety of bacteria, streptomycetes, yeasts, and fungi. The isolation of *Corynebacterium glutamicum* (synonym: *Micrococcus glutamicus*) was accomplished in 1957. It was immediately used industrially by Kyowa Hakko because of its high excretion of glutamic acid. Other industrially important strains with L-glutamic acid excretion of at least 30 g/l belong to the genera *Corynebacterium*, *Brevibacterium*, *Microbacterium*, or *Arthrobacter*. Morphologically and physiologically, these glutamic acid-producing strains resemble *C. glutamicum*. They are usually Gram-positive, nonsporulating, nonmotile bacteria. Moreover, all glutamic acid producers require biotin, lack or show little activity of α -ketoglutarate dehydrogenase, and show increased activity of glutamate dehydrogenase. In addition, some *Brevibacterium* and *Corynebacterium* mutants have lower isocitrate lyase activity. Further strain development has led to the isolation of mutants which overproduce glutamic acid even in the presence of high concentrations of biotin. For instance, a lyszyme-sensitive mutant of *C. glutamicum* is able to convert 40% of

the added carbon source to L-glutamic acid even in the presence of 100 $\mu\text{g/l}$ biotin.

Successful hybrids have also been constructed by cloning *Brevibacterium* or *Corynebacterium* DNA into *Brevibacterium* or *Corynebacterium* recipients.

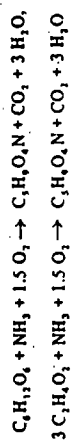
Biosynthesis of glutamic acid

The glucose carbon source is broken down into C_2 and C_3 fragments by glutamic-acid-producing microorganisms through the Embden-Meyerhof-Parnas (EMP) pathway and the pentose-phosphate-cycle, and the fragments are channeled into the tricarboxylic acid (TCA) cycle. The EMP pathway is more common under conditions of glutamic acid production. The key precursor of glutamic acid is α -ketoglutarate, which is formed in the TCA cycle via citrate and isocitrate and then converted into L-glutamic acid through reductive amination with free NH_4^+ ions (Figure 9.3). This last step is catalyzed by the NADP-dependent glutamate dehydrogenase. The NADPH, required at this stage of the reaction is furnished through the preceding oxidative decarboxylation of isocitrate to α -ketoglutarate by the enzyme isocitrate dehydrogenase. The NADPH is then regenerated by the reductive amination of α -ketoglutarate:



The strain used commercially for glutamic acid production has a block in α -ketoglutarate dehydrogenase. In the absence of NH_4^+ ions, α -ketoglutarate accumulates because of the interruption of the TCA cycle. Thus efficient anaplerotic sequences are necessary to provide TCA cycle intermediates which are required for other cell reactions. Studies on the pathway of biosynthesis have been carried out using labeled compounds.

During glutamic acid formation in the presence of $^{14}\text{CO}_2$, the α -carboxyl group of glutamate is labeled radioactively. Oxalacetate carboxylase and the NADP-dependent malate enzyme are involved with the CO_2 fixation process. The malate enzyme catalyzes the carboxylation of pyruvate to malate (Figure 9.3). These anaplerotic sequences complete the TCA cycle with C_4 dicarboxylic acids. Malate is then transformed via oxalacetate into citrate and isocitrate, which either serve as preliminary stages to the glutamic acid formation or are channeled into the glyoxylate cycle. Particularly with acetate as a carbon source, the energy gain and the formation of intermediates are carried out chiefly via the glyoxylate cycle for *C. glutamicum* (Figure 9.4). Thus there is competition between a) the isocitrate lyase reaction, which forms succinate and glyoxylate (necessary for optimal growth) and b) the isocitrate dehydrogenase reaction which leads to the key precursor α -ketoglutarate. The stoichiometry for glutamic acid formation from glucose or acetate as a carbon source would be as follows:



One mole of glutamic acid is produced from 1 mole of glucose or from 3 moles of acetate. Experiments with resting cells have shown that the actual conversion rate is between 50–70 mole%. Some part of the yield reduction is due to the reversibility of the malate enzyme reaction and the decarboxylation of oxalacetate to CO_2 .

Effect of permeability on glutamic acid production

Production and excretion of excess glutamic acid is dependent upon cell permeability. Increased permeability in glutamic-acid-producing bacteria can be attained in several different ways:

- Through biotin deficiency
- Through oleic acid deficiency in oleic acid auxotrophs

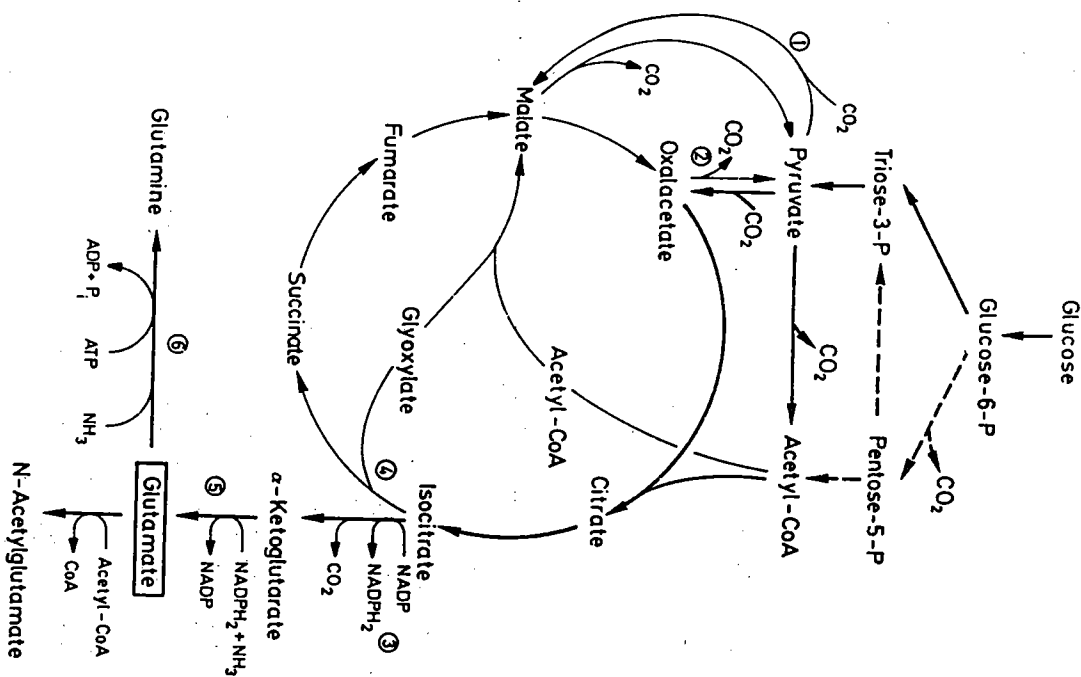


Figure 9.3 Biosynthesis of L-glutamic acid using glucose as the carbon source. Glyoxylic acid cycle, thin lines; pentose phosphate cycle, broken lines. 1, Malic enzyme; 2, Oxaloacetate carboxylase; 3, Isocitrate dehydrogenase; 4, Isocitrate (Modified according to Kinosha and Nakayama, 1978)

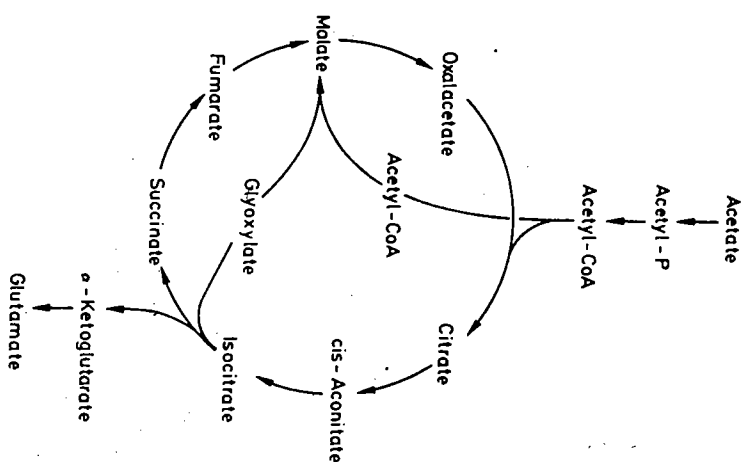


Figure 9.4 Biosynthesis of L-glutamic acid using acetate as the carbon source

- Through the addition of saturated fatty acids (C_{16} - C_{18}) or fatty acid derivatives
- Through the addition of penicillin
- Through glycerol deficiency in glycerol auxotrophs

All glutamic-acid-producing strains have a growth requirement for biotin, an essential coenzyme in fatty acid synthesis. In the presence of biotin concentrations $> 5 \mu\text{g/l}$, increased oleic acid synthesis results in a high phospholipid content of the cell membrane. Cells with high phospholipid content are incapable of excreting glutamic acid; up to 25–35 μg L-glutamic acid/mg

dry weight is accumulated intracellularly, so that biosynthesis is stopped because of feedback inhibition. On the other hand, growth in a biotin-deficient medium causes membrane damage through reduction of phospholipid synthesis, leading to a changed ratio of saturated to unsaturated fatty acids. Under these conditions, intracellular glutamic acid can be excreted. It has been postulated that a biotin-containing acetyl-CoA-carboxylase, which catalyzes the first stage of fatty acid synthesis from acetyl-CoA to malonyl-CoA, is involved in fatty acid biosynthesis. Saturated fatty acids (C_{16} - C_{18}) repress this acetyl-CoA-carboxylase. Hence it is understandable that oleic acid auxotrophs or strains to which saturated fatty acids have been added synthesize cell membranes with lower phospholipid content; such strains also excrete L-glutamic acid in the presence of biotin.

The addition of penicillin in the logarithmic growth phase promotes significant excretion of L-glutamic acid, even in the presence of biotin. Penicillin is added to fermentations in media containing large amounts of biotin 8–12 hours after inoculation of the fermenter. A penicillin concentration is selected (between 5–300 units/ml) so that the bacterial growth rate is reduced to a level corresponding to the rate in low-biotin media.

The use of penicillin or saturated fatty acids makes possible the commercial use of inexpensive culture medium components, such as sugar cane or sugar beet molasses, which otherwise cannot be utilized due to their high biotin content. The discovery of the role of cell permeability in the production of glutamic acid has thus made possible some rational approaches to the industrial production of this important amino acid.

Conditions of manufacture

Under optimal culture conditions, glutamic-acid-producing bacteria convert about 50–60% of the added carbon source to L-glutamic acid. If less favorable fermentation conditions are used so that low glutamic acid production is obtained, an

increase of cell mass and an excretion of lactate, succinate, α -ketoglutarate, glutamine and N-acetylglutamine have been observed. Factors which affect glutamic acid fermentation are described below.

Carbon sources A wide variety of carbohydrates can be used as carbon sources in the fermentation process. Among the monosaccharides, glucose and sucrose are frequently used, and fructose, maltose, ribose, and xylose find some role. Of the unrefined carbohydrate sources, sugar cane and sugar beet molasses are most important, but starch hydrolysates are also frequently employed. Since molasses has a high biotin content (0.4–1.2 mg/kg for cane molasses; 0.02–0.08 mg/kg for beet molasses), penicillin or fatty acid derivatives (e.g., Tween 60) must be added to the

fermentation when these inexpensive carbon sources are used. In Europe, only beet molasses is considered an inexpensive carbon source (proportion of manufacturing cost: 26%) but in other areas of the world cane molasses is chiefly used. In Japan, where acetate is inexpensive and readily available in large quantities, extensive studies have been carried out to use this carbon source but for industrial production, cane sugar molasses or starch hydrolysates are still the main carbon sources used.

Processes using methanol, ethanol, acetaldehyde, or n-alkanes have also been developed, but the cost-effectiveness of these processes depends largely on the price of petroleum. Figures 9.5 and 9.6 show the progress of typical fermentations with glucose and acetate as carbon sources.

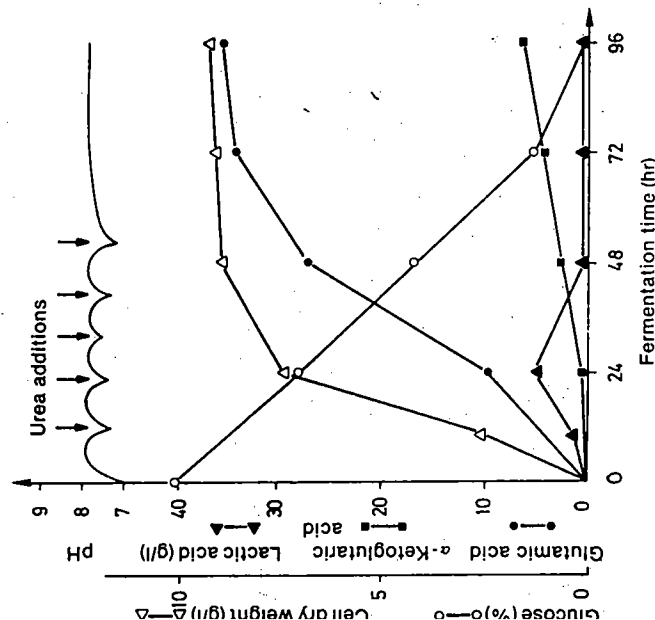


Figure 9.5 Production of L-glutamic acid with *Corynebacterium glutamicum* No. 541 using glucose as the carbon source

Nitrogen sources In addition to ammonium salts, ammonia (gaseous or in aqueous solution) can be used as a nitrogen source. In the industrial manufacture of glutamic acid, ammonia feeding permits pH control and obviates the problem of ammonia toxicity. Most glutamic acid-producing bacteria possess urease activity, so that urea is also frequently used as a nitrogen source. In the acidic pH range with excess ammonia, glutamine is produced instead of glutamic acid.

Growth factors The optimal biotin concentration is dependent on the carbon source used. In media with 10% glucose, it is 5 μ g/l; in media with lower glucose concentrations it is considerably lower, and for acetate it is between 0.2–1.0 μ g/l. Some strains require L-cysteine as an additional

growth factor; for media based on an n-alkane, supplementation with thiamine may be necessary.

O₂ Supply Optimal glutamic acid yields are obtained at a K_a value of 3.5×10^{-4} mole $\cdot$ O₂ / atm $\cdot$ min $\cdot$ ml. The oxygen concentration should be neither too low nor too high. Under oxygen deficiency, excretion of lactate and succinate occurs, whereas excess oxygen in the presence of an ammonium ion deficiency causes growth inhibition and production of α -ketoglutarate; in both cases glutamic acid yields are low.

Production processes

A typical fermentation from glucose with *Brev-*

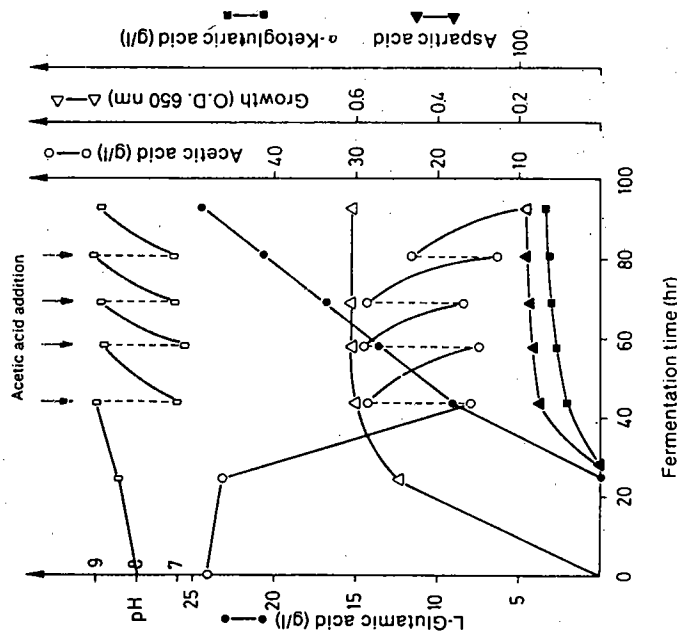


Figure 9.6 Production of L-glutamic acid with *Brevibacterium flavum* No. 2247 using acetate as the carbon source (Tsunoda et al., 1961)

ibacterium dinitratum (NRRL B-231) runs as follows (Miescher, 1975):

- Seed culture: glucose 40 g; K_2HPO_4 1.0 g; $MgSO_4 \cdot 7H_2O$ 0.5 g; yeast extract 1.0 g; urea 8 g; tap water 1 l; 16 h incubation at 35°C.
- Main culture: glucose 121 g; ammonium acetate 5 g; molasses from starch saccharification 6 g; KH_2PO_4 1.2 g; K_2SO_4 1.2 g; $MgSO_4$ (anhydrous) 6 g; $FeSO_4 \cdot 7H_2O$ 6 ppm; $MnSO_4 \cdot H_2O$ 6 ppm; antifoam agent Hodag K-67 0.1 ml; tap water 1 l. Inoculum volume: 6%.

At the beginning of the fermentation, 0.65 ml/l of oleic acid is added. The pH is set at 8.5 with ammonia and is automatically maintained at 7.8 during the course of the fermentation. After beginning growth of the culture (about 14 hours), the temperature is increased from 32–33°C to 38°C. After metabolism of the glucose down to a level of 0.5–2%, glucose feeding is done until the fermentation is completed. 160 g/l are fed on the average. Aeration is controlled so that the CO_2 content in the exhaust gas does not exceed 4.5 vol%. The glutamic acid content is analyzed hourly. As a rule, the fermentation is stopped after 30–35 hours with a glutamic acid yield of about 100 g/l. If molasses from starch saccharification is substituted for glucose, the glutamic acid yield is 94 g/l after 36 hours. The glutamic

acid yields with different carbon sources are listed in Table 9.5.

9.9 L-LYSINE

Lysine is an amino acid essential for animal and human nutrition. It occurs in plant proteins only in low concentrations; addition of lysine can therefore increase the quality of plant foods. The market for lysine is increasing. Lysine is produced today only by microbial processes and a variety of approaches for its production have been developed.

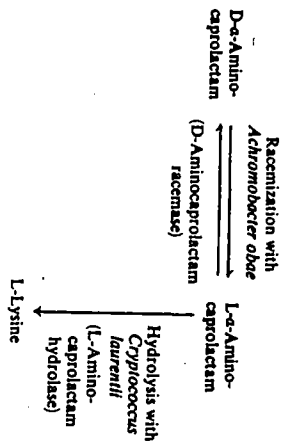
Lysine production via diaminopimelic acid

Lysine-histidine double auxotrophic mutants of *Escherichia coli* (ATCC 13002) produce diaminopimelic acid (DAP) on a molasses medium with a yield of 19–24 g/l. The entire fermentation solution, including the cell material, is subsequently incubated with *Aerobacter aerogenes* (ATCC 12409) at 35°C. After 20 hours, the DAP has been quantitatively decarboxylated to L-lysine. One complication with this procedure is that the DAP formed during the fermentation is a mixture of meso- and L-forms; since only meso-DAP can be decarboxylated into L-lysine,

the L-DAP must be transformed into the meso form by racemization before the decarboxylation step.

Conversion of DL- α -amino caprolactam

The enzymatic transformation of DL- α -amino caprolactam into L-lysine takes place according to the following scheme:



A 10% DL- α -amino caprolactam solution (pH 8.0) is added to 0.1% (w/v) acetone-dried cells of *Corynebacterium laurentii* and of *Achromobacter obae*. A conversion efficiency of 99.8% is obtained at 40°C after 24 hours. The enzymatic process has been commercialized in Japan by the Toray Company, which has a capacity of over 4000 tons per year. In the Toray process, the enzyme D- α -amino caprolactam racemase from *Achromobacter obae* has been expressed in a lysine auxotrophic mutant of *E. coli*.

Direct fermentation

Direct fermentation processes are now used world-wide for the production of L-lysine.

Production strains Efficient L-lysine producers are found among glutamic-acid-producing mutants of *Corynebacterium* and *Brevibacterium* which are homoserine auxotrophs or among methionine-threonine double auxotrophs. High-lysine-producing strains are also found among organisms resistant to the lysine antimetabolite S-(β -aminoethyl)-L-cysteine (AEC). The most im-

portant lysine-excreting strains are listed later in this chapter (Table 9.7). The development of high-yielding strains by mutation to auxotrophy and to antimetabolite resistance has been carried out with *B. lactofermentum*.

Protoplast fusion between high-yielding strains and wild strains of *B. lactofermentum*, *Corynebacterium*, and *Brevibacterium* mutants has led to strains with improved growth properties or higher efficiencies.

Cloning studies with *E. coli*, using plasmid pBR322 as a vector, have shown that only in transformed strains that contain the *dapA* gene does a significant increase in lysine production occur (6.5 g/l). The enzyme coded by *dapA*, dihydrodipicolinate synthase (DDPS) is therefore indicated as a rate-limiting step for lysine biosynthesis. Studies have also been carried out on the transformation of *C. glutamicum* with plasmid pAC2 as vector for the DDPS-encoding gene.

Yeasts, such as *Candida utilis*, *Saccharomyces cerevisiae* or *Saccharomycopsis lipolytica* have been studied extensively for lysine production. Lysine accumulates intracellularly in these organisms at concentrations up to 20% of the dry weight. However, these yeasts cannot be used successfully in industrial processes because lysine is not excreted into the medium.

Biosynthesis and regulation Lysine is synthesized in microorganisms either via the diaminopimelic acid pathway or the amino adipic acid pathway. However, in any single organism, only one of the two alternatives is used: Bacteria, actinomycetes, cyanobacteria (blue-green algae), some phycocyanes, and protozoa use the DAP pathway. Some phycocyanes, all ascomycetes, all basidiomycetes, and eucaryotic algae use the amino adipic acid pathway.

Biosynthesis via the DAP pathway in bacteria is shown in Figure 9.7. Although two organisms may use the same pathway, the manner in which this pathway is regulated may differ, as shown by the comparison of *Escherichia coli* and the lysine-producer *Corynebacterium glutamicum* in

Table 9.5 Processes for glutamic acid production with different carbon sources

Carbon source	Organism	Yield (g/l)
Sugar beet molasses	<i>C. glutamicum</i>	> 100
Glucose + Ammonium acetate	<i>Brevibacterium dinitratum</i>	100
Acetate	<i>B. flavum</i>	98
Ethanol	<i>Brevibacterium</i> sp. 136	59
n-Alkanes	<i>Arthrobacter paraffinicus</i>	62
	<i>Corynebacterium hydrocarbonolus</i>	84
	<i>C. alkanolyticum</i> (Glycerol)	72
Benzoic acid	<i>Brevibacterium</i> sp.	80
Methanol	<i>Methylobacterium methylotropha</i> M12-4	7

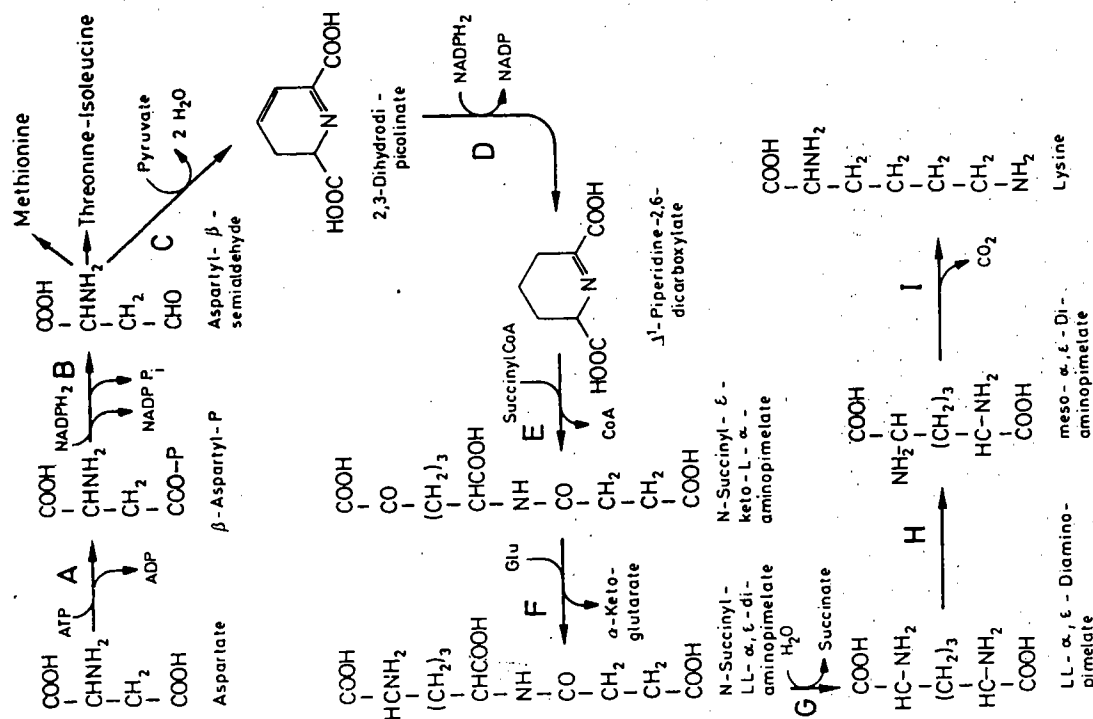


Figure 9.7 Biosynthesis of L-lysine via the DAP pathway. A, Aspartokinase; B, Aspartyl-β-semialdehyde dehydrogenase; C, Dihydrodipicolinate reductase; D, Dihydrodipicolinate synthase; E, Succinylketaminopimelate synthase; F, Succinylketaminopimelate aminotransferase; G, Succinylketaminopimelate desuccinylase; H, Diaminopimelate epimerase; I, Diaminopimelate decarboxylase.

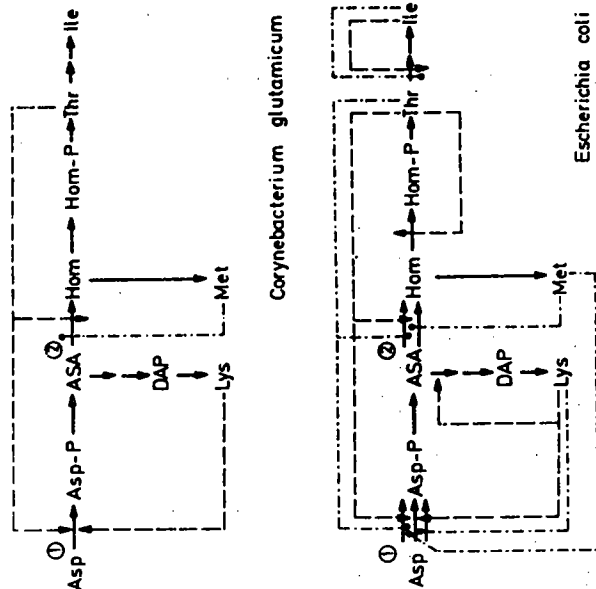


Figure 9.8 Regulation of L-lysine and L-threonine biosynthesis in *Escherichia coli* and *C. glutamicum*. 1, Aspartokinase; 2, Homoserine dehydrogenase. Dashed line, feedback inhibition; dotted line, repression. (From Kinoshita and Nakayama, 1978)

Figure 9.8. In *Escherichia coli* three distinct regulatory processes are involved:

- Two isoenzymes of homoserine dehydrogenase exist which are repressed by L-methionine or L-threonine.
- There are three isoenzymes of aspartokinase, the one showing repression by L-methionine, the second showing multivalent repression by L-threonine and L-isoleucine in addition to feedback inhibition by L-threonine, and the third showing feedback inhibition and repression by L-lysine.
- Dihydrodipicolinate synthase, the first specific enzyme of lysine biosynthesis, shows feedback inhibition due to L-lysine.

For all three of these enzymatic reactions, regulatory mechanisms must be eliminated to obtain the overproduction of L-lysine which is necessary for its commercial preparation.

In contrast to *Escherichia coli*, the regulatory mechanism for lysine-producing strains, such as *Corynebacterium glutamicum* or *Brevibacterium flavum*, is much simpler (Figure 9.8). There is only one aspartokinase and one homoserine dehydrogenase. Aspartokinase is regulated via multivalent feedback inhibition from L-threonine and L-lysine, as the experimental data in Table 9.6 show.

In both organisms, feedback inhibition due to L-threonine and repression due to L-methionine regulate the homoserine dehydrogenase. Biosynthesis from aspartate semialdehyde to L-lysine has already been accomplished in wild strains without regulatory control. Thus good lysine producers are classified among three mutant types:

- The flux of aspartate semialdehyde to threonine is reduced in homoserine auxotrophs which have a block in the homoserine de-

Table 9.6 Effect of various amino acids on aspartokinase activity

Added amino acid (10 μ mol/ml each, L-form)	Relative aspartokinase activity (%) <i>C. glutamicum</i> (Homoserine)	<i>B. flavum</i>
Control	100	100
Threonine	123	99
Lysine	94	80
Homoserine	94	102
Methionine	109	102
Threonine + Lysine	52	44
Homoserine + Lysine	95	74
Threonine + Methionine	129	114

(Nakayama, 1972)

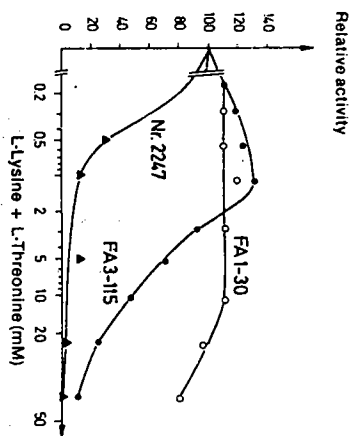
hydrogenase. Because of the low threonine content in the cell, the multivalent feedback inhibition of aspartokinase is prevented despite increased lysine formation.

The same effect is obtained with a mutant in which homoserine dehydrogenase is super-sensitive to feedback inhibition through threonine.

In comparison to the wild type, the aspartokinase of AEC-resistant mutants is insensitive to multivalent feedback inhibition, as Figure 9.9 shows. Excess aspartate semialdehyde flows into lysine biosynthesis, because the conversion to L-threonine is prevented through feedback inhibition. In this simply regulated system, strains with increased lysine formation can be developed in just a few mutation steps.

Lysine excretion is accomplished through active transport. In some organisms, a loss of lysine through decarboxylation to cadaverine occurs, but this destruction does not occur in *Corynebacterium* or *Brevibacterium* strains.

Conditions for commercial production Sugar cane molasses is primarily used as a carbon source in industrial production; acetate, ethanol or alkanes can also be used. Gaseous ammonia or ammonium salts are used as nitrogen sources; urea is also used if the producing microorganism has

**Figure 9.9** Effect of lysine and threonine on aspartokinase activity. No. 2247: wild type *Brevibacterium flavum*; FA 1-30 and FA 3-115: AEC mutants (From Hirose et al., 1978)

urease activity. Growth factors L-homoserine or L-threonine and L-methionine must be added, but in suboptimal concentration to avoid undesirable regulatory effects. Soy protein hydrolysates or other inexpensive protein sources are frequently used. The biotin content in the medium must be over 30 μ g/l for optimal lysine production. The biotin content in sugar cane molasses is usually high enough, but if sugar beet molasses or starch hydrolysates are used, biotin must be added. Table 9.7 shows lysine production from different carbon sources.

A typical fermentation is shown in Figure 9.10. The lysine yield is 44 g/l and the conversion rate in relation to the sugar used approaches a maximum of 30–40%.

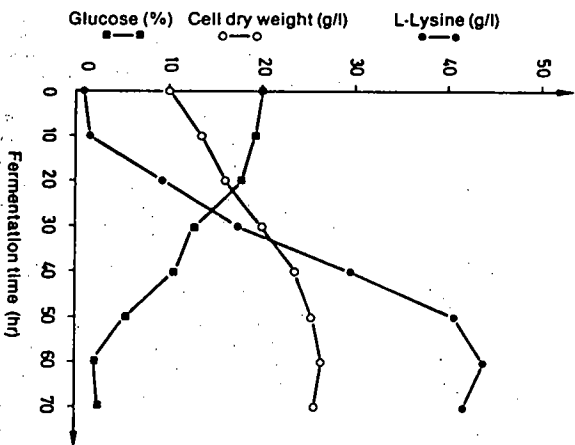
A typical fermentation based on sugar cane molasses with *C. glutamicum* (Strain No. 901; (Hom.) takes place as follows:

- First seed culture: glucose 20 g; peptone 10 g; meat extract 5 g; NaCl 2.5 g; tap water 1 liter.
- Second seed culture: sugar cane molasses 50 g; $(\text{NH}_4)_2\text{SO}_4$ 20 g; corn steep liquor 50 g; CaCO_3 10 g; tap water 1 l.
- Main culture: Sugar cane molasses 200 g

Table 9.7 Lysine production on media with different carbon sources

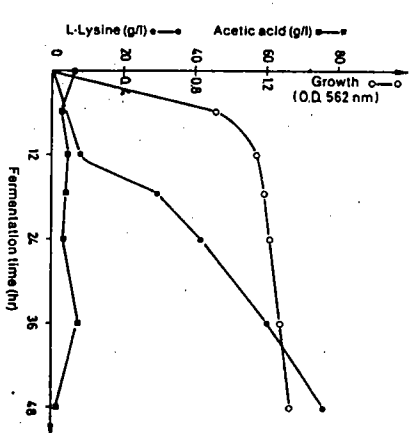
Carbon source	Organism	Genetic characteristics	Yield (g/l)
Glucose	<i>C. glutamicum</i>	Hom ⁺ Leu ⁻ AEC	34
Glucose	<i>C. glutamicum</i>	Hom ⁺ Leu ⁻ AEC	39
Glucose	<i>B. lactofermentum</i> A111204	AEC/Ala CCL/ML/FP	70
Acetate	<i>B. flavum</i>	Hom ⁺ Leu ⁻ Thr ⁻	75
Ethanol	<i>B. lactofermentum</i>	AEC	66
n-Alkanes (C_4 – C_{14})	<i>Nocardia alkanoglutinosa</i>	AEC	29

(AEC): Resistance to S-(β -aminoethyl)-L-cysteine; CCL, α -chlorocaprolactam; ML, γ -methyl-L-lysine; FP, β -fluoropyruvate sensitive. Auxotrophs: Ala, alanine; Hom, Homoserine; Leu, Leucine; Thr, Threonine)

**Figure 9.10** Production of L-lysine with *Corynebacterium glutamicum* No. 901 using glucose as the carbon source (From Nakayama, 1972)

(based on sugar content); soy protein hydrolysate 18 g; tap water 1 l. The pH is kept neutral with aqueous ammonia. Duration of fermentation, 60 hours. Impeller speed 150 rpm; aeration 0.6 vvm; temperature 28°C.

Figure 9.11 shows a fermentation by *Brevibacterium flavum* (Hom⁺, Thr⁻) with acetate as the carbon source. The composition of the pro-

**Figure 9.11** Production of L-lysine with *Brevibacterium flavum* using acetate as the carbon source (From Tanaka et al., 1971)

duction medium is as follows: acetic acid 7 g; KH_2PO_4 0.4 g; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ 0.4 g; $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ 0.01 g; soy protein hydrolysate 35 g; glucose 30 g; biotin 50 μ g; thiamine HCl 40 μ g; tap water 1 l. The pH is kept at 7.4 through acetic acid feeding (60% acetic acid and ammonium acetate in molar ratio of 100:25; 3% glucose). After 48 hours at 33°C the lysine content is 75 g/l; a conversion rate of 29% is obtained based on acetic acid and glucose used.

9.10 L-TRYPTOPHAN

Tryptophan has traditionally been produced through chemical synthesis or through fermenta-